Institut für Baustatik und Konstruktion, ETH Zürich

Fatigue performance of orthogonally reinforced concrete slabs

Christian Alexander Spathelf

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KEYWORDS:
reinforced concrete, fatigue, slabs, load-deformation response, orthogonal reinforcement

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FATIGUE PERFORMANCE OF ORTHOGONALLY REINFORCED CONCRETE SLABS

CHRISTIAN ALEXANDER SPATHELF

Institute of Structural Engineering

ETH Zurich

2018

Foreword

The evaluation of existing structural concrete bridges for currently prescribed traffic loads typically indicate a limited remaining fatigue life for the deck slabs. The statically indeterminate nature of slabs and their capacity to redistribute stresses generally results in bending action as combinations of bending and torsional moments. The direction of principal compressive stresses associated with bending action can vary throughout the concrete cross-section, whereas the reinforcing steel bars are typically assumed to resist purely uniaxial loading. Thus, the orientation of tensile stresses in a cracked cross-section is restricted to the reinforcement directions. Christian Spathelf investigated the influence of such deviations between the principal moment and reinforcement directions on cyclically loaded slabs in a series of large-scale tests. The redistribution of internal forces after fatigue rupture of individual reinforcing bars could be clearly demonstrated. A generalised physical-mechanical model, adapted from models previously developed at the Institute of Structural Engineering of ETH Zurich for quasi-static loading conditions, is proposed in his doctoral dissertation to analyse the fatigue performance of orthogonally reinforced concrete slabs.

The theoretical work of Christian Spathelf, substantiated by a favourable comparison to experimental results, contributes to a more realistic assessment of the progression of the fatigue damage process in reinforced concrete.

Zurich, February 2018 Thomas Vogel

Abstract

This thesis aims at contributing to a better understanding of the fatigue performance of orthogonally reinforced concrete slabs, focusing on the influence of a deviation between the principal moment and reinforcement directions with regards to fatigue strength. A new physical-mechanical model is presented that allows (i) a realistic assessment of the load-deformation behaviour of cyclically loaded slabs under various combinations of bending and torsional moments, (ii) the characterisation of changes in the response upon rupture of individual reinforcing bars, and (iii) a realistic prediction of fatigue life to supplement the current provisions for the design and evaluation of concrete structures.

In the first part of this thesis, a brief overview of the fatigue phenomenon is provided, previous work on fatigue in reinforced concrete is reviewed and the relevant material properties of concrete, reinforcing steel and their interaction through bond are examined.

The second part comprises the theoretical modelling of the response of reinforced concrete slabs to cyclic loading. Relevant kinematic and static relationships governing the load-deformation behaviour are outlined. The proposed model, based on a layered plate element formulation, is introduced. Herein, compressive stresses resulting from bending are assigned to an assemblage of plane stress concrete layers distributed over the element height. Tensile stresses in the reinforcement are considered in fictitious cracked membrane cover elements at the outer surfaces. Tension stiffening effects are considered in the cover elements, resulting in a realistic calculation of maximum reinforcing steel stresses at the cracks. A stochastic distribution of fatigue strengths is assigned to the reinforcement such as to allow an evaluation of the response after rupture of individual bars.

In the third part, details of an experimental campaign carried out at the Institute of Structural Engineering of ETH Zurich are described. The testing program comprised two series of four large-scale tests on concrete slabs exhibiting differently orientated orthogonal reinforcement layouts with respect to the direction of applied moment. Specimens were tested under constant force-amplitude cyclic loading until failure.

The fourth part of this thesis relates to the validation of the proposed new model through the comparison of calculations to own experimental results. The accuracy of the model assessment of fatigue performance is highlighted and the limitations are discussed.

Keywords: reinforced concrete, fatigue, slabs, load-deformation response, orthogonal reinforcement

Kurzfassung

Mit der vorliegenden Forschungsarbeit soll zu einem besseren Verständnis des Ermüdungsverhaltens von kreuzweise bewehrten Stahlbetonplatten beigetragen werden. Dabei wird insbesondere darauf eingegangen, welchen Einfluss eine Abweichung der Hauptmomentenrichtung zur Bewehrungsrichtung auf das Ermüdungsverhalten hat. Ein neues physikalisch-mechanisches Modell wurde entwickelt, welches (i) eine realistische Abschätzung des Last-Verformungsverhaltens von zyklisch belasteten Platten unter verschiedenen Kombinationen von Biege- und Drillmomenten, (ii) eine Beschreibung des veränderten Tragverhaltens nach Ermüdungsbrüchen in einzelnen Bewehrungsstäben, und (iii) eine realistische Prognose der Bruchlastspielzahl erlauben, um damit die heutigen Bestimmungen zur Bemessung und Beurteilung von Betonbauwerken zu ergänzen.

Der erste Teil der Arbeit beschreibt die Grundlagen zum Phänomen Ermüdung und fasst den Stand der Forschung zusammen. Hier finden sich ebenfalls die relevanten Parameter zu Beton, Bewehrungsstahl und dem Verbundverhalten beider Materialien.

Der zweite Teil umfasst die theoretische Modellierung des Verhaltens von Stahlbetonplatten unter zyklischer Last. Die nötigen kinematischen und statischen Beziehungen für das Last-Verformungsverhalten werden umrissen. Aufbauend auf ein geschichtetes Flächentragwerkelement wird das Modell hergeleitet. Die Druckspannungen infolge Biegung werden in Form von ebenen Spannungszuständen in den jeweiligen Betonschichten über die Elementhöhe verteilt; Zugspannungen der Bewehrung werden über idealisierte gerissene Membranelemente am oberen und unteren Plattenrand berücksichtigt. Um zu einem realistischen Wert für die maximalen Stahlspannungen in den Rissen zu gelangen, wird die zugversteifende Wirkung in den äusseren Membranelementen berücksichtigt. Die Ermüdungsfestigkeiten der Bewehrungsstäbe werden stochastisch zugewiesen, um das Verhalten der Platte nach dem Bruch einzelner Stäbe auswerten zu können.

Im dritten Teil wird die Versuchsreihe beschrieben, wie sie am Institut für Baustatik und Konstruktion an der ETH Zürich durchgeführt wurde. Das Versuchsprogramm bestand dabei aus zwei Serien von insgesamt vier grossmassstäblichen Stahlbetonplatten. Der Unterschied zwischen beiden Serien bestand in der Orientierung der orthogonalen Bewehrungsanordnung in Bezug zur Hauptmomentenrichtung unter Belastung. Die Prüfkörper wurden zyklisch bei konstanter Kraft-Amplitude bis zum Bruch belastet.

Der vierte Teil beschäftigt sich mit der Validierung des entwickelten Modells, indem dieses in einer Vergleichsrechnung mit den Resultaten der Versuchsreihe verglichen wird. Die Modellgenauigkeit zur Beschreibung des Ermüdungsverhaltens und die Grenzen des Modells werden diskutiert.

Résumé

Cette thèse vise à contribuer à une meilleure compréhension du comportement en fatigue des dalles en béton armé. L'accent est mis sur l'influence d'une déviation entre les moments principaux et la direction des barres d'armature sur le comportement en fatigue. Un nouveau modèle physique et mécanique est développé, permettant (i) d'évaluer de façon réaliste le comportement charge-déformation des dalles sollicitées en différentes combinaisons de flexion et torsion, (ii) de caractériser le changement de la réponse après la rupture de certaines barres d'armatures, et (iii) de prédire la résistance à la fatigue de manière réaliste, afin de compléter les règles de dimensionnement et d'évaluation contemporaines des structures en béton.

La première partie de la thèse donne un aperçu sur le phénomène de la fatigue, décrit quelques travaux précédents concernant la fatigue du béton armé et examine les propriétés mécaniques du béton, de l'acier de renforcement ainsi que de leur interaction.

La seconde partie comprend la modélisation théorique de la réponse des dalles en béton armé soumises à des sollicitations répétées. Les principales relations cinématiques et statiques définissant le comportement charge-déformation sont exposées. Un modèle qui base sur un élément de dalle stratifié est introduit. Les contraintes de compression dues à la flexion sont attribuées en contraintes planes à des panneaux de béton réparties sur la hauteur de l'élément. Dans les surfaces supérieures et inférieures de l'élément, les contraintes de traction dans les barres d'armature sont prises en compte par un modèle idéalisé de la membrane fissurée. La considération de la contribution du béton tendu dans les surfaces extérieures permet une estimation réaliste des contraintes maximales du renforcement au niveau des fissures. La résistance à la fatigue des barres d'armatures est définie de manière stochastique afin de permettre l'évaluation de la réponse au-delà de la rupture de quelques barres.

Les détails d'une campagne expérimentale effectuée à l'institut d'ingénierie des structures à ETH Zurich sont présentés dans la troisième partie de cette thèse. Les essais à grande échelle ont compris deux séries de dalles avec un ordre d'armature en croix variablement orienté par rapport à la direction des moments appliqués. Les spécimens ont été soumis à un chargement cyclique constant jusqu'à leurs ruptures.

La quatrième partie contient la validation du modèle présenté, en comparant ses résultats à ceux obtenus dans la campagne expérimentale. La précision du modèle à décrire le comportement en fatigue ainsi que ses limites sont discutées.

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Chapter 1

Introduction

1.1 Context

An estimated 75% of the reinforced concrete bridges in Europe were constructed more than 20 years ago (in 2008) [35]. In Switzerland, the majority of bridges were opened for operation in the 1960s and '70s and have been in service for more than 40 years [101]. These structures have been subjected to a steady increase in traffic over the last decades, in particular with respect to the proportion of heavy vehicles [12]. The first provisions for the design of structural concrete members under repeated loading, however, were only introduced in the Swiss standard SIA 162 [84] in 1989. Hence, a significant proportion of concrete bridges, underpasses and culverts on the Swiss highway network were designed without considering the hazard scenario "fatigue". Within the next 10 to 20 years, some of these structures may reach an age where fatigue-induced damage has to be considered [23].

Application of the current provisions for cyclically loaded structural members typically leads to larger concrete cross-sections and/or possibly higher reinforcement volumes compared to designs on the basis of the previous generation of Swiss standards [84]. For the design of new structures, a more conservative approach is justifiable in view of the increased service life such measures afford. In the evaluation of existing structures, the provisions for fatigue safety can typically not be fulfilled. However, no observable damage which can be attributed exclusively to fatigue processes, has been identified on concrete bridges in Switzerland [84]. Possible explanations for this discrepancy are (i) that most concrete structures have simply not been in service for long enough to exhibit damage, or (ii) some structures are already damaged due to fatigue processes, whereas this has remained undetected in routine inspections or have possibly been attributed to other actions such as overloading or corrosion of the reinforcement [84].

Probably the most important and commonly used structural concrete member subjected to cyclic loading is the deck slab of bridges, see Fig. 1.1. Slabs are typically slender and exhibit low ratios of own weight to live loads. In consequence, highway or railway traffic loads result in high fluctuations of bending and shear action. Corresponding stress ranges in the concrete and reinforcing steel are also high. Bending action in slabs is gen-

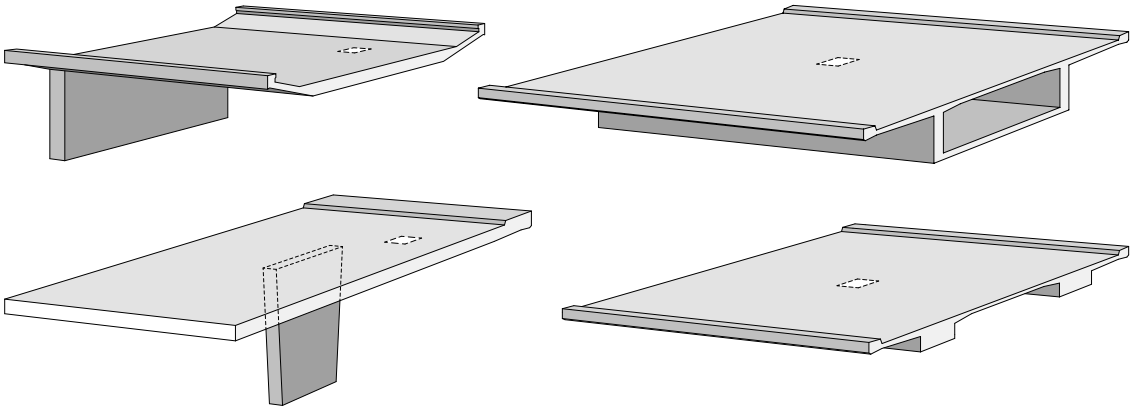


Fig. 1.1 – Fatigue sensitive regions (squares) in typical structural concrete bridges.

erally resisted by layers of orthogonally placed reinforcement. The flow of forces from the introduction of loads to the supports, however, often deviate from this reinforcement layout. Moreover, in regions where concentrated loads are introduced, in close proximity to wheel loads of heavy vehicles or at column supports, a significant torsional moment may occur [63]. Such conditions indicate that, in general, the directions of principal moment and that of the reinforcement do not coincide.

Describing the load-deformation behaviour of reinforced concrete slabs under arbitrary combinations of bending and torsional moments poses a considerable challenge due to the nonlinear nature of the response and the generally iterative, numerical approach required. Hence, the need for a model on the basis of consistent mechanical principles that enables a realistic determination of principal concrete and reinforcing steel stresses under cyclic loading was identified. Moreover, a scarcity of experimental data regarding the fatigue performance of slabs under clearly defined combinations of bending and torsional moments necessitated own experimental tests. In consequence, an experimental campaign consisting of two series of four large-scale fatigue tests was carried out at the structural laboratory of the Institute of Structural Engineering of ETH Zurich between 2015 and 2016.

1.2 Objective and scope

This thesis aims at contributing to a better understanding of the fatigue performance of orthogonally reinforced concrete slabs. In particular, the influence of a deviation between the principal moment and reinforcement directions with regards to the fatigue strength of such members shall be investigated. A new physical-mechanical model shall be developed that allows (i) a realistic assessment of the load-deformation behaviour of cyclically loaded slabs under various combinations of bending and torsional moments, (ii) the characterisation of changes in the response upon rupture of individual reinforcing bars, and (iii) a realistic prediction of fatigue life to supplement the current provisions for the design and evaluation of concrete structures.

1.3 Overview

In the first part of this thesis, a basic outline of the fatigue phenomenon is provided and the material properties are examined. In Chapter 2, the fundamental aspects related to fatigue are summarised and previous work on fatigue in reinforced concrete is reviewed. In Chapter 3, the characteristics of concrete and reinforcing steel as well as their interaction through bond are examined for both static and cyclic loading conditions.

The second part of this thesis covers the theoretical modelling of the response of reinforced concrete slabs to cyclic loading. Chapter 4 outlines the static and kinematic relationships governing the load-deformation behaviour of membrane and plate elements and introduces typical approaches for the analysis of structural concrete. Chapter 5 outlines the basic assumptions and idealisations underlying to the model formulation and describes the implemented constitutive relationships. The equilibrium conditions for the analysis of structural concrete membrane and plate elements are defined for uncracked and cracked conditions. A numerical method for the analysis of reinforced concrete slabs under cyclic loading is introduced and a general solution algorithm is presented, which allows the determination of complete element stress and strain states for a given spectrum of loads. Calculations of load-deformation response under monotonically increasing loads are validated with experimental results reported in literature and the consideration of cyclic loading is examined.

In the third part, Chapter 6, details of an experimental campaign in which the fatigue performance of orthogonally reinforced concrete slabs was investigated are provided. The test specimens, test procedure and detailed results from two series of four large-scale fatigue tests carried out at ETH Zurich are described.

The fourth part, Chapter 7, covers the validation of the proposed new model. Calculations are compared to measured results from the experimental campaign.

Finally, Chapter 8 summarises and discusses the results obtained in this work and concludes with recommendations for future research.

1.4 Limitations

The present work is limited to the investigation of the fatigue phenomenon in orthogonally reinforced concrete plates or slabs. The influences of post-tensioning or reinforcement layouts with skew layers are not considered.

Slabs are analysed as thin plates with *Kirchhoff* plate theory. The underlying assumptions imply that line elements normal to the mid-plane remain straight and normal to the mid-plane during deformation and that the thickness of the plate remains constant. In consequence, the influence of shear forces on the deformation is neglected. Only stresses resulting from bending and/or torsional moments are considered, excluding the influence of shear forces on the flexural reinforcement. Stresses are calculated from gross cross-sectional areas under the assumption of quasi-static loading conditions. A reduction in

concrete cross-section due to the reinforcement is neglected and a dynamic structural response or strain rate effects as a result of cyclic loading is not considered. Furthermore, reinforcing steel bars are assumed to resist only uniaxial loading; their flexural and shear stiffness is assumed to be negligible.

Failure of typically reinforced concrete slabs under repeated loading is generally dominated by the fatigue strength of the reinforcement. In this thesis, the concrete is assumed to play a secondary role. Apart from a consideration of the deterioration of concrete tensile strength under cyclic loading, no further accumulation of damage is assumed for the concrete. Furthermore, time-dependant effects such as creep and shrinkage of the concrete as well as relaxation are neglected.

Case studies of structures where fatigue damage is suspected typically mention an interaction with corrosion of the reinforcing steel. In practise, it is often not possible to distinguish between the extent of damage resulting from each of these actions. The influence of corrosion of the reinforcement on the fatigue strength of the bars is not considered.

Chapter 2

Fatigue of materials

2.1 General

For centuries, it has been known that wood or metal can be broken by repeatedly bending it back and forth with a large amplitude. Only relatively recently, through the investigations by *Albert* (1837) and *Wöhler* (1858), was it recognised that these fractures occur under repeated stressing even when the loading level fluctuates within the elastic range and remains below the static strength of the material [16]. Nowadays, the phenomenon of *fatigue* is well recognised in the discipline of engineering. It is rated as the primary cause of failure in metallic materials, constituting an estimated 90% of cases [13]. A further factor increasing the relevance of fatigue was the development of high tensile strength steels. These facilitated the design of more slender structures and/or possibly a reduction in reinforcement content. Increases in tensile strength, however, were not accompanied by a proportional improvement on the fatigue strength of the steel [79]. Thus, the benefits of high strength are limited in structures subjected to cyclic loading by the fatigue strength of the material

Rather than attempting to provide a complete review of the state of the art on fatigue research, the present chapter will focus on introducing the terminology in the field of research, describing the relevant processes involved and will discuss how they are dealt with in an engineering sense. *Schütz* provides a thorough historical review of work on fatigue during the period 1838 – 1994 in [83]. A broad overview of fatigue of metallic materials is provided in [26, 57]. *Fehlmann* summarises the major milestones in the historical background of work on fatigue of metallic materials and of structural concrete in [23]. *Tilly* reviews the fatigue properties of reinforcing steel bars in [96]. Thoroughly documented fatigue test results on reinforcing steel bars are provided by *Moss* in [66], by the American Transportation Board [34] as well as by *Fernández Canteli et al.* in [25]. An overview of fatigue of plain concrete is provided in the dissertations by *Hordijk* [39] and *Hohberg* [37] as well as in the state of the art reports [98, 103]. The *CEB* state of the art report (1988) [98], although almost 30 years since publishing, still provides a remarkably current outline of the major findings in previous work on fatigue of structural concrete.

2.2 Terminology and descriptions

In the present section, recurrent terms are introduced and defined. These include some general concepts on fatigue, the terminology related to cyclic loading and terms associated with fatigue testing of materials as well as the subsequent interpretation of the results.

2.2.1 General concepts

The term *fatigue*, probably first mentioned by *Braithwaite* [10] in 1854, suggests the notion that a material becomes "tired" under a continuous accumulation of damage; ultimately resulting in failure. A critical difference to the medical sense of the word is the fact that the damage sustained is cumulative and irreversible as removal of the cyclic loading does not lead to any significant recovery of the material [16]. Fatigue is defined by the American Society for Testing and Materials (ASTM) as

> "[...] the process of progressive localized permanent structural change occurring in a material subjected to conditions that produce fluctuating stresses and strains at some point or points and that may culminate in cracks or complete fracture after a sufficient number of fluctuations." [6]

Fluctuating stress is a prerequisite for fatigue in materials. The term *stress* is defined as the intensity and direction of internal forces acting at a given point on a particular plane of a body in response to and in equilibrium with a system of body or surface forces. A further classification of this intensity of internal forces is made with regards to the surface upon which the stress is defined. *Engineering* or *nominal stress* refers to the intensity of internal forces in relation to the original geometrical properties before undergoing deformation. *True* stress, on the other hand, is defined with respect to the specimen geometry at every instant of loading and offers a more accurate indication of this intensity. Stresses are typically determined from (i) the algebraic division of applied force by cross-sectional area, possibly including stress concentration factors for specific geometries, (ii) simulations with the Finite Element Method (FEM), or (iii) experimental investigations [77].

Fatigue damage is defined as the progressive deterioration of the internal structure of a material under increasing stress cycles. The initiation of fatigue damage can often be traced back to initial defects or micro-cracks on the specimen surface, which were either present in the material (possibly introduced during the manufacturing process) or formed in response to the applied loading. Such micro-cracks can grow continuously under normal service loads and result in unstable growth once a critical threshold is crossed [16]. The detection of fatigue damage during the early stages of crack initiation and propagation is difficult and failure often occurs with little or no warning. Such cases have been documented for high-pressure gas tanks, pipelines and airplanes [77].

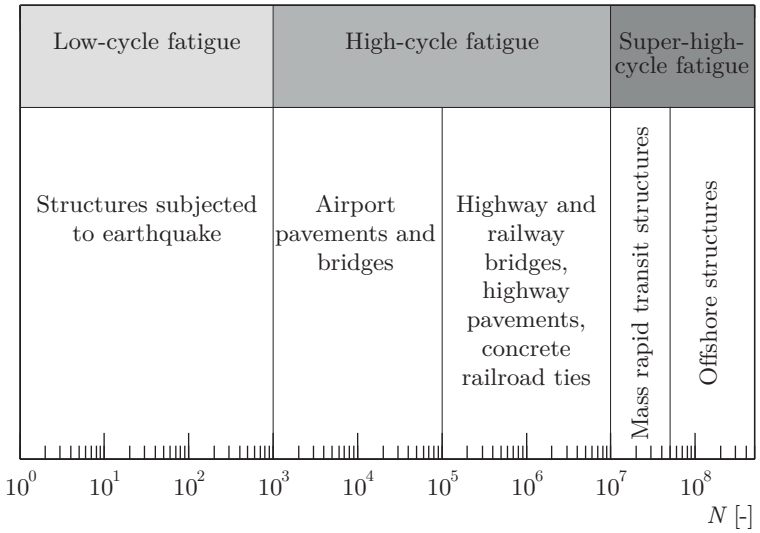


Fig. 2.1 – Spectrum of fatigue loading. Redrawn from [40].

2.2.2 Loading spectra

The *loading spectrum* or number of fatigue-inducing load cycles an engineering structure is subjected to may vary significantly. This may range from low-cycle, high-amplitude loading as observed during earthquakes to the extreme conditions relevant to offshore structures having to withstand in excess of 10^8 load cycles under wave action. A classification into *low-cycle*, *high-cycle* and *super-high-cycle* regimes was proposed by *Hsu* [40], see Fig. 2.1. However, there are no clear divisions within this wide spectrum of loading.

Structural members subjected to fluctuating or *cyclic loading* experience stress variations which result in an accumulation of fatigue damage. Such loading may be purely axial, flexural or torsional in nature or consist of a combination of these modes. The simplest form of cyclic loading is the sinusoidal or harmonic wave described by

$$\sigma(t) = \sigma_m + \frac{\Delta\sigma}{2}\sin(\omega t) \tag{2.1}$$

exhibiting a constant stress-amplitude and frequency where σ_m denotes the mean stress and $\Delta\sigma$ the range of stress varying at the circular frequency ω, see Fig. 2.2. The relationships between these parameters describing the stress cycles can be formulated as

$$\Delta\sigma = \sigma_{\max} - \sigma_{\min}, \qquad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}, \qquad \sigma_a = \frac{\Delta\sigma}{2} = \frac{\sigma_{\max} - \sigma_{\min}}{2} \tag{2.2}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ denote the maximum and minimum stress levels and σ_a refers to the stress amplitude. The stress ratio R is defined as the ratio of the minimum to the maximum stress level during a stress cycle

$$R = \frac{\sigma_{\min}}{\sigma_{\max}} \tag{2.3}$$

and is typically used to characterise the level of loading. A *fully reversed* stress spectra or pulsating tension-compression refers to the condition where the maximum and minimum

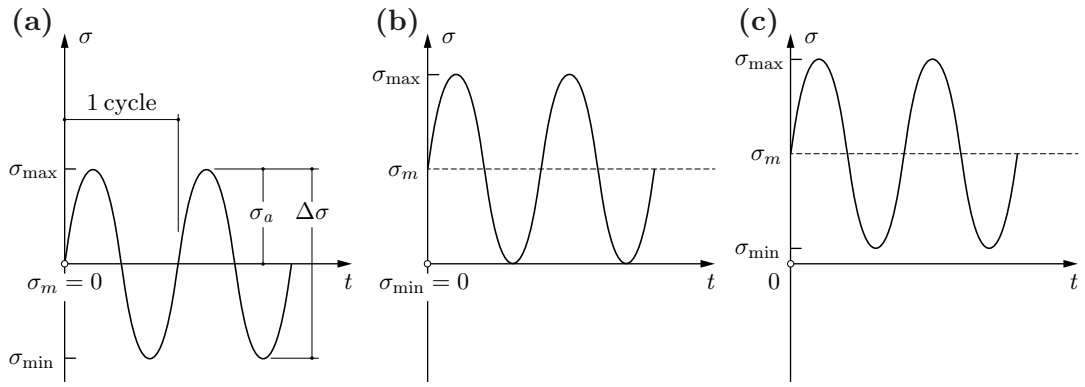


Fig. 2.2 – Typical constant amplitude stress spectra: (a) fully reversed; (b) released tension, $R = 0$; (c) non-zero mean stress. Adapted from [16].

stress values are equal and opposite in sign such that $R = -1$. Conditions of *released tension* indicate that the minimum stress level is zero, resulting in $R = 0$. The most general conditions are represented by spectra of *non-zero mean stress*.

2.2.3 Fatigue testing

The majority of fatigue tests are conducted by subjecting a specimen to stress cycles of constant amplitude around a specified mean stress until failure occurs. Such tests are referred to as *single-step* or Wöhler-tests after the revolutionary work performed by *August Wöhler* on railway axles. An alternative to the aforementioned is the *multi-step test* in which the loading is increased stepwise during the course of the test in a defined sequence. Testing sequences in which the stress fluctuations closely resemble those expected under actual service conditions are referred to as *operational fatigue strength tests*.

The *fatigue life* of a member is defined as the number of load or stress cycles that can be sustained for a given test condition [57]. *Fatigue strength* refers to the stress range corresponding to fatigue failure for a specific number of stress cycles.

The *failure* of a material or member is typically defined as the condition of actual rupture. The formal definition of failure according to ASTM is: "the complete separation, or the presence of a crack of specified length visible at a specified magnification" [6]. Failure, however, may also refer to the condition in which the member is no longer able to resist the applied loading. Fatigue tests are typically terminated either at failure of the specimen or once a defined number of load cycles has been exceeded at that stress range. In the latter case, the test is classified as a *run-out*.

The relation between the number of load cycles to failure and the corresponding stress range is typically represented in so-called S-N diagrams which will be described in Section 2.4. Considering a double-logarithmic representation, the general trend of a large number of fatigue test results can typically be described by two straight lines of different slopes. The initial range is defined as the *finite fatigue life* or *finite-life* region (*Zeitfestigkeit* in German) in which specimens are observed to fail after the application of a certain

number of load cycles. Early investigations have shown already that a decrease in stress range within this region results in a greater number of load cycles until failure. In certain steel types and Titanium alloys, this curve practically approaches an asymptote parallel to the abscissa at large numbers of load cycles. The ordinate value of this asymptote represents the stress range below which the specimen can withstand a practically unlimited number of load cycles and is defined as the *fatigue endurance limit* (*Dauerfestigkeit* in German). Moreover, this stress range represents the inflection point between the two parts of the S-N curve which typically lies in the range $35\% \ldots 60\%$ of the tensile strength for most steel types [13]. The second range at applied stress levels below the endurance limit of the material is defined as the *long-life* region.

2.3 Fatigue fractures

The present section introduces the processes of crack initiation and propagation under fluctuating stresses and compares the characteristics of brittle and ductile fractures. Finally, a characterisation of fatigue fractures is provided, indicating the influence of the mode of loading, the level of nominal stress and of notches or defects in the material.

2.3.1 Crack initiation and propagation

The fatigue process is typically characterised by the following three phases: (1) *crack initiation*, in which micro-cracks form on the surface of the specimen in regions of high stress and combine to form incipient cracks; (2) *crack propagation*, in which these incipient cracks continue to develop with subsequent stress cycles due to stress concentrations at the crack tip (stable crack growth); and finally (3) *brittle failure* of the remaining cross-section once the crack has reached critical dimensions (unstable crack growth) [13].

The mechanisms of crack initiation and propagation are still not fully understood. One of the most widely accepted explanations according to [77] is outlined with reference to Fig. 2.3. Crack initiation on a smooth surface takes place primarily on a microstructural level. An incipient crack, controlled by the principal shear stress, generally forms at an angle of 45° with respect to the direction of loading. Upon reaching a threshold crack length[1], the crack propagation direction changes towards an orientation perpendicular to the applied normal stress; thus, principal stress controlled. The presence of a notch on the surface of the specimen accelerates the initiation of fatigue cracks. In such cases, the crack initiates in the region of highest stress concentration and propagates perpendicular to the direction of principal stress. Surface defects can be considered as equivalent to the incipient cracks. From the onset of cyclic loading, crack propagation is perpendicular to the direction of principal stress – independent of the surface defect orientation. The latter two cases represent a reduction on the duration of the crack initiation phase and of the total fatigue life in relation to that of the specimen starting with a smooth surface.

[1] The threshold crack length is typically in the order of 300 µm [64].

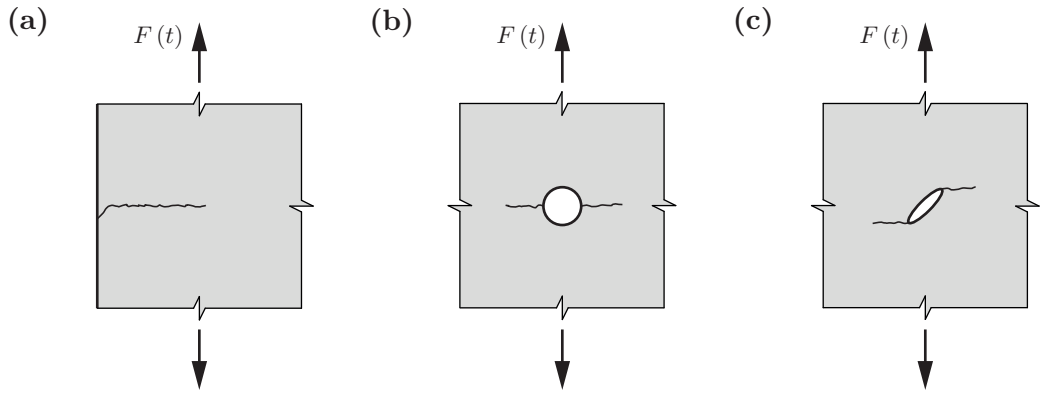


Fig. 2.3 – Typical modes of crack initiation and propagation: (a) smooth surface; (b) at notch; (c) at defect or imperfection. Redrawn from [77].

2.3.2 Fracture types

Fractures are generally classified as being either *brittle* or *ductile* in nature. Brittle fractures are characterised by rapid crack propagation[2] after little or no plastic deformation. Such cracks form approximately perpendicular to the direction of applied principal stress. Ductile fractures, on the other hand, follow extensive plastic deformation of the material. These fractures are characterised by a slow crack propagation speed due to the formation and merging of voids that form under tensile load [16]. The formation of a ductile fracture exhibits three characteristic stages: (1) local necking of the sample and the formation of small cavities within the necked region; (2) merging of these cavities into a single crack at the centre of the cross-section, approximately perpendicular to the direction of principal stress; and (3) subsequent crack propagation along shear planes at approximately 45° to the direction of applied tensile load [16]. The result is the so-called *cup-and-cone* failure typically observed in tensile tests on ductile materials such as steel.

Typical features of the fracture surfaces resulting from fatigue (brittle) and deformation (ductile) failures are illustrated in Fig. 2.4 as observed in tests on reinforcing steel bars. It is not always clear whether a fracture occurred due to fatigue from the macroscopic appearance of the fracture surface alone. Some generalisation is necessary to discern between a fatigue fracture and one that is attributed to overloading with high plastic deformation as the former does not always present itself with the same classic characteristics. There are, however, certain common features which will mostly be present in any fatigue fracture. A fatigue fracture surface can generally be divided into two zones, namely, a *fatigue* and a *rupture zone*. The fatigue zone represents the portion of the cross-section where crack propagation occurs under fluctuating stress; the rupture zone – as the name implies – comprises the remaining portion of the cross-section where the final failure occurs. The fatigue zone is generally smooth and exhibits a dull, rubbed and sometimes velvety appearance. This zone includes the crack origin or origins and is often marked with a system of con-

[2]The crack propagation speed increases rapidly from zero at initiation up to a limiting value of approximately one third of the speed of sound within the material [16].

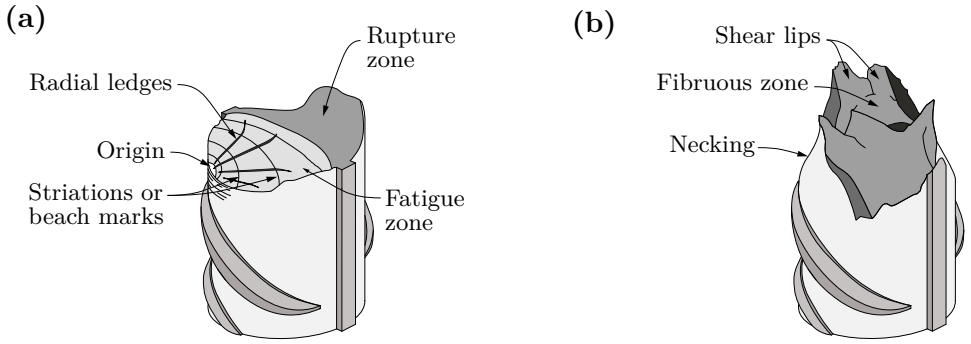


Fig. 2.4 – Fracture surface characteristics for reinforcing steel bars: (a) fatigue, description from [26]; (b) ductile failure.

centric striations called *beach* or *tide marks*, which indicate a variation in the crack growth rate. These striations are the true hallmark feature of fatigue fractures. Furthermore, radial markings or ledges on the fracture surface are typically observed which converge on the origin or crack initiation point [26]. Upon failure, the rupture zone can generally be observed as a crescent-shaped region surrounding the fatigue crack. This zone typically exhibits a rough, crystalline surface with evidence of plastic deformation [26].

2.3.3 Characterisation of fatigue fractures

The macroscopic characteristics of fatigue fractures are predominantly determined by the mode of loading, the level of nominal stress and the smoothness of the specimen. The influence of these factors on the shape and relative proportions of the fatigue and rupture zone surfaces at failure is illustrated in Fig. 2.5. In specimens subjected to tension, tension-compression or rotating bending, the fatigue crack initiation is largely dominated by the position of the critical surface flaw. In specimens subjected to unidirectional and reversed bending, however, the crack initiation originates in the region of maximal bending stress. Fatigue cracks resulting from axial loading usually develop on a plane perpendicular to direction of principal stress. Members subjected to high shear or torsional loading exhibit an initial cracking parallel to the direction of maximum shear stress with subsequent crack development perpendicular to the principal stresses. The level of nominal stress or degree of overstress in the specimen determines the relative size of the fatigue zone on the fracture surface. A small fatigue zone generally signifies high nominal stresses during the service life of the specimen, whereas a small rupture zone results from low nominal stresses [57]. The surface smoothness of the specimen determines whether the position of crack initiation is localised around existing defects on the surface or more random in nature. In specimens with a smooth surface, the interface between the fatigue and the rupture zone is often observed to be concave towards the crack initiation site or origin. This interface typically becomes less curved of "flatter" with increasing notch sharpness such that it might even become convex towards the origin for the case of a sharp notch.

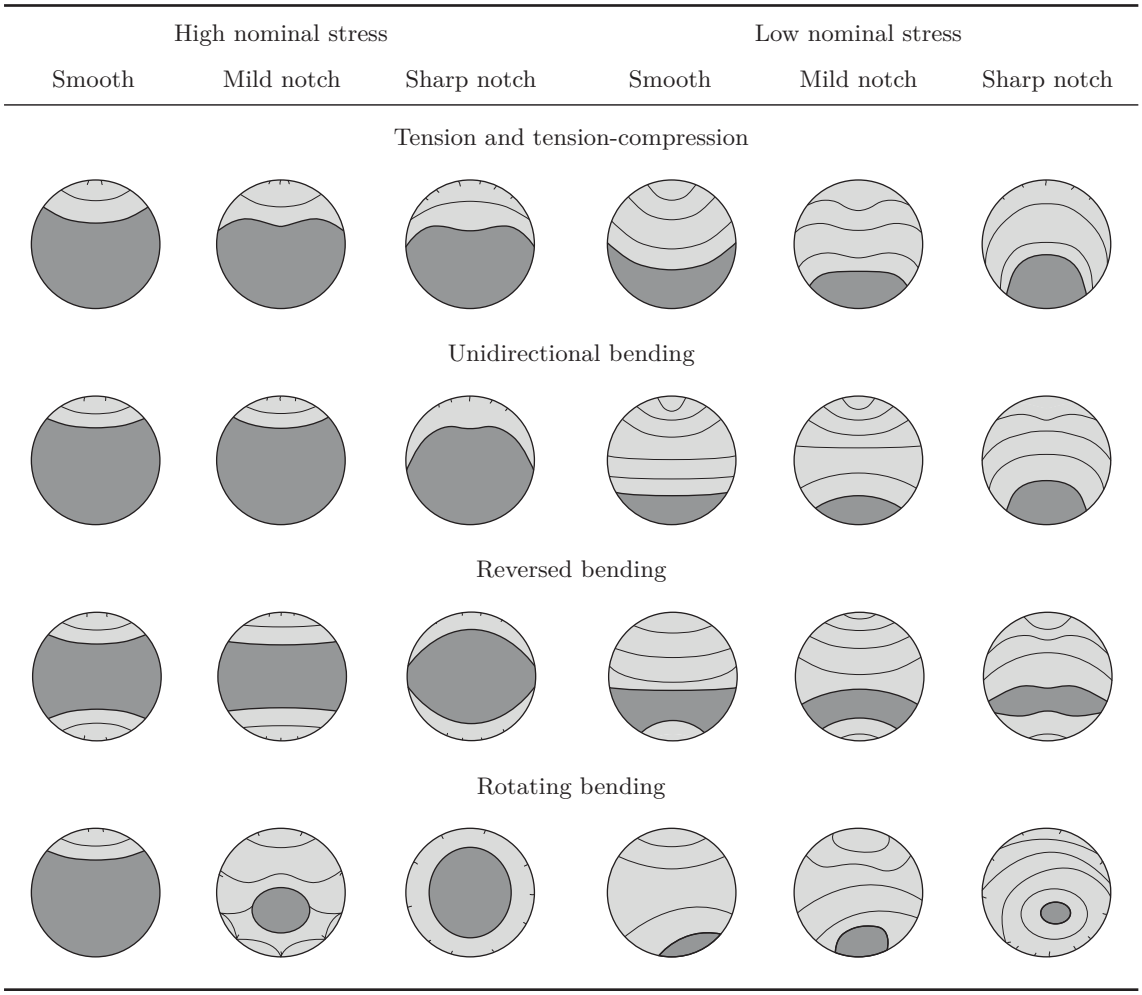


Fig. 2.5 – Macroscopic characterisation of fatigue fractures. Redrawn from [44].

2.4 Representation of fatigue data

One of the major problems designers are faced with is the fact that the fatigue characteristics of a material cannot be deduced from the other mechanical properties; they have to be measured directly in fatigue tests [16]. The purpose of fatigue testing is twofold, (i) to establish the relationship between stress range (load, strain or deflection) and number of stress repetitions to failure for a given material or component, and (ii) to enable a comparison of the fatigue strengths of different materials or components [5]. The assessment of fatigue life for steel and concrete structures relies predominantly on a realistic description of the relationship between stress range and number of load cycles. For this purpose, *S-N* diagrams or *Wöhler* curves are commonly used representations.

S-N curves for specific elements or materials are established from large series of fatigue test at different stress ranges. Typically, a semi- or double-logarithmic representation is used with the stress range of testing displayed on the ordinate and the corresponding number of stress cycles to failure on the abscissa, see Fig. 2.6 (a). Fatigue test results

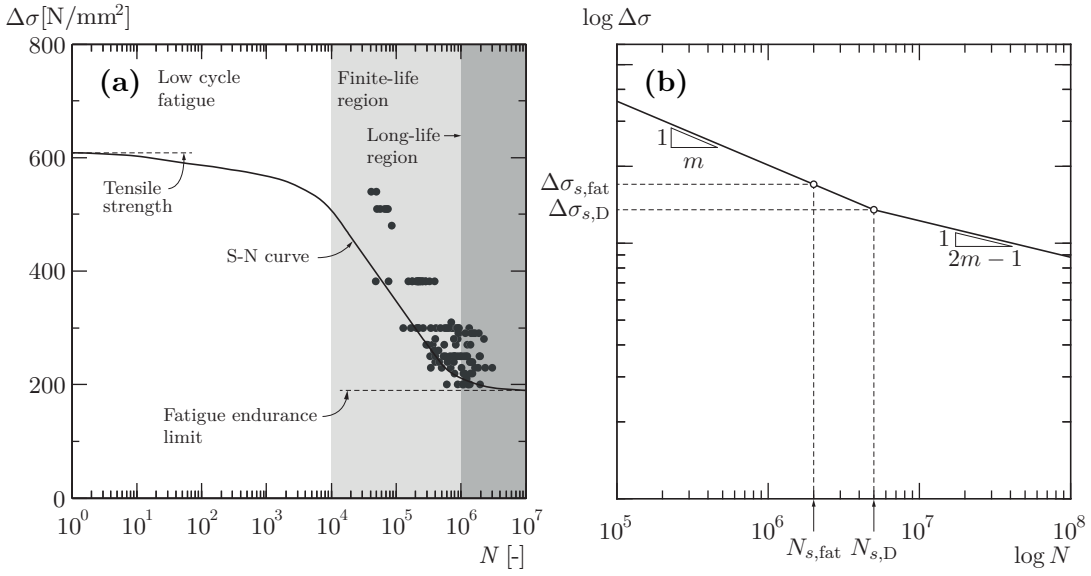


Fig. 2.6 – S-N diagram for reinforcing steel bars: (a) semi-logarithmic representation adapted from [34], test results from [24, 25]; (b) double-logarithmic representation of fatigue strength curve according to Swiss standard [84].

exhibit a significant scatter such that number of load repetitions to failure at the same stress range may vary by an order of magnitude [60]. A statistical interpretation of such test results is therefore necessary. In 1910, *Basquin* [8] formulated a simple relation between the applied stress and the number of stress repetitions to failure for the *finite-life* region. In double-logarithmic representation, this reduces to the equation of a straight line

$$S = CR^n \tag{2.4}$$

where S denotes the maximum applied stress and R represents the corresponding number of load repetitions that leads to failure. The constant C represents the stress obtained by extending the straight line until the intersection with $R = 1$, whereas the exponent n defines the slope of the line. The values of C and n are typically determined from a statistical regression with the method of least squares for a particular set of data. Such fatigue strength curves are subsequently constructed as single lines or composed of multiple straight lines, possibly even offset by a number of deviations from the mean for design purposes.

The fatigue strength curve recommended in the Swiss standard [84] for reinforcing steel bars is formulated as a variation of Eq. (2.4) in the form

$$N = C \cdot \Delta\sigma^{-m} \tag{2.5}$$

wherein N denotes the number of stress cycles until failure, $\Delta\sigma$ is the stress range in a cycle, C is a regression constant and m is the slope of the curve, see Fig. 2.6 (b). The standard allows the use of the same slope for all steel types and bar diameters. This has the advantage that the same values of the *damage equivalence factor* (factor to relate the

effects of the fatigue load model to those from actual traffic loading) and fatigue endurance limit can be used in a fatigue evaluation. The characteristic fatigue life of reinforcing steel is defined at $N_{s,\text{fat}} = 2 \cdot 10^6$ and the corresponding stress range $\Delta\sigma_{s,\text{fat}}$ is defined as the *nominal fatigue strength*. The transition from the finite-life to long-life regions is defined at $N_{s,\text{D}} = 5 \cdot 10^6$. This point on the curve represents the fatigue endurance limit which results from

$$\Delta\sigma_{s,\text{D}} = \Delta\sigma_{s,\text{fat}} \cdot \sqrt[4]{\frac{N_{s,\text{fat}}}{N_{s,\text{D}}}} \approx 0.8 \cdot \Delta\sigma_{s,\text{fat}} \tag{2.6}$$

for a given value of the nominal fatigue strength. The second portion of the fatigue strength curve is extended past the endurance limit, albeit at a reduced slope of $(2m - 1)$.

2.5 Theory of damage accumulation

Most engineering structures are subjected to stress ranges that do not remain constant, but vary throughout their service life. The use of conventional S-N diagrams are therefore limited as they are only valid for constant stress ranges. In order to overcome this limitation, it is assumed that fluctuating stresses at any amplitude causes fatigue damage. The extent of this damage is proportional to the number of stress repetitions at a given amplitude in relation to the total number of repetitions required to cause failure of the undamaged specimen at that stress level. The damage sustained during cyclic loading accumulates until a critical value is reached, which represents failure. In this section, only the linear damage rule and an alternative, nonlinear approach for the prediction of damage accumulation is described. A review of cumulative fatigue damage and life prediction theories is provided in [22].

2.5.1 Linear damage rule

A member subjected to a simple sinusoidal stress spectrum is considered, which fails after N_u load cycles at constant stress range. When only N cycles are applied at this stress range, where $N < N_u$, the member remains intact, however, some extent of damage was sustained. In an attempt to quantify this damage, *Palmgren* [70] proposed the *linear damage rule* (1924) which was further developed by *Miner* (1945) [65]. Herein, the damage increment resulting from N_i load cycles at a specific stress range is expressed as

$$d_i = \frac{N_i}{N_{u,i}} \tag{2.7}$$

where d_i is also known as the *cycle ratio*. The total damage resulting from a loading spectrum consisting of different stress ranges is expressed as the sum of the damage increments incurred at each stress range

$$D = \sum_{i=1}^{n} \frac{N_i}{N_{u,i}} = \sum_{i=1}^{n} d_i \leq 1 \tag{2.8}$$

wherein failure is assumed to occur when the accumulated damage reaches a value of unity. The evaluation of fatigue test results with the linear damage rule, however, indicates that an accumulated sum of damage reaching a value of one does not necessarily constitute failure. In addition to a more statistical explanation, a further possible reason for this might lie in the fact that stress cycles below the fatigue endurance limit were previously not considered in the summation. *Haibach* proposed a solution to this problem by prescribing a reduced slope of $(2m-1)$ for the second portion of the S-N curve following the fatigue endurance limit in order to consider the lower stress ranges. This proposal, even though somewhat pragmatic and only partly justifiable on the basis of fracture mechanics, has been widely accepted in the current design standards for steel and concrete construction [106]. The summation of damage increments sustained at different stress range levels is illustrated graphically on the S-N curve in Fig. 2.7 (a). A further drawback to Miner's law is the fact that the rate of damage accumulation is constant, see Fig. 2.7 (b). The damage increment resulting from each group of stress cycles is independent of the sequence of the loading groups in the stress history. A high initial stress range followed by lower stress range load cycles is more detrimental than the inverse. The outcome of both these spectra, however, is considered to be the same according to the linear damage rule.

The major advantage of this rule clearly lies in its simplicity. This is probably the primary reason why it is still widely used in fatigue life predictions. The use of more complex damage accumulation techniques is often not justifiable when considering the many uncertainties in the definition of the expected loading spectrum, the structural response to the loading and with regards to the fatigue strength of the material.

2.5.2 Alternative prediction of damage accumulation

An alternative, nonlinear approach for fatigue damage accumulation is presented in [60] and discussed in the dissertation by *Eskola* [20]. In this approach, a specimen with cross-section A under constant cyclic loading of magnitude $\Delta\sigma_0 A$ is considered. The remaining cross-sectional area after N_i load cycles is defined as $A\,(1-D)$, resulting in the effective stress range of

$$\Delta\sigma = \frac{\Delta\sigma_0}{(1-D)} \tag{2.9}$$

The rate of damage accumulation with increasing load cycles is assumed to be proportional to $\Delta\sigma^m$, which can be expressed as

$$\frac{\mathrm{d}D}{\mathrm{d}N} = \frac{c\Delta\sigma_0^m}{(1-D)^m} \tag{2.10}$$

under consideration of Eq. (2.9) and by including the proportionality constant c. The resulting differential equation is solved through integration over the number of load cycles and by satisfying the boundary conditions $D\,(N=0) = 0$ and $D\,(N=N_u) = 1$. The damage accumulated after N load cycles can subsequently be expressed as

$$D = 1 - \left(1 - \frac{N}{N_u}\right)^{\frac{1}{m+1}} \tag{2.11}$$

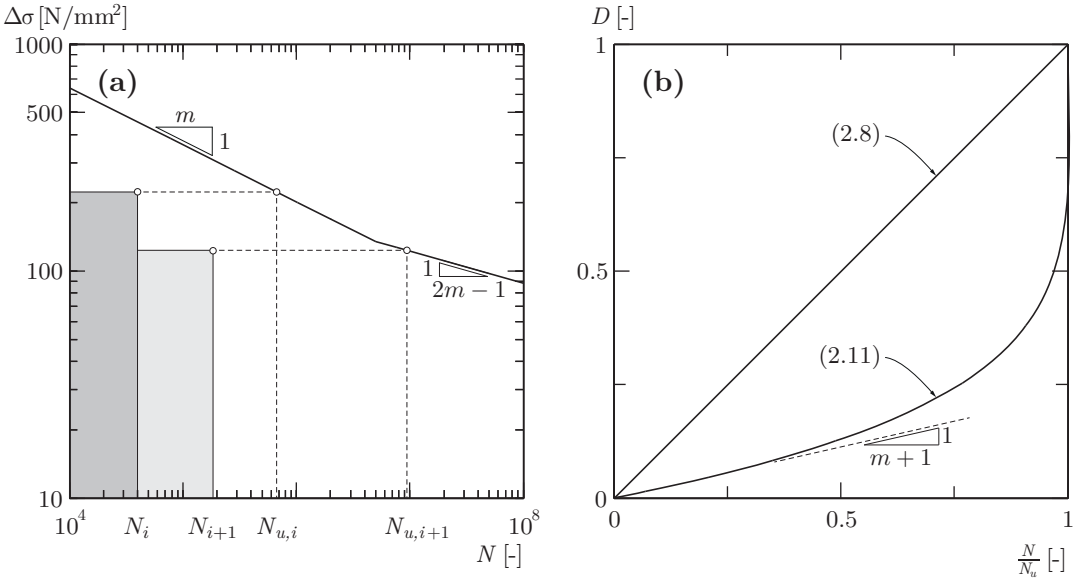


Fig. 2.7 – Fatigue damage accumulation: (a) consideration of spectrum loading with S-N diagram; (b) comparison of approaches, redrawn from [60].

which implies that a definition of the fatigue endurance limit is not required and represents an increasing rate of damage accumulation, see Fig. 2.7 (b). The sequence of stress cycles of different intensities, however, is not considered in this approach.

2.6 Fatigue of reinforced concrete

The load-carrying capacity of the composite material *reinforced concrete* lies in the ability of the concrete to resist (primarily) compressive forces and of the reinforcement to resist tensile forces after cracking of the concrete. Both materials as well as their interaction through bond have been shown to be susceptible to fatigue. The *CEB* state of the art report [98] describes 17 case studies of concrete structures and structural members such as bridges, pavements, slabs, crane tracks and machine foundations where structural damage could be attributed to fatigue processes. The deterioration of structural condition, in particular for the bridges, pavements and slabs, was mostly due to increases in the magnitude and frequency of traffic loads. The interaction of fatigue and corrosion of the reinforcement was also found to be particularly detrimental to the structural condition.

The deck slabs of bridges have been identified as structural elements being especially susceptible to fatigue damage [11]. This can be attributed to a direct contact with the wheel loads, a typically light reinforcement content and a small ratio of own weight to live load. Numerical calculations of the stress range in various bridge cross-sections under traffic loads have confirmed that the transverse deck slab direction at the cantilevers and between the webs is fatigue critical in the absence of prestressing [31].

Schläfli and Brühwiler [82] differentiate between flexural and shear modes of fatigue

failure of structural concrete members. In elements subjected predominantly to flexural loading, a progressive deterioration of bond causes existing cracks to increase in width as the tension stiffening effect is lost. Disproportionally large deflections are observed and failure ensues through rupture of individual reinforcing bars or spalling of the concrete in the compressive zone. Such failure modes have been verified in experimental investigations [23, 48, 81]. A further classification of concrete structures subjected to high fluctuating shear loads differentiates between elements with shear reinforcement and those without. Beams with shear reinforcement have been shown to behave in a ductile manner under shear loading; failure occurs due to fatigue fracture of the stirrups and/or spalling of the surrounding cover concrete [27]. The cyclic loading of beams without shear reinforcement leads to the development of a critical shear crack; failure is typically observed as fatigue of the concrete compression strut at the supports [27, 81]. Fatigue tests on cantilever deck slabs without shear reinforcement indicate that similar cracking patterns are observed to those obtained under static loading [56]. The majority of these slabs failed due to shear-fatigue without fracture of the flexural reinforcement, albeit only at very high levels of maximum load. Punching shear type failures have also been observed in slabs subjected to cyclic loading [71, 91].

The majority of experimental investigations on fatigue of reinforced concrete members in bending constitutes constant amplitude loading of statically determinate beam or slab strip specimens [27, 48, 81, 104]. Stress cycles are typically introduced through a fixed-pulsating load applied at one or more cross-sections of the specimen. A number of tests have been performed on slabs with moving constant wheel loads [71, 91]. The fatigue strengths obtained under such loading conditions were lower than with fixed-pulsating loading. Furthermore, the typical tortoise-shell cracking pattern on the slab soffit, described as being representative of traffic-induced fatigue, could be reproduced with these moving loads. The number of such tests, however, remains limited due to the complexity of the test setup and the obvious restrictions on the loading frequency. Concrete members subjected to cyclic loading have also been shown to exhibit different failure modes to those predicted under static loading. *Chang* and *Kesler* [14] conducted a large number of tests on concrete beams; specimens failed in flexure under static loading and due to fatigue shear modes under cyclic loading. Similar observations were made by *Lambotte* and *Baus* in tests on prestressed railway sleepers [53]. The statically loaded specimens exhibited concrete compression, bond or shear failures, whereas the cyclically loaded specimens all failed due to fatigue of the prestressing tendons. *Fehlmann* [23] investigated the fatigue performance of a typical frame type bridge in a large-scale test. The level of cyclic loading corresponded to the provisions of the Swiss standard for the fatigue load model. The investigation showed that fatigue damage in such structures may be expected in future and confirmed the validity of this hazard scenario for design and in the evaluation of existing structures. Virtually no changes in the load-carrying response, apart from the formation of some new cracks, could be observed during approximately 90% of the fatigue life. Fatigue damage to the structure, therefore, remained mostly undetected by conventional methods

until shortly before failure.

Notwithstanding the substantial body of work on fatigue with regards to the fatigue strength of the individual materials, their interaction and on actual structural concrete members, designers still rely on conservative methods in the evaluation of concrete structures for safety against fatigue. It is currently neither possible to establish the present state of damage in a structural concrete member, nor to predict the remaining fatigue life for future loads [23].

Chapter 3

Material behaviour and bond interaction

3.1 General

The load-carrying behaviour of structural concrete members is directly related to the mechanical properties of plain concrete, the reinforcement as well as the interaction of these materials through bond. Similarly, the behaviour under repeated loading is typically described in terms of the response of the constituent materials and bond to stress fluctuations. The present chapter will provide an overview of the relevant material characteristics, in particular with regards to fatigue behaviour.

In the first part, Section 3.2, the material characteristics of plain concrete will be discussed. A description of concrete under conditions of static loading will be provided; this will include uniaxial compression and tension as well as conditions of biaxial loading. The transfer of stresses across cracks through aggregate interlock and the fatigue behaviour of concrete will be examined.

The second part (Section 3.3) deals with the mechanical properties of reinforcing steel. An overview of the production process of high strength steels and its influence on the characteristics of strength and ductility will be provided. The behaviour of reinforcing steel under static loading will be discussed in terms of its stress-strain relationship. The historical development of reinforcing steel types in Switzerland will be presented, illustrating the mechanical properties of the reinforcement placed in many concrete structures still in service today. Finally, the behaviour of reinforcing steel under cyclic loading will be examined with regards to fatigue strength.

In the final part, Section 3.4, the concept of bond between the reinforcing steel bars and surrounding concrete will be introduced. Firstly, bond under static loading will be discussed, examining the bond stress-slip relationship and introducing the concept of tensile stiffening. Last, the deterioration of bond under repeated loads will be reviewed.

3.2 Behaviour of concrete

Concrete consists of aggregate, portland cement and water with the possible addition of chemical admixtures in order to obtain certain desirable properties. The cement reacts with the water in a chemical process termed *hydration* upon which it hardens and binds the aggregate to form a stone-like material with high compressive strengths.

Much of the following description regarding the properties of concrete under static loading is based on the dissertations of *Etter* [21], *Galmarini* [28] and *Kaufmann* [49]. The description of the fatigue behaviour of concrete is based on the dissertation of *Fehlmann* [23], the review by *Murdock* [67] and the *CEB* state of the art report [98].

3.2.1 Uniaxial compression

The most relevant mechanical properties of plain concrete are obtained from standard compressive strength tests under conditions of uniaxial compression. These include the compressive strength f_c, the corresponding strain at peak load ε_{cu} and possibly the modulus of elasticity E_c, which is calculated as a tangent modulus at low stress levels. The stress-strain relationship is usually only measured for academic purposes in order to verify constitutive models for analysis. Test specimens are typically cubes with side lengths of 150 mm or cylinders with a diameter of 150 mm and height of 300 mm. The peak load measured in these tests divided by the plan cross-sectional area constitutes the cube $f_{c,\text{cube}}$ and cylinder strength f_{cc} of the concrete for these specimen types. Friction between the test specimen and the loading plates introduce a confinement effect, which increases the apparent strength. This effect is particularly severe for short/squat specimens. For this reason, a ratio of specimen height to the greatest plan dimension in the order of 2.0...2.5 is recommended in order to limit this effect [28]. The relationship between cube strength and the typically lower cylinder compressive strength is approximately

$$f_{c,\text{cube}} = 1.15 f_{cc} \quad \text{in } [\text{N/mm}^2]. \tag{3.1}$$

An increase in concrete strength with increasing age is observed due to the progressive hydration of the cement; the magnitude of which is dependent on the type of cement and the conditions during curing such as temperature and humidity [21]. The strain at peak load increases slightly for higher concrete strengths, however, this property is subject to a large degree of scatter. Based on the proposal by *Popovics* [74], an approximation of the strain at peak load is obtained with

$$\varepsilon_{cu} = 0.9 {f_{cc}}^{1/4} \quad \text{in } [\text{N/mm}^2] \tag{3.2}$$

as a function of the cylinder compressive strength. The modulus of elasticity can be approximated according to *fib* Model Code (2010) [42] with the expression

$$E_c = k_E {f_{cc}}^{1/3} \quad \text{in } [\text{N/mm}^2] \tag{3.3}$$

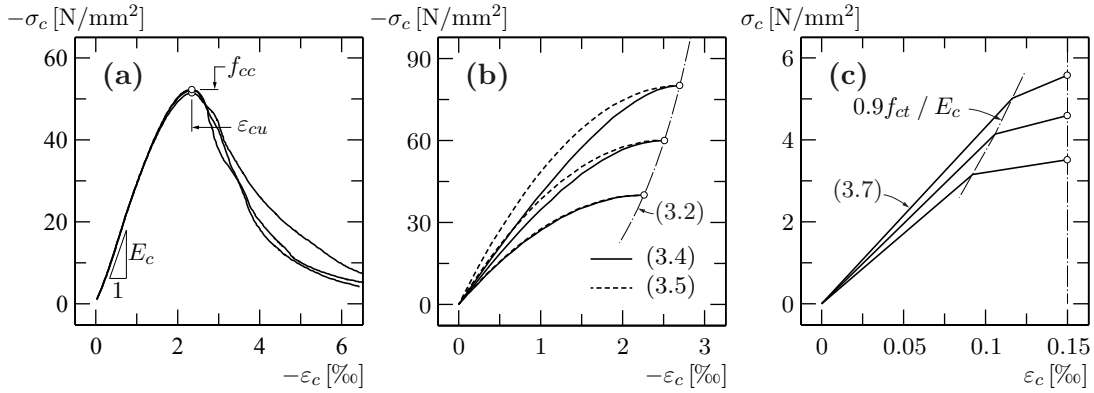


Fig. 3.1 – Stress-strain relationships of concrete under uniaxial loading: (a) deformation-controlled test in compression, author's data; (b) idealised response in compression; (c) idealised response in tension.

wherein the parameter k_E varies within the range $7 \cdot 10^3 \ldots 12 \cdot 10^3$ according to the type of aggregate. *Poisson's* ratio for plain concrete ν_c varies between 0.14 and 0.26 for typical ranges of loading [42]. However, an assumption of $\nu_c = 0.2$ is generally deemed appropriate for the purposes of analysis.

The stress-strain relationship of concrete in uniaxial compression exhibits roughly three stages: (1) an initial, approximately linear response is observed up to roughly one third of the compressive strength; (2) micro-cracks start forming upon further loading which results in a reduction in stiffness and increasingly nonlinear response until reaching the compressive strength; (3) a subsequent decrease in strength can be observed in displacement-controlled tests upon further increase in deformation [106], see Fig. 3.1 (a). The stress-strain relationship proposed by *Sargin* [80] provides a good approximation of concrete under uniaxial compression up to the peak load

$$\sigma_c(\varepsilon_c) = -f_{cc}\frac{k_\sigma\zeta - \zeta^2}{1 + (k_\sigma - 2)\cdot\zeta} \quad \text{with} \quad \zeta(\varepsilon_c) = \frac{-\varepsilon_c}{\varepsilon_{cu}} \quad \text{and} \quad k_\sigma = \frac{E_c\varepsilon_{cu}}{f_{cc}} \tag{3.4}$$

which is also recommended in *fib* Model Code (2010) [42]. Alternatively, the stress-strain response in compression up to the peak load can be idealised with a parabolic function as

$$\sigma_c(\varepsilon_c) = f_c\frac{\left({\varepsilon_c}^2 + 2\varepsilon_c\varepsilon_{cu}\right)}{{\varepsilon_{cu}}^2} \quad \text{for} \quad -\varepsilon_c \leq \varepsilon_{cu} \tag{3.5}$$

where f_c denotes the peak compressive strength. A comparison of the proposed stress-strain relationships in compression represented by Eqs. (3.4) and (3.5) is illustrated in Fig. 3.1 (b). The stress-strain curve according to Model Code is almost identical to that of the parabolic proposal for low concrete strengths. At higher concrete strengths, however, the relationship recommended in Model Code exhibits a more linear form and a lower stiffness than that of Eq. 3.5).

3.2.2 Uniaxial tension

The tensile strength of concrete f_{ct} is significantly smaller than its compressive strength[1] and typically subject to a larger scatter than observed in compressive strength tests. Direct tensile tests are demanding as the loading is often not purely uniaxial and the stiffness of the testing machine plays a more significant role than in compressive tests. As a result, the tensile strength is typically determined by indirect means such as with splitting cylinder, double-punch or bending tests. An approximation of the tensile strength results from

$$f_{ct} = 0.3 {f_{cc}}^{2/3} \quad \text{in } [\text{N/mm}^2] \tag{3.6}$$

for a known cylinder compressive strength, as proposed in [75]. The stress-strain relationship of concrete in tension is initially linear-elastic with roughly the same modulus of elasticity as observed in compression. The stiffness reduces at tensile stresses of approximately $0.9 f_{ct}$ due to the formation of micro-cracks within a small failure zone [42]. A bilinear relationship is prescribed in *fib* Model code [42] for uncracked concrete subjected to tension

$$\begin{aligned} \sigma_{ct} &= E_c \cdot \varepsilon_c && \text{for} \quad 0 \le \varepsilon_c \le 0.9 f_{ct}/E_c \\ \sigma_{ct} &= f_{ct} \cdot \left(1 - 0.1 \frac{0.00015 - \varepsilon_c}{0.00015 - 0.9 f_{ct}/E_c}\right) && \text{for} \quad 0.9 f_{ct}/E_c < \varepsilon_c \le 0.15‰ \end{aligned} \tag{3.7}$$

as illustrated in Fig. 3.1 (c). A relatively gradual decrease in strength is observed in deformation-controlled tests after exceeding peak tensile load. This softening behaviour, however, can be neglected for most practical purposes.

3.2.3 Biaxial loading

Concrete under biaxial loading or in a state of plane stress, where $\sigma_{c2} = 0$, is typically investigated using cylindrical or plate specimens. Special consideration of the confinement of transverse strains due to friction with the bearing platens is necessary. Such effects are reduced with the use of lubricants, Teflon-covered surfaces or with brush bearing platens.

Concrete under biaxial loading exhibits a similar stress-strain relationship as observed in uniaxial compression [28]. Concrete strengths are higher when a compressive stress acts perpendicular to the principal compressive direction ($0 \ge \sigma_{c1} \ge \sigma_{c3}$) compared to the case of uniaxial loading. This effect is observed to be most pronounced at a stress ratio of $\sigma_{c1} / \sigma_{c3} = 0.5$, resulting in compressive strength increases of approximately $27 \ldots 38\%$. For equal biaxial compression $\sigma_{c1} / \sigma_{c3} = 1$ increases in compressive strength of only $14 \ldots 19\%$ are observed [21]. *Kupfer and Hilsdorf* [52] observed an increase in compressive strength of approximately 25% for a stress ratio ratio of $\sigma_{c1} / \sigma_{c3} = 0.52$, while increasing the strain at peak load by 40%, see Fig. 3.2 (a). Conversely, lower compressive strengths are observed when a tensile stress acts perpendicular to the principal compressive direction ($\sigma_{c1} \ge 0 \ge \sigma_{c3}$) compared to uniaxial compression. The compressive strength decreases

[1] The concrete tensile strength is typically only about in the range $8 \ldots 15\%$ of f_c [28].

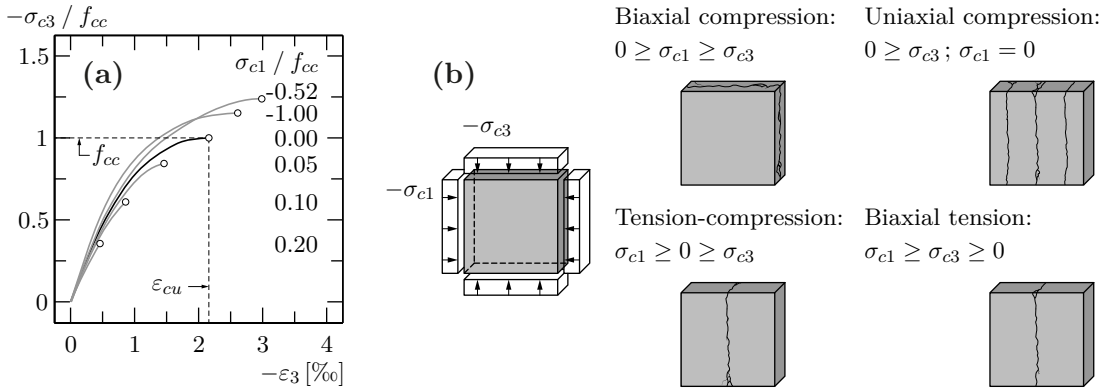


Fig. 3.2 – Biaxial loading of concrete: (a) stress-strain relationship for biaxial compression and tension-compression, data from [52]; (b) typical failure modes to [92].

approximately linearly with increasing tensile stress [92] and a significant reduction of the strain at peak load is observed.

The failure mechanism of concrete specimens under biaxial or uniaxial loading conditions is observed as a tensile splitting with the cracks forming perpendicular to the direction of maximum tensile strain. Different failure modes are observed with regards to the combination of biaxial loading, see Fig. 3.2 (b). In *biaxial compression*, failure occurs due to cracking along a plane parallel to the unloaded surface. *Uniaxial compression* leads to the formation of cracks parallel to the direction of applied load and perpendicular to the larger unloaded surface of the specimen. Combined *tension-compression* causes a continuous crack to form normal to the principal tensile stress direction. Finally, *biaxial* and *uniaxial tension* result in a single crack forming perpendicular to direction of maximum tensile stress [92].

3.2.4 Aggregate interlock

Cracks in normal strength concrete typically develop within the cement matrix and pass around the individual grains of the aggregate. The rough interfaces resulting at the crack faces enable a transfer of normal σ_n and shear stresses τ_{tn} in response to crack displacements δ_n and δ_t; this is termed *aggregate interlock*. This process is characterised by small shear displacements on non-uniform crack surfaces as a result of high loads. Hence, the test setup required to establish such relations is generally very demanding [49]. Moreover, the scatter observed in test results is correspondingly large [88]. The *fib* Model Code (2010) [42] prescribes relationships on the basis of the model established by *Walraven* [102] between the mean shear and normal stress for a given shear displacement and crack width. The influence of the aggregate is included in a factor which is typically established from push-off tests. A strong nonlinearity is observed between the displacements and activated stresses [106]. Moreover, an analytical expression of the aggregate interlock relation would have to include the loading history as the behaviour is path-dependant [49].

3.2.5 Fatigue behaviour

The first investigations on the fatigue response of plain concrete in compression, starting in 1903, are attributed to *Van Ornum* [99, 100], as reported in [67]. *Nordby* (1958) provides a review of previous work on fatigue of plain concrete as well as on reinforced and prestressed concrete in [68]. *Murdock* wrote an insightful evaluation of fatigue research on plain concrete for the period 1903 – 1961 in [67]. Furthermore, the state of the art reports by *Wesche et al.* (1973) [103] as well as the *CEB* [98] provide a good overview of the primary findings in the work on fatigue of plain concrete.

General considerations

Concrete is susceptible to an accumulation of damage under repeated loading. Microcracks within the cement matrix, which commonly form due to shrinkage, coalesce into macroscopic cracks under repeated stress fluctuations. Ultimately, such cyclic compressive loading results in failure of the concrete. The fatigue strength of concrete is typically expressed as a ratio of the static strength. Such a normalisation of the fatigue strength is said to account for the influence of a wide range of variables such as water-cement ratio, cement content, amount of entrained air, curing conditions and age at loading as they seem to influence both the static and fatigue strength to the same degree [98]. From the evaluation of many test results it can be concluded that the fatigue strength of concrete, expressed as a ratio to the static compressive strength, must be roughly the same regardless whether the mode of loading is compressive, tensile or flexural [98]. *Fehlmann* [23] expressed the fatigue strength of concrete for different modes of loading in relation to the static strength as

$$f_{c,\mathrm{fat}} = 0.4 \ldots 0.7 f_{c,\mathrm{stat}} \tag{3.8}$$

in summary of a large number of reported fatigue test results from different researchers.

Fatigue strength in compression

No fatigue endurance limit has been established for concrete, even though fatigue tests up to 10^7 load cycles have been carried out [67]. In contrast to steel, the fatigue strength of plain concrete depends on both the maximal and minimal load level in each cycle [98]. In general, an increase in fatigue strength is observed at higher levels of minimum stress. For this reason, fatigue test data of plain concrete is typically presented in an S-N diagram for the maximum stress level and constant ratio of minimum to maximum stress level or stress ratio R. Alternative representations of the relation between stress level and number of load cycles to failure have been developed, namely, those by *Goodman* (1926) and *Smith* (1910). Goodman diagrams portray the relationship between the maximum and minimum stress level during cyclic loading, whereas Smith diagrams indicate the relationship between either the maximum or minimum stress level to the mean stress level. *Aas-Jakobsen* (1970) [1] found that the relationship between the maximum and minimum

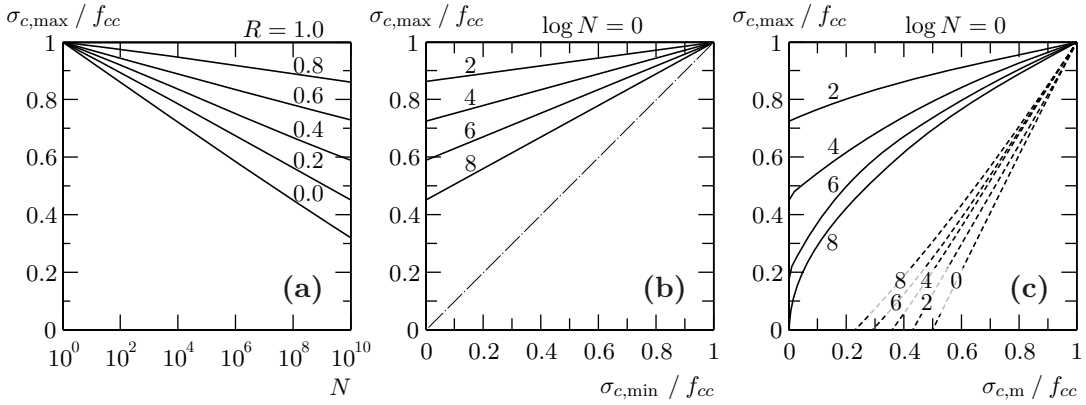


Fig. 3.3 – Fatigue strength of plain concrete: (a) S-N diagram; (b) Goodman diagram; (c) Smith diagram. Redrawn from [94].

stress level is approximately linear for fatigue failures at $N = 2 \cdot 10^6$ and proposed the formulation

$$\frac{\sigma_{c,\max}}{f_{cc}} = 1 - \beta\,(1 - R)\log N_u \quad \text{where} \quad R = \frac{\sigma_{c,\min}}{\sigma_{c,\max}} \tag{3.9}$$

where R denotes the stress ratio for the range $[0 \leq R \leq 1]$ and β is a material coefficient for which a value of 0.064 was recommended. In this way, the susceptibility of concrete under cyclic compressive loading to fatigue can be evaluated. Eq. (3.9) describes the fatigue strength curve of an S-N diagram for a constant level of maximum stress and stress ratio R, see Fig. 3.3 (a). This relationship can also be presented as Goodman or Smith diagrams for a constant number of load cycles to failure, see Fig. 3.3 (b) and (c). *Tepfers and Kutti* [94] investigated the validity of Eq. (3.9) with fatigue test data on ordinary and lightweight concrete. A data set of 475 fatigue tests, resulting from their own tests and results reported in literature, was used to evaluate the value of the material coefficient. In conclusion, a value of $\beta = 0.0685$ was proposed for both ordinary and lightweight concrete.

Influence of eccentric loading

The influence of a non-uniform strain distribution on the fatigue strength was investigated by *Ople and Hulsbos* [69] in order to simulate the stress gradient characteristic to the compression zone of a structural concrete member in bending. Both static and fatigue tests were conducted on plain concrete prisms for different strain distributions, namely, uniform with $e = 0$, zero to maximum strain with $e = 25.4\,\text{mm}$ and half the maximum to maximum strain with $e = 8.5\,\text{mm}$. The fatigue strength was observed to increase at the higher stress gradients or larger eccentricity of the loading, see Fig. 3.4 (b). The fatigue strength of the specimens with a strain distribution which varied between zero and the maximum strain was on average approximately 17% higher than that of the uniformly stressed specimens.

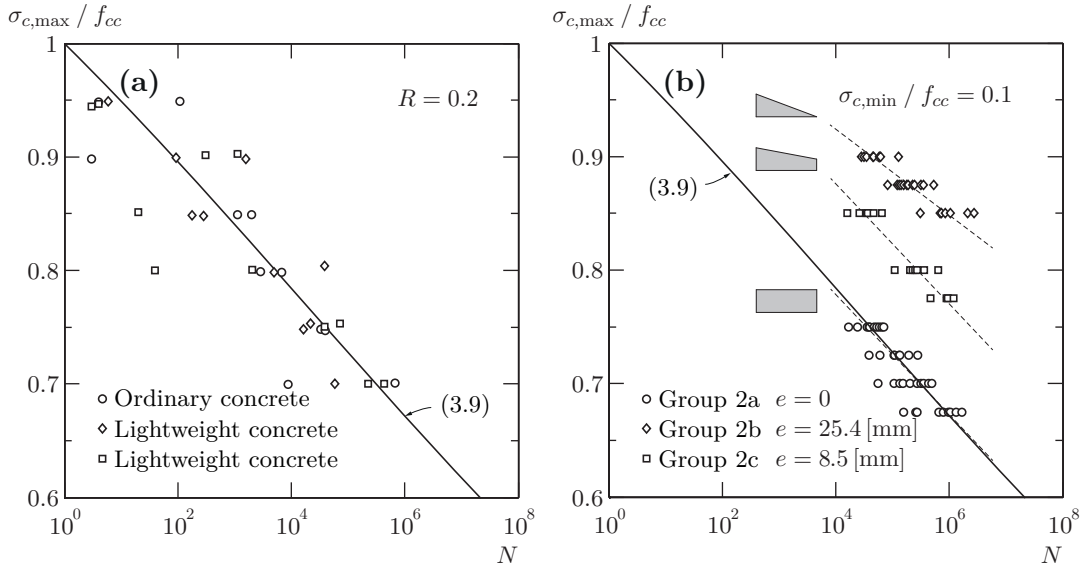


Fig. 3.4 – S-N diagram for plain concrete: (a) uniaxial compression, data from *Tepfers and Kutti* [94]; (b) compression with eccentricity, data from *Ople and Hulsbos* [69].

Fatigue strength in tension

The tensile fatigue strength of plain concrete was investigated by *Tepfers* [93] in cube splitting tests. The results indicate that Eq. (3.9) also provides a reasonable description of the susceptibility of concrete under cyclic tensile loading to fatigue. The original relationship established by *Aas-Jakobsen* was correspondingly expressed as

$$\frac{\sigma_{ct,\max}}{f_{ct}} = 1 - \beta\,(1 - R)\log N_u \quad \text{where} \quad R = \frac{\sigma_{ct,\min}}{\sigma_{ct,\max}} \tag{3.10}$$

where $\sigma_{ct,\max}$ denotes the maximum tensile stress under repeated loading and f_{ct} denotes the concrete tensile strength. A statistical evaluation of the material coefficient indicated that the same value, namely $\beta = 0.0685$, may be assumed as for cyclic compressive loading.

Influence of variable-amplitude loading

The first investigation of concrete under repeated compressive loading at variable levels of maximum and minimum load is attributed to *Hilsdorf and Kesler* (1966) [36]. They conducted tests on plain concrete beams in flexure under varying repeated loads and observed that both the fatigue strength and the fatigue life was influenced by the sequence in which these loads were applied. Two characteristics of variable-amplitude loading spectra were identified as having a significant influence on the fatigue response:

- loading spectra with many low minimum stress levels $\sigma_{c,\min}$ seem to cause stronger sequence effects and are typically more damaging than constant amplitude loading
- the presence of small amplitudes in the loading spectrum seems to reduce sequence effects and exhibit a favourable influence on the fatigue life [38].

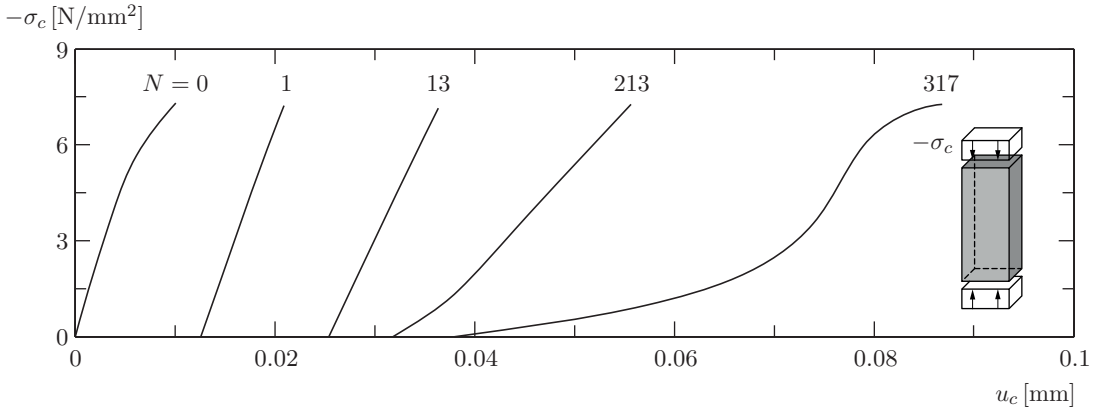


Fig. 3.5 – Development of compressive stress-strain relationship of plain concrete under repeated load. Redrawn from [100].

The assessment of fatigue damage with the linear damage rule typically provides unsafe results at high loading levels, whereas overly conservative results may be obtained for low levels of loading [36]. The linear damage rule, as described in Section 2.5.1, does not account for the sequence of loading at different levels of stress range and, therefore, only provides a rough approximation of the actual accumulated damage. Various standards prescribe different values of the accumulated damage at which fatigue failure is predicted; ranging from 0.2 to 1.0 [42].

However, despite the known limitations of the linear damage rule, it is still recommended in *fib* Model Code [42] for the prediction of fatigue life of plain concrete under variable-amplitude loading. Herein, failure is expected to occur at $D = 1$.

Deformation characteristics

Van Ornum discovered an interesting feature of the fatigue response of plain concrete, as reported in [67, 68], which was later confirmed in other investigations. Fig. 3.5 indicates how the shape of the stress-strain curve in compression develops with the number of load repetitions. The loading path, initially concave with respect to the strain axis, straightens after only a few repetitions of load and finally becomes convex with respect to the strain axis or even S-shaped, shortly before fatigue failure. The degree of convexity in the stress-strain response during loading can therefore provide an indication of remaining fatigue life. Furthermore, *Murdock* [67] reports that the peak concrete strain was observed to increase continuously with cycling. Such characteristic changes in the stress-strain curve were not observed for low stress levels, i.e. those where fatigue failure did not take place during a finite number of load cycles. In addition to the change in shape, the secant modulus of elasticity was observed to decrease during the fatigue life under both constant and variable amplitude cyclic loading [38].

3.3 Behaviour of reinforcement

Reinforced concrete is a composite material, efficiently using the high compressive strength of concrete in combination with the high tensile strength and large strain capacity of steel reinforcement. The latter component typically takes the form of profiled steel bars with an approximately circular cross-section and rolled-on ribs for improved bond properties. Typical diameters produced range from 8...40 mm. Alternative steel reinforcement types include welded meshes, prestressing wires and strands.

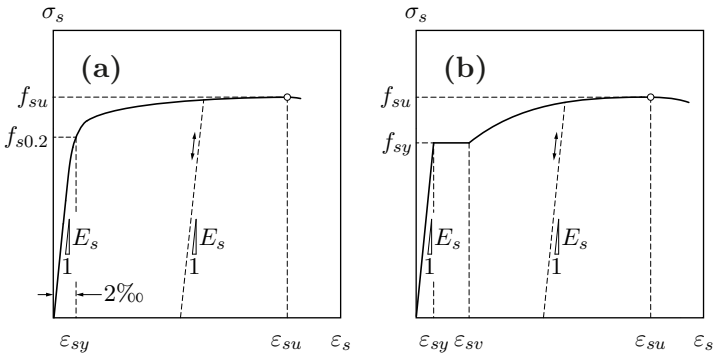
3.3.1 Production of reinforcing steel

The steel used to produce reinforcing bars is typically of a higher strength than the material used for steel construction [28]. Such high strengths can be achieved through the processes of hot-rolling and hardening through tempering and quenching, cold-working or micro-alloying. These processes are described in more detail in [21, 32].

In the so-called *Tempcore*-process, reinforcing steel bars are quenched directly after hot-rolling for a few seconds in water such that only a hard outer sheath is formed [21]. A self-tempering of this outer layer results due to the residual heat from the warm core. A change in the steel microstructure is achieved through the process of hardening, which requires that the material is heated above an upper critical temperature and subsequently *quenched* or cooled quickly, typically by immersing in water. These metallurgical changes in the microscopic structure of the steel result in improved strength, but are also associated with a reduction in ductility. The subsequent *tempering* is achieved through heating the steel again which results in an improvement of the ductility and reduction in strength. The cold-working of steel, on the other hand, is a mechanical process that involves the twisting or stretching of mild steel, thereby straining the material beyond the yield plateau. Similar to hardening, this results in an improvement on the ratio of yield to ultimate strength. A significant increase in the ultimate strength, however, is not achieved. A steadily increasing trend since the 1980s has been to coil the reinforcing steel onto spools in order to simplify transport and further processing. It is estimated that reinforcement steel in the coiled form constitutes roughly 70% of the total current usage in Switzerland. Mostly smaller diameter bars, but also those up to a diameter of 16 mm, are typically coiled onto spools [21]. Such steel should be considered to exhibit similar mechanical properties as cold-worked steels. Micro-alloying refers to the process in which small amounts of alloying elements are integrated into the crystal structure in order to achieve greater tensile strengths.

3.3.2 Static behaviour

The mechanical properties of reinforcing steel are typically determined from uniaxial tensile tests. The two main types of steel can be distinguished with regards to their characteristic stress-strain relationships, namely, steel exhibiting a well-defined yield plateau such as is observed in hot-rolled or micro-alloyed bars, and steel without a definite yield plateau such



Mechanical properties of typical reinforcing steels:

$f_{sy} = 450 \ldots 550$ N/mm^2
$E_s = 200 \ldots 210$ kN/mm^2
$\varepsilon_{su} = 20 \ldots 200$‰
f_{su} / $f_{sy} = 1.05 \ldots 1.40$

Fig. 3.6 – Stress-strain characteristic of reinforcing steel in uniaxial tension: (a) cold-worked and high-carbon steels; (b) hot-rolled, heat-treated and micro-alloyed steels; mechanical properties of typical reinforcing steel types summarised from [21, 28, 62].

as is the case for cold-worked bars. The stress-strain characteristics of these steel types is illustrated in Fig. 3.6 with an indication of the typical ranges of mechanical properties. Both steel types exhibit a linear increase in strain with loading; characterised by the modulus of elasticity E_s. The modulus of elasticity is independent of the type of steel alloy or whether it has been subjected to hardening or cold-working [21]. Upon reaching the yield strength f_{sy} in reinforcing steel with a distinct yield plateau, an increase in deformation at practically no further load is observed up to the onset of strain-hardening at ε_{sv}. In steel types without a well-defined yield plateau, the stress corresponding to a strain of 2‰ or $f_{s,0.2}$ is used to characterise the equivalent yield strength. *Strain-hardening* denotes the additional increase in stress associated with a subsequent increase in strain after the onset of yielding. This, however, occurs at a significantly lower rate than the elastic modulus. No change in volume is associated with the elongation within the strain-hardening range; hence, a continuous reduction in the cross-sectional area of the bar must take place. Steel stresses and strengths are typically calculated as engineering stress on the basis of the nominal cross-section. The true stress acting on the reduced cross-section may be considerably higher. Unloading at any point along the stress-strain curve is elastic and exhibits roughly the same stiffness as that observed during the initial loading [49]. The tangent stiffness of the stress-strain curve reduces to zero upon reaching the tensile strength f_{su} at the corresponding ultimate tensile strain ε_{su}. A further increase in deformation leads to a localisation of strain within the bar, characterised by visible necking shortly before failure.

The stress-strain relationship of reinforcing steel is idealised in different manners according to the required accuracy. A comparison of constitutive relations typically used for analysis to the stress-strain relationship measured in a deformation-controlled test is illustrated in Fig. 3.7. The simplest approaches constitute the *rigid-perfectly plastic* and *linear elastic models*. In the first approach, a constant steel stress (typically the yield strength) is assumed regardless of the strain; in the second, the stress is assumed to increase linearly

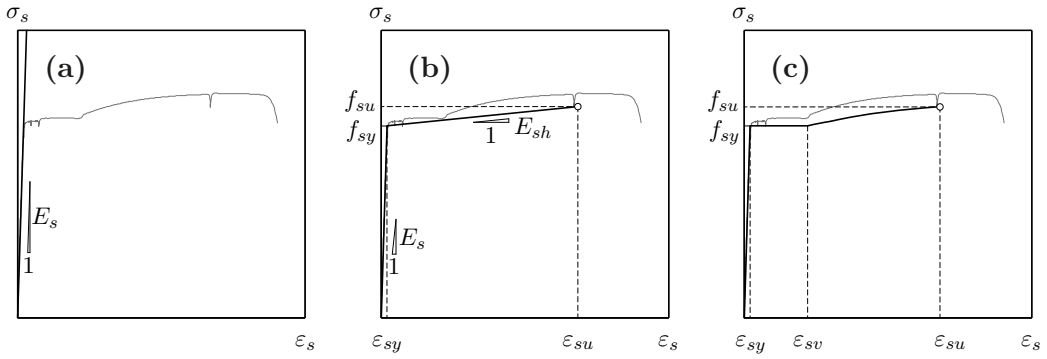


Fig. 3.7 – Idealisations of the stress-strain characteristic of reinforcement: (a) linear elastic; (b) bilinear; (c) hot-rolled steel, according to [58]. Comparison to author's test data.

with increasing strain at a rate equal to the modulus of elasticity. This latter approach is often sufficient for simple calculations under low service loads and is simply formulated as

$$\sigma_s\left(\varepsilon_s\right) = E_s \varepsilon_s. \tag{3.11}$$

In the analysis of structural concrete members in bending, yielding of the reinforcement constitutes an important structural condition. In such cases, a *bilinear* idealisation of the stress-strain curve is mostly adequate; formulated piecewise in the form

$$\begin{aligned} \sigma_s\left(\varepsilon_s\right) &= E_s \varepsilon_s && \text{for} \quad 0 \le \varepsilon_s \le \frac{f_{sy}}{E_s} \\ \sigma_s\left(\varepsilon_s\right) &= f_{sy} + E_{sh} \cdot \left(\varepsilon_s - \frac{f_{sy}}{E_s}\right) && \text{for} \quad \frac{f_{sy}}{E_s} < \varepsilon_s \le \varepsilon_{su} \end{aligned} \tag{3.12}$$

wherein the strain capacity is typically limited to the ultimate strain ε_{su}. The residual stiffness after the onset of yielding is referred to as the modulus of strain-hardening

$$E_{sh} = \frac{f_{su} - f_{sy}}{\varepsilon_{su} - \varepsilon_{sy}} \tag{3.13}$$

which is defined as the secant modulus between yielding and the ultimate strength. For the sake of completeness, a model proposed by *Mander* (1983) [58] for hot-rolled reinforcing steel bars, is described. This model represents the initial linear-elastic response and yield plateau with a bilinear function and idealises the strain-hardening region in the form of an exponential curve with the point at ultimate strength as origin, expressed as

$$\begin{aligned} \sigma_s\left(\varepsilon_s\right) &= E_s \varepsilon_s && \text{for} \quad 0 \le \varepsilon_s \le \frac{f_{sy}}{E_s} \\ \sigma_s\left(\varepsilon_s\right) &= f_{sy} && \text{for} \quad \frac{f_{sy}}{E_s} < \varepsilon_s \le \varepsilon_{sv} \\ \sigma_s\left(\varepsilon_s\right) &= f_{su} + \left(f_{sy} - f_{su}\right)\left[\frac{\varepsilon_{su} - \varepsilon_s}{\varepsilon_{su} - \varepsilon_{sv}}\right]^{\frac{1}{k}} && \text{for} \quad \varepsilon_{sv} < \varepsilon_s \le \varepsilon_{su} \\ \text{where} \quad k &= E_{sh}\left(\frac{\varepsilon_{su} - \varepsilon_{sv}}{f_{su} - f_{sy}}\right), \end{aligned} \tag{3.14}$$

the strain ε_{sv} indicates the onset of strain-hardening and E_{sh} is as defined in Eq. (3.13). Fig. 3.7 indicates that a realistic idealisation of the stress-strain relationship of reinforcing steel is possible using only four or five material properties. In the present thesis, only the bilinear model will be used with some modifications to consider the effects of tension stiffening, such as will be discussed in Section 3.4.1.

3.3.3 Development of reinforcing steel in Switzerland

High quality, cold-worked reinforcing steel has been developed and produced in Switzerland since before the Second World War. Further significant developments were again made in the 1950s and '60s [72]. During this time, the standards prescribing minimal requirements for the mechanical properties of reinforcement were frequently outdated. As a result, the steel producers typically published minimum guaranteed mechanical properties for design purposes. These were often prescribed in the form of allowable stresses, such as is summarised in Table 3.1. The most important producers of reinforcement during this time were: *Von Moos* and *Von Roll* (now *Swiss Steel* and *Stahl Gerlafingen*), producers of the cold-worked bars Tor-Stahl and Caron-Stahl; and *Monteforno*, producer of Box-Stahl and Box-Ultra, the only micro-alloyed steel at that time [21, 72]. Typical stress-strain characteristics of these steel types are illustrated in Fig. 3.8.

Initially, these high strength steels were either produced through micro-alloying or cold-working. The latter method was frequently implemented through a twisting of the bars such as in the production of Tor-Stahl. The primary reinforcement type used in Switzerland during the 1950s and 1960s was the micro-alloyed Box-Stahl and Box-Ultra. These products exhibit a significantly higher ratio of ultimate to yield strength as well as a higher ultimate strain compared to the cold-worked steels [72]. Many concrete structures in Switzerland were constructed with cold-worked steel; typically in the form of Caron-Stahl and Tor-Stahl [21, 24].

Currently, the use of hot-rolled bars is widely propagated [28]; whereas, the cold-worked product topar-Rc from *Stahl Gerlafingen* is commonly used for small diameter bars [21]. The majority of reinforcing steel used in Switzerland today is derived from scrap metal, which is predominantly produced by the company *Stahl Gerlafingen*. One of their commonly used steel types, topar-S, is produced with the Tempcore-process; a process which has been shown to result in reinforcement with superior fatigue strengths compared to other types of high strength steel [105].

3.3.4 Fatigue behaviour

In recent years, intensified worldwide research on the ultimate load-bearing capacity of structural concrete has led to the *limit state design* approach. This replaced the previous *working stress* or *permissible stress design* approach in which structural elements were designed such that stresses resulting from normal service loads do not exceed certain limits, typically specified as a proportion of the material strength. This new design approach

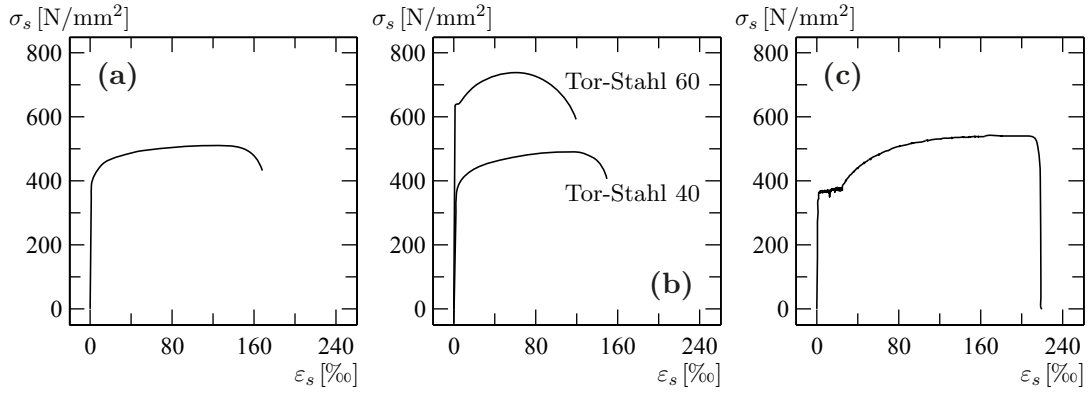


Fig. 3.8 – Stress-strain characteristics of reinforcement types produced in Switzerland during the 1950s and '60s: (a) Caron-Stahl; (b) Tor-Stahl; (c) Box-Stahl. Data from [24, 78].

allows a more efficient utilisation of materials. In reinforced concrete, this change in design philosophy, in combination with the development of high strength steel types, resulted in reduced reinforcement contents. The hidden reserves in the load-bearing capacity have been reduced to the extent that the still comparatively unpredictable hazard scenarios, such as fatigue, require more frequent verification. Herein, the fatigue strength of the reinforcing steel is often the critical parameter.

General considerations

The reinforcing steel bars used in structural concrete are susceptible to an accumulation of damage under repeated loading, which may result in fatigue fracture at sufficiently high stress levels. The fatigue strength of the reinforcement is typically determined from large series of constant amplitude cyclic loading tests. The stress range and corresponding number of load cycles to failure obtained from test series at different stress ranges are used to construct S-N diagrams for the reinforcement under consideration. The current standards prescribe fatigue strength curves, which are used for design purposes as well as in the evaluation of existing concrete structures.

Fatigue tests are either conducted as cyclic uniaxial tests of the bars in air or as bending tests with the bars embedded in the flexural tensile zone of concrete beams. The advantages of axial fatigue testing in air is that high frequencies of cyclic loading can be applied to investigate the fatigue strength in the long-life region and the fact that stresses can be calculated directly from the applied load. On the other hand, high transverse stresses are applied locally in the region of the grips, which often leads to premature failure of the specimen. In addition, slight misalignments in the testing machine can introduce unwanted secondary bending stresses, which are detrimental to the fatigue strength. Bending fatigue tests are typically conducted on either rectangular or T-shaped concrete beams containing a single bar of reinforcement. The application of loading on the reinforcement is more distributed as it is introduced through its bond with the concrete. Such tests can simulate

Production process	Steel type or product name	Produced from	f_{sy} [a] [N/mm^2]	f_{su} [a] [N/mm^2]
Cold-worked	Caron-Stahl	1948	350	410
	Caron-Stahl	1953	450	490
	Caron-Stahl	1965	450	490
	Tor-Stahl 40	1964	390[b]	490[b]
	Tor-Stahl 42	1966	410	460
	Tor-Stahl 50	1966	490	550
	B500A[c]	-	500	525
Micro-alloyed	Box-Stahl	1960	390	510
	Box-Ultra	1960	490	590
Hot-rolled or Tempcore	B500B[c]	-	500	540
	B500C[c]	-	500	575

[a] minimum values prescribed by producer.
[b] mean values.
[c] classification to Swiss standard SIA 262(2013) [87] indicated for comparison.

Tab. 3.1 – Mechanical properties of typical reinforcement types produced in Switzerland during the 1950s and '60s. Adapted from [72].

the conditions in a real structure more accurately. However, the loading frequency is limited due to the large actuator stroke required. Furthermore, the steel stress is calculated on the basis of an assumed concrete stress profile within the compression zone and is highly sensitive to an accurate placement of the bar.

The experimentally determined fatigue strength of reinforcing bars embedded in concrete is supposedly higher than in uniaxial tests in air. This is suspected to be due to a reduced probability of the critical defect on the bar surface coinciding with the highest stresses at the concrete cracks. However, fatigue tests in bending introduce additional effects that are difficult to quantify: a strain gradient results over the cross-section of the bar due to bending, whereas a uniform stressing of the cross-section is achieved in uniaxial tests; and a frictional action between concrete and steel arises due to bond slip. A masking of influences from the above-mentioned effects may explain why, according to [84], there are currently no conclusive results indicating a significant difference in the fatigue strength obtained from these two types of tests.

Previous investigations

During the 1960s and '80s, the fatigue strength of reinforcing steel was researched intensively [23]. This interest was initiated by new applications of structural concrete in cyclically loaded members such as railway sleepers, crane girders, highway and railway bridges and offshore structures. Furthermore, the development of new reinforcement types exhibiting higher tensile strengths enabled reduced reinforcement ratios such that higher

stress variations result under normal service loading.

Tilly provides an extensive review of previous work on fatigue of steel reinforcing bars in [96]. At around the same time, both *Helgason et al.* (1976) [34] and *Moss* (1980) [66] conducted fatigue tests on high yield strength reinforcement. In the former experimental campaign, a total of 353 concrete beams, each containing a single reinforcing bar, were tested under cyclic loading. A large number of test variables, such as stress range, bar diameter, static strength and geometry of the ribs, were investigated. None of the investigated variables were found to interact with the effect of stress range. The representation of fatigue data in an S-N diagram with the coordinates of stress range and number of load cycles to failure was, therefore, confirmed to be appropriate. The effect of cyclic loading on the remaining intact portions of bars after fatigue fracture had occurred was also investigated in uniaxial tests. Subsequent static uniaxial tests, conducted on these still intact lengths of bar, exhibited a reduced tensile strength as well as ultimate strain compared to tests on the same bars in their undamaged state. The bars typically failed at secondary fatigue cracks in the same brittle manner as the original fatigue fracture. *Helgason et al.* [34] concluded that once fatigue crack growth is initiated in the reinforcement, a sudden overload may cause brittle premature failure of the bar. *Moss* conducted uniaxial fatigue tests on both cold-worked and hot-rolled bars in air. He observed that fatigue crack initiation was commonly associated with small defects in the surface oxide layer. The rib geometry was found to have no significant influence on the fatigue strength.

Tilly [97] conducted tests with different types of small and larger diameter bars in air and as bending tests in concrete. He generally obtained higher fatigue strengths in the bending tests than in air. Fractographic inspection of the bars indicated that the critical fatigue crack under uniaxial loading initiated at surface defects, whereas these cracks typically formed at the root of the ribs when tested in bending. Furthermore, the larger diameter bars exhibited lower fatigue strengths compared to those of smaller diameter.

In 1984, *Fernández Canteli et al.* [25] conducted a large number of tests with different types of prestressing and reinforcing steel. The latter consisted of both cold-worked and micro-alloyed types from different producers in Switzerland. Significant differences were observed in the fatigue strengths between bars from different producers. A linear relation was proposed between stress range and number of load cycles to failure for the finite-life region, similar to a proposal by *Helgason et al.* [34], which appropriately describes a lower limit of fatigue strength in the S-N diagram for a large set of test results.

More recently, *Fehlmann and Vogel* (2009) [24] compared the fatigue strengths of older types of reinforcement used in Switzerland to that of currently used types. Both the micro-alloyed Box-Stahl and cold-worked bars of type Tor-Stahl 40 were removed from two bridges constructed in 1952 and 1955. No significant difference was found between the fatigue strengths of the old reinforcement types and those currently produced (comparison to types topar-R and BSW Tempcore) with the exception of some low values obtained for the micro-alloyed Box-Stahl.

Fatigue strength

The systematic statistical analysis of a large number of fatigue test results was performed by *Helgason et al.* [34]. In this way, the primary factors influencing the fatigue strength of reinforcing steel was ranked in decreasing order of statistical significance:

1. absolute stress range
2. minimum stress level
3. static strength
4. bar diameter
5. geometry of ribs

The following discussion is primarily based on the report by *Helgason et al.* [34] unless otherwise noted.

It is a widely confirmed observation that the fatigue strength of reinforcement in the finite-life region depends primarily on the level of *absolute stress range* $\Delta\sigma$ caused by the loading. This correlation is weaker at low levels of stress range in the long-life region as well as at high levels of stress range approaching the static strength of the steel. Stress-histories where both the minimum and maximum stress level is compressive do not lead to fatigue of the steel. The requirement for fatigue to occur is that either the maximum stress level or both the minimum and maximum stress is positive (tensile) [34].

The influence of the *minimum stress level* $\sigma_{s,\min}$ on fatigue strength is less clearly defined and subject to conflicting results from the literature. Generally, an increase in minimum stress level results in lower fatigue strengths. This reduction in fatigue strength appears to be of equal magnitude in both the finite-life and long-life regions.

The influence of reinforcing steel *static strength* on the fatigue strength has been inconclusive in many investigations [24, 29, 66, 97]. The results reported by *Helgason et al.* [34], however, do indicate an increase in fatigue strength at higher static strengths. The trend, however, was not regular and cannot be assumed to be generally true.

Bar diameter influences the fatigue strength of reinforcement in the sense that smaller diameter bars exhibit a higher fatigue strength than those of a larger diameter [66, 97]. This reduction in fatigue strength with increasing bar diameter was found to be especially significant in the lower stress range levels approaching the fatigue endurance limit [25]. This effect is commonly attributed to the additional working of the material required to produce smaller diameter bars as well as a statistical size effect. In the latter mentioned size effect, it is presumed that the probability of finding a fatigue-critical defect in a bar is proportional to the surface area of the bar. The influence of the bar diameter is recognised in the classification of fatigue strengths in current design standards, see Table 3.2. No clear distinction between "large" and "small" diameter bars can be made. The documentation to the Swiss standard SIA D 0133 [84] distinguishes between straight bars with $\varnothing_s \leq 20\,\text{mm}$ and $\varnothing_s > 20\,\text{mm}$. The *fib* Model Code [42] defines small diameter bars as those where $\varnothing_s \leq 16\,\text{mm}$. Due to a scarcity of fatigue test results for large diameter bars, only the curve for $\varnothing_s = 40\,\text{mm}$ is prescribed with the recommendation that the fatigue strength

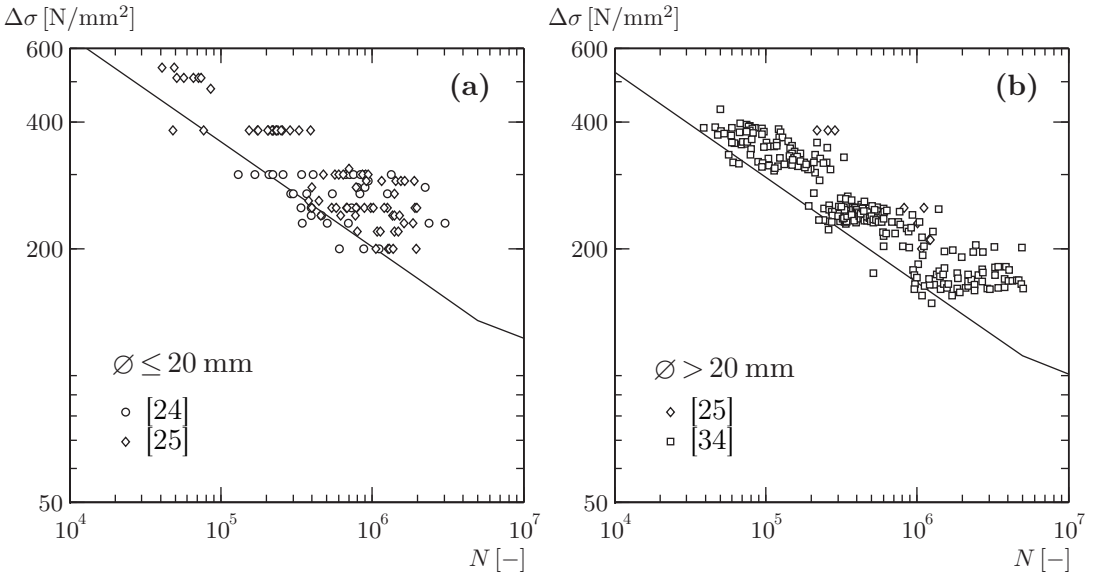


Fig. 3.9 – Fatigue strength of reinforcement: (a) smaller diameter bars; (b) larger diameter bars. Note: nominal fatigue strength curves indicated for each category to [84].

Design standard	$\varnothing_s$ [mm]	N^* [-]	$\Delta\sigma_{s,N^*}$ [N/mm^2]	m [-]	$N_{s,\mathrm{D}}$ [-]	$2m-1$ [-]
SIA D 0133 [84]	$\varnothing_s \leq 20\,\mathrm{mm}$	10^6	202.2	4	$5 \cdot 10^6$	7
	$20\,\mathrm{mm} < \varnothing_s \leq 40\,\mathrm{mm}$	10^6	166.5	4	$5 \cdot 10^6$	7
fib Model Code (2010) [42]	$\varnothing_s \leq 16\,\mathrm{mm}$	10^6	210.0	5	$1 \cdot 10^6$	9
	$\varnothing_s = 40\,\mathrm{mm}$	10^6	160.0	5	$1 \cdot 10^6$	9
EN 1992-1-1:2004 [85]	–	10^6	162.5	5	$5 \cdot 10^6$	9

Tab. 3.2 – Parameters describing the fatigue strength curves for straight reinforcing steel bars.

for bars between these limits can be interpolated linearly. EN 1992-1-1:2004 [85] does not recognise an improvement in the nominal fatigue strength for small diameter bars.

The *ribs* on deformed reinforcing steel bars, designed to improve bond properties, are known to have a detrimental effect on the fatigue strength. Significantly lower fatigue strengths have been observed in bars with ribs compared to smooth specimens machined from the same steel [34]. The abrupt changes in cross-section associated with the geometry of the ribs cause a concentration of fatigue-inducing stresses on the bar surface. The stresses at these so-called *hot-spots* may be a multiple of the nominal stress acting on the cross-section. Stress concentration factors, defined as the ratio of the maximum elastic stress at the root of a transverse rib to the nominal stress, as high as 1.82 have been measured experimentally with strain gauges and verified with numerical results [47]. The sharpness of the rib base radius, or ratio of rib base radius to overall rib height, was shown to be the most critical parameter. The geometry of the ribs with respect to the rib width and

the angle of transverse ribs with respect to the longitudinal bar axis were also shown to influence the fatigue strength.

Finally, the results of a total of 375 fatigue tests on reinforcing steel bars (run-outs were not considered) were combined into a single database. These results are based on the tests conducted by *Helgason et al.* [34], *Fernández Canteli et al.* [25] and *Fehlmann and Vogel* [24]. The reinforcement tested varied with respect to the age of bars, originated from different producers, some were produced with cold-working whereas others were micro-alloyed or hot-rolled and they exhibited a wide range of different rib geometries. Fig. 3.9 represents the test results in S-N diagrams for small and large diameter bars using the definition according to [84]. For comparison, the corresponding fatigue strength curves according to SIA D 0133 are indicated with the nominal fatigue strengths as described in Table 3.2.

3.4 Interaction of concrete and reinforcement

The majority of the following description of the static behaviour of bond is based on the dissertations of *Alvarez* [3] and *Kaufmann* [49]. The *CEB* state of the art report [98] provides an overview of the fatigue behaviour of bond, whereas *Lindorf* [55] describes more recent work on bond under repeated loads.

3.4.1 Static behaviour of bond

General considerations

The relative displacement $\delta = u_s - u_c$ between concrete and steel due to bond slip, where u_s and u_c denote the steel and concrete displacements, activates bond stresses at the interface between the reinforcement and the concrete. Bond enables a transfer of forces between the reinforcing bars and the surrounding concrete and is imperative for the anchorage of bars. Moreover, bond controls the width as well as the spacing of the concrete cracks and increases the stiffness of cracked reinforced concrete members in direct tension or bending by activating the intact concrete between cracks. Bond stresses arise primarily from a mechanical interlocking between the ribs of profiled reinforcing bars and the surrounding concrete. Further factors that have an influence on bond properties are concrete strength, position and orientation of the bar during casting, concrete cover to the reinforcement and the stress state in the concrete and reinforcement [42].

Bond stress-slip relationships τ_b-δ_b are typically obtained from pull-out tests in which a reinforcing bar is embedded over a short length[2] within a concrete prism and loaded in tension. A significant initial increase in bond stress is observed in such tests with only marginal slip. The response, however, is highly nonlinear due to the progressive deterioration of the concrete bearing against the ribs of the reinforcement, a smoothing of the friction surface and the formation of secondary cracks [3]. The ultimate bond shear

[2]The embedment length l_b in pull-out tests is typically selected in the range of $2 \ldots 5\varnothing$.

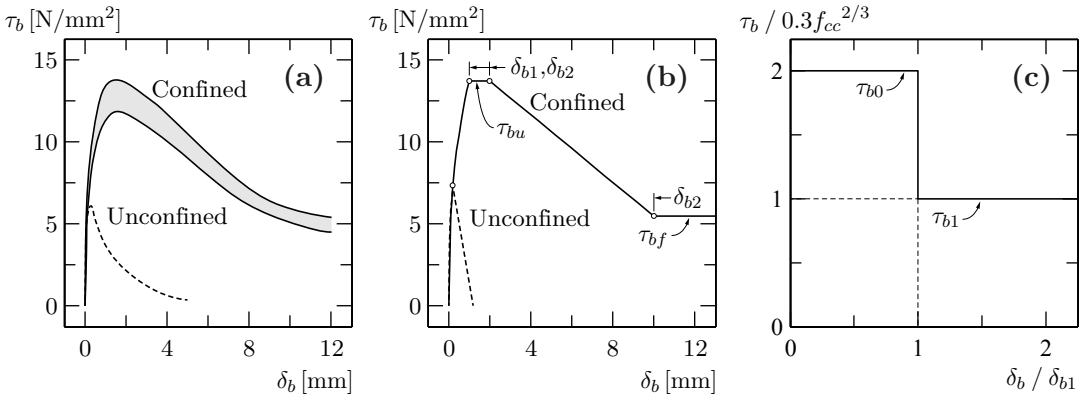


Fig. 3.10 – Bond stress-slip characteristic of reinforcing bars in concrete: (a) influence of confinement of concrete, data from [19]; (b) to *fib* Model Code [42] for confined concrete with pull-out type failure and unconfined concrete with splitting failure; (c) stepped, rigid-perfectly plastic relationship to [90].

stress or bond strength τ_{bu} is typically reached at a slip in the order of $0.5 \ldots 1.0$ mm [49]. The bond stress decreases thereafter with increasing slip, the rate of which depends on the mode of failure. Confinement of the concrete through the placement of stirrups or transverse reinforcement results in a significant improvement of bond strength. On the other hand, an insufficient concrete cover to the reinforcement can lead to premature splitting failure.

Bond stress-slip relationship

The *fib* Model Code [42] recommends a local bond stress-slip relationship for profiled reinforcing bars according to the expected failure mode, namely *pull-out* failure for well-confined concrete or *splitting* failure for unconfined concrete. Bond stresses are calculated according to the relation established by *Eligehausen et al.* [19] as

$$\begin{aligned}
\tau_b &= \tau_{bu} \left(\frac{\delta_b}{\delta_{b1}} \right)^{0.4} && \text{for} \quad 0 \le \delta_b \le \delta_{b1} \\
\tau_b &= \tau_{bu} && \text{for} \quad \delta_{b1} < \delta_b \le \delta_{b2} \\
\tau_b &= \tau_{bu} - (\tau_{bu} - \tau_{bf}) \frac{(\delta_b - \delta_{b2})}{(\delta_{b3} - \delta_{b2})} && \text{for} \quad \delta_{b2} < \delta_b \le \delta_{b3} \\
\tau_b &= \tau_{bf} && \text{for} \quad \delta_{b3} < \delta_b
\end{aligned} \tag{3.15}$$

with $\tau_{bu} = 2.5 {f_{cc}}^{1/2}$ and $\tau_{bf} = 0.4\tau_{bu}$. The slip boundaries are defined according to the expected failure mode and the bond conditions. Fig. 3.10 (b) illustrates the implementation of Eq. (3.15) for confined and unconfined concrete, calculated using the concrete compressive strength reported by *Eligehausen et al.* in [19], see Fig. 3.10 (a).

Sigrist [90] established a stepped, rigid-perfectly plastic bond stress-slip relationship as illustrated in Fig. 3.10 (c) The initial value of bond stress τ_{b0} is assumed to remain constant until the bond slip $\delta_b = \delta_{b1}$ is reached corresponding to onset of yielding of the

reinforcement. *Shima et al.* [89] observed that the bond stress in the post-yield range was significantly lower than in the elastic range; the reduction occurring suddenly upon yielding. After the onset of yielding, a constant value of bond stress $\tau_{b1} = 0.5\tau_{b0}$ is assumed. This relationship is expressed as

$$\begin{aligned} \tau_b &= \tau_{b0} = 0.6{f_{cc}}^{2/3} \quad \text{for} \quad 0 \leq \delta_b \leq \delta_{b1} \\ \tau_b &= \tau_{b1} = 0.3{f_{cc}}^{2/3} \quad \text{for} \quad \delta_{b1} < \delta_b \end{aligned} \tag{3.16}$$

where $\delta_{b1} = \delta_b\,(\sigma_s = f_{sy})$. Although this simple formulation represents a considerable simplification of the real bond stress-slip relationship, it facilitates the analytical derivation of the distribution of bond stress as well as steel and concrete stresses. The proposed relationship was implemented in the so-called *tension chord model* and provides promising results compared to experimental data.

Tension Chord Model

The Tension Chord Model (TCM), established by *Marti et al.* [61], enables a consistent description of the load-carrying behaviour of cracked reinforced concrete elements; retaining its validity from initial crack formation to rupture of the reinforcement. The TCM is a purely mechanical model in which the only degrees of freedom are the choice of bond stress-slip relationship, typically using the formulation in Eq. (3.16), and the assumed crack spacing [62]. The model provides an analytical means for describing the increased stiffness of cracked reinforced concrete elements under tensile load as opposed to that of the reinforcing bar alone; referred to as *tension stiffening*. This effect is attributed to the bond interaction between reinforcement and the surrounding intact concrete.

A reinforced concrete chord element between two consecutive cracks, which is subjected to a constant tensile force such that yielding of the reinforcement occurs is considered. The distribution of bond stress, steel and concrete stresses and strains as well as the bond slip along the length of the bar result as indicated in Fig. 3.11. The concrete stress is zero at the crack where its tensile strength has been exceeded; the load is resisted exclusively by the reinforcement. As a result, the steel stress reaches a maximum value at the crack and diminishes with increasing distance away from the crack due to a transfer of stress to the concrete through bond. A change of sign in bond stress is observed between consecutive cracks, taking place at the midpoint between the cracks for equal tensile forces at both ends of the element [49]. The slip at this point is zero, whereas the concrete stress reaches a maximum value. The distribution of these parameters between two cracks can be determined with assumed stress-strain relationships for steel and concrete as well as the bond stress-slip relationship. Such calculations, however, are demanding and generally require iterative numerical methods. The use of linear or bilinear material models and the stepped, rigid-perfectly plastic bond stress-slip relationship proposed by *Sigrist*, enables an analytical solution with the TCM, resulting directly from equilibrium considerations [88].

The maximum crack spacing s_{r0} can subsequently be derived from a summation of bond stress over the length of the cracked chord element. Recognising that the concrete

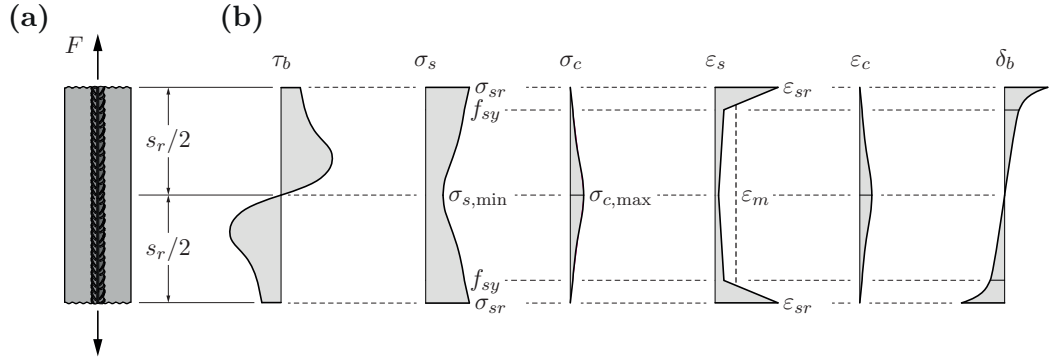


Fig. 3.11 – Tension stiffening: (a) chord element between consecutive cracks; (b) distribution of bond stress, steel and concrete stresses and strains, and bond slip. Redrawn from [88].

tensile stress cannot exceed its tensile strength $\sigma_c \leq f_{ct}$, this results in the expression

$$s_{r0} = \frac{f_{ct} \varnothing}{2\tau_{b0}} \frac{(1-\rho)}{\rho} \tag{3.17}$$

where ρ denotes the geometric reinforcement content in terms of the cross-sectional area of the reinforcing bar to that of the gross concrete chord. The actual crack spacing s_r, however, can vary freely between this maximum value $s_r = s_{r0}$ and a lower boundary of $s_r = s_{r0} \,/\, 2$, which implies the formation of a further crack. For this reason, the factor λ is introduced such that

$$s_r = \lambda \cdot s_{r0} \quad \text{where} \quad 0.5 \leq \lambda \leq 1.0 \tag{3.18}$$

where s_r denotes the actual or selected crack spacing. In reality, however, the crack spacing is often controlled by the position of transverse reinforcement as the concrete is disrupted at such cross-sections.

The maximum steel stress resulting at the crack σ_{sr} can be calculated for a given average strain ε_m which depends on the level of applied tensile force. Under loading conditions where the steel stress remains below the yield strength of the reinforcement along the entire length of the chord element $\sigma_{sr} \leq f_{sy}$, the maximum steel stress is calculated as

$$\sigma_{sr}\left(\varepsilon_m\right) = E_s \varepsilon_m + \frac{\tau_{b0} s_r}{\varnothing_s} \tag{3.19}$$

and for a steel stress, which is partially higher than as well as lower than the yield strength $\sigma_{s,\min} \leq f_{sy} \leq \sigma_{sr}$ the maximum steel stress is expressed as

$$\sigma_{sr}\left(\varepsilon_m\right) = f_{sy} + 2\frac{\frac{\tau_{b0} s_r}{\varnothing_s} - \sqrt{\left(f_{sy} - E_s\varepsilon_m\right)\frac{\tau_{b1} s_r}{\varnothing_s}\left(\frac{\tau_{b0}}{\tau_{b1}} - \frac{E_s}{E_{sh}}\right) + \frac{E_s}{E_{sh}}\tau_{b0}\tau_{b1}\frac{{s_r}^2}{{\varnothing_s}^2}}}{\frac{\tau_{b0}}{\tau_{b1}} - \frac{E_s}{E_{sh}}}, \tag{3.20}$$

whereas for steel stresses exceeding the yield strength of the reinforcement along the entire chord element $f_{sy} \leq \sigma_{s,\min}$

$$\sigma_{sr}\left(\varepsilon_m\right) = f_{sy} + E_{sh}\left(\varepsilon_m - \frac{f_{sy}}{E_s}\right) + \frac{\tau_{b1} s_r}{\varnothing_s}. \tag{3.21}$$

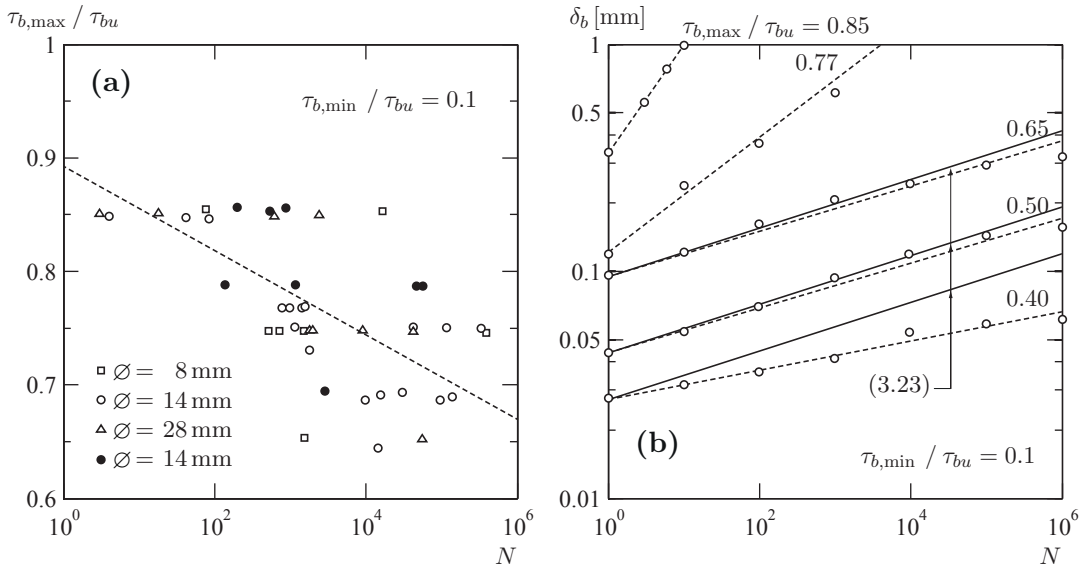


Fig. 3.12 – Bond under repeated loading: (a) S-N diagram of bond with $f_{cc} = 23.5\,\text{N/mm}^2$ except for filled markers in which $f_{cc} = 48.0\,\text{N/mm}^2$; (b) bond slip at different maximum load levels, with $f_{cc} = 23.5\,\text{N/mm}^2$, $\varnothing = 14\,\text{mm}$ and $l_b = 3\varnothing$. Adapted from [76].

The crack width w_r can be expressed as the sum of all slip between concrete and reinforcement occurring within half of two adjacent chord elements on either side of a crack. An approximation of crack width can be expressed as

$$w_r\,(\lambda) = s_r \varepsilon_m \tag{3.22}$$

under the assumption that the concrete deformation is negligible.

3.4.2 Fatigue behaviour of concrete-steel bond

Cyclic loading results in a progressive increase in bond slip between the reinforcement and the surrounding concrete which may cause failure at bond stress levels lower than the static bond strength [42]. The slip increments under repeated loading decrease during the initial load repetitions and remain roughly constant thereafter until the slip approaches the value at which the monotonic bond strength is reached $\delta_b\,(\tau_{bu})$ [7]. Further load cycles lead to a rapid increase in slip and subsequent fatigue failure of bond through pull-out of the reinforcing bar.

Rehm and Eligehausen [76] observed that the number of load repetitions to bond failure increases with decreasing maximum load for constant minimum lower load levels, see Fig. 3.12 (a). Fatigue test results represented in a S-N diagram can be described by a straight line and exhibit the scatter typically associated with fatigue testing. No significant influence of the bar diameter or the concrete strength has been observed on the fatigue strength of bond [76]. Repeated loading results in a "softening" of the bond relationship, indicated by an increase in bond slip [7]. Fig. 3.12 (b) indicates the increase in slip under repeated loading. The main parameters influencing the extent of slip are the upper load

level and the bond length used in tests. The increase in bond slip with load repetitions, in which bond failure did not occur, is described by approximately parallel lines in a double logarithmic representation. At high upper load levels leading to fatigue failure, the gradient of these lines was higher. From their observations on a large number of fatigue pull-out type tests, *Rehm and Eligehausen* [76] proposed an expression to approximate the expected slip after a number of load repetitions in the form

$$\delta_b\,(N) = \delta_{b0}(1+N)^{0.107} - 1 \tag{3.23}$$

where δ_{b0} denotes the initial slip value, as recommended in *fib* Model Code [42].

Chapter 4

Membrane and plate elements

4.1 General

In preparation for the description of the proposed model in Chapter 5, the kinematic and static relations governing the load-deformation behaviour of membrane and plate elements will be introduced. The *cracked membrane model*, an approach for the analysis of cracked reinforced concrete membranes in a state of plane stress, will be presented. Finally, the analysis of reinforced concrete plate elements will be described on the basis of *classical laminate theory*. The concept of a layered element formulation will be introduced. The present chapter is limited to the essential theoretical background and relationships relevant to this work. Much of the following outline of the kinematic and static relationships is based on the dissertation of *Seelhofer* [88], which deals extensively with plane stress problems in structural concrete. The description of the Cracked Membrane Model is based primarily on the dissertation of *Kaufmann* [49].

The following element definition will be adopted throughout this thesis: *membranes* or *panels* are elements subjected purely to in-plane forces such that $m_x = m_y = m_{xy} = 0$ and $v_x = v_y = 0$; *slabs* are elements subjected to pure bending $n_x = n_y = n_{xy} = 0$; and *plates* are elements subjected to a combination of both membrane forces and bending moments. Furthermore, the tensorial convention for stresses and stress resultants will be used. Positive stresses are defined as those acting in the direction of the positive coordinates on boundaries with a positive outer normal. In the case of normal stresses, therefore, tension is by definition positive. Positive membrane forces are those corresponding to positive normal stresses, whereas positive bending and torsional moments are those resulting in positive stresses for positive values of the z-coordinate. In variables with double subscript indices such as a_{ij}, the first index denotes the direction and the second refers to the normal direction of the boundary on which it is acting. The subscript indices i and j typically refer to generalised directions, whereas the subscript k is reserved to indicate individual layers of reinforcement.

4.2 Membrane elements

4.2.1 Kinematic relationships

The strain state in a membrane element is defined by the normal strains ε_x, ε_y and the shear strain γ_{xy}. Considering an infinitesimal element with side lengths of dx and dy and the displacements u_x and u_y in the x- and y-directions, as shown in Fig. 4.1 (a), the kinematic relationships can be formulated as

$$\varepsilon_x = \frac{\partial u_x}{\partial x} = u_{x,x}, \quad \varepsilon_y = \frac{\partial u_y}{\partial y} = u_{y,y}, \quad \gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = u_{x,y} + u_{y,x} \tag{4.1}$$

with shear strains $\gamma_{xy} = \gamma_{yx} = 2\varepsilon_{xy} = 2\varepsilon_{yx}$. The strain tensor is therefore symmetrical and can be written in contracted, vector form as $\varepsilon_j = [\varepsilon_x, \varepsilon_y, \gamma_{xy}]$. The strain state is defined directly for given displacements; whereas, obtaining the displacements from a given strain state generally requires an iterative approach. Strains in the generalised n-t-z coordinate system, orientated at an angle φ with respect to the x-axis, can be obtained through a rotation of the strain vector with the transformation relationship

$$\begin{bmatrix} \varepsilon_n \\ \varepsilon_t \\ \gamma_{nt} \end{bmatrix} = \begin{bmatrix} \cos^2\varphi & \sin^2\varphi & \sin\varphi\cos\varphi \\ \sin^2\varphi & \cos^2\varphi & -\sin\varphi\cos\varphi \\ -2\sin\varphi\cos\varphi & 2\sin\varphi\cos\varphi & (\cos^2\varphi - \sin^2\varphi) \end{bmatrix} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix}. \tag{4.2}$$

The direction of principal strain, defined by the angle φ_1 at the orientation where the shear strain becomes zero $\gamma_{nt} = \gamma_{tn} = 0$, and the corresponding principal strains $\varepsilon_{1,2}$ result from

$$\varphi_1 = \frac{1}{2}\arctan\left(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}\right) \tag{4.3}$$

$$\varepsilon_{1,2} = \frac{1}{2}\left[\varepsilon_x + \varepsilon_y \pm \sqrt{(\varepsilon_x - \varepsilon_y)^2 + {\gamma_{xy}}^2}\right] \tag{4.4}$$

which can be represented graphically on *Mohr's* circle for strains, see Fig. 4.1 (b). The angle β describing the direction of the smaller principal strain ε_2 is expressed as

$$\begin{aligned} \beta &= \frac{1}{2}\operatorname{arccot}\left(\frac{\varepsilon_y - \varepsilon_x}{\gamma_{xy}}\right) && \text{for} \quad \gamma_{xy} \geq 0 \\ \beta &= \frac{1}{2}\operatorname{arccot}\left(\frac{\varepsilon_y - \varepsilon_x}{\gamma_{xy}}\right) + \frac{\pi}{2} && \text{for} \quad \gamma_{xy} \leq 0 \end{aligned} \tag{4.5}$$

using the notation of *Seelhofer* [88]. A positive shear strain $\gamma_{xy} \geq 0$ results in $[0 \leq \beta \leq \pi/2]$ and a negative shear strain $\gamma_{xy} \leq 0$ results in $[\pi/2 \leq \beta \leq \pi]$.

4.2.2 Static relationships

Similar to the kinematic relationships, the stress state in a membrane element is defined by the symmetrical stress tensor of normal stresses σ_x, σ_y and shear stress τ_{xy}. In vector

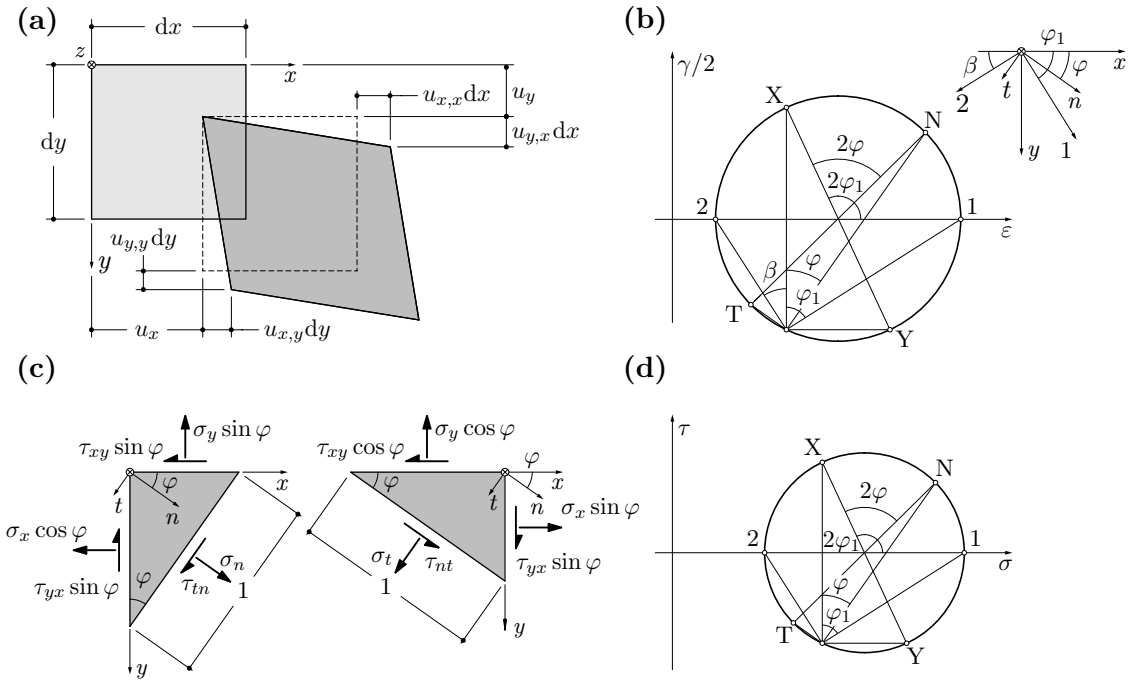


Fig. 4.1 – Kinematic and static relationships for membrane elements: (a) displacement and distortion; (b) *Mohr's* circle for strains; (c) stress transformation; (d) *Mohr's* circle for stresses.

form, the stress tensor can be written as $\sigma_i = [\sigma_x, \sigma_y, \tau_{xy}]$. Generalised stresses in the n-t-z coordinate system, rotated at an angle φ with respect to the x-axis, result directly from equilibrium considerations, see Fig. 4.1 (c). The same transformation relationship as defined in Eq. (4.2) applies such that

$$\begin{bmatrix} \sigma_n \\ \sigma_t \\ \tau_{tn} \end{bmatrix} = \begin{bmatrix} \cos^2\varphi & \sin^2\varphi & 2\sin\varphi\cos\varphi \\ \sin^2\varphi & \cos^2\varphi & -2\sin\varphi\cos\varphi \\ -\sin\varphi\cos\varphi & \sin\varphi\cos\varphi & \left(\cos^2\varphi - \sin^2\varphi\right) \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} . \tag{4.6}$$

The direction of principal stress, defined by the angle φ_1, describes the orientation where the shear stress becomes zero $\tau_{xy} = \tau_{yx} = 0$ and the corresponding normal stresses reach their maximum and minimum values, namely, the principal stresses $\sigma_{1,2}$. The graphical representation on *Mohr's* circle for stresses, see Fig. 4.1 (d), enables a derivation of the direction of principal stress and corresponding principal stresses as

$$\varphi_1 = \frac{1}{2}\arctan\left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right) \tag{4.7}$$

$$\sigma_{1,2} = \frac{1}{2}\left[\sigma_x + \sigma_y \pm \sqrt{(\sigma_x - \sigma_y)^2 + 4{\tau_{xy}}^2}\right] . \tag{4.8}$$

4.2.3 Analysis of reinforced concrete membranes

Fundamentals of compression field approaches

The *classical compression field* approach enables the analysis of cracked reinforced concrete membranes by satisfying the compatibility (kinematic) requirements and equilibrium (static) conditions. Complete load-deformation analyses can be performed prior to and after the onset of yielding of the reinforcement until failure is predicted. The load-carrying behaviour of the uncracked element is similar to that of unreinforced concrete. At principal stresses exceeding the concrete tensile strength, cracks form at an orientation roughly perpendicular to the direction of principal tensile stress. A redistribution of forces occurs at this point, which is generally accompanied by a change in the direction of principal stress [62]. This direction remains constant thereafter under increasing load provided that sufficient nominal reinforcement is present at the point where either the concrete crushes or after the onset of yielding in one of the reinforcement directions [62].

Most classical compression field approaches consider fictitious, free-rotating cracks with infinitely small crack spacing and that the cracks are stress-free, forming parallel to the principal concrete compressive direction [49]. Moreover, cracks are assumed to open perpendicular to their own orientation. These assumptions imply that the principal compressive direction of the concrete stress coincides with the principal compressive direction of the concrete strains [49]. Furthermore, the assumption that the crack spacing is infinitely small implies a uniform, uniaxial compressive state for the concrete between adjacent cracks [88].

Stresses in cracked concrete membranes

Fig. 4.2 (a) shows an orthogonally reinforced concrete membrane subjected to normal and shear stresses with a resulting set of parallel, uniformly spaced cracks at an inclination θ_r with respect to the x-axis. Considering equilibrium of the externally applied stresses and those activated in the element at the cracks results in the conditions

$$\begin{aligned} \sigma_x &= \rho_x \sigma_{sxr} + \sigma_{cnr} \sin^2 \theta_r + \sigma_{ctr} \cos^2 \theta_r - \tau_{ctnr} \sin(2\theta_r) \\ \sigma_y &= \rho_y \sigma_{syr} + \sigma_{cnr} \cos^2 \theta_r + \sigma_{ctr} \sin^2 \theta_r + \tau_{ctnr} \sin(2\theta_r) \\ \tau_{xy} &= (\sigma_{cnr} - \sigma_{ctr}) \sin \theta_r \cos \theta_r - \tau_{ctnr} \cos(2\theta_r) \end{aligned} \tag{4.9}$$

where σ_{sxr} and σ_{syr} denote the reinforcement stresses at the cracks in the x- and y-directions, σ_{cnr} and σ_{ctr} represent the concrete stresses at the cracks and τ_{ctnr} is the shear stress activated at the crack. Eq. (4.9) allows a complete description of the stress and strain states in a cracked element for known values of the crack inclination θ_r, the crack spacing s_r and for selected constitutive relationships for the concrete, reinforcement as well as the bond stress-slip relationship [49]. In general, rather than considering interlocked cracks through complicated, empirical aggregate-interlock relationships, the cracks are assumed to be stress-free such that $\sigma_{cnr} = \tau_{ctnr} = 0$. Furthermore, the concrete compressive stress parallel to the crack direction is assumed to be equal to the principal concrete compressive stress $\sigma_{ctr} = \sigma_{c2r}$.

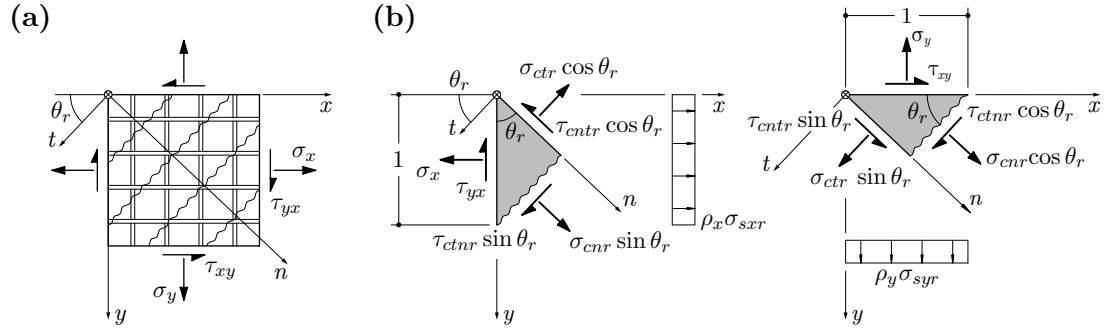


Fig. 4.2 – Reinforced concrete membrane element: (a) notation; (b) equilibrium at cracks.

Cracked Membrane Model

The Cracked Membrane Model (CMM), developed by *Kaufmann* and *Marti* [49, 50], enables the analysis of cracked, orthogonally reinforced concrete panels in a homogeneous state of plane stress. The model constitutes a combination of compression field approaches and the Tension Chord Model, see Section 3.4.1, extended to membrane elements. The major advantage of this formulation lies in the calculation of crack spacings and steel stresses from equilibrium conditions, and thus considering the tension stiffening effect of the concrete. Furthermore, by considering equilibrium of the stresses at the cracks, instead of the use of average stresses between cracks, the connection to limit analysis is maintained [49].

Similar to the classical compression field approaches, the underlying assumptions of the model imply free-rotating, stress-free cracks that form perpendicular to the principal tensile direction of the average strains. Hence, the angle of principal compressive strain coincides with the crack direction $\beta = \theta_r$ and no concrete stresses are transferred across the cracks such that $\sigma_{cnr} = \tau_{ctnr} = 0$.

Fig. 4.3 (a) shows a cracked membrane element for which the generalised equilibrium conditions of the compression field approaches, presented in Eq. (4.9), simplify to

$$\begin{aligned} \sigma_x &= \rho_x \sigma_{sxr} + \sigma_{c2r} \cos^2 \theta_r \\ \sigma_y &= \rho_y \sigma_{syr} + \sigma_{c2r} \sin^2 \theta_r \\ \tau_{xy} &= -\sigma_{c2r} \sin \theta_r \cos \theta_r \end{aligned} \tag{4.10}$$

assuming that $\sigma_{c2r} = \sigma_{ctr}$ and that cracking of the concrete in the principal tensile direction results in $\sigma_{c1r} = \sigma_{cnr} = 0$. A graphical representation of these conditions is illustrated on *Mohr's* circle for stress in Fig. 4.3 (c).

The compatibility requirement relating the average strains in the element to the crack direction can similarly be derived from *Mohr's* circle for strains, see Fig. 4.3 (b), resulting in the expression

$$\cot^2 \theta_r = \frac{\varepsilon_y - \varepsilon_2}{\varepsilon_x - \varepsilon_2}. \tag{4.11}$$

In the original formulation [49], a parabolic stress-strain relationship for concrete in uniaxial compression, see Eq. (3.5), is implemented to model the concrete compression

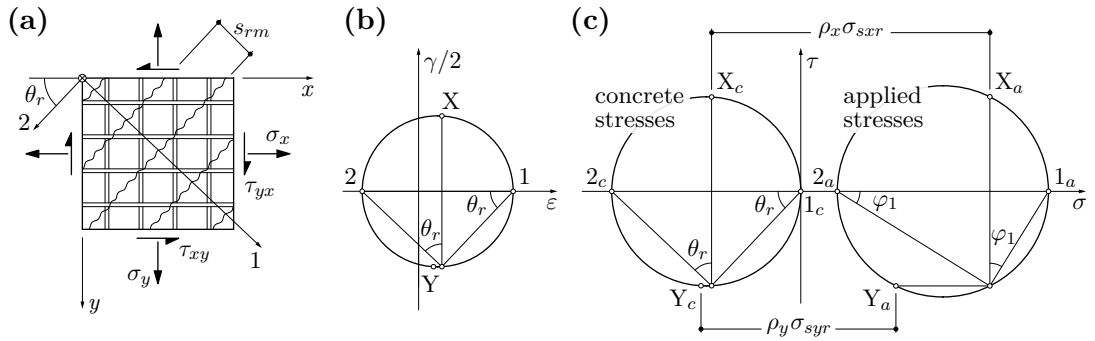


Fig. 4.3 – Cracked membrane element: (a) notation; (b) *Mohr's* circle for strains; (c) *Mohr's* circle for stresses. Redrawn from [49].

struts parallel to the cracks. A reduction in peak compressive concrete strength due to the compression-softening effect of strains perpendicular to the principal compressive direction is considered, see Section 5.3. Furthermore, a bilinear stress-strain relationship is selected for the reinforcing steel considering tension stiffening effects, see Eqs. (3.19), (3.20) and (3.21). An approximation of the maximum diagonal crack spacing s_{r0} for an orthogonal reinforcement layout is proposed in [49] as

$$s_{r0} = \left(\frac{\sin\theta_r}{s_{rx0}} + \frac{\cos\theta_r}{s_{ry0}} \right)^{-1} \tag{4.12}$$

and the actual crack spacing s_r follows from multiplication with the factor λ, varying in the range $0.5 \leq \lambda \leq 1$. The maximum crack spacings in the x- and y-directions are determined independently, on the basis of the Tension Chord Model, see Eq. (3.17).

The determination of complete load-deformation response curves generally requires a numerical method or iterative approach. The solution procedure involves solving four primary unknowns, namely the strains ε_x, ε_y and ε_2 as well as the crack inclination θ_r with the three equilibrium conditions in Eq. (4.10) and the compatibility requirement in Eq. (4.11). For given constitutive relationships, all terms on the right-hand side of Eq. 4.10 can be expressed in terms of the unknown strains. Hence, the strain state can be determined for any given set of applied stress components. The calculated load-deformation response is shown to be in good agreement with experimental results from a large number of tests on orthogonally reinforced concrete panels [49].

4.3 Plate elements

4.3.1 Kinematic relationships

In plate elements subjected to membrane forces and bending action, line elements normal to the mid-plane are assumed to remain straight and normal to the mid-plane during deformation. As a result, the variation of strains over the element thickness can be described

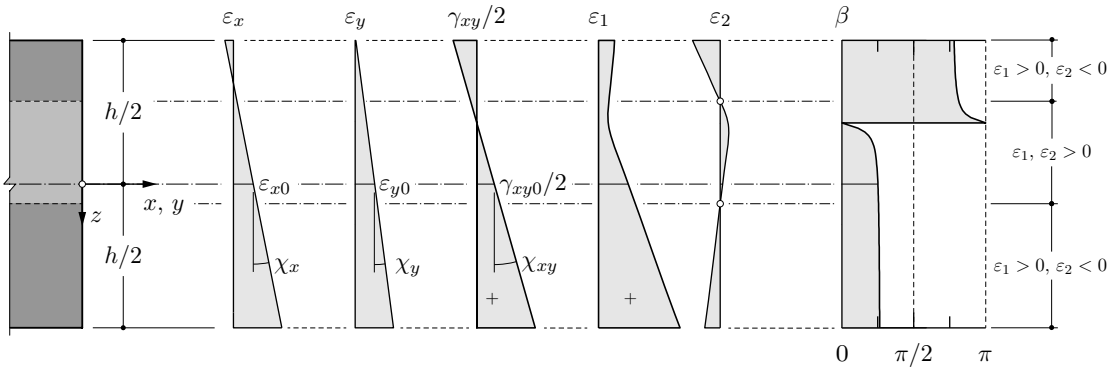


Fig. 4.4 – Typical distribution of normal, shear and principal strains as well as the direction of the smaller principal strain over the element thickness. Adapted from [88].

by only six variables, namely, the normal and shear strains at the mid-plane of the plate $\varepsilon_{0j} = [\varepsilon_{x0},\, \varepsilon_{y0},\, \gamma_{xy0}]$ and the components of curvature $\chi_j = [\chi_x,\, \chi_y,\, \chi_{xy}]$ such that

$$\varepsilon_x(z) = \varepsilon_{x0} + z \cdot \chi_x, \quad \varepsilon_y(z) = \varepsilon_{y0} + z \cdot \chi_y, \quad \gamma_{xy}(z) = \gamma_{xy0} + z \cdot 2\chi_{xy} \tag{4.13}$$

as illustrated in Fig. 4.4. The abovementioned assumption, however, implies a shear-stiff theory corresponding to the *Euler-Bernoulli* formulation for beams, wherein the influence of shear forces on the plate deformation is neglected.

In general, the principal strains and the corresponding principal direction will also vary over the element thickness. The principal strains, expressed in terms of the z-coordinate with respect to the mid-plane, result from the substitution of Eq. (4.13) into (4.4)

$$\varepsilon_{1,2}\,(z) = \frac{1}{2}\left\{\varepsilon_{x0} + z\chi_x + \varepsilon_{y0} + z\chi_y \pm \sqrt{\left[\varepsilon_{x0} + z\chi_x - (\varepsilon_{y0} + z\chi_y)\right]^2 + (\gamma_{xy0} + 2z\chi_{xy})^2}\right\} \tag{4.14}$$

whereas the variation of the angle β, describing the direction of smaller principal strain, results from the substitution of Eq. (4.13) into (4.5)

$$\begin{aligned} \beta\,(z) &= \frac{1}{2}\,\mathrm{arccot}\left[\frac{\varepsilon_{y0} + z\chi_x - (\varepsilon_{x0} + z\chi_y)}{\gamma_{xy0} + 2z\chi_{xy}}\right] && \text{for} \quad (\gamma_{xy0} + 2z\chi_{xy}) \geq 0 \\ \beta\,(z) &= \frac{1}{2}\,\mathrm{arccot}\left[\frac{\varepsilon_{y0} + z\chi_x - (\varepsilon_{x0} + z\chi_y)}{\gamma_{xy0} + 2z\chi_{xy}}\right] + \frac{\pi}{2} && \text{for} \quad (\gamma_{xy0} + 2z\chi_{xy}) \leq 0. \end{aligned} \tag{4.15}$$

4.3.2 Static relationships

The *stress resultants* or bending and torsional moments $m_i = [m_x,\, m_y,\, m_{xy}]$ and membrane forces $n_i = [n_x,\, n_y,\, n_{xy}]$ acting on a plate are determined by integration of the stress components over the element thickness, see Fig. 4.5 (a). The bending and torsional moments per unit length correspond to the curvatures

$$m_x = \int_{-h/2}^{h/2} \sigma_x \cdot z\,\mathrm{d}z, \quad m_y = \int_{-h/2}^{h/2} \sigma_y \cdot z\,\mathrm{d}z, \quad m_{xy} = m_{yx} = \int_{-h/2}^{h/2} \tau_{xy} \cdot z\,\mathrm{d}z \tag{4.16}$$

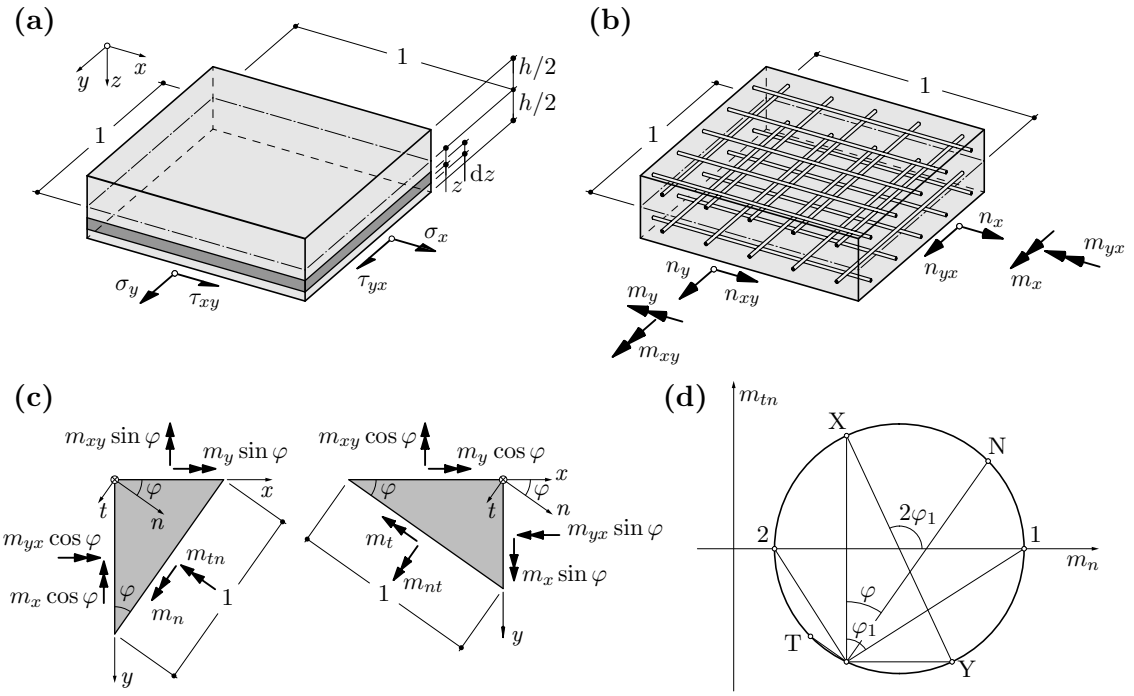


Fig. 4.5 – Plate element: (a) normal and shear stress components; (b) stress resultants (not considering shear forces); (c) moment transformation; (d) *Mohr's* circle representing bending and torsional moments per unit length.

whereas the membrane forces per unit length result from the mid-plane strains

$$n_x = \int_{-h/2}^{h/2} \sigma_x \, \mathrm{d}z, \qquad n_y = \int_{-h/2}^{h/2} \sigma_y \, \mathrm{d}z, \qquad n_{xy} = n_{yx} = \int_{-h/2}^{h/2} \tau_{xy} \, \mathrm{d}z. \tag{4.17}$$

The transformation relationship to determine the generalised bending moments in the *n-t-z* coordinate system is derived from equilibrium considerations as illustrated in Fig. 4.5 (c), such that

$$\begin{bmatrix} m_n \\ m_t \\ m_{nt} \end{bmatrix} = \begin{bmatrix} \cos^2\varphi & \sin^2\varphi & 2\sin\varphi\cos\varphi \\ \sin^2\varphi & \cos^2\varphi & -2\sin\varphi\cos\varphi \\ -\sin\varphi\cos\varphi & \sin\varphi\cos\varphi & (\cos\varphi^2 - \sin\varphi^2) \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_{xy} \end{bmatrix} \tag{4.18}$$

for a symmetrical tensor with $m_{nt} = m_{tn}$ and $m_{xy} = m_{yx}$. A graphical representation of the bending and torsional moments is shown in the corresponding *Mohr's* circle in Fig. 4.5 (d). The similar transformation relationship can be derived for the generalised membrane forces by considering equilibrium of forces.

The direction of principal moment, defined by the angle φ_1 at which the torsional moment becomes zero $m_{nt} = m_{tn} = 0$, is expressed as

$$\tan 2\varphi_1 = \frac{2m_{xy}}{m_x - m_y} \tag{4.19}$$

such that a positive torsional moment $m_{xy} > 0$ results in $[0 \leq \varphi_1 \leq \pi/2]$, and a negative torsional moment $m_{xy} < 0$ results in $[-\pi/2 \leq \varphi_1 \leq 0]$. The corresponding principal moments result to

$$m_{1,2} = \frac{1}{2}\left[m_x + m_y \pm \sqrt{(m_x - m_y)^2 + 4m_{xy}^2}\right] \tag{4.20}$$

with $m_1 > m_2$.

4.3.3 Analysis of reinforced concrete plates

Fundamentals of classical laminate theory

According to *classical laminate theory*, a plate element may be represented as an assemblage of *lamina* or layers in which each constitutes an individual membrane element. Different constitutive relationships may be implemented in the individual layers such as to represent uncracked or cracked concrete with the addition of layers of reinforcement at arbitrary orientations. The significant increase in the computing performance of personal computers allowed *Kollegger* (1991) [51] to present a general numerical procedure to determine the complete load-deformation response of a reinforced concrete shell element subjected to bending and torsional moments and membrane forces. Such so-called *layered model* formulations have been widely used in research applications to analyse the load-deformation response of reinforced concrete plates and shells. The calculations generally show a good correlation with experimental results [33, 41, 43, 73, 88]. The theoretical background of such models is provided in [88].

In the analysis of reinforced concrete plates, the element is typically discretised into m concrete layers over the element thickness with the addition of k layers of reinforcing steel. The underlying assumptions of classical laminate theory imply that each layer is in a state of plane stress and that the stresses are uniform within each layer. The variation of strains over the element thickness is a function of the strain state at the element mid-plane and the curvatures, see Eq, (4.13), and is thus continuous. Stresses, however, are calculated at the centre of each individual layer from the constitutive relationship and given strains. Hence, the stress profile is generally not continuous, but rather exhibits a stepped profile over the individual layers. A large number of layers may be required to adequately represent the stress profile for strongly varying principal compressive directions at high concrete stresses. For typical cases, however, the use of at least 10 layers over the element height is recommended [88].

System of equations

The integrals in the determination of the stress resultants n_i and m_i, see Eqs. (4.17) and (4.16), reduce to a summation of the stress contributions from each plane stress layer. Generally, both the membrane forces and the bending moments are composed of concrete and reinforcing steel contributions in the form

$$n_i = n_{ci} + n_{sk}, \quad m_i = m_{ci} + m_{sk}. \tag{4.21}$$

The system of equations describing the relationship between the stress resultants and the strains at the mid-plane of the plate as well as the curvatures is expressed as

$$\begin{bmatrix} n_i \\ m_i \end{bmatrix} = \begin{bmatrix} c_{1ij} & c_{12ij} \\ c_{21ij} & c_{2ij} \end{bmatrix} \begin{bmatrix} \varepsilon_{0j} \\ \chi_j \end{bmatrix} \tag{4.22}$$

where each term in the 6×6 stiffness matrix c_{ij} is the sum of the contributions from all layers. Cracking of the concrete results in a significant reduction of the terms in the stiffness matrix and corresponding increase in both the strains at the mid-plane of the plate and the curvatures. The contribution of the reinforcement, however, remains constant prior to yielding. The stiffness matrix is typically partitioned into four sub-matrices, which relate the applied loading to the deformations. The sub-matrix c_{1ij} is referred to as the *extensional stiffness matrix* and provides the relation between membrane forces and strains at mid-plane of the element. The sub-matrix c_{2ij} constitutes the *bending stiffness matrix*, relating the bending and torsional moments to the curvatures. Finally, the symmetrical sub-matrix $c_{12ij} = c_{21ij}$, referred to as the *extension-bending coupling matrix*, provides a coupling of membrane forces and bending moments to both strains at the mid-plane of the element and the curvatures. For cases of geometrical or material symmetry with respect to the mid-plane, this coupling effect is detached. In summary, the mathematical problem is reduced to the solution of the six equations represented by Eq. (4.22) for known strains at the mid-plane of the element ε_{0j} and the curvatures χ_j.

Chapter 5

Modelling fatigue of reinforced concrete slabs

5.1 General

The analysis of reinforced concrete slabs, or herein interchangeably referred to as *plates* when both bending action and membrane forces are considered, under arbitrary combinations of bending and torsional moments poses a considerable challenge due to the nonlinear nature of the response and the generally iterative, numerical approach required. The flexural resistance of such elements can be evaluated with the normal moment yield criterion. However, while this approach is useful for the dimensioning of reinforcement and evaluating the resistance of existing structures, it does not provide any insight into the strain and stress states in the concrete and reinforcement under service conditions.

The evaluation of reinforced concrete structures for safety against fatigue is typically stress-based. Hence, these methods rely on a realistic prediction of the expected stress range under repeated loading. In the analysis of beam elements, the tensile and compressive components of the force-pair representing applied bending moment are typically assigned to equivalent tensile and compressive chords within the cross-section. This analogy is extended to slabs by discretising the element into an assemblage of concrete layers and considering fictitious, reinforced concrete cover elements at the top and bottom surfaces. Compressive stresses resulting from bending or torsional moments (or membrane forces) are assigned to the concrete layers, whereas tensile stresses exceeding the concrete tensile strength are kept in equilibrium by the reinforced cover elements.

The present chapter will provide a description of the physical-mechanical model that was developed for the analysis of orthogonally reinforced slabs under cyclic loading. Complete load-deformation curves can be determined for arbitrary combinations of bending or torsional moments. Furthermore, a prediction of failure, either through crushing of the concrete or rupture of the reinforcement, is implicitly considered. The deterioration of tension stiffening effects as well as the concrete tensile strength is considered under cyclic loading and the accumulation of damage is defined for individual reinforcing bars.

Firstly, an overview of the assumptions and idealisations implied in the model formulation will be presented, followed by a definition of the constitutive relationships considered for the concrete and reinforcement. The consideration of the uncracked and cracked response in the analysis of reinforced concrete membranes will be defined. The discussion will be extended to the analysis of reinforced concrete plates, introducing the element geometry and the system of equations describing the response. Subsequently, the solution procedure of the numerical method will be presented. Calculations of load-deformation response will be illustrated and compared to experimental results from tests on orthogonally reinforced concrete slabs. Finally, the calculations for cyclic loading response prediction will be examined and discussed.

5.2 Basic assumptions and idealisations

Thin plates or slabs in bending are analysed on the basis of *classical* or *Kirchhoff plate theory*. The fundamental assumptions stipulate that line elements normal to the mid-plane remain straight and normal to the mid-plane during deformation and that the element thickness remains constant. Hence, a deformation of the slab due to shear loads is not explicitly included in the formulation. A continuous, linearly varying strain profile results over the element height, see Eq. (4.13). The stress at any point within the cross-section is uniquely defined by the strains, the constitutive relationship and possibly the loading history. The stress profile over the height is approximated by discretising the element into an assemblage of layers and assuming a constant state of plane stress within each layer. The principal stress perpendicular to the mid-plane of each layer is assumed to be zero $\sigma_3 = \sigma_z = 0$. Stresses are calculated at the centre of each layer.

The transformation relationships for stresses are formulated more concisely by introducing the angle-dependant vector $t_i\,(\alpha)$, defined as

$$t_i\,(\alpha) = \left[\cos^2\alpha,\, \sin^2\alpha,\, -\sin\alpha\cos\alpha\right] \tag{5.1}$$

using the notation introduced in [88]. Furthermore, the angle ψ_k is defined to describe the direction of a reinforcement layer of arbitrary orientation with respect to the reinforcement in the x-direction. In the present work, only orthogonal reinforcement layouts are considered such that $\psi_x = 0$ and $\psi_y = \pi/2$ for the x- and y-directions of reinforcement, respectively. The element formulation allows the definition of four layers of reinforcement. The bars are assumed to be finely spaced such that stresses are uniformly distributed over the cross-sectional area. Furthermore, reinforcing bars are assumed to resist only uniaxial loading; their flexural and shear stiffness is neglected. A reduction in the concrete cross-sectional area due to the placement of reinforcing bars is not considered.

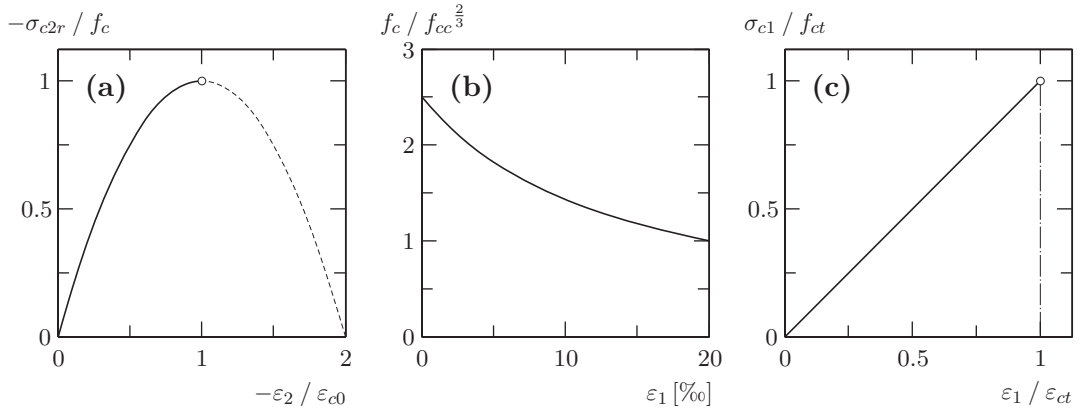


Fig. 5.1 – Constitutive relationships for concrete: (a) compression; (b) compression-softening effect of transverse strains; (c) tension.

5.3 Constitutive relationships

5.3.1 Concrete

A parabolic stress-strain relationship, as introduced in Section 3.2.1, is considered for the principal concrete compressive stress at the cracks of the form

$$\sigma_{c2r}(\varepsilon_2) = f_c \cdot \frac{{\varepsilon_2}^2 + 2\varepsilon_2\varepsilon_{c0}}{{\varepsilon_{c0}}^2} \tag{5.2}$$

wherein the peak compressive strength f_c accounts for the compression-softening effect of transverse strains perpendicular to the principal compressive direction. The relationship established by *Kaufmann* [49] is implemented, which is expressed as

$$f_c(\varepsilon_1) = \frac{{f_{cc}}^{2/3}}{0.4 + 30\varepsilon_1} \leq f_{cc} \quad \text{in } \left[\text{N/mm}^2\right] \tag{5.3}$$

as shown in Fig. 5.1 (b). For the case of biaxial compression wherein both principal strains are negative $\varepsilon_{1,2} < 0$, the maximum value of the concrete compressive strength is conservatively limited to the cylinder compressive strength f_{cc}. Furthermore, the post-peak response of the concrete in uniaxial compression is neglected; hence, crushing of the concrete is assumed at principal compressive strains $\varepsilon_2 < -\varepsilon_{c0}$. The principal concrete compressive stress is no longer considered in the affected layers $\sigma_{c2r} = 0$.

A linear elastic stress-strain relationship is considered for the uncracked concrete in tension within the range $0 \leq \varepsilon_1 \leq f_{ct} \,/\, E_c$. The modulus of elasticity in tension is approximated with Eq.(3.3) and the tensile stress is limited to the concrete tensile strength f_{ct}, which is approximated with Eq. (3.6). Cracking is assumed to occur at principal strains exceeding the strain corresponding to the concrete tensile strength $\varepsilon_1 \geq f_{ct} \,/\, E_c$.

5.3.2 Reinforcing steel

A bilinear stress-strain relationship with strain-hardening and limited strain capacity, see Eq. (3.12), is considered for the reinforcement in compression as illustrated in Fig. 5.2 (a).

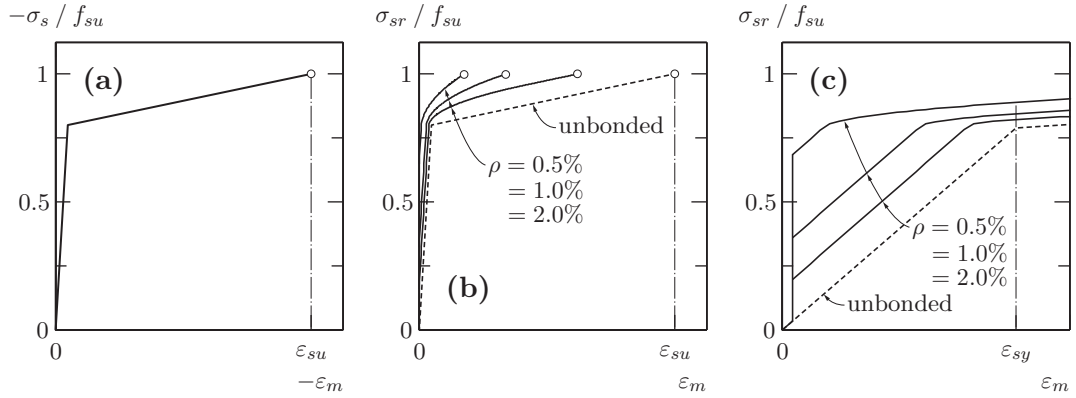


Fig. 5.2 – Constitutive relationships for reinforcing steel: (a) compression; (b) tension; (c) elastic range including stiffening effects and unbonded bars. Note: $\lambda = 1$.

At tensile strains resulting in cracking of the cover elements containing the reinforcement, maximum steel stresses at the cracks σ_{sr} are calculated using the constitutive relationships of the Tension Chord Model, see Eqs. (3.19), (3.20) and (3.21). Herein, the tension stiffening effect of the concrete is considered. The maximum crack spacing s_r is generally calculated assuming $\lambda = 1$, which results in the highest theoretical values of steel stress. The stress-strain relationship in tension is illustrated in Fig. 5.2 (b). Fig. 5.2 (c) illustrates the significant influence of the reinforcement content ρ on the maximum steel stress at the cracks in comparison to the relationship for unbonded bars. Alternatively, the bilinear relationship for unbonded bars can also be selected to model the tensile response of the reinforcement without tension stiffening effects.

5.4 Analysis of reinforced concrete membranes

The equilibrium conditions for membrane elements are defined in the present section for the treatment of both the uncracked and cracked response.

Reinforced concrete elements in a homogeneous state of plane stress $\sigma_i = [\sigma_x, \sigma_y, \tau_{xy}]$ can be analysed as membranes assuming the mid-plane strains $\varepsilon_{0j} = [\varepsilon_{x0}, \varepsilon_{y0}, \varepsilon_{xy0}]$ are uniformly distributed over the thickness. The load-deformation response of such elements is dominated by cracking as this results in permanent changes in the constitutive relations for the concrete and activates the reinforcement. Hence, a clear differentiation between the *uncracked response*, which includes the case of *biaxial compression*, and the *cracked response* is appropriate. The behaviour of uncracked concrete in tension is approximated with the theory of elasticity. Biaxial compression is modelled considering the stress-strain relationship formulated in Eq. (5.2) for both principal directions independently. Cracked elements are analysed with the Cracked Membrane Model introduced in Section 4.2.3. The consideration of the uncracked and cracked response is presented with an example of an orthogonally reinforced concrete membrane element as illustrated in Fig. 5.3.

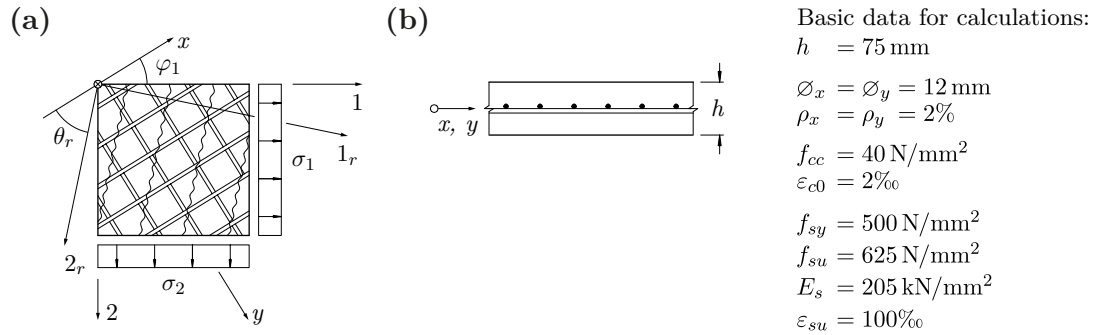


Fig. 5.3 – Orthogonally reinforced concrete membrane: (a) plan view with notation; (b) cross-sectional view.

5.4.1 Uncracked response

Concrete is assumed to remain uncracked at maximum principal stresses below the concrete tensile strength $\sigma_1 \leq f_{ct}$. Consequently, linear-elastic behaviour is assumed for principal strains below the strain corresponding to the concrete tensile strength $\varepsilon_1 \leq f_{ct}/E_c$. The stress tensor σ_i can be calculated directly for known strains ε_j using *Hooke's* law

$$\sigma_i = c_{\mathrm{I}ij} \cdot \varepsilon_j \tag{5.4}$$

where $c_{\mathrm{I}ij}$ denotes the stiffness matrix of the uncracked concrete. In general, the stiffness matrix consists of the sum of both the concrete $c_{\mathrm{cI}ij}$ and reinforcement contributions c_{sij}

$$c_{\mathrm{cI}ij} = \frac{E_c}{1-\nu_c}\begin{bmatrix} 1 & \nu_c & 0 \\ \nu_c & 1 & 0 \\ 0 & 0 & (1-\nu_c)/2 \end{bmatrix}, \qquad c_{sij} = \begin{bmatrix} E_{sx}\cdot\rho_x & 0 & 0 \\ 0 & E_{sy}\cdot\rho_y & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{5.5}$$

in which ν_c represents *Poisson's* ratio of the concrete and assuming that the orthogonally placed reinforcement in the x- and y-directions behave elastically. Prior to the onset of cracking, the contribution of the reinforcing steel is small in relation to that of the concrete. This can be neglected for practical purposes such that $c_{\mathrm{I}ij} \approx c_{\mathrm{cI}ij}$. However, the steel contribution is also considered in the present formulation for the sake of consistency.

Three states of uncracked response are distinguishable in terms of the principal strains, see Fig. 5.4. In state ①, both principal strains are positive, yet less than the strain corresponding to the concrete tensile strength. In state ②, the larger principal tensile strain is less than the strain corresponding to the concrete tensile strength, whereas the smaller principal strain is negative. State ③ refers to biaxial compression wherein both principal strains are negative. In the latter, the principal concrete compressive stresses are considered independently with Eq. (5.2) such that $\sigma_{c1} = \sigma_{c1}(\varepsilon_1)$ and $\sigma_{c2} = \sigma_{c2}(\varepsilon_2)$, as proposed in [88]. Furthermore, no confinement effects are considered and the concrete compressive strength is limited to f_{cc}. A summary of the constitutive relationships and corresponding definition of the tangent moduli of elasticity for each state follows.

State ①: uncracked, only tensile strains with $0 \leq \varepsilon_1 \leq f_{ct} \,/\, E_c$ and $0 \leq \varepsilon_2$

Concrete	Reinforcement
$\sigma_{ci} = c_{\mathrm{I}ij} \cdot \varepsilon_j$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$\sigma_{c1r} = \sigma_{cx} \sin^2 \theta_r + \sigma_{cy} \cos^2 \theta_r + 2\tau_{cxy} \sin \theta_r \cos \theta_r$	
$\sigma_{c2r} = \sigma_{cx} \cos^2 \theta_r + \sigma_{cy} \sin^2 \theta_r + 2\tau_{cxy} \sin \theta_r \cos \theta_r$	
$E_{c1} = E_{c2} = E_c$	$E_{sx} = E_{sy} = E_s$

State ②: uncracked, tension and compression with $0 \leq \varepsilon_1 \leq f_{ct} \,/\, E_c$ and $\varepsilon_2 < 0$

Concrete	Reinforcement
$f_c = \dfrac{{f_{cc}}^{2/3}}{0.4 + 30\varepsilon_1} \leq f_{cc}$	
$\sigma_{c1r} = E_c \varepsilon_1, \quad \sigma_{c2r} = f_c \cdot \dfrac{\varepsilon_2^2 + 2\varepsilon_2 \varepsilon_{c0}}{{\varepsilon_{c0}}^2}$	
$\sigma_{ci} = \sigma_{c1r} \cdot t_i\,(\theta_r - \pi/2) + \sigma_{c2r} \cdot t_i\,(\theta_r)$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$E_{c1} = E_c, \quad E_{c2} = \dfrac{\partial \sigma_{c2r}}{\partial \varepsilon_2} = f_c \cdot \dfrac{2\varepsilon_2 + 2\varepsilon_{c0}}{{\varepsilon_{c0}}^2}$	$E_{sx} = E_{sy} = E_s$

State ③: uncracked, only compressive strains with $\varepsilon_{1,2} \leq 0$

Concrete	Reinforcement
$f_c = f_{cc}$	
$\sigma_{c1} = f_c \cdot \dfrac{\varepsilon_1^2 + 2\varepsilon_1 \varepsilon_{c0}}{{\varepsilon_{c0}}^2}, \quad \sigma_{c2} = f_c \cdot \dfrac{\varepsilon_2^2 + 2\varepsilon_2 \varepsilon_{c0}}{{\varepsilon_{c0}}^2}$	
$\sigma_{ci} = \sigma_{c1} \cdot t_i\,(\theta_r - \pi/2) + \sigma_{c2} \cdot t_i\,(\theta_r)$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$E_{c1} = \dfrac{\partial\,(\sigma_{c1})}{\partial(\varepsilon_1)} = f_c \cdot \dfrac{2\varepsilon_1 + 2\varepsilon_{c0}}{{\varepsilon_{c0}}^2},$	$E_{sx} = E_{sy} = E_s$
$E_{c2} = \dfrac{\partial\,(\sigma_{c2})}{\partial(\varepsilon_2)} = f_c \cdot \dfrac{2\varepsilon_2 + 2\varepsilon_{c0}}{{\varepsilon_{c0}}^2}$	

5.4.2 Cracked response

Cracking of the concrete is assumed at a principal strain exceeding the strain corresponding to the concrete tensile strength $\varepsilon_1 \leq f_{ct} \,/\, E_c$. The response to applied stresses becomes nonlinear as the terms of the cracked stiffness matrix are a function of the strain state $c_{\mathrm{II}ij}(\varepsilon_j, \theta_r)$. The equilibrium conditions can be expressed as

$$\sigma_i = \sum_{k=1}^{r} \rho_k \sigma_{skr}\,(\varepsilon_j, \theta_r) \cdot t_i\,(-\psi_k) + \sigma_{c2r}\,(\varepsilon_j, \theta_r) \cdot t_i\,(\theta_r) \tag{5.6}$$

using the notation of *Seelhofer* [88] for k layers of reinforcement placed at the angle ψ_k with respect to the x-direction of reinforcement. In an orthogonal reinforcement layout, the angle-dependant vectors reduce to $t_i\,(0) = [1,\, 0,\, 0]$ and $t_i\,(-\pi/2) = [0,\, 1,\, 0]$ for the x- and y-directions of reinforcement, respectively. Hence, the equilibrium conditions of

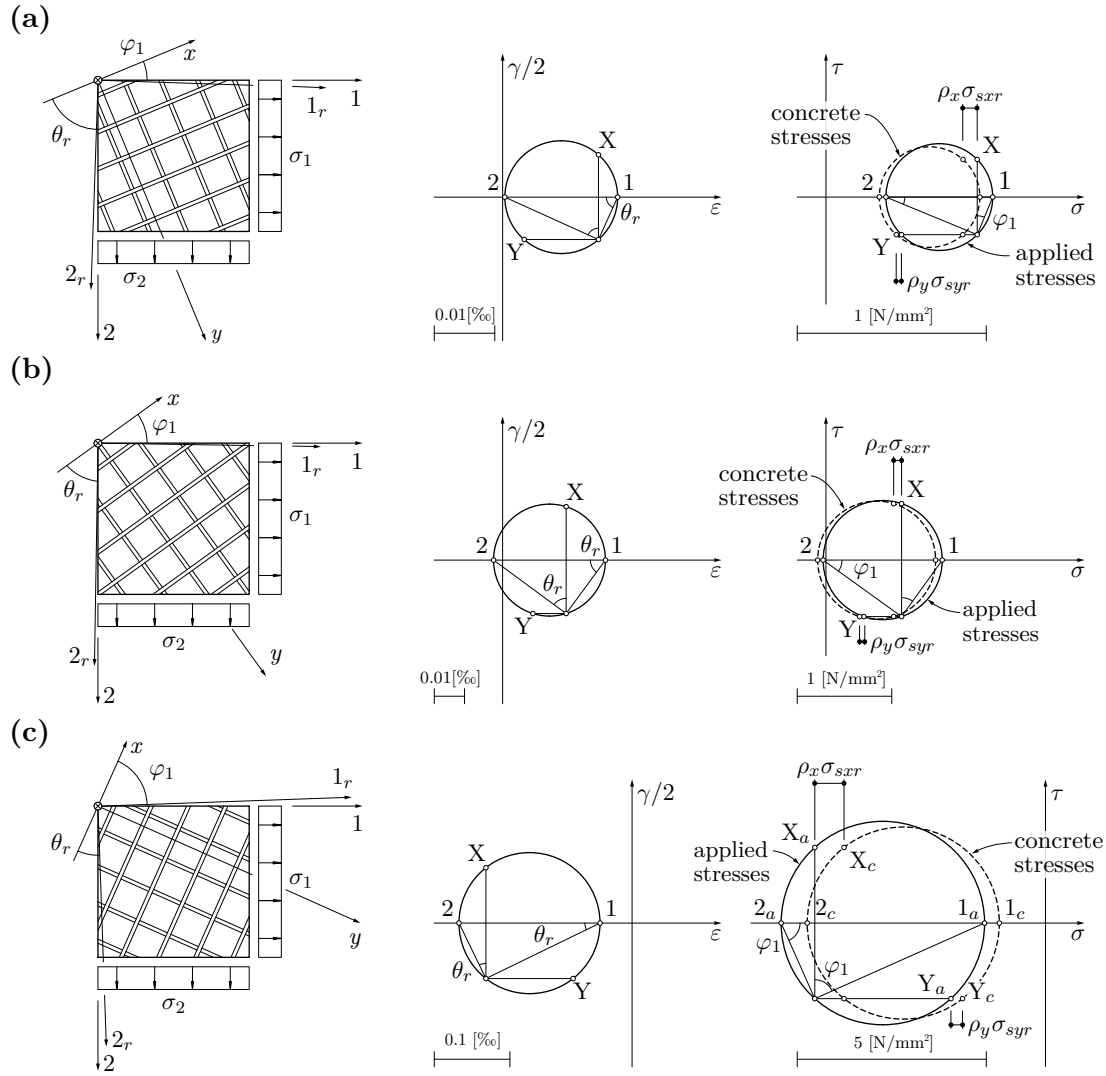


Fig. 5.4 – Uncracked membrane response: (a) state ①, only tensile strains; (b) state ②, tensile and compressive strains; (c) state ③, only compressive strains.

Eq. (5.6) for orthogonally reinforced concrete membranes reduce to

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \rho_x \sigma_{sxr} \\ \rho_y \sigma_{syr} \\ 0 \end{bmatrix} + \sigma_{c2r} \cdot \begin{bmatrix} \cos^2\theta_r \\ \sin^2\theta_r \\ -\sin\theta_r \cos\theta_r \end{bmatrix} \tag{5.7}$$

which corresponds to the formulation for the Cracked Membrane Model.

In addition to the uncracked response states, three further states of cracked response are distinguishable, see Fig. 5.5. State ④ refers to a membrane that is cracked in one principal direction while the smaller principal strain is negative and results in compressive stresses. In state ⑤, the larger principal strain exceeds the strain corresponding to the concrete tensile strength, whereas the smaller principal strain is positive without resulting in cracking. In state ⑥, both principal strains exceed the strain corresponding to the

concrete tensile strength; the element is cracked in both principal directions and the concrete provides no further contribution to the response in compression. A summary of the constitutive relationships and corresponding definition of the tangent moduli of elasticity for each state follows.

State ④: cracked in principal direction, compressive diagonal with $\varepsilon_1 > f_{ct} \,/\, E_c$ and $\varepsilon_2 \leq 0$

Concrete	Reinforcement
$f_c = \dfrac{{f_{cc}}^{2/3}}{0.4 + 30\varepsilon_1} \leq f_{cc}$	
$\sigma_{c1r} = 0, \quad \sigma_{c2r} = f_c \cdot \dfrac{\varepsilon_2^2 + 2\varepsilon_2\varepsilon_{c0}}{{\varepsilon_{c0}}^2}$	
$\sigma_{ci} = \sigma_{c2r} \cdot t_i\,(\theta_r)$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$E_{c1} = 0, \quad E_{c2} = \dfrac{\partial \sigma_{c2r}}{\partial \varepsilon_2} = f_c \cdot \dfrac{2\varepsilon_2 + 2\varepsilon_{c0}}{{\varepsilon_{c0}}^2} \leq E_c$	$E_{sx} = \dfrac{\partial \sigma_{sxr}}{\partial \varepsilon_x}, \quad E_{sy} = \dfrac{\partial \sigma_{syr}}{\partial \varepsilon_y}$

State ⑤: cracked in principal direction, tensile diagonal $\varepsilon_1 > f_{ct} \,/\, E_{cm}$ and $0 < \varepsilon_2 \leq f_{ct} \,/\, E_c$

Concrete	Reinforcement
$\sigma_{c1r} = 0, \quad \sigma_{c2r} = E_c \cdot \varepsilon_2$	
$\sigma_{ci} = \sigma_{c2r} \cdot t_i\,(\theta_r)$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$E_{c1} = 0, \quad E_{c2} = E_c$	$E_{sx} = \dfrac{\partial \sigma_{sxr}}{\partial \varepsilon_x}, \quad E_{sy} = \dfrac{\partial \sigma_{syr}}{\partial \varepsilon_y}$

State ⑥: cracked in both principal directions with $\varepsilon_{1,2} > f_{ct} \,/\, E_c$

Concrete	Reinforcement
$\sigma_{c1r} = \sigma_{c2r} = 0$	
$\sigma_{ci} = 0$	$\sigma_{si} = c_{sij} \cdot \varepsilon_j$
$E_{c1} = E_{c2} = 0$	$E_{sx} = \dfrac{\partial \sigma_{sxr}}{\partial \varepsilon_x}, \quad E_{sy} = \dfrac{\partial \sigma_{syr}}{\partial \varepsilon_y}$

5.5 Analysis of reinforced concrete plates

The discretisation of the layered element and notation for the geometry is introduced in the present section. Subsequently, the relationships between externally applied membrane forces $n_i = [n_x, n_y, n_{xy}]$ and bending as well as torsional moments $m_i = [m_x, m_y, m_{xy}]$ per unit length and the mid-plane strains $\varepsilon_{0j} = [\varepsilon_x, \varepsilon_y, \gamma_{xy}]$ as well as the curvatures $\chi_j = [\chi_x, \chi_y, \chi_{xy}]$ are presented for the uncracked and cracked response. The stiffness matrix is defined in terms of the concrete and reinforcing steel contributions to the response.

5.5.1 Plate element geometry

The discretisation of the layered model is indicated schematically in Fig. 5.6 (a). The distribution of strains over the element height is continuous, whereas the concrete stresses are uniform within each layer. The fictitious cover elements are indicated at the top and

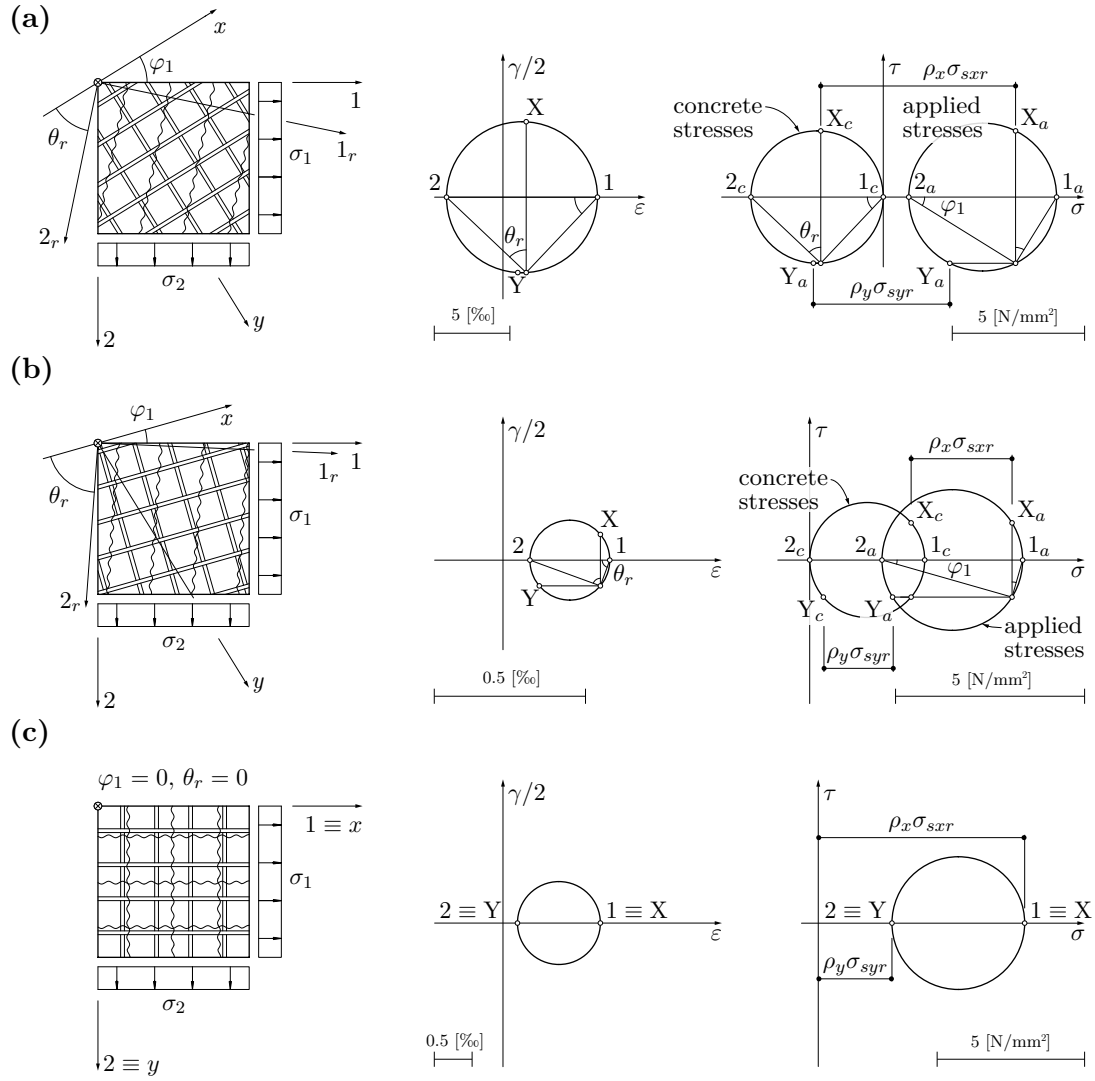


Fig. 5.5 – Cracked membrane response: (a) state ④, compressive diagonal; (b) state ⑤, tensile diagonal; (c) state ⑥, cracked in both principal directions.

bottom of the element, which allow the consideration of the reinforcement and the influence of cracking on the outer surfaces.

The origin of the x-y-z coordinate system is defined to be within the mid-plane of the element. The x- and y-axes are defined to correspond to the reinforcement directions of the orthogonal layout. The concrete cross-section of the element is discretised into m panels over the height as illustrated in Fig. 5.6 (b). The layer thickness and distance from the element mid-plane to the centre of the j^{th} panel are expressed as

$$\Delta z = \frac{h}{m}, \quad \text{and} \qquad z_j = \frac{\Delta z\,(2j-1) - h}{2}. \tag{5.8}$$

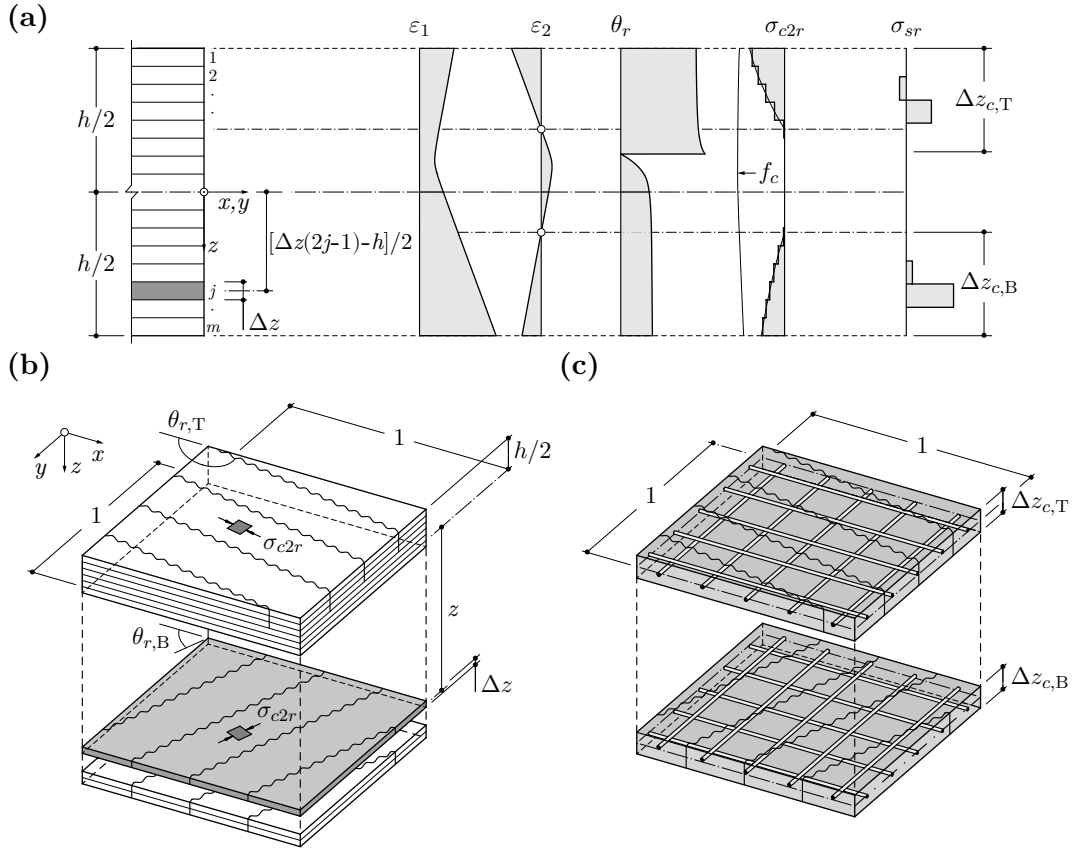


Fig. 5.6 – Layered element: (a) discretisation into layers with resulting concrete stress profile and reinforcing steel stresses, redrawn from [88]; (b) concrete layers; (c) fictitious cracked membrane cover elements.

The reinforcement is considered within two fictitious cover elements at the top and bottom of the plate, modelled as Cracked Membranes, see Fig. 5.6 (c). These cover elements are superimposed onto the concrete layers. The mid-plane of the cover elements is defined to correspond to the plane between the layers of reinforcement, as recommended in [45]. Hence, the thickness of the top and bottom cover elements and z-coordinate from the mid-plane of the plate to the centre of these elements are

$$\Delta z_{c,\mathrm{T}} = 2\left(c_{\mathrm{nom}} + \varnothing_{x,\mathrm{T}}\right), \qquad z_{c,\mathrm{T}} = \frac{\Delta z_{c,\mathrm{T}} - h}{2} \tag{5.9}$$

$$\Delta z_{c,\mathrm{B}} = 2\left(c_{\mathrm{nom}} + \varnothing_{x,\mathrm{B}}\right), \qquad z_{c,\mathrm{B}} = \frac{\Delta z_{c,\mathrm{B}} + h}{2} \tag{5.10}$$

assuming that the outer layers of reinforcement in both the top and bottom cover elements are placed in the x-direction.

5.5.2 Uncracked response

The distribution of both principal strains follow hyperbolic functions over the element height, see Eq. (4.14), such that the second derivatives are positive for $\varepsilon_1\,(z)$ and negative

for $\varepsilon_2(z)$ [88]. The function of the larger principal strain $\varepsilon_1 = \varepsilon(z)$ is therefore concave and displays maximum values at either the top, the bottom or at both outer surfaces of the plate. As would be expected, cracking of the concrete initiates in the outer layers. The plate is considered to be uncracked if the larger principal strains in both outer layers do not exceed the strain corresponding to the concrete tensile strength.

For uncracked concrete, the stiffness matrix $c_{\mathrm{I}ij}$ is constant and independent of the strain state such that the stress resultants can be calculated directly for a given strain state, see Eq. (4.22). The inverse procedure requires inversion of the stiffness matrix for a given set of stress resultants in order to determine the strain state. Using the terminology of classical laminate theory, as introduced in Section 4.3.3, the extensional stiffness matrix for the uncracked element is formulated as

$$c_{\mathrm{I}1ij} = \frac{E_c h}{1-{\nu_c}^2} \begin{bmatrix} 1 & \nu_c & 0 \\ \nu_c & 1 & 0 \\ 0 & 0 & (1-\nu_c)/2 \end{bmatrix} + E_s \sum_{k=1}^{r} a_{sk} \cdot t_i(-\psi_k) t_j(-\psi_k), \tag{5.11}$$

the extension-bending coupling matrix is written as

$$c_{\mathrm{I}12ij} = c_{\mathrm{I}21ij} = E_s \sum_{k=1}^{r} a_{sk} \cdot z_k \cdot t_i(-\psi_k) t_j(-\psi_k), \tag{5.12}$$

and the bending stiffness matrix results to

$$c_{\mathrm{I}2ij} = \frac{E_c h^3}{12(1-{\nu_c}^2)} \begin{bmatrix} 1 & \nu_c & 0 \\ \nu_c & 1 & 0 \\ 0 & 0 & (1-\nu_c)/2 \end{bmatrix} + E_s \sum_{k=1}^{r} a_{sk} \cdot {z_k}^2 \cdot t_i(-\psi_k) t_j(-\psi_k). \tag{5.13}$$

5.5.3 Cracked response

The element is considered to be cracked if the larger principal strain at either of the outer concrete layers exceeds the strain corresponding to the concrete tensile strength $\varepsilon_1 > f_{ct}/E_c$. The depth of flexural cracks is assumed to extend up to the layer in which this strain limit is not exceeded.

The stress resultants are calculated as the sum of concrete contributions from all m layers in addition to the sum of reinforcement contributions in all k layers of reinforcement

$$\begin{aligned} n_i &= \sum_{j=1}^{m} \sigma_{cij} \cdot \Delta z_j + \sum_{k=1}^{r} \rho_k \sigma_{skr} \cdot \Delta z_c \cdot t_i(-\psi_k) \\ m_i &= \sum_{j=1}^{m} \sigma_{cij} \cdot \Delta z_j z_j + \sum_{k=1}^{r} \rho_k \sigma_{skr} \cdot \Delta z_c \cdot z_c \cdot t_i(-\psi_k) \end{aligned} \tag{5.14}$$

where Δz_c refers to the thickness of the top $\Delta z_{c,\mathrm{T}}$ and bottom $\Delta z_{c,\mathrm{B}}$ fictitious cover elements. Similarly, the z-coordinate z_c refers to the distance between the element mid-plane

and the centre of the top $z_{c,\mathrm{T}}$ and bottom $z_{c,\mathrm{B}}$ cover elements. The general formulation of the concrete contribution to the response σ_{cij} is

$$\sigma_{cij} = \sigma_{c1r} \cdot t_i \left[\theta_r \left(z_j\right) - \pi/2\right] + \sigma_{c2r} \cdot t_i \left[\theta_r \left(z_j\right)\right] \tag{5.15}$$

where the larger principal concrete stress σ_{c1r} is typically equal to zero after cracking in the principal direction.

The total stiffness matrix for the cracked response $c_{\mathrm{II}ij}$ is composed of the concrete and reinforcing steel contributions in all layers

$$c_{\mathrm{II}ij} = \sum_{j=1}^{m} c_{c\mathrm{II}ij} + \sum_{k=1}^{r} c_{sk}. \tag{5.16}$$

Herein, the cracked concrete stiffness matrix $c_{c\mathrm{II}ij}$ is calculated with the approach recommended in [9]. The concrete stiffness matrix is initially calculated in the principal directions $c_{c1,2}$, consisting of the tangent moduli of elasticity of the concrete $E_{c1,2}$ on the diagonals in the first two rows in the form

$$c_{c1,2} = \begin{bmatrix} E_{c1} & 0 & 0 \\ 0 & E_{c2} & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{5.17}$$

and rotated back to the model coordinate system through the transformation relationship

$$c_{c\mathrm{II}ij} = T_{cji} c_{c1,2} T_{cij} \tag{5.18}$$

where

$$T_{cij} = \begin{bmatrix} \cos^2\varphi_1 & \sin^2\varphi_1 & \sin\varphi_1\cos\varphi_1 \\ \sin^2\varphi_1 & \cos^2\varphi_1 & -\sin\varphi_1\cos\varphi_1 \\ -2\sin\varphi_1\cos\varphi_1 & 2\sin\varphi_1\cos\varphi_1 & (\cos^2\varphi_1 - \sin^2\varphi_1) \end{bmatrix}. \tag{5.19}$$

The effects of a possible change in the crack direction between subsequent load steps is considered by adding a *rotational stiffness matrix* $c_{c\theta}$, as derived by *Akbar* in [2], to the concrete stiffness matrix. This additional contribution to the total stiffness matrix, formulated as

$$c_{c\theta} = \frac{-\sigma_{c2r}}{2\sqrt{\left(\varepsilon_x - \varepsilon_y\right)^2 + \gamma_{xy}^2}} \begin{bmatrix} \sin^2(2\varphi_1) & -\sin^2(2\varphi_1) & -\sin(2\varphi_1)\cos(2\varphi_1) \\ -\sin^2(2\varphi_1) & \sin^2(2\varphi_1) & \sin(2\varphi_1)\cos(2\varphi_1) \\ -\sin(2\varphi_1)\cos(2\varphi_1) & \sin(2\varphi_1)\cos(2\varphi_1) & \cos^2(2\varphi_1) \end{bmatrix} \tag{5.20}$$

is included in order to improve the convergence of the numerical solution [51]. The four sub-matrices of the total stiffness matrix for the cracked response result to

$$
\begin{aligned}
c_{\mathrm{II}1ij} &= \sum_{j=1}^{m} \left[(c_{c\mathrm{II}ij} + c_{c\theta}) \cdot \Delta z_j \right] + \sum_{k=1}^{r} \rho_k E_s \cdot \Delta z_c \cdot t_i \left(-\psi_k \right) \\
c_{\mathrm{II}12ij} &= c_{\mathrm{II}21ij} = \sum_{j=1}^{m} \left[(c_{c\mathrm{II}ij} + c_{c\theta}) \cdot \Delta z_j \cdot z_j \right] + \sum_{k=1}^{r} \rho_k E_s \cdot \Delta z_c \cdot z_c \cdot t_i \left(-\psi_k \right) \\
c_{\mathrm{II}2ij} &= \sum_{j=1}^{m} \left[(c_{c\mathrm{II}ij} + c_{c\theta}) \cdot \Delta z_j \cdot {z_j}^2 \right] + \sum_{k=1}^{r} \rho_k E_s \cdot \Delta z_c \cdot {z_c}^2 \cdot t_i \left(-\psi_k \right).
\end{aligned}
\tag{5.21}
$$

5.6 Numerical method

The numerical method developed for the analysis of orthogonally reinforced concrete slabs under cyclic loading is outlined in the present section. Fig. 5.7 illustrates the sequence of operations as implemented in *Mathworks* MATLAB®.

Initialisation of the model requires some user-defined inputs in order to describe the element geometry, reinforcement layout, mechanical properties of the concrete and reinforcing steel and the characteristics of cyclic loading. Subsequently, the fatigue strengths of individual bars are assigned for the calculation and all concrete layers are defined as uncracked. Initial values of reinforcement content and crack spacing are calculated.

The load cycle counter is initialised at the first load cycle with $N = 1$. Calculations are performed at discrete load cycles N_{LC}; the accumulation of damage is considered to progress linearly between subsequent calculations. The reinforcement content is reduced upon failure of individual bars. Corresponding crack spacings are calculated anew; altering the influence of tension stiffening. Subsequently, the concrete tensile strength and tension stiffening effects due to bond is modified to account for the effects of cyclic loading.

The solution algorithm is called upon to solve the system of equations at the defined maximum and minimum load. Herein, the strain state as well as the corresponding concrete and steel stresses are calculated. The stress range is calculated in all layers of reinforcement and the linear damage rule is applied to calculate the damage increment sustained in each bar with respect to the previously calculated load cycle. The total accumulated damage is updated for each bar followed by a check on their integrity. Bars in which the accumulated damage is greater than unity $D_{\mathrm{LC}+1} \geq 1$ are excluded from the calculation. This is achieved by reducing the cross-sectional area of the layer of reinforcement with the amount corresponding to the cross-section of the fractured bar.

If the equilibrium conditions are satisfied, the load cycle counter is incremented and the procedure starts anew for $N = N_{\mathrm{LC}+1}$. In the case that the solution algorithm fails to find a state of equilibrium, global failure of the element is predicted. Due to the fact that calculations are only performed at discrete load cycles, the number of load cycles in the previous calculation step is defined as the maximum number of load cycles to global failure $N_u = N_{\mathrm{LC}-1}$.

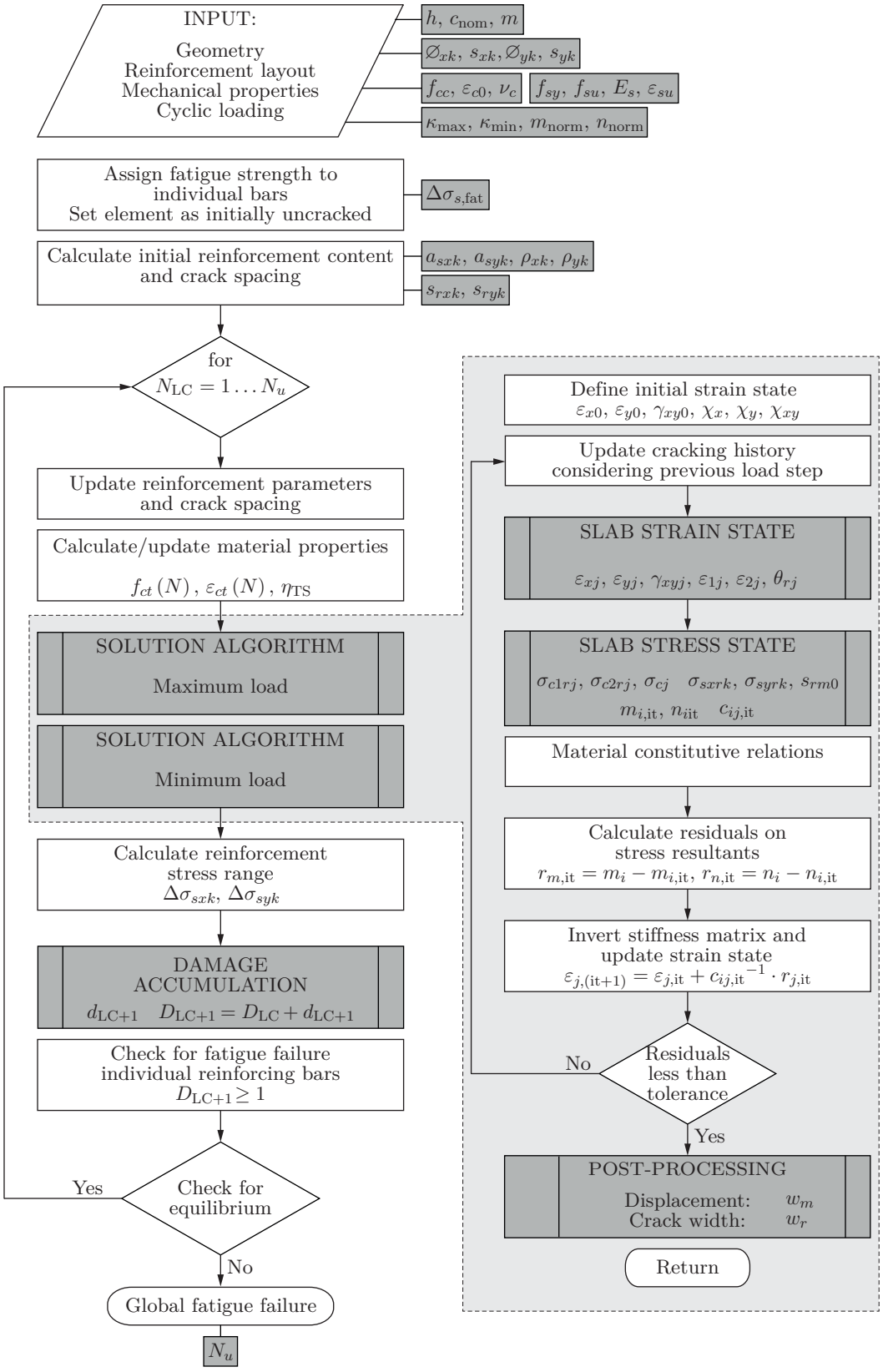


Fig. 5.7 – Flow diagram of the numerical method.

5.6.1 Input parameters

The input parameters required for the calculations were intentionally kept to a minimum for illustrative purposes. The programming structure, however, enables an extension of the model formulation with minimal effort. The essential input parameters are summarised as

Geometry:	h, c_{nom}, m
Bottom cover element reinforcement:	$\varnothing_{x\mathrm{B}}$, $s_{x\mathrm{B}}$, $\varnothing_{y\mathrm{B}}$, $s_{y\mathrm{B}}$
Top cover element reinforcement:	$\varnothing_{x\mathrm{T}}$, $s_{x\mathrm{T}}$, $\varnothing_{y\mathrm{T}}$, $s_{y\mathrm{T}}$
Concrete properties:	f_{cc}, ε_{c0}, ν_c
Reinforcement properties:	f_{sy}, f_{su}, E_s, ε_{su}
Fatigue parameters:	$N_{s,\mathrm{fat}}$, $N_{s,\mathrm{D}}$, m
Loading parameters:	$\kappa_{\max}$, $\kappa_{\min}$, $m_{i,\mathrm{norm}}$, $n_{i,\mathrm{norm}}$

The geometry of the slab is defined by the element thickness h, the concrete cover to the reinforcement c_{nom} and the discretisation into m concrete layers. The reinforcement layout is defined through a choice of bar diameter and spacing within each of the four layers. Subsequently, the essential mechanical properties of the concrete and reinforcing steel are defined. The fatigue strength of the bars is assigned from a database of test results as will be described in the following section. Furthermore, the fatigue parameters of the S-N diagram for reinforcing steel is defined. Finally, the loading parameters are defined with the use of the scaling factor κ for the maximum and minimum load, where $m_{i,\mathrm{norm}}$ and $n_{i,\mathrm{norm}}$ denote the normalised vectors of bending moment and membrane forces.

5.6.2 Fatigue strength of reinforcement

Slabs are typically reinforced with layers of closely spaced reinforcing steel bars. Such elements possess a large degree of redundancy in the resistance of loads under service conditions and will generally have sufficient residual load-carrying capacity after fatigue failure of individual bars. However, a redistribution of internal forces will take place in order to find a new state of equilibrium. The remaining bars experience higher stresses, resulting in an acceleration of damage accumulation.

Similar to the approach implemented by *Fehlmann* [23] in the analysis of beams, the fatigue strength of the reinforcing bars is considered as a stochastic parameter in order to account for the high degree of variability observed in fatigue tests. The fatigue strength of individual reinforcing bars is assigned according to a normal distribution fitted to fatigue test results, see Fig. 5.8 (a). The database of fatigue tests includes 126 constant amplitude tests on reinforcing bars from the investigations by *Fernández Canteli et al.* [25] and *Fehlmann and Vogel* [24]. These tests were conducted at different stress ranges and include both small and large diameter bars, varying in the range $\varnothing = 10 \ldots 30\,\mathrm{mm}$, see

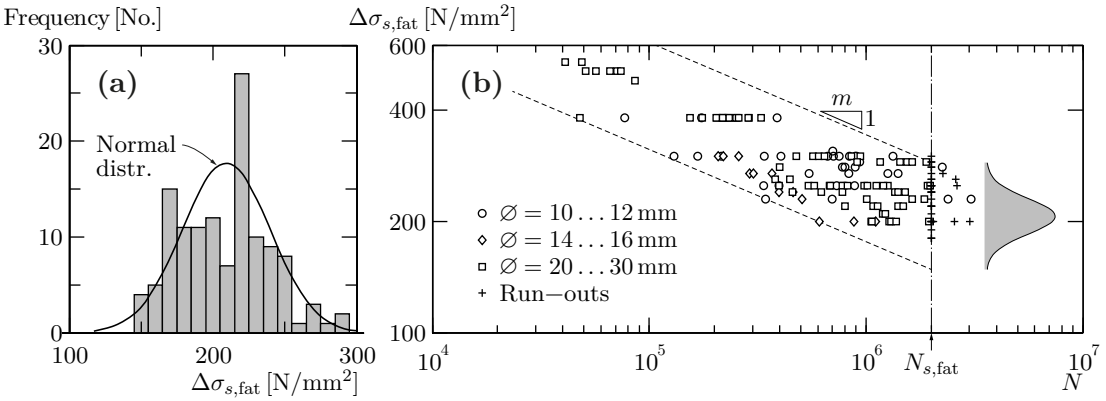


Fig. 5.8 – Database of reinforcement fatigue strengths: (a) histogram with fitted normal distribution ($n = 126$, $\mu = 208\,\mathrm{N/mm^2}$, $\sigma = 31.2\,\mathrm{N/mm^2}$); (b) S-N diagram of fatigue tests. Data from [24, 25].

Fig. 5.8 (b). The consideration of outliers of fatigue strength from the tails of the distribution significantly alters the number of load of cycles until global fatigue failure of the element, see Section 5.8.4. For this reason, the normal distribution is truncated such that only fatigue strength values between the 5$^{\mathrm{th}}$ and 95$^{\mathrm{th}}$ percentiles are considered. In contrast to typical fatigue evaluation approaches, which assume a conservative value for the fatigue strength for all bars, the present formulation enables an investigation of the changes in load-deformation response after fatigue failure of individual bars.

5.6.3 Loading history

The flexural response of a slab under cyclic loading depends on the loading history. Cracking of the concrete results in irreversible structural changes that have to be considered in the constitutive relationships governing the response. Hence, upon initialisation of the calculations, a vector describing the cracking state of each concrete layer is defined with all values set to zero. This corresponds to an element that is uncracked over the entire height. In subsequent load cycles, a check is performed in each layer with respect to the principal strains, see Section 5.4, such as to consider the constitutive relationships of either the uncracked or cracked response. In layers where the principal strains are below the level indicating cracking of the concrete, the corresponding entry in the vector of the previous load cycle is regarded. If the layer was uncracked in the previous load cycle, the corresponding entry of the vector remains unchanged and the constitutive relationships of states ① – ③ are considered. However, if the vector indicates cracking in a previous load cycle, the applicable constitutive relationships for the cracked response are considered. Inversely, in layers where either one or both principal strains exceed the strain corresponding to cracking, the corresponding entry of the vector is set to a value of one and the applicable cracked response as defined by states ④ – ⑥ is considered.

This method of programming simulates the irreversible cracking process. Upon finding

a state of equilibrium, this new vector of ones and zeros is appended to the previous one in order to be considered in subsequent calculations. In this way, the depth of flexural cracks depends on the level of loading and a deterioration of concrete tensile strength with increasing load cycles can be considered.

5.6.4 Deterioration of concrete tensile strength and bond

Reinforced concrete elements subjected to repeated loads exhibit a progressive propagation of cracks, which leads to a redistribution of stresses and large deformations [98]. Fatigue tests of reinforced concrete members indicate that the increase in existing crack widths and formation of new cracks is especially pronounced within the first 100 to 1′000 load cycles [23]. Subsequent load cycles result in a greatly decreased, albeit invariably positive, rate of increase in existing crack widths until the rupture of individual reinforcing bars significantly changes the equilibrium conditions. Such changes in the load-deformation response can be attributed to a deterioration of the concrete tensile strength and of bond between the reinforcing bars and surrounding concrete. Both these effects are considered in the present formulation, albeit in a simplified manner. The concrete tensile strength as well as the terms in the stress-strain relationship of the reinforcing steel corresponding to tension stiffening effects are modified as a function of the number of load cycles.

The deterioration of concrete tensile strength under cyclic loading $f_{ct}(N)$ is considered according to the recommendation by *fib* Model Code [42] in the form

$$\begin{aligned} f_{ct}(N) &= \left(1 - \frac{\log N}{12}\right) \cdot f_{ct0} \quad \text{for} \quad N \leq 10^6 \\ f_{ct}(N) &= 0.5 \cdot f_{ct0} \qquad\qquad\quad \text{for} \quad N > 10^6 \end{aligned} \tag{5.22}$$

where f_{ct0} represents the static tensile strength of the concrete, see Fig. 5.9 (a). The introduction of a lower limit of half the static tensile strength after 10^6 load cycles is adopted from the formulation by *Fehlmann* [23].

The progressive deterioration of bond with repeated loading results in a reduction in the tension stiffening influence of the concrete. This process is considered by multiplying the term in Eq. (3.19) corresponding to the difference between the average steel stress between cracks and the maximum value at the crack with a load cycle dependent coefficient η_{TS}. The proposed reduction according to *Fehlmann* in [23] is implemented which varies logarithmically with the number of load cycles according to

$$\begin{aligned} \eta_{\mathrm{TS}}(N) &= \left(1 - \frac{\log N}{6}\right) \quad \text{for} \quad N \leq 10^6 \\ \eta_{\mathrm{TS}} &= 0 \qquad\qquad\quad \text{for} \quad N > 10^6 \end{aligned} \tag{5.23}$$

as illustrated in Fig. 5.9 (b). The influence of this modification on the stress-strain relationship of the reinforcement is illustrated in Fig. 5.9 (c) within the elastic range. For a number of load repetitions greater than 10^6, tension stiffening is neglected and the stress-strain relationship of the reinforcement reduces to that of an unbonded bar.

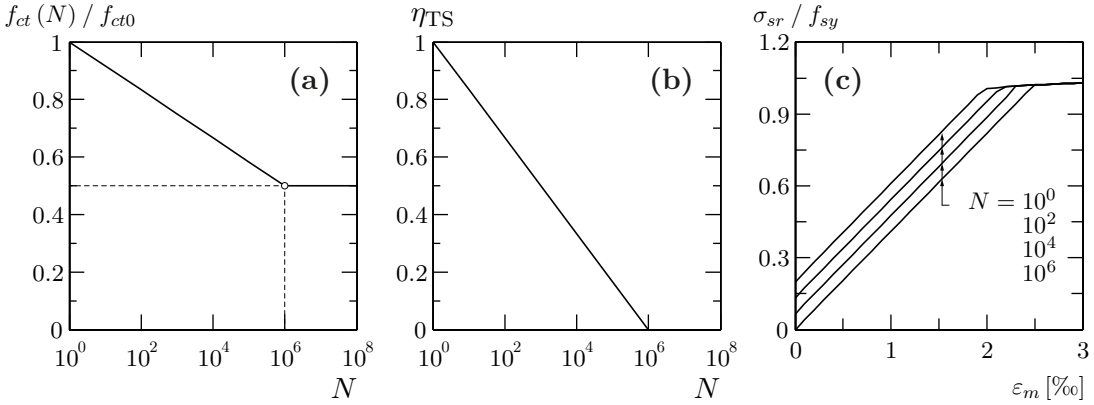


Fig. 5.9 – Modification of material parameters for cyclic loading: (a) concrete tensile strength; (b) reduction in tension stiffening; (c) modification of the stress-strain relationship of reinforcement in the elastic range, assuming $\rho = 2\%$.

5.6.5 Solution algorithm

The solution algorithm refers to the main part of the numerical method, i.e. the numerical solver which finds unique states of equilibrium in terms of the element mid-plane strains and curvatures for given stress resultants through Eq. (4.22). Once cracking occurs, the solution can only be found numerically in an iterative manner.

The procedure implemented for the solution algorithm is outlined with reference to the flow diagram illustrated in Fig. 5.7 in the following steps:

Define initial strain state An initial estimate of the mid-plane strains ε_{0j} as well as the curvatures χ_j is required to start the solution algorithm. It was found that the use of constant starting values for every calculation generally results in stable numerical convergence. One limitation, however, is that the initial values correspond to a combination of stress resultants below the ultimate resistance of the slab.

Slab strain state In each iteration, a continuous strain profile is calculated over the element height according to Eq. (4.13). This is subsequently reduced to discrete strain states at the centre of each concrete layer as well as for both cover elements. Moreover, the principal strains and their corresponding directions are calculated in each layer according to Eqs. (4.4), (4.3) and (4.5).

Slab stress state Using the principle strains determined in the previous step, principal concrete stresses are calculated in each layer using the constitutive relationships defined in states ① – ⑥, see Section 5.4. The reinforcing steel stresses are calculated at the cracks in both fictitious cover elements according to the Cracked Membrane Model. Furthermore, the contribution of each layer to the global stiffness matrix is calculated.

Calculate residuals on stress resultants The stress resultants $m_{i,\mathrm{it}}$ and $n_{i,\mathrm{it}}$ in each iteration are calculated through a summation of the concrete and reinforcing steel stresses in each layer according to Eq. (5.14). These are compared to the applied stress resultants and the vectors of residuals for bending moments and membrane forces are determined as

$$r_{m,\mathrm{it}} = m_i - m_{i,\mathrm{it}}, \qquad r_{n,\mathrm{it}} = n_i - n_{i,\mathrm{it}} \tag{5.24}$$

after which a *Square Root of the Sum of Squares* or SRSS on the residuals is calculated as

$$\mathrm{SRSS}_r = \sqrt{{r_{m,\mathrm{it}}}^2 + {r_{n,\mathrm{it}}}^2} \tag{5.25}$$

in order to quantify the error on the solution as a single value[1].

Invert stiffness matrix and update strain state The system of equations relating the stress resultants to the strains at mid-plane of the slab as well as the curvatures, see Eq. (4.22), is formulated in terms of the residuals. Pre-multiplication with the inverse of the stiffness matrix enables an estimation of the strain $\Delta\varepsilon_{0j,\mathrm{it}}$ and curvature increments $\Delta\chi_{j,\mathrm{it}}$ to be considered in the subsequent iteration

$$\begin{bmatrix} c_{1ij,\mathrm{it}} & c_{12ij,\mathrm{it}} \\ c_{21ij,\mathrm{it}} & c_{2ij,\mathrm{it}} \end{bmatrix}^{-1} \begin{bmatrix} r_{n,\mathrm{it}} \\ r_{m,\mathrm{it}} \end{bmatrix} = \begin{bmatrix} \Delta\varepsilon_{0j,\mathrm{it}} \\ \Delta\chi_{j,\mathrm{it}} \end{bmatrix}. \tag{5.26}$$

Updated mid-plane strains and curvatures for iteration $(\mathrm{it}+1)$ result from

$$\begin{bmatrix} \varepsilon_{oj,(\mathrm{it}+1)} \\ \chi_{j,(\mathrm{it}+1)} \end{bmatrix} = \begin{bmatrix} \varepsilon_{0j,\mathrm{it}} \\ \chi_{j,\mathrm{it}} \end{bmatrix} + \zeta \begin{bmatrix} \Delta\varepsilon_{0j,\mathrm{it}} \\ \Delta\chi_{j,\mathrm{it}} \end{bmatrix} \tag{5.27}$$

where the factor ζ is introduced to improve the numerical stability of convergence in a technique known as *under-relaxation* [73]. Essentially, the rate of advance of the solution process is slowed down through a scalar multiplication of the vector of increments by a factor ranging between 0 and 1. A value of $\zeta = 0.07$ generally resulted in robust convergence; however, at high computational cost due to an increased number of iterations.

Subsequently, the error SRSS_r is compared to a user-defined tolerance with respect to the solution accuracy in order to establish whether the strain state corresponding to the given stress resultants has been found. If this is not the case, the process is repeated with the updated mid-plane strains and curvatures until the error becomes acceptably small[2].

Post-processing Once an acceptable solution has been found, the slab deflection w_m is calculated for defined support conditions, span and loading using the *principal of virtual work*. In the present formulation, the midspan deflection of a slab in four-point-bending is calculated. Herein, the principal moment and corresponding curvature at the onset of cracking is determined. The principal curvature for the constant moment region results

[1] Square root and power functions in Eq. (5.25) refer to element-wise operations on the 3×1 residuals.

[2] Acceptable solutions were found in less than 300 iterations for elements discretised with 100 layers.

from the calculation for the defined principal moments. Furthermore, an estimate of crack width w_r on the slab surfaces is provided considering the larger principal concrete strain in the outer layers and the calculated crack spacing such that

$$w_r = s_r \varepsilon_1 \tag{5.28}$$

as considered in the Tension Chord Model, see Eq. (3.22).

5.6.6 Fatigue damage accumulation

The progressive accumulation of damage in the reinforcing steel due to repeated loading is considered with the linear damage rule, as described in Section 2.5.1. Herein, the damage increment d_i sustained during one load cycle at the stress range $\Delta\sigma_s \geq \Delta\sigma_{s,\mathrm{D}}$ within the finite-life region results to

$$d_i = \left(\frac{1}{N_{s,\mathrm{fat}}}\right)\left(\frac{\Delta\sigma_s}{\Delta\sigma_{s,\mathrm{fat}}}\right)^m \tag{5.29}$$

and for $\Delta\sigma_s < \Delta\sigma_{s,\mathrm{D}}$ in the long-life region to

$$d_i = \left(\frac{1}{N_{s,\mathrm{D}}}\right)\left(\frac{\Delta\sigma_s}{\Delta\sigma_{s,\mathrm{D}}}\right)^{2m-1} \tag{5.30}$$

considering $N_{s,\mathrm{fat}} = 2 \cdot 10^6$ and $N_{s,\mathrm{D}} = 5 \cdot 10^6$ load cycles with a slope in the finite-life region of $m = 4$. The corresponding fatigue strength curve is defined according to the documentation to the Swiss standard D 0133 [84], see Fig. 2.6 (b). The fatigue strength of individual bars $\Delta\sigma_{s,\mathrm{fat}}$ and the corresponding fatigue endurance limit $\Delta\sigma_{s,\mathrm{D}}$ is assigned as described in Section 5.6.2. The total accumulated damage D in a reinforcing bar results from a summation of the individual damage increments according to Eq. (2.8). Fatigue failure is assumed to occur once the sum of damage increments reaches unity $D = 1$.

5.6.7 Monte Carlo simulation

The *Monte Carlo method* makes use of random sampling, typically of results from computer simulations, in order to approximate the range of solutions to physical problems. This method is particularly useful to model phenomena with significant variability in the input values such as is the case with experimental results from fatigue tests.

The fatigue strength of individual reinforcing bars in the element formulation is considered as a stochastic parameter. In consequence, the predicted number of load cycles to fatigue failure of the first bar – and global failure of the slab – will vary between calculations. The Monte Carlo method is used to obtain an indication of the degree of variation in the predicted number of load cycles to fatigue failure. A large number of calculations are performed, often simultaneously on multiple computer processing units; the cause of non-convergence is verified, i.e. yielding of the reinforcement; and the outcomes are stored for further statistical analysis.

5.7 Monotonic loading response

In the present section, the results obtained from calculations considering constant or monotonically increasing load are presented and discussed. The calculated response of a slab subjected to different combinations of bending and torsional moments is illustrated with the aid of an example. The prediction of failure mode, firstly due to compressive failure of the concrete and secondly due to yielding of the reinforcement, is presented. Finally, the calculations are compared to experimental results from investigations on orthogonally reinforced concrete slabs reported in the literature.

5.7.1 Cross-sectional analysis

Fig. 5.10 (a) indicates an orthogonally reinforced concrete slab subjected to an arbitrary state of bending, defined by the principal moments $m_{1,2}$ and corresponding principal direction φ_1. The state of bending is expressed in terms of a normalised vector of bending and torsional moments with the scaling factor $\kappa = 100\,\mathrm{kNm/m}$. Calculations are performed for cases of uniaxial bending, pure torsion, biaxial bending as well as a general combination of bending and torsional moments. The distribution of principal strains, the crack inclination as well as principal concrete stresses over the slab height is illustrated in each case. Furthermore, the maximum reinforcing steel stresses at the cracks as well as the stresses in the compression reinforcement are indicated.

Uniaxial bending refers to the bending state wherein the direction of principal moment corresponds to a major axis of the slab, typically coinciding with a reinforcement direction, i.e. $m_x = m_1$, see Fig. 5.10 (b). The larger principal strain ε_1 reaches a maximum value on the slab soffit, which results in cracking perpendicular to the reinforcement in x-direction with $\theta_r = 90°$. The bottom layer of reinforcement in the x-direction is activated in tension. The smaller principal strain ε_2 reaches a minimum (negative) value on the top surface, leading to the principal concrete compressive stress σ_{c2r} parallel to the x-axis. Moreover, the reinforcement within the top layer in x-direction is activated in compression. The concrete tensile strength is considered as indicated by σ_{c1r} below the neutral axis.

A state of *pure torsion* is defined where $m_{xy} = m_1$, see Fig. 5.10 (c). The larger principal strain reaches maximum values at both outer surfaces of the slab, resulting in cracking at the angles $\theta_r = 135°$ and $45°$ for the top surface and soffit, respectively. As a result, both the top and bottom reinforcement in both orthogonal directions are activated in tension. Furthermore, principal concrete compressive stresses act on both the top and bottom surfaces at orientations parallel to the cracks.

In Fig. 5.10 (d), a case of *biaxial bending* is defined in which $m_x = m_1 = 2m_y$. Both the larger and smaller principal strains reach maximum values at the slab soffit such that cracking occurs in both principal directions at $\theta_r = 90°$ and $\theta_r = 0$. In consequence, the reinforcement in both the x- and y-direction is activated in tension in the bottom layers. On the slab top surface, both principal strains reach minimum values such that a state of biaxial compression develops in terms of the principal concrete stresses σ_{c1r} and σ_{c2r}.

A general state of bending is considered in Fig. 5.10 (e) for illustrative purposes. The moment resultants are defined as $m_i = \kappa\,[0.8,\, 0.2,\, 0.4]$ indicating a combination of bending and torsional moments. The larger principal strain reaches a maximum value at the slab soffit, resulting in cracking. Hence, both directions of reinforcement in the bottom layers are activated in tension. The smaller principal strain remains negative over the entire height of the cross-section such that a continuous principal compressive stress results in the direction indicated by θ_r; reaching a maximum value at the slab top surface.

5.7.2 Prediction of failure

The assumption that a breakdown in numerical convergence corresponds to failure of the slab is validated in two calculations indicating different modes of failure in bending. These constitute (1) a case of compression failure of the concrete in the flexural compression zone and (2) conditions resulting in yielding of the flexural reinforcement. The same orthogonally reinforced concrete slab is considered in both calculations. The combination of bending and torsional moments corresponds to $m_i = \kappa\,[0.5,\, 0.5,\, 0.5]$ where the scaling factor κ is increased from zero to a level where equilibrium can no longer be found. A concrete cylinder strength of $f_{cc} = 40\,\mathrm{N/mm^2}$ is considered in the calculations illustrating crushing of the concrete. In order to investigate failure through yielding of the reinforcement, the concrete strength is increased to $f_{cc} = 50\,\mathrm{N/mm^2}$ and the reinforcement content reduced. Tension stiffening effects are considered in both calculations with $\lambda = 1$.

Compressive failure of concrete

Fig. 5.11 indicates the results of calculations predicting flexural failure due to compression failure of the outer layers of concrete. The calculated load-deformation response is shown from the uncracked condition up to the prediction of failure. Principal concrete stresses and maximum reinforcement stresses at the cracks are illustrated at three levels of loading corresponding to $\kappa = 100$, 150 and 195 kNm/m. The calculations are able to simulate the stiff, uncracked response as well as the abrupt increase in the response quantities corresponding to cracking of the cross-section. The principal concrete compressive stress on the slab top surface indicates a nonlinear response after cracking. This is due to the parabolic stress-strain relationship implemented for the concrete in compression. The profile of the principal concrete compressive stress over the slab thickness becomes increasingly nonlinear under increasing load, see Fig. 5.11 (d). In the top layer, this value approaches the peak concrete compressive strength f_c at $\kappa = 195\,\mathrm{kNm/m}$. Failure is predicted shortly thereafter at $\kappa_u = 204\,\mathrm{kNm/m}$ where the principal concrete compressive stress σ_{c2r} within the top surface layer reaches the peak compressive strength. The reinforcement stress at this level of loading remains below that at the onset of yielding, see Fig. 5.11 (e). The prediction of failure is justified as no further states of equilibrium can be found.

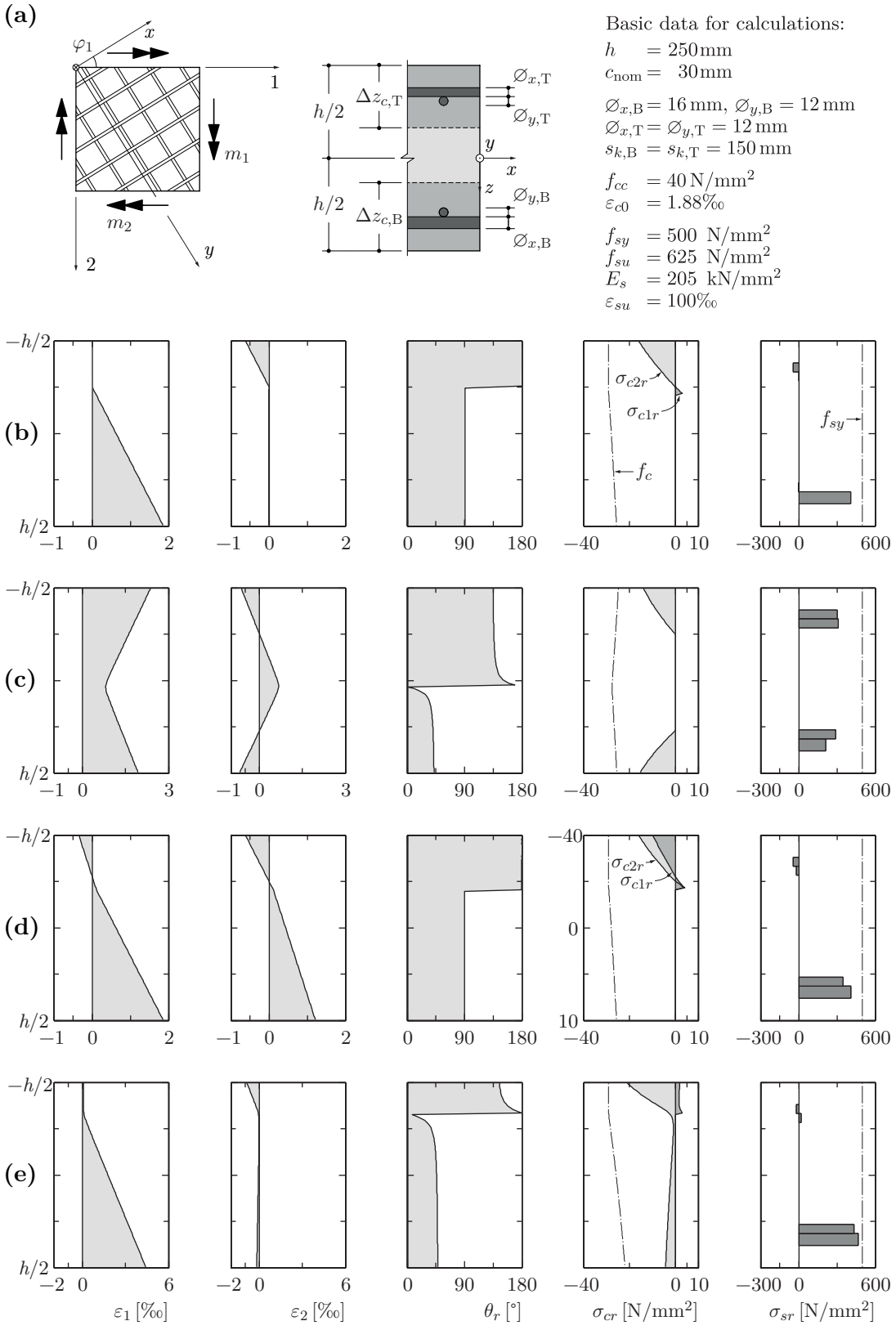


Fig. 5.10 – Cross-sectional analysis of slab in bending: (a) notation; (b) uniaxial bending, $m_i = \kappa\,[1,\,0,\,0]$; (c) pure torsion, $m_i = \kappa\,[0,\,0,\,1]$; (d) biaxial bending, $m_i = \kappa\,[1,\,0.5,\,0]$; (e) general state of bending, $m_i = \kappa\,[0.8,\,0.2,\,0.4]$. Note: $\kappa = 100\,\mathrm{kNm/m}$.

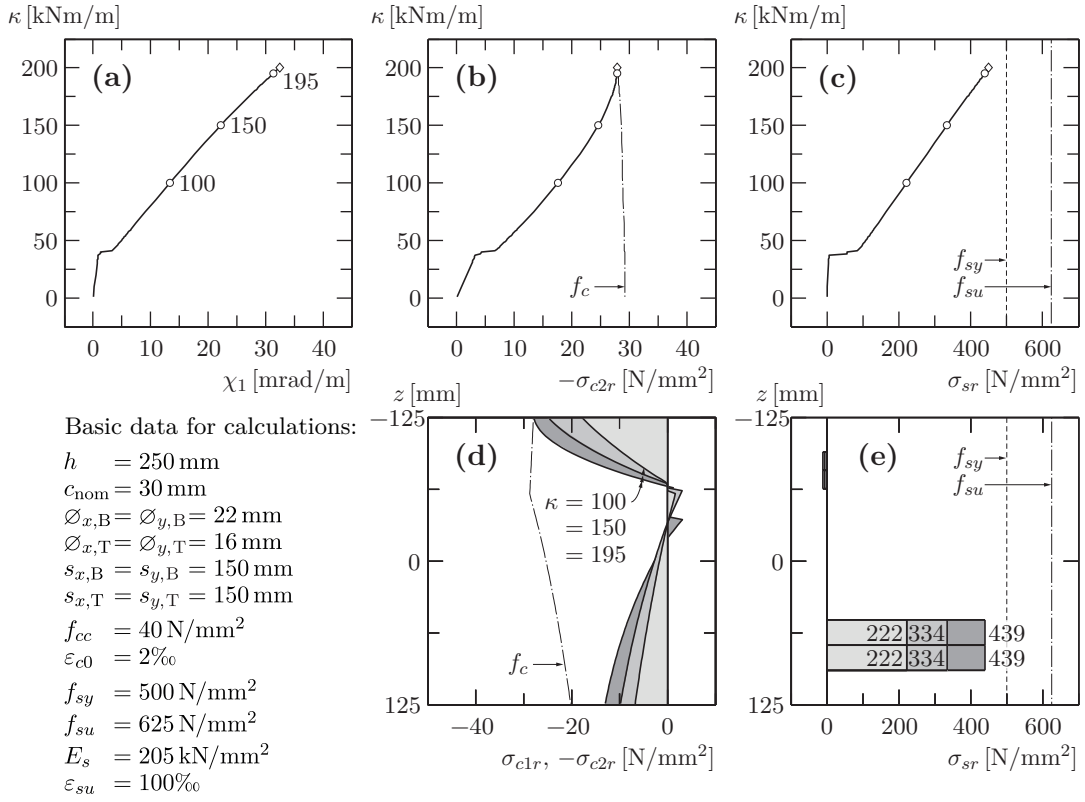


Fig. 5.11 – Calculations indicating failure of the flexural compression zone: (a) moment-principal curvature response; (b) moment-principal concrete stress on top surface; (c) moment-steel stress in bottom layers; (d) and (e) cross-sectional analysis of principal concrete compressive stress and steel stresses. Note: $m_i = \kappa\,[0.5,\ 0.5,\ 0.5]$.

Yielding of reinforcement

The prediction of failure due to yielding of the reinforcement is indicated in Fig. 5.12. The response quantities after cracking indicate a roughly linear increase up to a value of $\kappa = 110\,\text{kNm/m}$. A further increase in load is predicted to result in the simultaneous yielding of both bottom layers of reinforcement. The reinforcement stresses continue to increase thereafter in order to maintain equilibrium at the higher loading. The reinforcement modulus of strain-hardening, however, is roughly 160 times smaller than the initial modulus of elasticity. As a result, the overall stiffness is greatly reduced, see Fig. 5.12 (a). A significant increase in the principal concrete compressive stresses on the top surface of the slab is observed at $\kappa = 118.5\,\text{kNm/m}$ after the onset of yielding of the reinforcement. The reinforcing steel stress cannot increase significantly after yielding; hence, higher levels of bending action can only be resisted by an increase in the lever arm of internal forces. This necessitates high compressive concrete stresses within the topmost layer. Failure is predicted to occur as the principal concrete stress in the outer layer reaches the peak compressive strength f_c. The calculations are able to consider yielding of the reinforcement; correctly predicting failure of the slab due to the breakdown in numerical convergence.

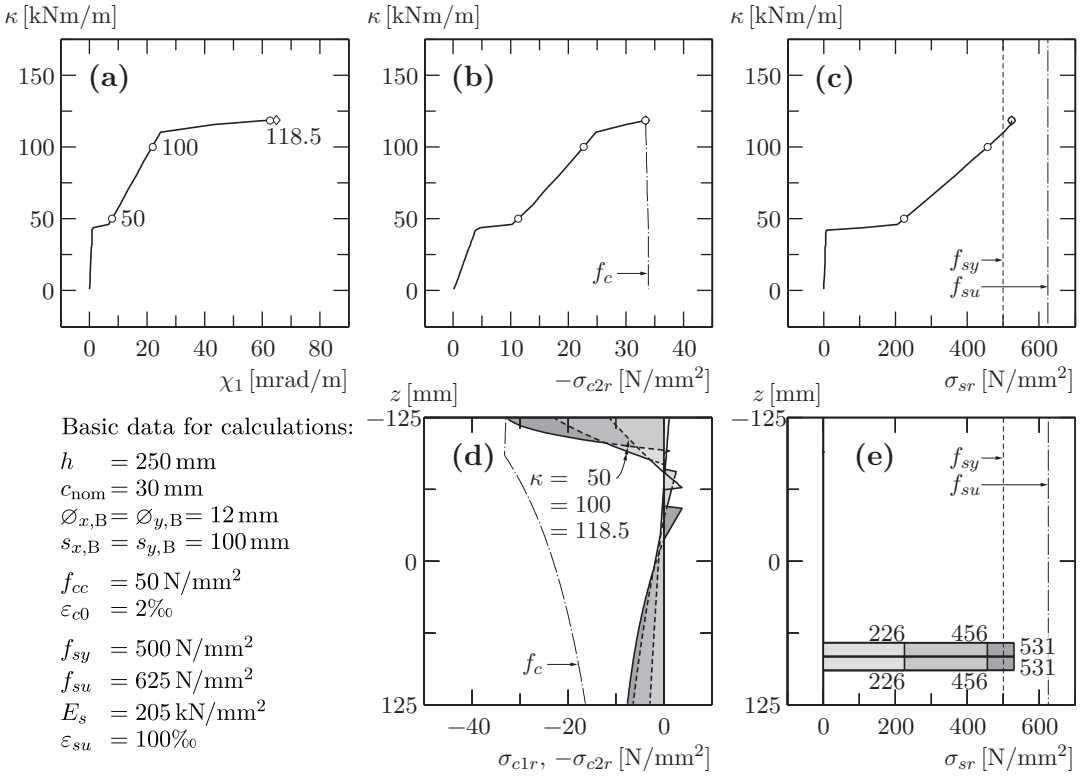


Fig. 5.12 – Calculations indicating yielding of the reinforcement: (a) moment-principal curvature response; (b) moment-principal concrete stress on top surface; (c) moment-steel stress in bottom layers; (d) and (e) cross-sectional analysis of principal concrete compressive stress and steel stresses. Note: $m_i = \kappa\,[0.5,\ 0.5,\ 0.5]$.

5.7.3 Comparison to experimental results

Calculations are compared to experimental results from tests on orthogonally reinforced concrete slabs reported in the literature. The test parameters represent various combinations of bending and torsional moment for a wide range of reinforcement contents. The results from calculations in which tension stiffening effects are considered with $\lambda = 1$ as well as those in which tension stiffening is neglected are indicated.

Tests by *Lenschow and Sozen*

The first set of experimental results refers to an investigation by *Lenschow* and *Sozen*, the details of which are provided in [54]. Slabs with a thickness of 105 mm were tested in a four-point bending setup such that a region of constant bending moment of dimensions 1067 × 1524 mm resulted. The primary variable in the three considered tests was the direction of reinforcement with respect to the direction of principal moment. Deviations between these directions of $\varphi_n = 0$, 45° and 22.5° were investigated in tests B4, B5 and B6, respectively. All three specimens had the same isotropic reinforcement layout consisting of imperial #2 bars (⌀ 6.4 mm) at a spacing of 38 mm. The curvature was measured with

dial gauges, whereas concrete strains on the top surface and soffit of the slabs as well as reinforcing steel strains were measured with electrical strain gauges. In the latter case, the strain gauges were applied directly to the surface of individual reinforcing bars within the test region.

The comparison of the calculations for test B4 to the experimental results is illustrated in Fig. 5.13. The x-direction of reinforcement coincided with the principal moment direction, such that $m_i = \kappa\,[1,\,0,\,0]$. Overall, a good correlation is obtained between the calculations and experimental results. The steel strains measured in four strain gauges on the x-direction of reinforcement correlate well with the calculations and vary within the theoretical limits. According to the test report, poor performance of the type of strain gauge prevented further measurement of steel strains after the onset of yielding.

Fig. 5.14 indicates the comparison of the calculations to the experimental results of test B5. The orthogonal reinforcement layout was rotated by 45° with respect to the principal moment direction. This corresponds to a combination of bending and torsional moments of $m_i = \kappa\,[0.5,\,0.5,\,0.5]$ with respect to the reinforcement directions. The calculations show a good correlation for this combination of bending moments; especially when considering tension stiffening effects. Furthermore, the onset of yielding of the reinforcement is predicted correctly. The only discrepancy is observed in the calculation of the concrete strain on the slab top surface $\varepsilon_{c1,\mathrm{sup}}$, perpendicular to the direction of principal moment, see Fig. 5.14 (b). The calculations overestimate this concrete strain resulting from *Poisson* effects; even predicting cracking in the top concrete layers. On the other hand, the calculated reinforcing steel strains show a good correlation with the measurements from two strain gauges on the reinforcing steel in both the x- and y-directions, see Fig. 5.14 (d) and (e).

The results of test B6 are compared to calculations in Fig. 5.15. Herein, the orthogonal reinforcement layout was rotated such that a deviation of 22.5° and −67.5° resulted between the x- and y-directions of reinforcement with respect to the principal moment direction, respectively. The combination of bending and torsional moments result in $m_i = \kappa\left[\cos^2\varphi_1,\ \sin^2\varphi_1\,;\sin\varphi_1\cos\varphi_1\right]$ with $\varphi_1 = 22.5°$. Again, the calculations generally correlate well with the experimental results. Furthermore, the comparison suggests that the consideration of tension stiffening effects generally provides a more realistic representation of the load-deformation response. The onset of yielding in the more highly stressed x-direction of reinforcement is predicted correctly. The strains in the y-direction of reinforcement, however, are underestimated after the onset of yielding of the x-direction. Similar to test B5, the principal concrete tensile strain perpendicular to the direction of principal moment on the compressive surface is overestimated in the calculations.

Tests by *Marti, Leesti and Khalifa*

A series of nine orthogonally reinforced concrete slabs were tested in a state of pure torsion $m_i = \kappa\,[0,\,0,\,1]$ by *Marti, Leesti and Khalifa* [63]. The slabs were 200 mm thick with square plan dimensions of 1700×1700 mm and geometric reinforcement contents varying between

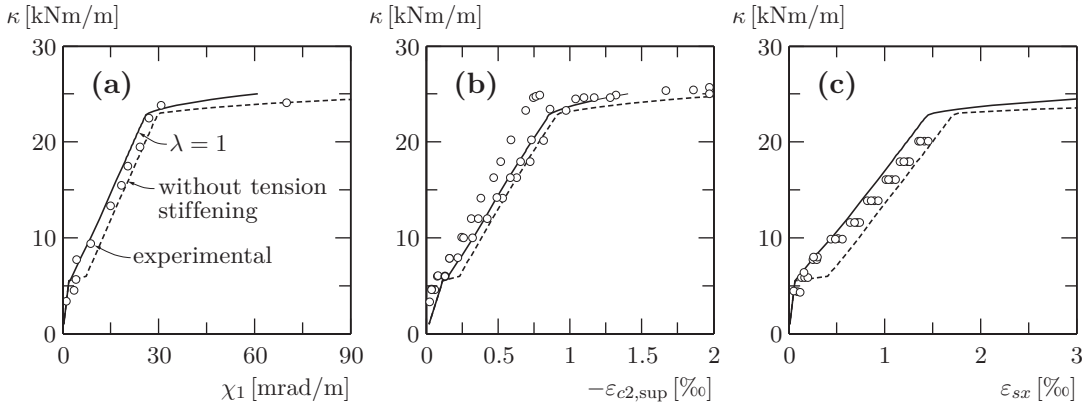


Fig. 5.13 – Comparison to test B4 by *Lenschow and Sozen* [54]: (a) moment-curvature; (b) reinforcing steel; and (c) principal concrete compressive strains. Basic data for calculations: $h = 105\,\text{mm}$, $c_{\text{nom}} = 9.5\,\text{mm}$, $\varnothing_{x\text{B}} = 6.4\,\text{mm}$, $s_{x\text{B}} = 38\,\text{mm}$, $\varphi_n = 0$, $f_{cc} = 32.7\,\text{N/mm}^2$, $\varepsilon_{c0} = 2‰$, $f_{sy} = 345\,\text{N/mm}^2$, $f_{su} = 500\,\text{N/mm}^2$, $\varepsilon_{su} = 100‰$, $E_s = 200\,\text{kN/mm}^2$. Note: $m_i = \kappa\,[1,\ 0,\ 0]$.

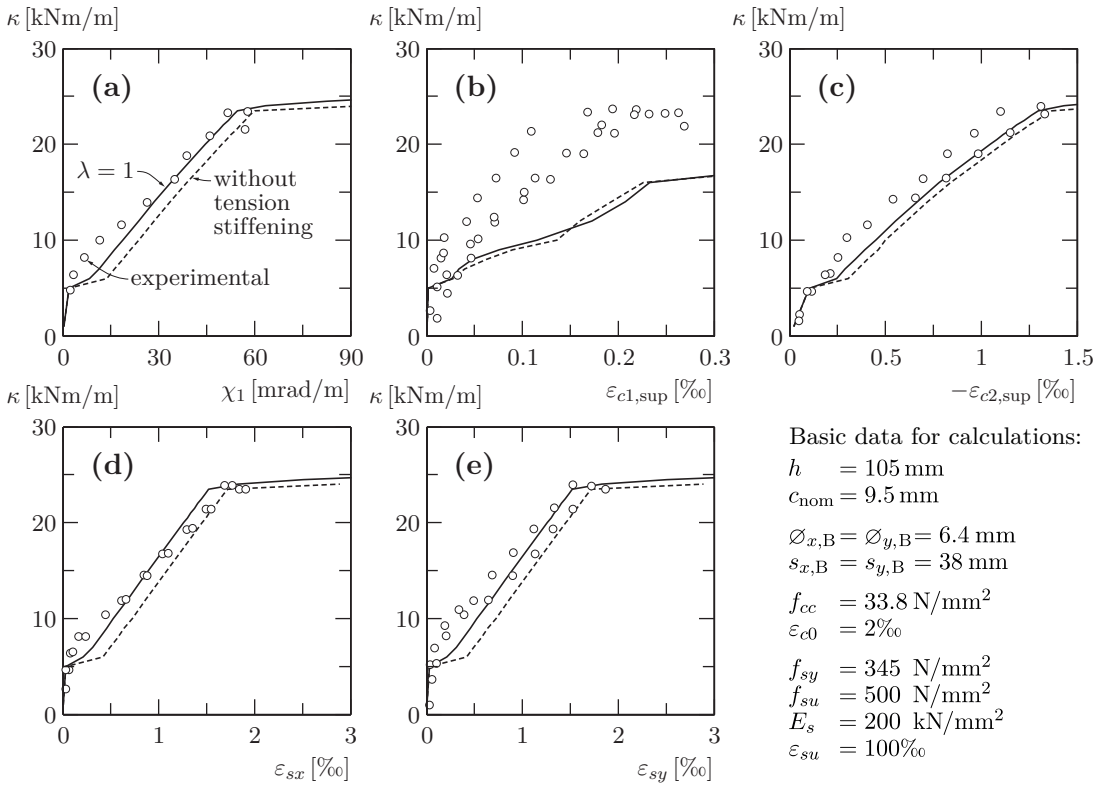


Fig. 5.14 – Comparison to test B5 by *Lenschow and Sozen* [54]: (a) moment-curvature; (b) and (c) moment-principal concrete strains on the compressive surface; (d) and (e) reinforcing steel strains in x- and y-direction. Note: $\kappa\,[0.5,\ 0.5,\ 0.5]$.

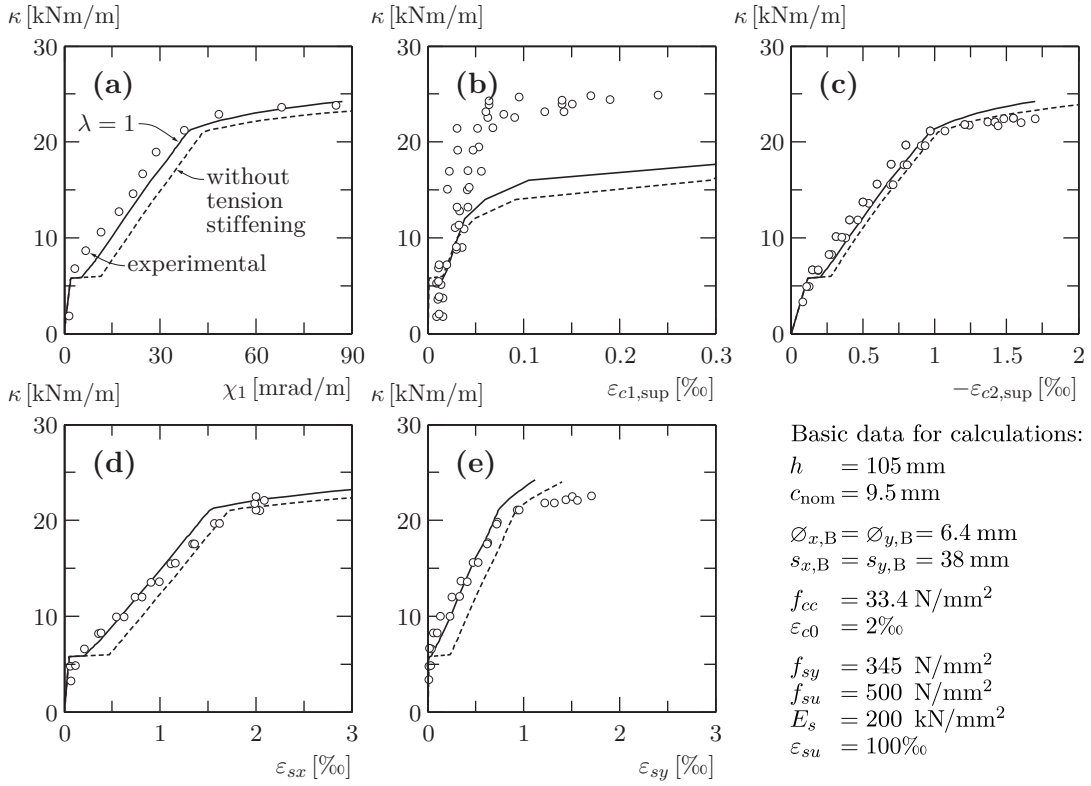
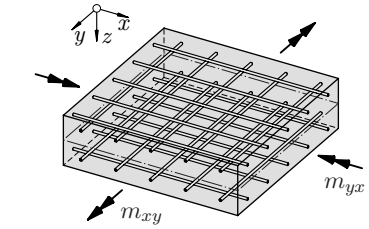


Fig. 5.15 – Comparison to test B6 by *Lenschow and Sozen* [54]: (a) moment-curvature; (b) and (c) moment-principal concrete strains on the compressive surface; (d) and (e) reinforcing steel strains in x- and y-directions. Note: $m_i = \kappa \left[\cos^2\varphi_1,\, \sin^2\varphi_1,\, \sin\varphi_1 \cos\varphi_1\right]$ with $\varphi_1 = 22.5°$.

0.25% and 1%. Five of the tested slabs were reinforced isotropically. In the remaining four tests, the geometric reinforcement content in the x-direction was doubled or increased fourfold compared to the reinforcement in the y-direction. A premature failure of the corner region occurred in all except for the tests ML6, ML8 and ML9 due to a concentrated application of the load. The final measurements are therefore generally not representative of the ultimate resistance to torsional moment.

A comparison of the calculations to the experimental results of tests ML1 to ML9 is shown in Fig. 5.16. The calculations generally provide a realistic prediction of the torsional moment-curvature response for both isotropically and orthotropically reinforced specimens. In tests ML1 and ML4, exhibiting the lowest geometric reinforcement contents, the calculations neglecting the effects of tension stiffening provide a better prediction of the response. The tests with high reinforcement contents, such as ML2, ML3, ML6 and ML9, indicate a more pronounced nonlinear response as the stress-strain relationship of the concrete in the outer layers becomes decisive. This effect is predicted satisfactorily in the calculations.



Basic data for calculations:

h	$= 200\,\text{mm}$	f_{cc}	$= 23.3 \ldots 49.1\,\text{N/mm}^2$
c_{nom}	$= 16 \ldots 18\,\text{mm}$	ε_{c0}	$= 2.12 \ldots 2.50‰$
$\varnothing_s$	$= 11.28\,\text{mm}\,(10\text{M}) \ldots$ $15.96\,\text{mm}\,(15\text{M})$	f_{sy}	$= 412 \ldots 551\,\text{N/mm}^2$
		f_{su}	$= 660\,\text{N/mm}^2$
		E_s	$= 200\,\text{kN/mm}^2$
$\rho_{x,y}$	$= 0.25 \ldots 1\%$	ε_{su}	$= 100‰$

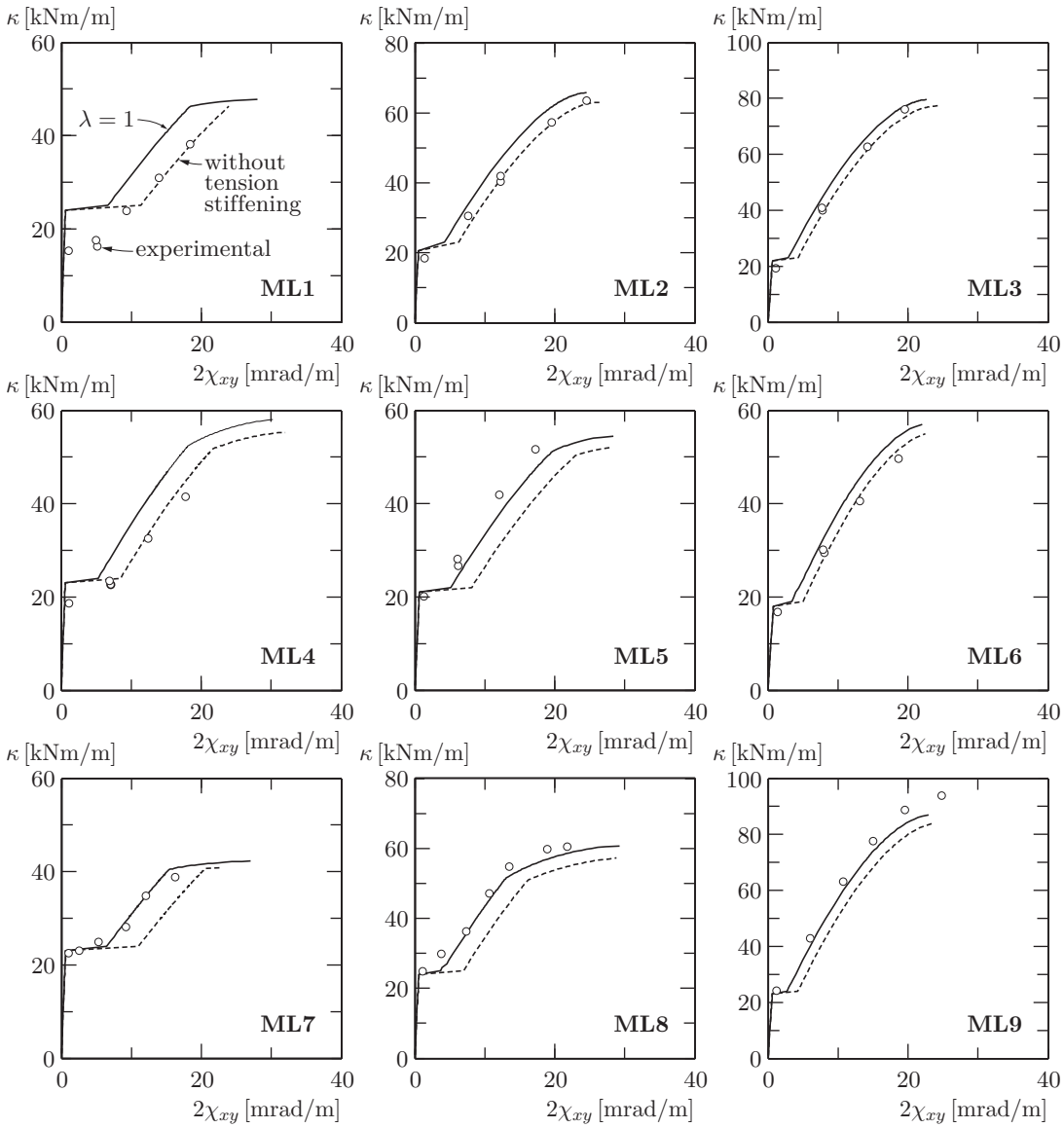


Fig. 5.16 – Comparison to torsion tests ML1 to ML9 by *Marti, Leesti and Khalifa* [63]. Note: $m_i = \kappa\,[0,\ 0,\ 1]$.

5.8 Cyclic loading response

In the present section, the calculations corresponding to cyclic loading are discussed. Firstly, the accumulation of fatigue damage in individual reinforcing bars is outlined. The assessment of load-deformation response at maximum and minimum load is examined with regard to the loading and unloading paths. Subsequently, the influence of the stress range in relation to the fatigue endurance limit of the reinforcement on the global response is examined. In conclusion, the allocation of fatigue strengths to individual reinforcing bars and the influence thereof on the fatigue life prediction is presented.

5.8.1 Fatigue failure of reinforcing bars

The accumulation of fatigue damage in the reinforcement is considered with the linear damage rule as described in Section 5.6.6. The magnitude of individual damage increments sustained in each load cycle is a function of the stress range in the bar $\Delta\sigma_s$ and its corresponding fatigue strength $\Delta\sigma_{s,\mathrm{fat}}$, see Eqs. (5.29) and (5.30). The stress range is assumed to be equal in all reinforcing bars of a layer of reinforcement. The assigned fatigue strengths, however, vary among the individual bars. A higher rate of damage accumulation results in the "weaker" bars than in those for which a higher fatigue strength is assigned.

Fig. 5.17 (a) indicates a typical damage accumulation history of the bars in a layer of reinforcement. The bar for which the lowest fatigue strength is assigned exhibits the highest rate of damage accumulation; the total accumulated damage $D_{sk,1}$ reaching unity at roughly 70% of the number of load cycles until global failure. This bar is assumed to fail and is excluded from further calculations. Higher stresses ensue in the remaining bars in order to maintain equilibrium for the same load level. The increased stress range in the remaining bars alter the magnitude of the damage increment sustained in each load cycle. This can be observed as an increase in the slope of the damage accumulation curves.

The total damage accumulated within the reinforcement layer is represented as the sum of accumulated damage in the individual bars $\sum D_{sk}$, see Fig. 5.17 (b). The dashed line represents a constant rate of damage accumulation such as would be observed assuming the same fatigue strength for all bars. The implemented damage accumulation is piecewise linear; increasing initially at a lower rate than represented by the dashed line. The rate of damage accumulation increases markedly at load cycles approaching the fatigue life of the element. This, however, only occurs after failure of the bar with the lowest fatigue strength. This approach represents a more realistic and less conservative treatment of fatigue damage accumulation than typical calculations assuming a single – often conservatively defined – value of the fatigue strength for all bars. The major drawbacks of this approach are the increased computing time required to consider individual bars and the need for random sampling algorithms, such as the Monte Carlo method, to generate numerical results.

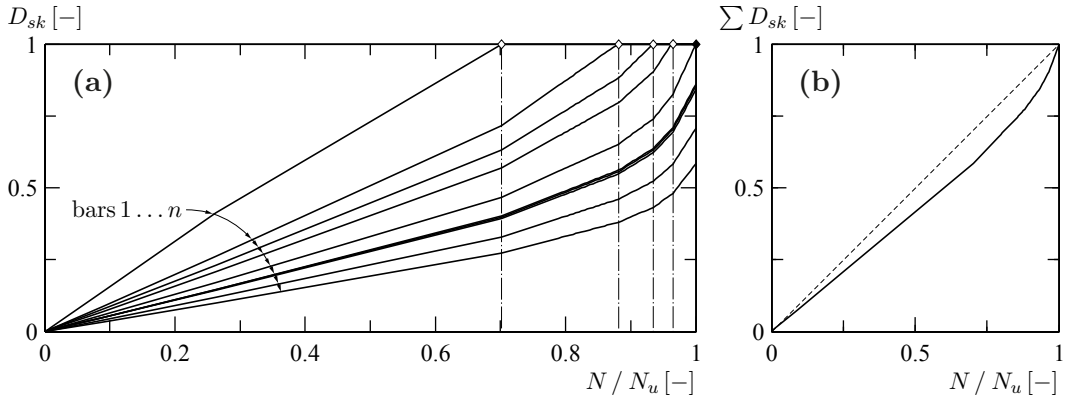


Fig. 5.17 – Damage accumulation in a layer of reinforcement: (a) individual reinforcing bars; (b) sum of accumulated damage in all bars.

5.8.2 Calculations at maximum and minimum load

The formation of flexural cracks in a reinforced concrete element leads to irreversible deformations, which remain even after complete unloading. Mathematical descriptions of the loading and unloading paths under cyclic action, however, are still unreliable and would most likely require some empirical modifications.

Loading history in the present formulation is considered by recording the formation of cracks in the concrete layers and altering the constitutive relationships for these layers in the calculations at subsequent load cycles. Path-dependant constitutive relationships, however, are not considered. Calculations at minimum load, corresponding to the unloading of the slab, result in lower predictions of the response quantities as the residual deformations are not considered. Residual deformations in the current formulation arise solely from the specified deterioration of concrete tensile strength and bond.

This limitation is illustrated in the calculated moment-curvature response at discrete load cycles shown in Fig. 5.18. For comparison, the corresponding principal concrete compressive stress on the slab compressive surface as well as the reinforcement stresses are shown. The initial curve progressing from the origin through the uncracked and cracked response to the maximum load level represents the initial loading at $N = 1$. The unloading branch, however, follows the same path as considered during loading to the minimum load level. The subsequent curves represent the straight line connection between the calculations at maximum and minimum load for increasing load cycles. The stress range in the reinforcing steel $\Delta\sigma_s$ is overestimated in these calculations; thus, resulting in a conservative prediction of fatigue life. Furthermore, the deterioration of the concrete tensile strength and the influence of tension stiffening results in a horizontal translation of the curve for all the response quantities. In the final load cycles, failure of individual bars results in the significant changes observed in the response. Particularly the calculated curvature as well as principal concrete and reinforcing steel stresses at maximum load increase disproportionally as the response becomes increasingly nonlinear.

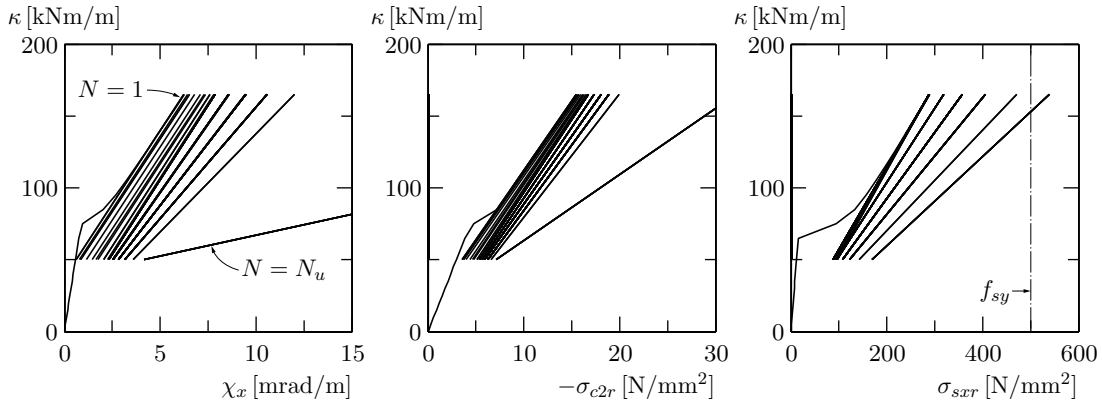


Fig. 5.18 – Calculations of moment-curvature, principal concrete stress on the slab surface and maximum reinforcing steel stress at maximum and minimum load of cyclic loading. Note: $\kappa_{\max} = 165\,\text{kNm/m}$ and $\kappa_{\min} = 50\,\text{kNm/m}$.

5.8.3 Influence of stress range in reinforcement

Fig. 5.19 illustrates a sensitivity study on the influence of the stress range in the reinforcement $\Delta\sigma_{sk}$ on the calculated fatigue response. Herein, the same orthogonally reinforced concrete slab is considered and the fatigue strengths of individual reinforcing bars, assigned according to the normal distribution of values, are kept constant for all calculations. Three combinations of bending and torsional moments are considered, namely, uniaxial, biaxial as well as a general state of bending. In the case of uniaxial bending, only the x-direction of reinforcement in the bottom layer is activated, see Fig. 5.19 (a). For biaxial bending, both the x- and y-directions of reinforcement are activated, however, the torsional moment is zero, see Fig. 5.19 (b). Fig. 5.19 (c) represents a general state of bending where both the x- and y-directions of reinforcement are activated due to a combination of bending and torsional moment. Three levels of reinforcing steel stress range are considered for each of the three cases. The steel stress range at the first level corresponds roughly to the mean fatigue endurance limit $\Delta\sigma_{s,\text{D}}$ of the bars. In the second set of calculations, the steel stress range in the x-direction of reinforcement is slightly higher than the fatigue endurance limit of the bars. High stress ranges are considered at the third level; exceeding the fatigue endurance limit of the bars for all layers of reinforcement. Moreover, the corresponding sum of accumulated fatigue damage $\sum D_s$ in all bars of a layer of reinforcement is indicated.

The remaining phase or number of load cycles between failure of the first bar and global failure of the element is defined herein as the *remaining fatigue life*. This phase is found to be greater at the lower stress range levels in the order of the fatigue endurance limit of the bars. This observation is valid for all three combinations of bending and torsional moment considered. A similar observation was made by *Fehlmann* in the investigation of beams reinforced with multiple bars [23]. At stress range levels below the fatigue endurance limit damage accumulation occurs within the long-life region according to Eq. (5.30). The rate of damage accumulation is correspondingly low and increases only marginally upon

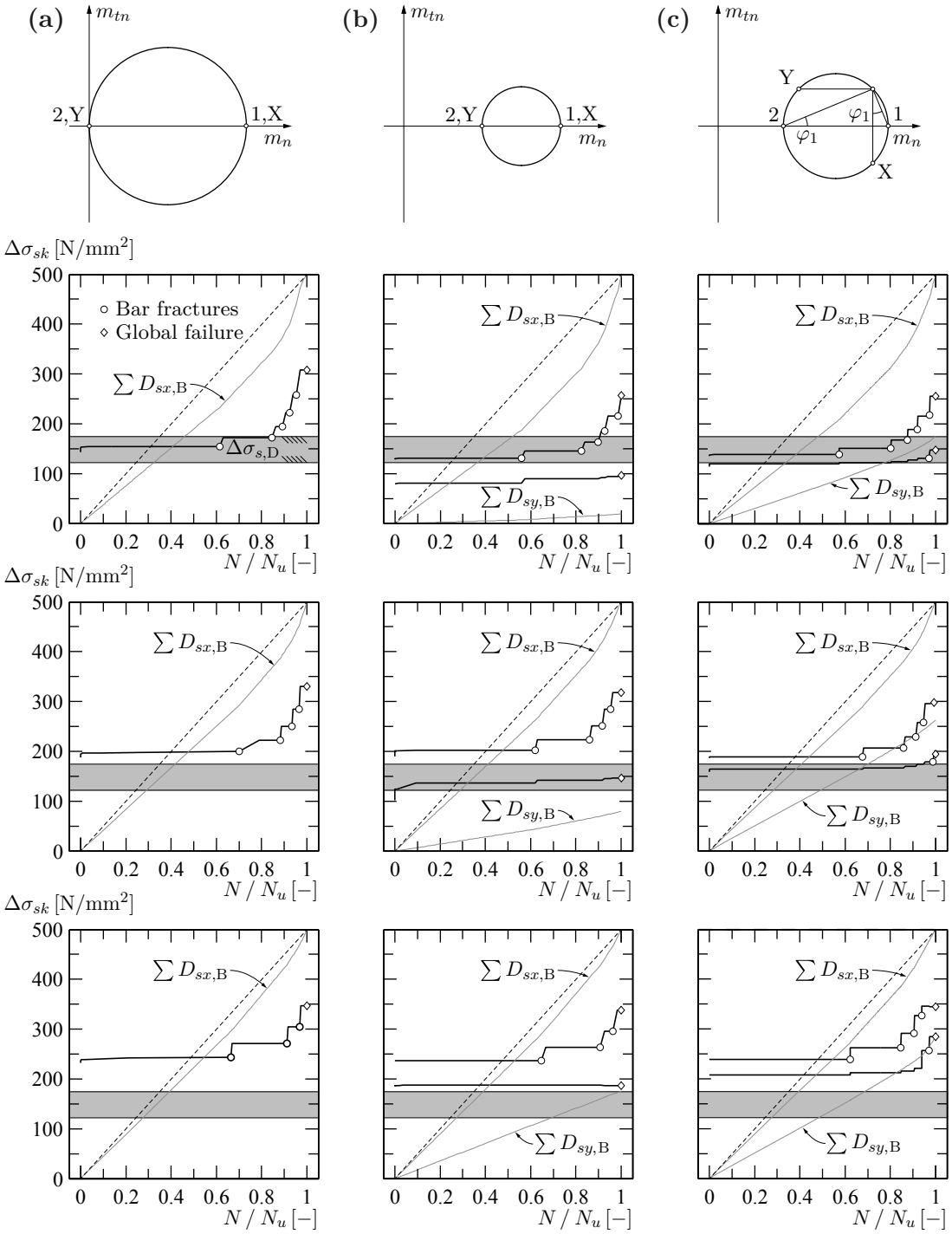


Fig. 5.19 – Development of reinforcing steel stress range at different intensities of cyclic loading: (a) uniaxial bending; (b) biaxial bending; (c) general state of bending.

fracture of individual bars. A transition occurs when the redistribution of internal forces results in a stress range exceeding the fatigue endurance limit. Damage accumulation subsequently occurs in the finite-life region according to Eq. (5.29), resulting in a rapid increase in the rate of accumulation. At stress range levels exceeding the fatigue endurance limit the accumulation of damage occurs entirely within the finite-life region. The rate of damage accumulation approaches the rate which would be obtained with the linear damage rule (indicated by the dashed line) considering the same fatigue strength for all bars. In conclusion, the use of the linear damage rule for high expected stress range levels may represent a sufficiently realistic approach. In the case of low stress ranges, however, the nonlinear approach proposed in Section 2.5.2 might be more appropriate.

5.8.4 Assignment of reinforcing bar fatigue strengths

The fatigue strength of the reinforcement is the primary parameter limiting the fatigue life of lightly reinforced concrete elements in bending. The present formulation considers the fatigue strength of individual bars as a stochastic parameter, which is assigned according to a normal distribution of values, see Section 5.6.2. Hence, factors such as the specific set of considered fatigue test results and the statistical distribution fitted to this database will influence the fatigue life prediction. The sensitivity of such predictions to the fatigue strengths assigned to the individual bars warrants investigation. The Monte Carlo method is used in order to obtain an indication of the mean and variation of fatigue life predictions from a large number of calculations.

Fig. 5.20 (a) indicates 25 calculations of fatigue life considering the unmodified normal distribution of fatigue strengths. A mean fatigue life of $N = 1'572'000$ with corresponding variation of $323'100$ load cycles is obtained. In the subsequent 25 calculations, the normal distribution of fatigue strengths is truncated to contain only values between the 5^{th} and 95^{th} percentiles, see Fig. 5.20 (b). In this way, the probability of assigning an outlier value of fatigue strength to the bars is limited. The obtained mean fatigue life is only slightly lower, resulting in $N_u = 1'397'600$. However, a significant reduction in the variation to $226'900$ load cycles is observed.

Fig. 5.20 (c) indicates calculations in which the fatigue strengths of all reinforcing bars are assumed to be the same. Considering a fatigue strength equal to the mean value of the normal distribution, i.e. $\Delta\sigma_{s\mu,\text{fat}} = 208\,\text{N/mm}^2$, results in $N_u = 2'290'000$. Assigning the nominal fatigue strength of $\Delta\sigma_{s,\text{fat}} = 170\,\text{N/mm}^2$ for straight, small diameter bars according to [84] results in a fatigue life prediction of $N_u = 1'019'000$. This corresponds favourably to the mean values obtained with both the normal and the truncated distribution of fatigue strength. It is, however, slightly more conservative as would be expected. Considering the design value of fatigue strength, i.e. $\Delta\sigma_{sd,\text{fat}} = 145\,\text{N/mm}^2$ according to [87], results in a fatigue life of only $N_u = 540'000$.

Fatigue life predictions varying by more than a factor of two result from consideration of the unmodified normal distribution of fatigue strength. The interpretation of such results is difficult and may be misleading in the evaluation of structures. Assigning fatigue strengths

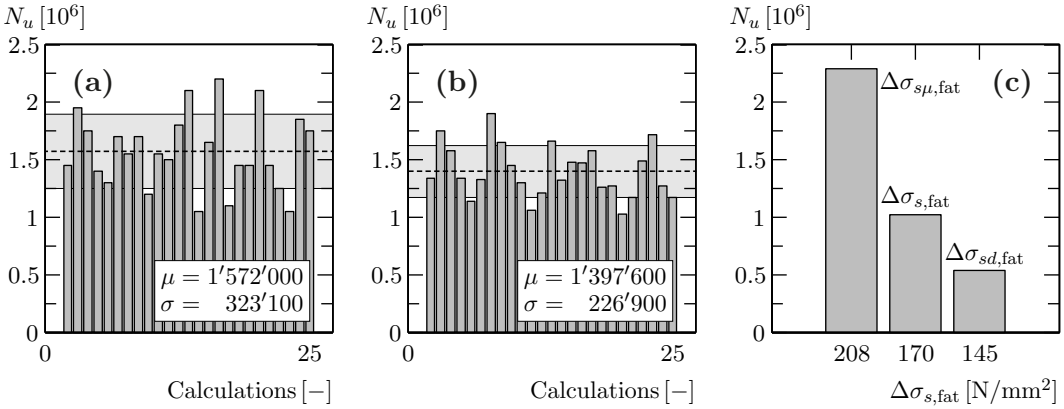


Fig. 5.20 – Influence of assigned reinforcement fatigue strength on fatigue life prediction: (a) normal distribution fitted to test results; (b) normal distribution truncated to values between the 5$^{\text{th}}$ and 95$^{\text{th}}$ percentiles; (c) mean of all test results, nominal and design fatigue strength according to [84, 87].

to individual reinforcing bars according to the truncated normal distribution results in a smaller variation in fatigue life predictions. Therefore, despite a slightly more conservative prediction for the case considered, the use of the truncated normal distribution of fatigue strength is favoured above that of the unmodified normal distribution.

Chapter 6

Experimental campaign

6.1 General

In the present chapter, details of two series of large-scale tests performed in the structural laboratory of ETH Zurich, Hönggerberg between 2015 and 2016 will be provided. Four slab specimens were tested under constant force-amplitude cyclic loading. The specimens were reinforced with standard profiled reinforcing steel bars in an orthogonal layout. Two different orientations of the reinforcement layout were investigated: (1) with the principal moment and reinforcement direction coinciding; and (2) wherein a deviation of 45° was introduced between the principal moment and reinforcement directions.

Firstly, the objectives of the experimental campaign will be defined. Thereafter, the testing program will be introduced explaining the selected test parameters. A description of the specimens and testing procedure will be provided. The results of the individual tests in each series will be presented in Section 6.6, which will be followed by a comparison of the tests in each series (Section 6.7). Finally, some additional considerations with regard to the test results will be discussed and the conclusions summarised.

6.2 Objectives

An experimental campaign was carried out to investigate the fatigue performance of orthogonally reinforced concrete slabs wherein the principal moment and reinforcement direction do not coincide. The following objectives were defined:

- to investigate whether an inclination of reinforcing bars with respect to the principal moment/stress direction detrimentally influences their fatigue strength;
- to study the redistribution of internal forces within the reinforcement after failure of individual bars under cyclic loading;
- to characterise the development in response in the remaining failure life between failure of the first reinforcing bar and global failure of the slab.

6.3 Testing program

Two slab strips in each of the two series of tests were constructed with the reinforcement layouts indicated in Fig. 6.1 (a). The tests of series A constituted the reference wherein only one layer of reinforcement was activated due to bending action. Both layers of reinforcement were activated in the tests of series B through a rotation of the orthogonal layout by $\varphi_n = 45°$ with respect to the principal moment direction. Specimens were subjected to constant amplitude cyclic loading wherein the *maximum load* $F_{\max}$ corresponded to a bending demand of approximately 65% of the yield moment. The *minimum load* $F_{\min}$ was defined such that the expected reinforcing steel stress range exceeded the fatigue endurance limit within all layers of reinforcement activated in bending. The stress range in the reinforcement is the primary parameter limiting the number of load cycles at which failure occurs. Therefore, the testing program was defined with the aim to induce similar stress ranges in the reinforcement of both series of tests.

In the tests of series A, the orientation of the activated reinforcement in x-direction corresponds to the principal moment direction such that $m_x = m_1$ and $m_y = m_{xy} = 0$. In the tests of series B, the activated reinforcement in both the x- and y-directions deviates from the principal moment direction. A combination of bending and torsional moments results, which corresponds to $m_x = m_y = m_{xy} = 0.5 \cdot m_1$ with respect to the reinforcement directions. A graphical representation of the bending conditions is illustrated on *Mohr's* circle for bending and torsional moments in Fig. 6.1 (c).

Considering the tensile zone of the slabs as equivalent reinforced concrete cover elements, see Fig. 6.1 (b), enables a description of the corresponding stress and strain states as shown in Fig. 6.1 (d) and (e), respectively. The flexural cracks form approximately perpendicular to the principal stress direction. In the tests of series A, the concrete between cracks remains unstressed; the principal stress at the crack is resisted exclusively by the reinforcement in x-direction $\sigma_1 = \sigma_{sxr} \cdot \rho_x$. Considering equilibrium at the cracks in the tests of series B indicates that the applied stress is resisted by both directions of reinforcement. This results in the equilibrium condition $\sigma_1 = \sigma_{sxr} \cdot \rho_x = \sigma_{syr} \cdot \rho_y$. In theory, application of the same stresses in the flexural tension zone of the slabs with equal reinforcement contents ρ would result in similar reinforcement stresses in both series of tests.

6.4 Test specimens

6.4.1 Geometry and reinforcement

The geometry and reinforcement layout of the specimens of series A and B are illustrated in Figs. 6.2 and 6.3, respectively. The test parameters, summarised in Table 6.1, are identical for both series with the exception of the direction of the flexural reinforcement φ_n with respect to the n-axis of the specimen. Two layers of reinforcement were provided in the flexural tensile zone. The outer layer in x-direction and inner layer in y-direction were placed in an orthogonal layout at a spacing of 100 mm. The concrete cover to the flexural

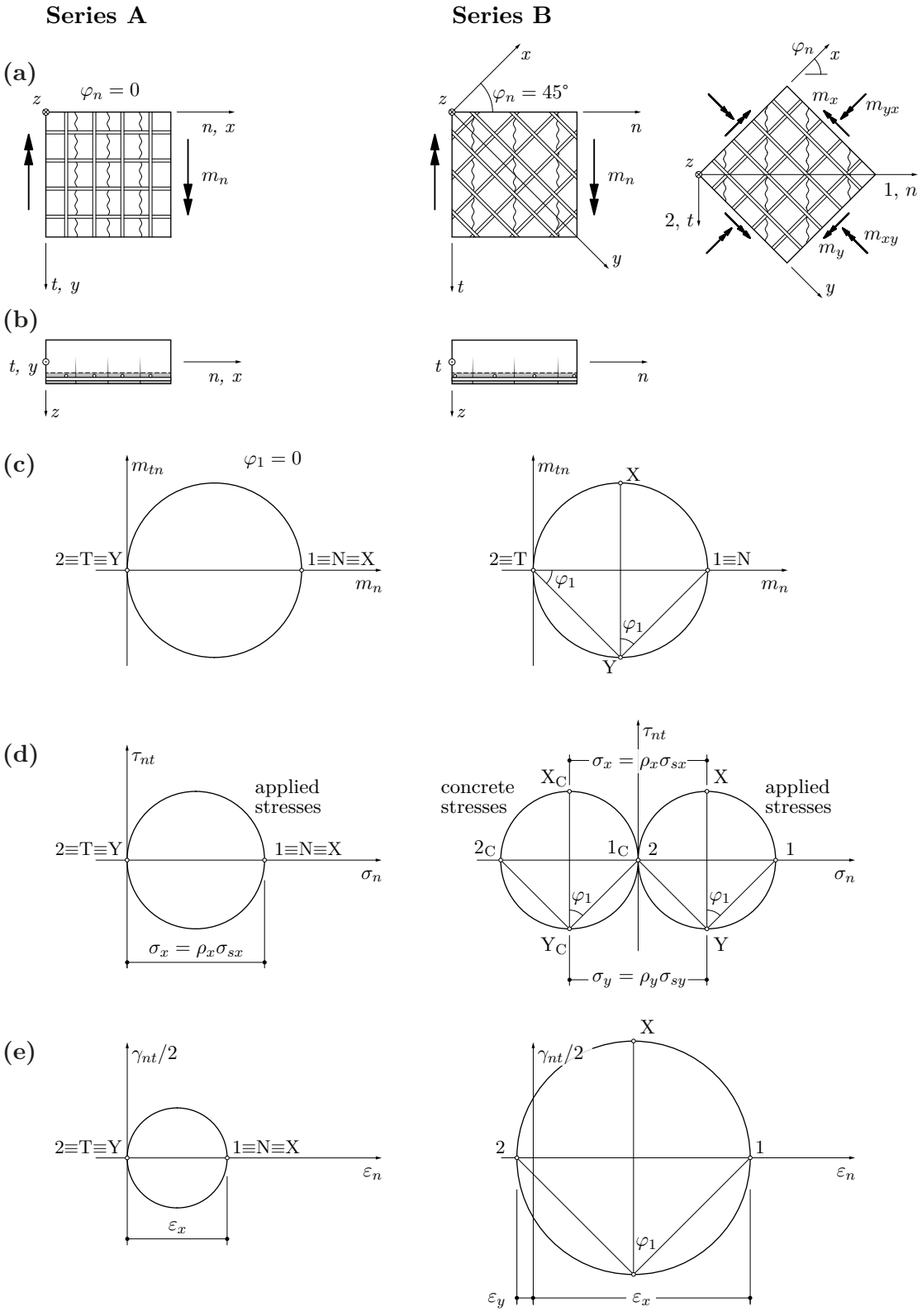


Fig. 6.1 – Overview of testing program for series A and B: (a) plan view with notation; (b) cross-sectional view; (c) *Mohr's* circle for bending and torsional moments; (d) cover element stress state; (e) cover element strain state. Representation adapted from [46].

reinforcement was $c_{\text{nom}} = 20\,\text{mm}$. All bars activated in bending were ordered with steel end anchorages of the type ancoFIX®, see documentation in [4], in order to ensure a short bond development within the edge regions of the specimen. The flexural compression zone remained unreinforced. A vertical shear reinforcement was provided between support and the cross-section of load application in order to prevent a premature shear failure of the concrete.

The reinforcement in the specimens of series A consisted of 8 $\varnothing$ 12 mm bars, placed parallel to the longitudinal or n-axis of the specimen in the first layer, and 26 $\varnothing$ 12 mm bars, constituting the second layer in the transverse direction (t-axis). The flexural reinforcement in the specimens of series B consisted of two layers of 22 $\varnothing$ 12 mm bars each, placed with deviations of $-45°$ and $45°$ with respect to the specimen n-axis. An approximately isotropic reinforcement layout resulted in all specimens with geometric reinforcement contents of $\rho_x = 0.65\%$ and $\rho_y = 0.70\%$.

6.4.2 Construction of specimens

All four specimens were constructed in the structural laboratory. Concrete was delivered by truck mixer in a single batch and placed by crane in the wooden formwork. The position and spacing of all reinforcing bars as well as the concrete cover were carefully controlled. A hand-held vibrator was used to consolidate the concrete and a float finish was produced on the top surface. Special care was taken to achieve an even surface at the regions of load application in order to ensure uniform bearing stresses. The specimens were damp-cured for seven days by covering them with a plastic sheet and sprinkling regularly with water. Subsequently, the specimens and simultaneously cast test cylinders and cubes were removed from their formwork and placed on the strong floor of the laboratory until testing. Key dates for casting of the concrete, removal of the formwork and testing are provided in Table 6.2.

Series			A	B
Slab thickness	h	[mm]	200	200
Reinforcement direction	φ_n	[°]	0	45
Geometric reinforcement content x-direction	ρ_x	[%]	0.65	0.65
Geometric reinforcement content y-direction	ρ_y	[%]	0.70	0.70
Moment at maximum load	$M_{n,\max}$	[kNm]	46.64	46.64
Moment at minimum load	$M_{n,\min}$	[kNm]	12.80	12.80

Tab. 6.1 – Test parameters.

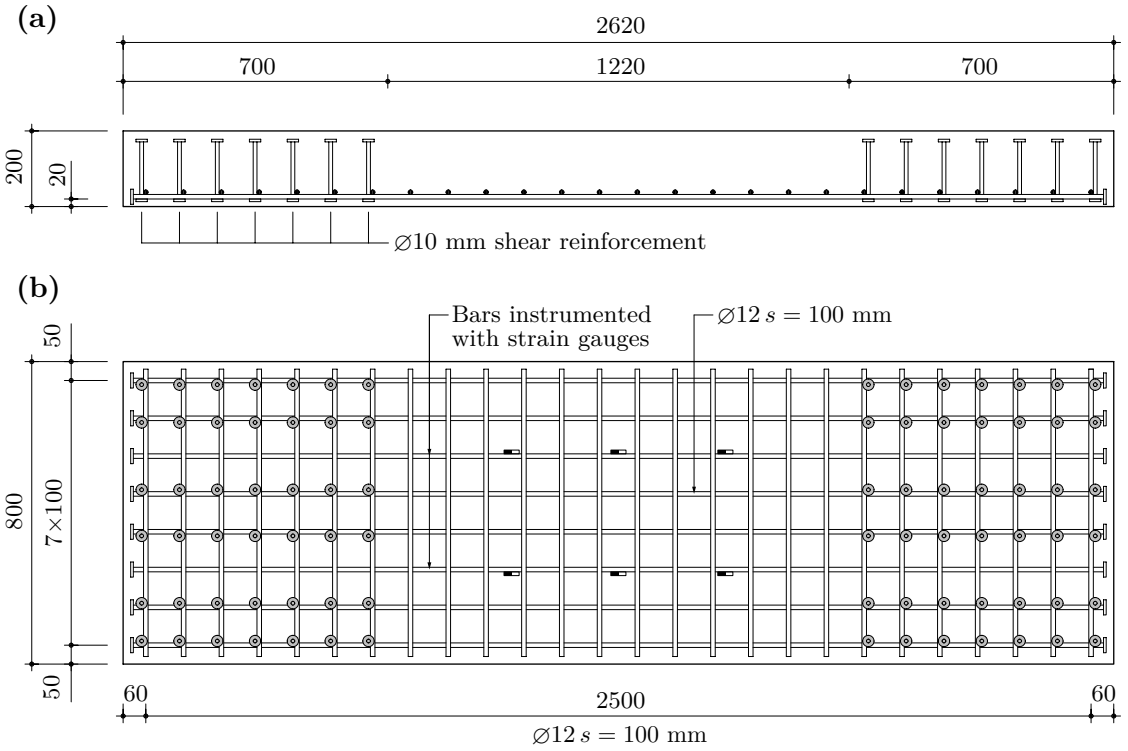


Fig. 6.2 – Series A – Geometry and reinforcement layout: (a) view; (b) plan. Dimensions in mm.

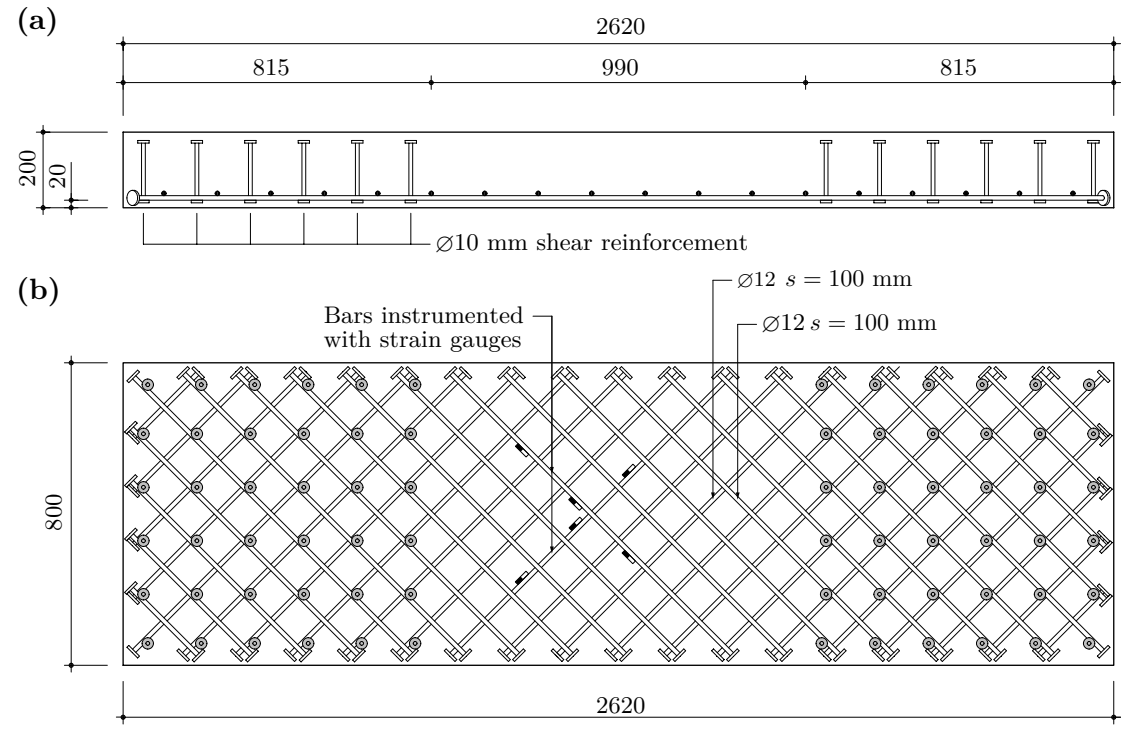


Fig. 6.3 – Series B – Geometry and reinforcement layout: (a) view; (b) plan. Dimensions in mm.

Specimen	A1	A2	B1	B2
Casting	19.02.2015	19.02.2015	19.02.2015	19.02.2015
Removal of formwork	23.02.2015	23.02.2015	23.02.2015	23.02.2015
Begin of testing	10.06.2015	29.01.2016	09.09.2015	01.04.2016
End of testing	25.08.2015	26.02.2016	23.11.2015	28.06.2016
Concrete tests	26.05.2015	28.01.2016	16.09.2015	09.03.2016

Tab. 6.2 – Key dates for construction and testing.

6.4.3 Materials

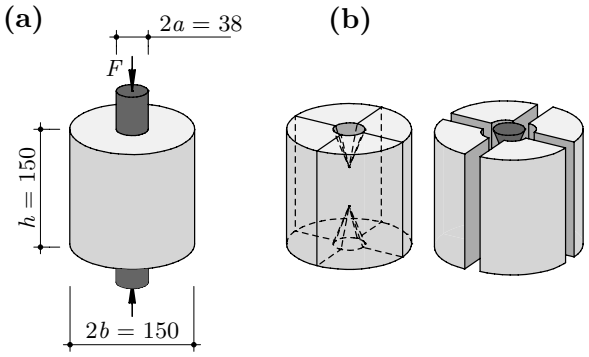
Concrete

All test specimens were produced with the same concrete mix. Washed aggregate with a maximum diameter of 16 mm was used with CEM II 42.5 cement; a water-cement ratio of 0.5 was required from the supplier. An air void content of 1.3% was measured for the wet concrete. The mechanical properties were determined by testing a total of 20 cylinders ($\varnothing = 150\,\text{mm}$, $h = 300\,\text{mm}$) and 12 cubes (side lengths 150 mm).

The compressive strength and the modulus of elasticity were determined by testing all 12 cubes and 12 of the cylinders. The remaining eight cylinders were cut in half and used to determine the splitting tensile strength in a double-punch test. Material tests were performed prior to the testing of each specimen in order to record changes in the mechanical properties with increasing concrete age. The cube compressive tests were conducted under force control at a loading velocity of $0.6\,\text{N}/(\text{mm}^2 \cdot \text{s})$. The modulus of elasticity and cylinder compressive strength were determined under displacement control at a velocity of $0.12\,\text{mm}/\,\text{min}$. The measured stress-strain relationships of the cylinder specimens for all four sets of tests are illustrated in Fig. 6.5. The modulus of elasticity was calculated in accordance with [86] as the secant value between a lower stress of $0.5\,\text{N}/\text{mm}^2$ and an upper stress corresponding to a third of the mean cube compressive strength. Concrete strains were measured with two Linear Variable Displacement Transformers (LVDTs), attached on opposing sides of the cylinder. These measurements were taken across the central region of the cylinders over a base length of 150 mm. The splitting tensile strength was determined in double-punch tests according to [15, 59]. These tests were performed under force control with a fictitious tensile stress increase of approximately $0.5\,\text{N}/(\text{mm}^2 \cdot \text{min})$. The splitting tensile strength was calculated with Eq. (6.1) using the test setup dimensions illustrated in Fig. 6.4. Concrete strain at peak load as well as the stress-strain relationship was determined from the actuator stroke. The stiffness of the testing machine was considered in these calculations. A summary of the mechanical properties is provided in Table 6.3.

Specimen		A1	A2	B1	B2
Age of concrete	[d]	96	343	209	384
Density	[kg/m^3]	2381	2366	2386	2364
ρ_c		±0.9%	±0.6%	±0.6%	±0.6%
Cube compressive strength	[N/mm^2]	51.8	56.4	56.9	55.6
$f_{c,\mathrm{cube}}$		±6.3%	±3.5%	±6.3%	±3.3%
Cylinder compressive strength	[N/mm^2]	51.3	51.9	52.1	51.9
f_{cc}		±2.5%	±0.8%	±6.9%	±4.4%
Splitting tensile strength	[N/mm^2]	3.92	3.61	3.59	3.67
f_{ct}		±5.3%	±6.1%	±5.7%	±10.1%
Strain at peak load	[‰]	2.51	2.35	2.32	2.34
ε_{cu}		±1.4%	±0.2%	±9.8%	±3.2%
Modulus of elasticity	[kN/mm^2]	34.9	34.7	34.5	34.1
E_c		±2.0%	±1.2%	±1.2%	±1.0%

Tab. 6.3 – Mechanical properties of concrete; mean values and coefficient of variation.



$$f_{ct} = \frac{F_u}{\pi\,(1.2bh - a^2)} \quad (6.1)$$

Fig. 6.4 – Double-punch test for concrete: (a) test setup; (b) typical failure mode. Dimensions in mm.

Reinforcing steel

Standard profiled steel bars with a nominal diameter of 12 mm were used for the flexural reinforcement in all four specimens. All bars classify as type B500B according to the Swiss standard SIA 262(2013) [87]. The reinforcing bars were procured from the same heat in order to ensure uniform material properties with respect to their chemical composition and the manufacturing conditions. The mechanical properties of the steel were determined by testing 15 bars randomly selected from the batch.

Tensile tests were performed under deformation control in a servo-hydraulic testing machine. The applied force and cylinder stroke was measured continuously. In addition, the elongation of the bar was measured with an extensometer attached at mid-height over a base length of 300 mm. The loading velocity prior to the onset of yielding amounted to a tensile stress increase of approximately 10 N/(mm^2 · s). At the onset of yielding, the stroke was kept constant for two minutes; the loading velocity was increased ten-fold thereafter. Prior to reaching the peak load, the stroke was kept constant again for a period of two

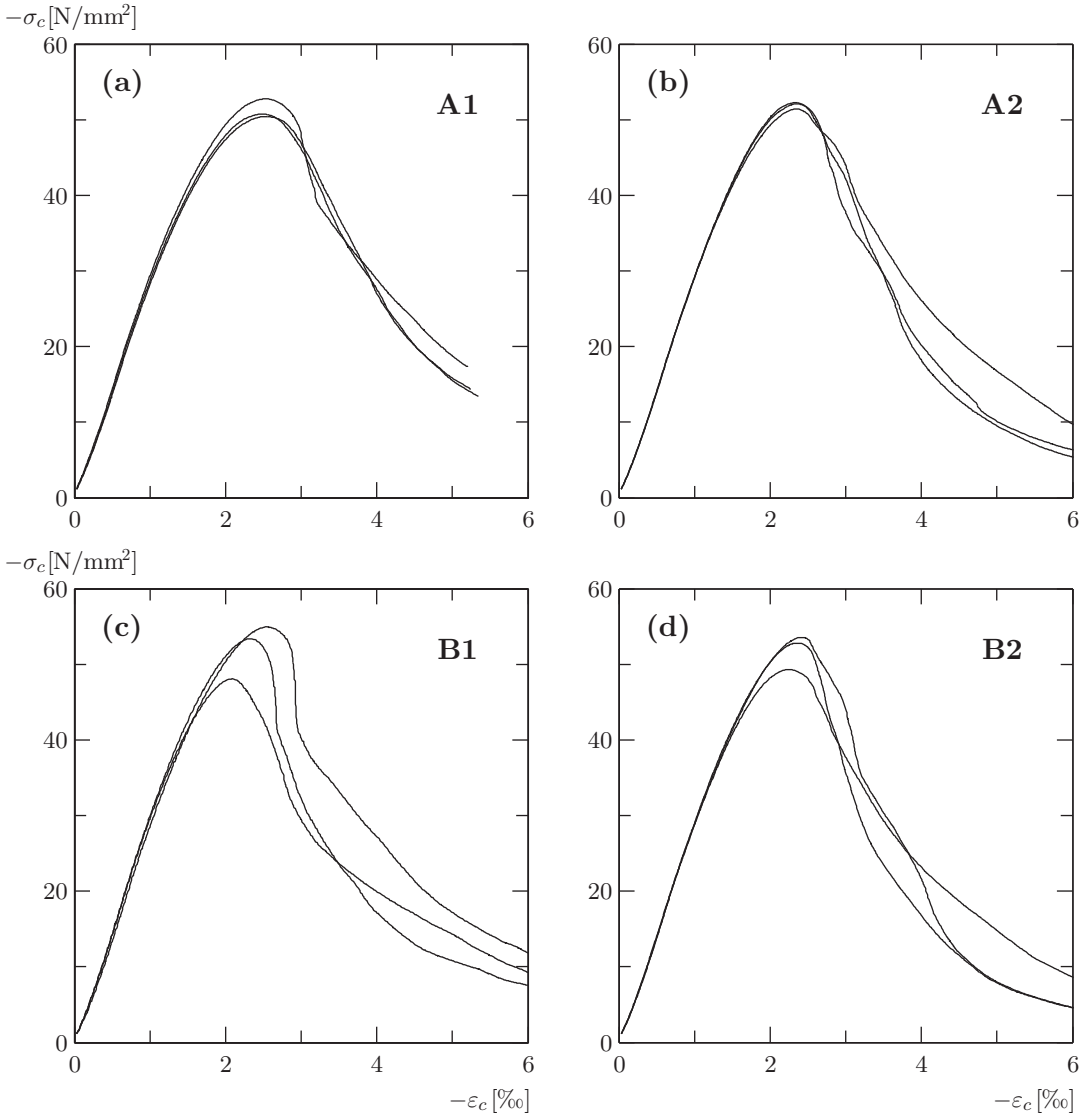


Fig. 6.5 – Stress-strain characteristic of concrete: (a) A1; (b) A2; (c) B1; (d) B2.

minutes, whereupon the bar was loaded to failure at the same velocity. The stress-strain relationship of the reinforcing steel is illustrated in Fig 6.6. Strains indicated during the elastic phase $\varepsilon_{s(300)}$ were calculated on the basis of extensometer measurements; whereas the subsequent elongation up to failure $\varepsilon_{s(750)}$ was calculated from the cylinder stroke over the entire free length of 750 mm. All strengths were calculated considering the nominal bar diameter. The static values of the yield and tensile strength were determined by reducing the dynamic strengths by the corresponding stress reduction after two minutes at constant stroke. The strain at peak load and the ultimate strain were determined from the stroke of the testing machine; the strain at onset of strain-hardening corresponds to the extensometer measurement. The stiffness of the testing machine was considered in these calculations. Furthermore, the modulus of elasticity was determined as the secant modulus between a lower stress of 100 N/mm^2 and an upper stress of 400 N/mm^2. The

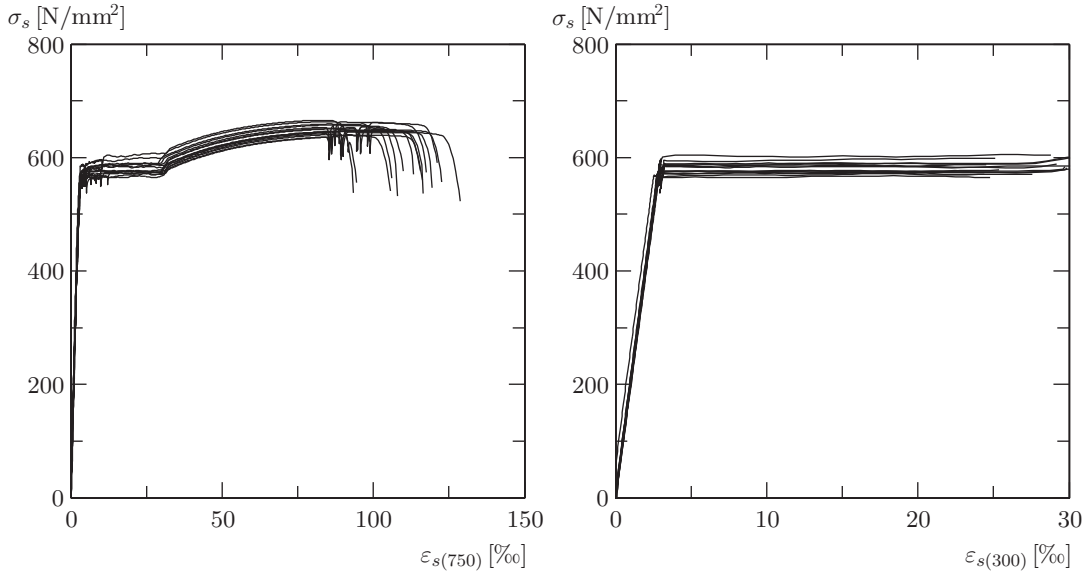


Fig. 6.6 – Stress-strain characteristic of reinforcing steel.

effective diameter of the bars was determined through weighing and measuring the bars, assuming a density of 7850 kg/m^3. A summary of the mechanical properties is provided in Table 6.4.

6.5 Test procedure

6.5.1 Test setup

The test setup was constructed in the structural laboratory, see Figs. 6.7 and 6.8. A steel frame, spanning the length of the specimen, was constructed and prestressed to the strong floor. This acted as a reaction structure for a centrally-placed hydraulic actuator with a capacity of 400 kN in compression and a stroke of 200 mm. A servo-hydraulic system was used to control the actuator. A loading yoke, consisting of a load-transfer beam and two spreader-beams, enabled a uniform distribution of the force across the specimen width at two load application cross-sections. The force was applied on the top surface of the specimen through two rocker bearings spaced at a distance of 1.12 m.

The supports constituted statically determinate boundary conditions; hence, the sectional forces depended only on the setup geometry and applied loads and could be kept constant for the duration of the test. The specimen was supported over a span of 2.40 m on two rocker bearings; one of which was bolted to the support, whilst the other was mounted on a steel-Teflon sliding surface that enabled translation in the longitudinal direction. Both the reaction frame and loading yoke were dimensioned for a high stiffness in order to limit deformations during loading. Furthermore, all bolted connections were fully prestressed to reduce fatigue-inducing stress variations and prevent the loosening of connections under cyclic loading.

Nominal bar diameter		[mm]	12
Number of bars		[-]	15
Effective diameter	$\varnothing_{\text{eff}}$	[mm]	12.08 ±0.1%
Dynamic yield strength	$f_{sy,\text{dyn}}$	[N/mm^2]	571.5 ±1.4%
Static yield strength	$f_{sy,\text{stat}}$	[N/mm^2]	554.6 ±1.5%
Dynamic ultimate strength	$f_{su,\text{dyn}}$	[N/mm^2]	652.6 ±1.3%
Static ultimate strength	$f_{su,\text{stat}}$	[N/mm^2]	612.5 ±2.2%
Strain at onset of strain-hardening	ε_{sv}	[‰]	30.1 ±2.2%
Strain at peak load	A_{gt}	[‰]	91.6 ±6.5%
Ultimate strain	ε_{su}	[‰]	112.6 ±8.8%
Modulus of elasticity	E_s	[kN/mm^2]	201.3 ±1.4%

Tab. 6.4 – Mechanical properties of reinforcing steel; mean values and coefficient of variation.

6.5.2 Measurements

Manual measurements

Manual measurements were made during the initial loading and periodically at maximum and minimum load between intervals of cyclic loading. These measurements included (i) recording concrete deformations on all surfaces of the specimen, (ii) measuring the deflection at multiple locations on the slab soffit, (iii) marking newly formed concrete cracks and measuring crack widths, (iv) photographing the specimen, and (v) detecting damage in the reinforcing bars. Local concrete deformations and deflections on the soffit were recorded periodically with demountable deformeters on a measurement grid, see Figs. 6.9 and 6.10. All deformeter measurements were made under displacement control after allowing the applied force to stabilise. A total of 669 measurements were made at each interval at maximum and minimum load, 21 of which were deflection measurements on the slab soffit relative to the strong floor, see Appendix. A for details. Cracks were marked using a heavy black marker. New cracks were numbered according to the Load Stage (LS) in which they were first observed. Maximum crack widths were measured with a crack width ruler on the front and back surfaces of the specimen. The cracking pattern was also documented by photographing the slab surfaces from fixed positions.

A reliable detection of fractures in the reinforcing bars was important for the interpretation of the results. A major challenge, however, was to detect these fractures in a nondestructive manner. For this purpose, the *magnetic flux leakage method*, which enables damage detection in ferromagnetic reinforcement through constant magnetic fields, was used. Similar to the other manual measurements, the detection of damage in the reinforcing bars was performed at maximum and minimum load in each layer of reinforcement activated in bending. In this way, the fracture position and corresponding number of load cycles in which individual bars failed could be identified in the majority of cases. Details on this method and the detection of damage will be provided in Section 6.6.1.

Fig. 6.7 – Test setup with installed specimen.

Continuous measurements

Continuous measurements of the applied actuator force, midspan deflection, concrete deformations on the top surface and soffit as well as reinforcing steel strains were made with 15 permanently installed sensors. The sensor placement is illustrated in Fig. 6.12. Measurements were recorded in a data acquisition software on a computer. A data acquisition frequency of 2 Hz was used during the initial and quasi-static loading of the specimen, while performing manual measurements and during unloading. A high-frequency, quasi-continuous measurement was defined during cyclic loading in order to accurately capture the response while limiting the amount of data. For this purpose, a data acquisition frequency of 50 Hz was selected to record for a duration of five minutes after the completion of every 500 load cycles.

The applied actuator force was recorded with a calibrated load cell integrated into the actuator housing. Vertical deflection of the slab at midspan was measured with an LVDT at the front and at the back surface of the specimen. Concrete deformations on the top surface and soffit were each measured with an LVDT over a base length of 350 mm within the constant moment region. Furthermore, reinforcing steel strains were measured with three strain gauges applied to two reinforcing bars in each specimen. The bars were milled at these locations in order to produce even, flat surfaces of roughly 6×20 mm on which the strain gauges were glued, see Fig. 6.11. The strain gauges were subsequently coated with silicone for mechanical protection and insulation from moisture. Milling of the bars

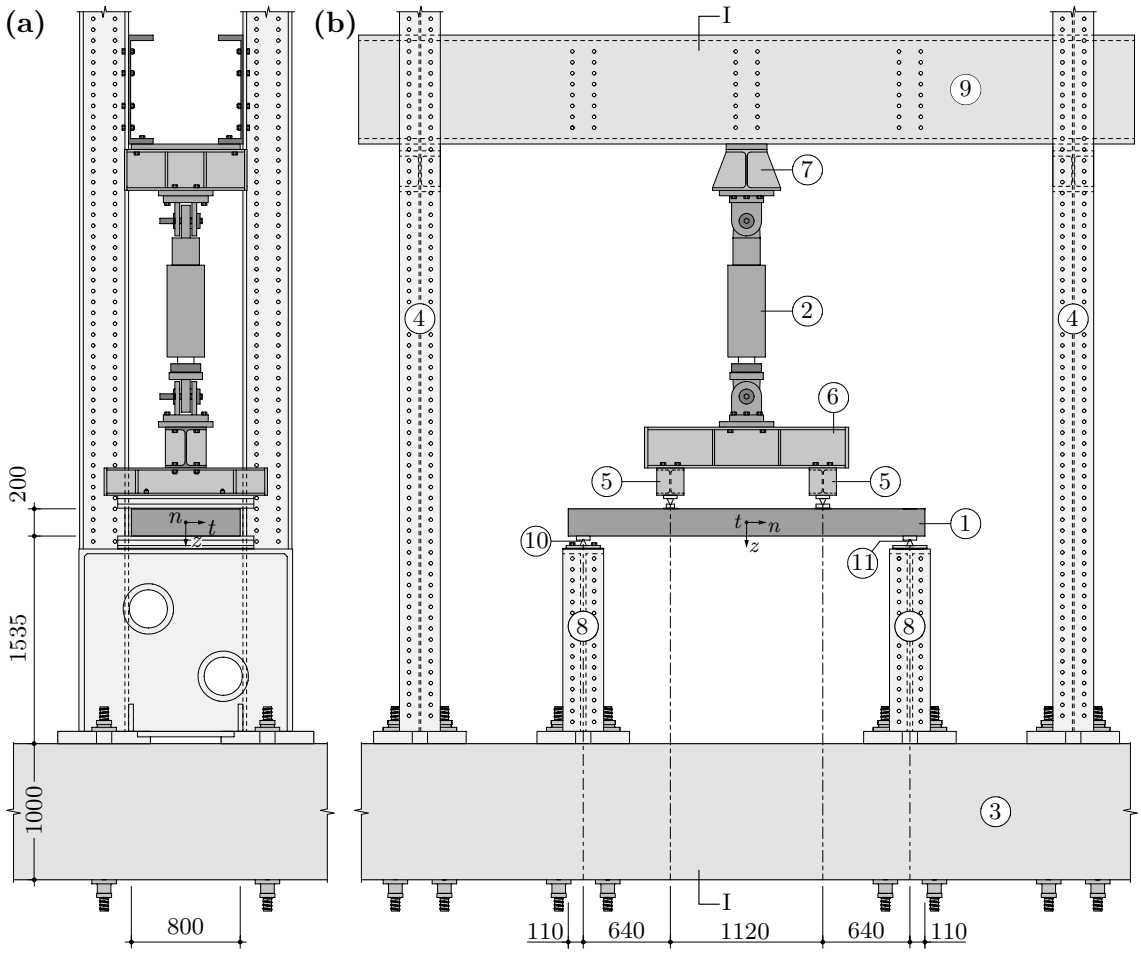


Fig. 6.8 – Test setup details: (a) section I-I; (b) front view. Dimensions in mm.

resulted in a local cross-sectional reduction of approximately 4%. The instrumented bars were rotated along their axis such that the strain gauges were oriented outwards to the sides of the specimen. Gauge cables were guided towards the centre of the specimen and taken vertically out of the cross-section at two locations. In addition, the influence of temperature fluctuations on the strain gauge measurements during testing was compensated with a reference measurement. For this purpose, an additional strain gauge was glued to a short, similarly prepared bar which was cast into an unloaded region of the specimen. Each individual strain gauge was calibrated before placement of the bars in the formwork. This was achieved through loading the instrumented bars within their elastic range and comparing the electrical resistances to an extensometer measurement on the bar. Details on the calibration is provided in Appendix A.

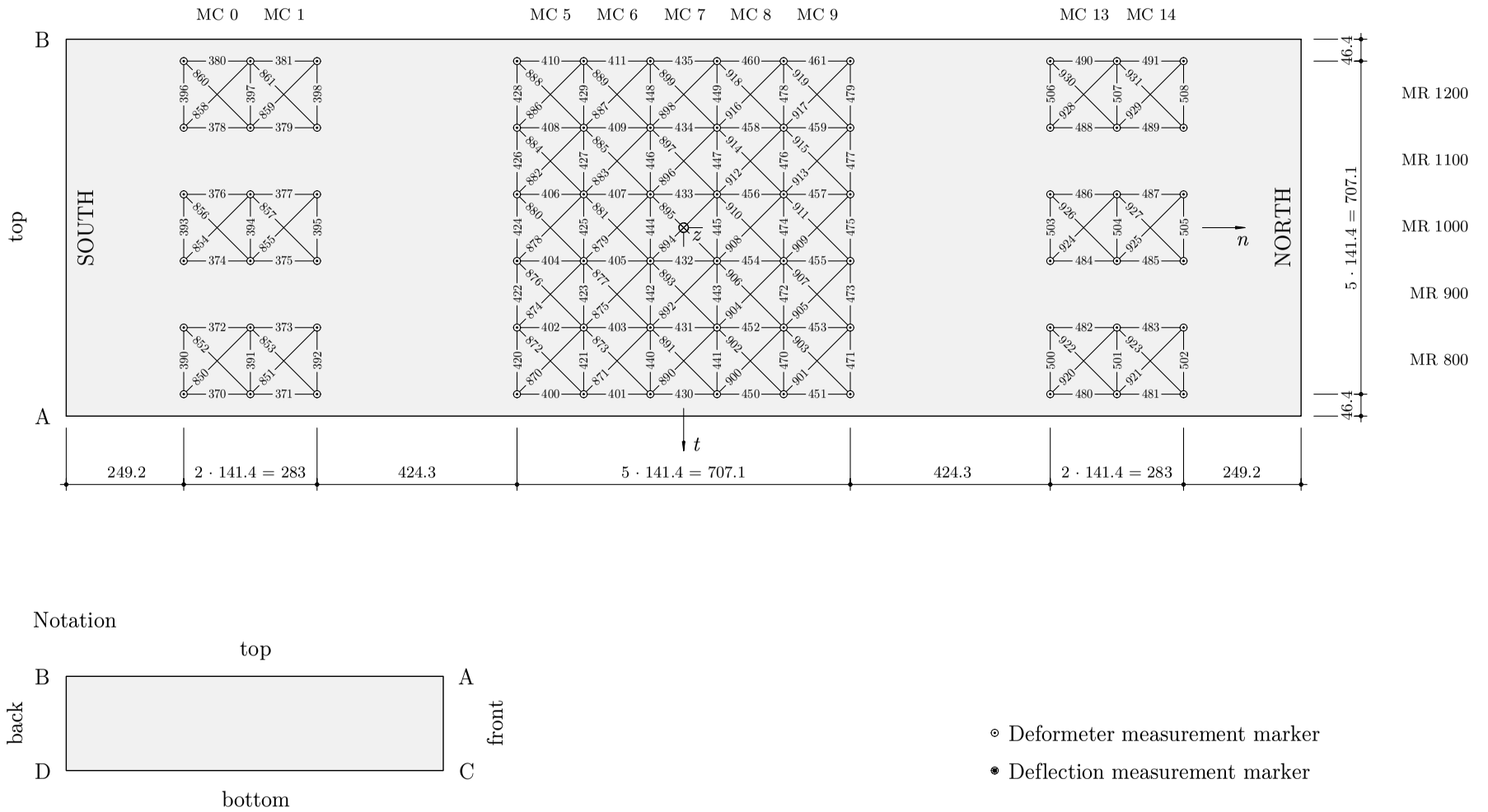


Fig. 6.9 – Grid layout and numbering for deformeter measurements on specimen top surface. Dimensions in mm.

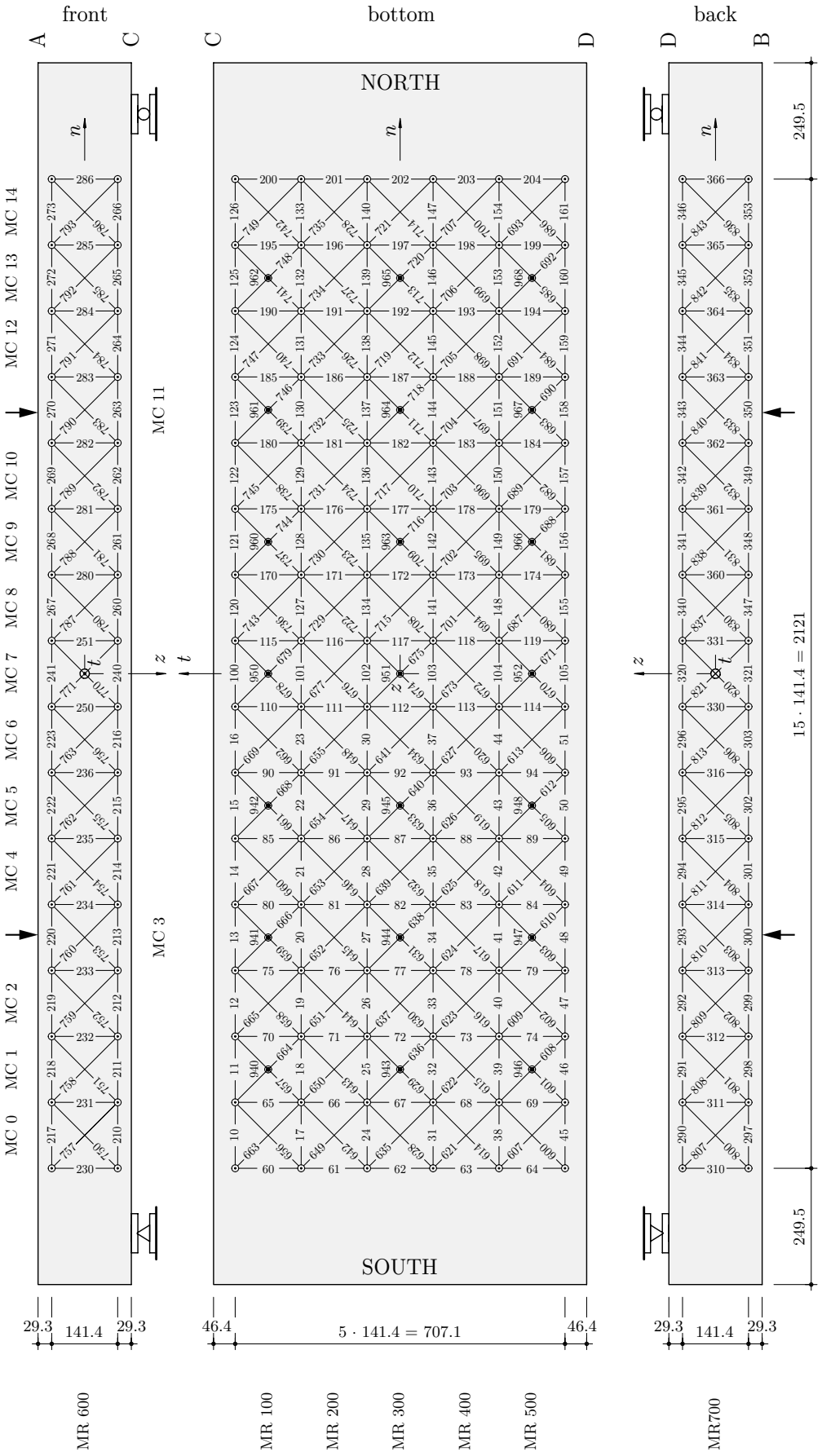


Fig. 6.10 – Grid layout and numbering for deformeter and deflection measurements on specimen soffit and sides. Dimensions in mm.

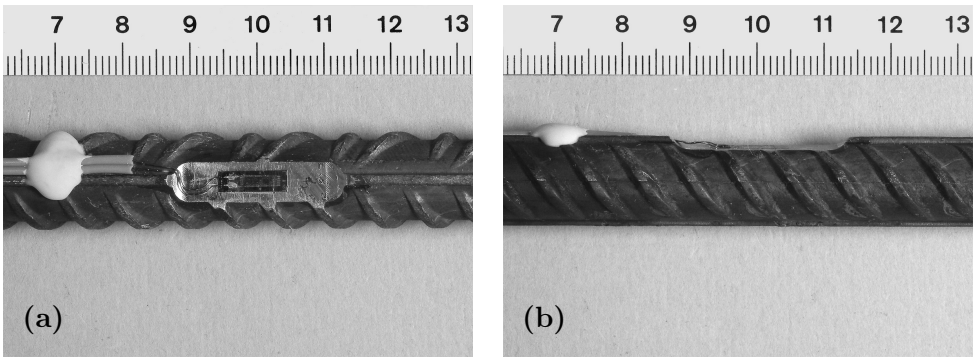


Fig. 6.11 – Reinforcing bar instrumented with a strain gauge: (a) plan view; (b) side view.

6.5.3 Testing sequence

The preparation time for each test was approximately one week and included the following tasks: preparation of the measurement grid for deformeter measurements; mounting of the bearing plates for the rocker bearings; painting the specimen white in order to facilitate crack detection; placement of the specimen in the setup; and installation of the sensors. A zero balance was performed on all permanently installed sensors and the offset from their physical zero point was stored. Offsets were kept constant for the duration of each test. Two deformeter reference measurements were made over the entire measurement grid as well as the deflections on the slab soffit. The second measurement was used to confirm the base lengths obtained in the first measurement. In this way, it was possible at each stage of loading to determine the difference between a measurement and the condition for the undeformed specimen. Each test started with an *initial loading* followed by intervals of *cyclic loading* with progressively increasing load cycles. Periodic measurements were made during *quasi-static loading* phases between intervals of cyclic loading.

Initial loading

The initial loading was performed in displacement control in four load stages up to the maximum load. The final load stage, corresponding to an actuator force of 131.13 kN, also represented the maximum load during cyclic loading. Loading commenced at constant stroke velocities of 0.5 mm/ min and 1.0 mm/ min in the tests of series A and B, respectively. The actuator stroke was kept constant after reaching the target load and the force was allowed to stabilise before starting the manual measurements. In the initial load stages LS 1 to LS 3, only the vertical deflection was measured with a demountable deformeter. All manual measurements, as described in Section 6.5.2, were made upon reaching the maximum load in LS 4. Upon completion of the measurements, the specimen was unloaded to a *pre-load* of roughly 1 kN before commencing loading again to the minimum load. Manual measurements were again made, similar to those at maximum load, before unloading the specimen.

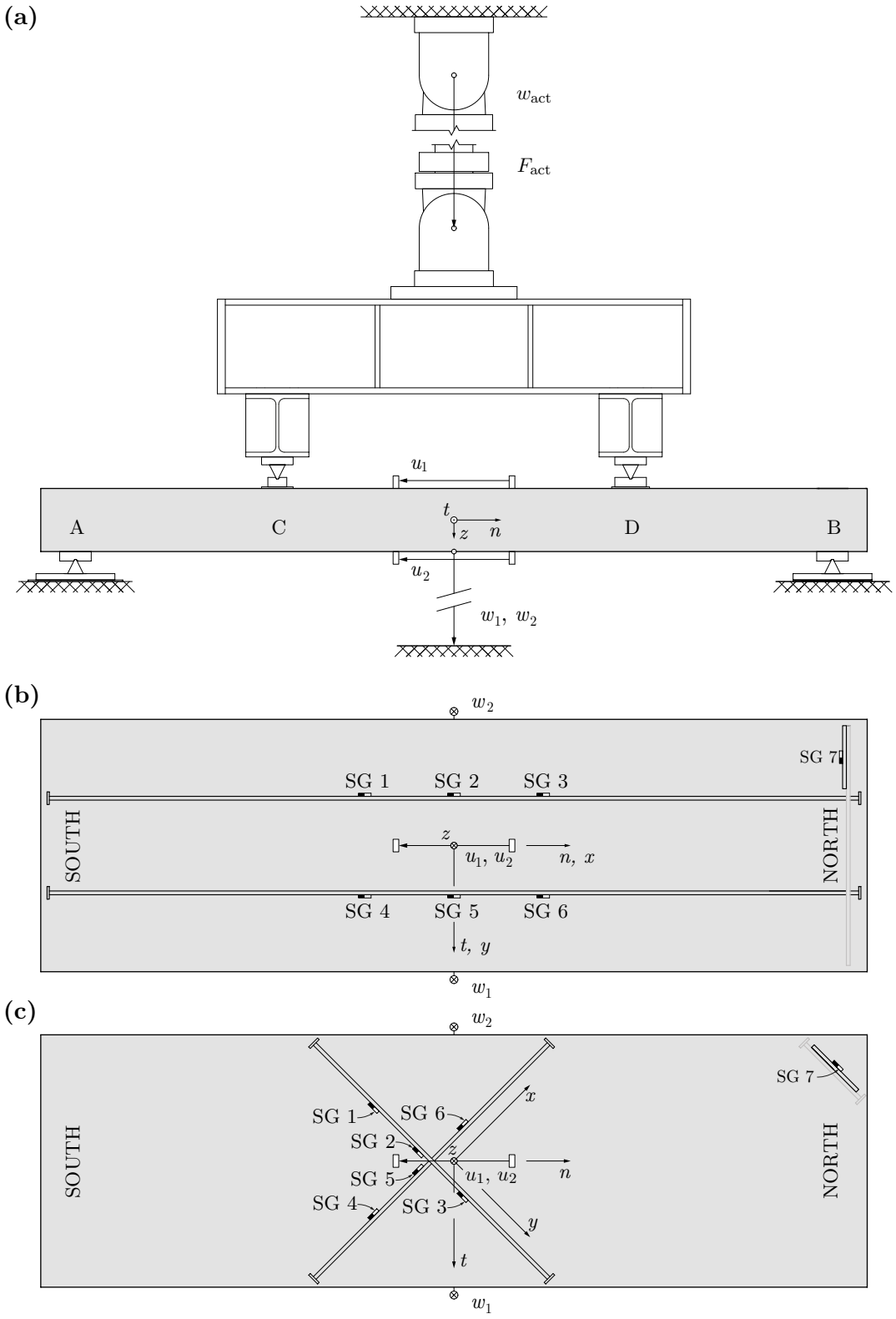


Fig. 6.12 – Permanently installed sensors: (a) front view; (b) plan view for series A; (c) plan view for series B.

Cyclic loading

Cyclic loading was applied in displacement control according to the protocol illustrated in Fig. 6.13 (a). This allowed a constant loading velocity even at large displacements close to failure of the specimen. The actuator load was increased monotonically up to $F_{\text{act}} = 75\,\text{kN}$ before initiating the cyclic loading with a triangularly-shaped spectrum. The applied force was monitored continuously in the controller unit which reversed the stroke velocity upon reaching the target loads. The sensitivity of the servo-hydraulic valve was tested in order to determine stroke velocities at which the maximum and minimum load was not exceeded. Optimal results were obtained by defining intermediate target loads at 115 kN and 40 kN in combination with rapid and slow loading rates. The rate of change in applied force varied in displacement control according to the stiffness of the specimen. This resulted in slightly different loading frequencies as increasing damage led to larger displacements. The average loading frequency in the tests of series A changed from 1.0 Hz in the initial load cycles to 0.7 Hz slightly before failure. The loading frequency in the tests of series B was lower due to the larger deflections. A frequency of 0.7 Hz was observed initially which changed only slightly to 0.6 Hz shortly before failure. A digital cycle counter was updated upon completion of each load cycle by means of a digital signal sent between the controller unit and the data acquisition system. In this way, the number of completed load cycles could be recorded continuously for the duration of the test. Cyclic loading was discontinued once the defined number of load cycles was completed or if a predefined limit on the deflection or applied force was exceeded. Upon completion, the actuator remained in displacement control until unloading commenced manually.

Quasi-static loading

The quasi-static loading sequence commenced similar to that described for the initial static loading, see Fig. 6.13 (b). Upon completion of a phase of cyclic loading, the applied force was increased at a constant rate up to the maximum load. The force was allowed to stabilise, whereupon all manual measurements, as described in Section 6.5.2, were made. The specimen was subsequently unloaded to the pre-load. Thereafter, the force was increased again to the minimum load and allowed to stabilise under displacement control. Manual measurements, similar to those at maximum load, were made. Finally, the specimen was unloaded completely. The two LVDTs measuring the midspan deflection of the specimen were removed after allowing the deformation to stabilise in the unloaded state. This was done in order to facilitate a complete measurement of the slab soffit with the magnetic flux leakage method such as to detect new bar fractures or damage to the reinforcement. These measurements were again made at maximum and minimum load before unloading the specimen completely, thus concluding the quasi-static loading. The sensors were installed again and their zero offsets redefined with respect to the previous measurement in the unloaded state.

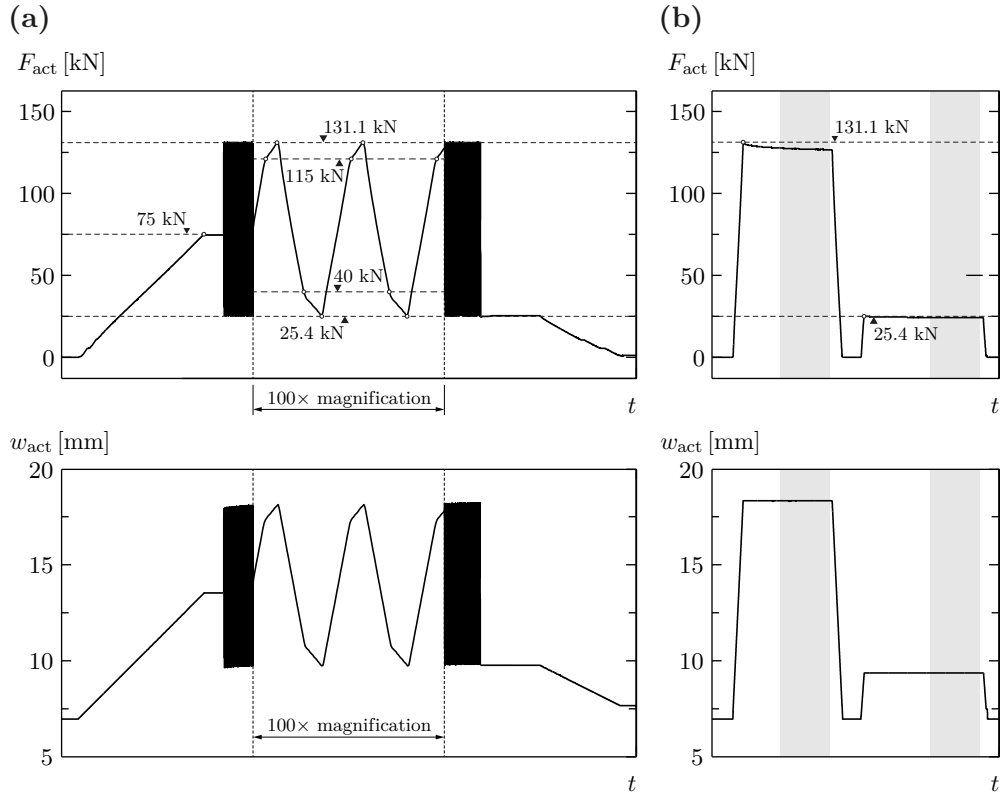


Fig. 6.13 – Loading protocol for actuator force and stroke: (a) cyclic loading; (b) quasi-static loading. Note: grey regions indicate periods of manual measurements.

6.6 Test results

6.6.1 Analysis and representation of data

Notation and conventions

The coordinate system introduced in Fig. 6.1 is used throughout the present chapter for the representation of results. The x- and y-axes define the orientation of the outer and inner layers of reinforcement, respectively. Furthermore, the n- and t-axes indicate the longitudinal and transverse axes of the specimen, respectively. Positive forces, displacements and stress resultants are as illustrated in Fig. 6.14.

Analysis of manual measurements

The demountable deformeters used to measure concrete deformations and deflections were calibrated prior to and after a number of readings on an Invar length reference bar. In this way, systematic errors on the measurement due to thermal expansion of the deformeter housing and drift in the sensors could be compensated. A linearised correction term was added to the results between subsequent reference measurements. Accidental measurement errors could be reduced due to the over-determined grid (all four sides of each square as

well as both diagonals) with an error balancing procedure corresponding to the method of least squares. This approach required the measurement grid to be modelled as an internally statically indeterminate, linear elastic truss system with rotationally-unrestrained nodes. Deformations were introduced on the truss elements in the form of initial strains. Individual measurements were weighted according to the axial stiffness of each bar in the truss system. The stiffness was defined to be proportional to the bar lengths, leading to equal values of axial stiffness $EA \,/\, l$ for all elements. Subsequently, the strains were calculated for each measured length as the difference between the deformeter reading and the base length from the reference measurement. The curvature χ was calculated as the difference between the bottom and top longitudinal measurement on the specimen sides with the corresponding vertical distance. The vertical deflection w is indicated under the assumption that the supporting shear wall elements do not deform under the applied force.

Analysis of continuous measurements

The applied force $F\,[\mathrm{kN}]$ per load application position, the support reaction $R\,[\mathrm{kN}]$ and bending moment at midspan $M_n\,(0)\,[\mathrm{kNm}]$ shown in Fig. 6.14 was calculated with

$$F = 0.5 \cdot F_{\mathrm{act}} + G \tag{6.2}$$

$$R = 0.75 \cdot g_{c0} + 0.56 \cdot g_{m0} + F \tag{6.3}$$

$$M_n\,(0) = 1.2 \cdot R - 0.56 \cdot F - 0.75 \cdot 0.935 \cdot g_{c0} - \tfrac{1}{2} \cdot 0.56^2 \cdot g_{m0} \tag{6.4}$$

assuming an equal distribution of force across the loading yoke $F_{\mathrm{C}} = F_{\mathrm{D}} = F$ and support reactions $R_{\mathrm{A}} = R_{\mathrm{B}} = R$ from symmetry of the test setup. The weight of the loading yoke and rocker bearings on the top surface of $G = 3.06\,\mathrm{kN}$ per load application position and the own weight of the specimen was considered in the calculations. The latter was calculated with assumed unit densities of 7850 and $2370\,\mathrm{kg/m^3}$ for the steel and concrete, respectively. This resulted in a weight per unit length of $g_{c0} = 3.822\,\mathrm{kN/m}$ for the region with shear reinforcement and $g_{m0} = 3.810\,\mathrm{kN/m}$ for the central region without shear reinforcement. The displacement at midspan w_m is defined as the average of LVDT measurements w_1 and w_2 on the front and back sides of the specimen.

Specimen failure and detection of bar fractures

Global *failure* of the specimen was defined as the condition in which the maximum load was no longer reached during quasi-static loading in displacement control. The structural condition upon reaching this criterion was generally characterised by large deflections and often accompanied by spalling of the cover concrete to the reinforcement. The rupture of reinforcing bars due to large plastic deformations could be heard in most specimens after exceeding the residual flexural resistance. The loading was interrupted soon after the applied force indicated a continuous decrease in order to prevent further potentially damaged bars from rupturing plastically due to excessively large strains.

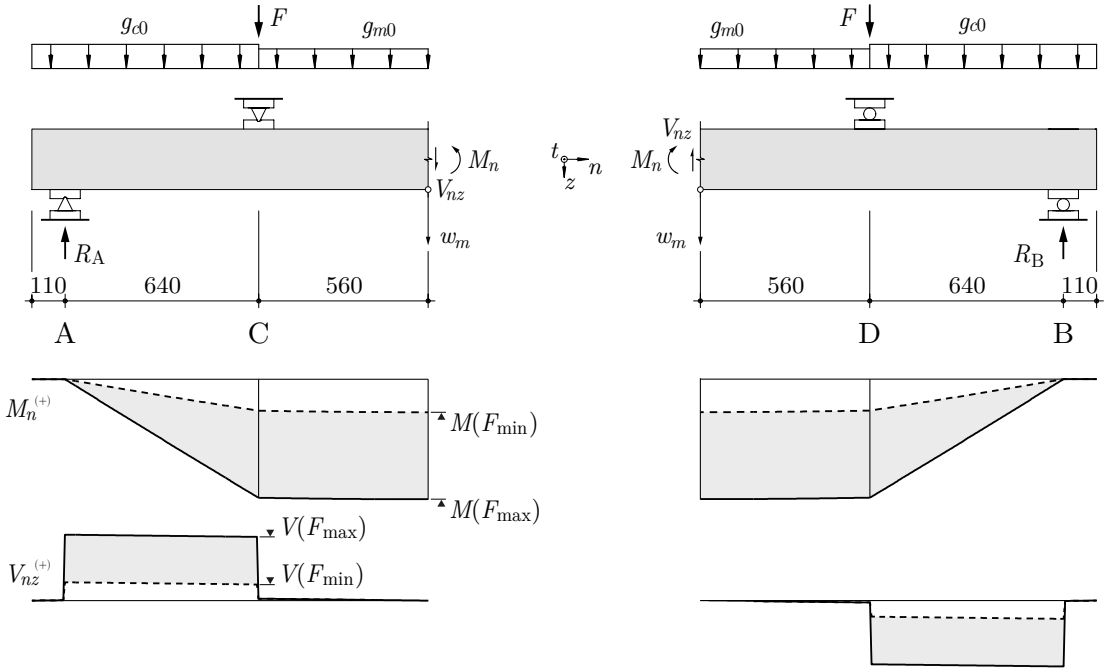


Fig. 6.14 – Static system with relevant forces and distributions of moment and shear force at maximum and minimum load. Dimensions in mm.

Reinforcing bar fractures were detected on the basis of measurements with the magnetic flux leakage method. Inspection of the measured signals and comparison with the results from a detection algorithm developed for this purpose resulted in a reliable identification of bar fractures in most cases. Details of this method and the measurements made in the present experimental campaign is provided in the dissertation of *Diederich* [18]. The majority of fractures, and even small cracks in some bars, could be detected in the tests of series A. The fracture detection in the tests of series B, however, proved to be less successful with this method. A number of false negative detections were obtained for cases of closely spaced fractures and in the detection of fractures within the inner layer of reinforcement.

Interpretation of the quasi-continuous measurements of steel strains on the instrumented bars provided a further indication of the rupture of individual bars. The two following characteristics were observed: (1) a sudden, disproportionally large increase in steel strain indicated failure of a bar other than the one wherein the measurement was made, which was close to or within the cross-section of the measurement position; and (2) a sudden decrease in steel strain indicated rupture of the bar in which the measurement was made, typically at a position close to the strain gauge. The number of load cycles at which the bars ruptured could be estimated by considering such developments in steel strain as well as the fracture position relative to the strain gauges. Fatigue failure of reinforcing bars were not heard during the cyclic loading. This may, however, have been due to a relatively high level of background noise in the laboratory.

6.6.2 Test A1

The development of deflection at maximum and minimum load for increasing load cycles is shown in Fig. 6.15. The midspan deflection at maximum load more than doubled from 6.1 to 14.2 mm between the initial loading and the final measurement at failure. First cracks were observed during the initial loading at LS 3 with a significant stiffness reduction occurring at $M_n = 23.5$ kNm. A practically complete cracking pattern was observed after the initial loading, see Fig. 6.16 (a). Crack depths increased only marginally with subsequent load cycles. Further flexural cracks developed after $N = 311'133$ between support and load application, see Fig. 6.17 (f). Significant crack width increases were observed between the maximum load measurements at $N = 906'347$ and $961'995$, changing from 0.25 to 0.6 mm at midspan (MC 7) and from 0.45 to 0.65 mm at MC 10 on the measurement grid. The latter, nearly vertical crack extended diagonally across the compression zone towards the load application after $N = 906'347$; coinciding with the first bar fractures. The critical crack measured 3.0 mm in width under maximum load at failure.

Significant changes were observed in all measurements between the initial loading and $N = 124$; developing only slightly thereafter up to $N = 711'133$, see Figs. 6.18 and 6.19. Marked changes were again observed after the rupture of individual bars which continued up to failure. The concrete strains in n-direction, corresponding to deformeter measurements on the top surface and soffit, also indicated large developments within the initial load cycles followed by relatively little change until fatigue failure of the bars, see Figs. 6.20 and 6.21. Low concrete strains were measured in t-direction on the top surface and slab soffit throughout the test. These did not indicate a clear trend under increasing load cycles. The curvature also increased rapidly within the first 100 load cycles and developed only marginally thereafter, see Figs. 6.22 and 6.23. Strong localisations of curvature were observed at the cracks; however, these only became evident in the measurements after failure of several bars.

A total of 11 bar fractures were observed at failure, see Fig. 6.24. A disproportionally large strain increase (SG 6) was observed at approximately $N = 834'100$. This indicated the occurrence of fracture No. 103, resulting in a redistribution of force among the remaining bars within the cross-section. A subsequent fracture occurred at approximately $N = 885'000$, adjacent to and in the same bar as strain gauge SG 1. This is illustrated by the sudden strain decrease in Fig. 6.19 (c). A further fracture was detected at $N = 906'347$ in the same bar. Three additional fractures were detected with the magnetic flux leakage method in the measurement at $N = 961'995$. One of these fractures was clearly visible in the increased strain in SG 2 and SG 5 at approximately $N = 940'600$. A large strain increase was again observed in SG 3 at $N = 1'033'200$. Two additional fractures were detected at $N = 1'035'895$. A further two fractures were seen at $N = 1'131'296$. The final partial fracture No. 104 was only identified after removal of the concrete cover. Five bars failed within the cross-section between MC 10 and MC 11; the three remaining intact bars were unable to resist the maximum load. Failure occurred at $N = 1'140'040$.

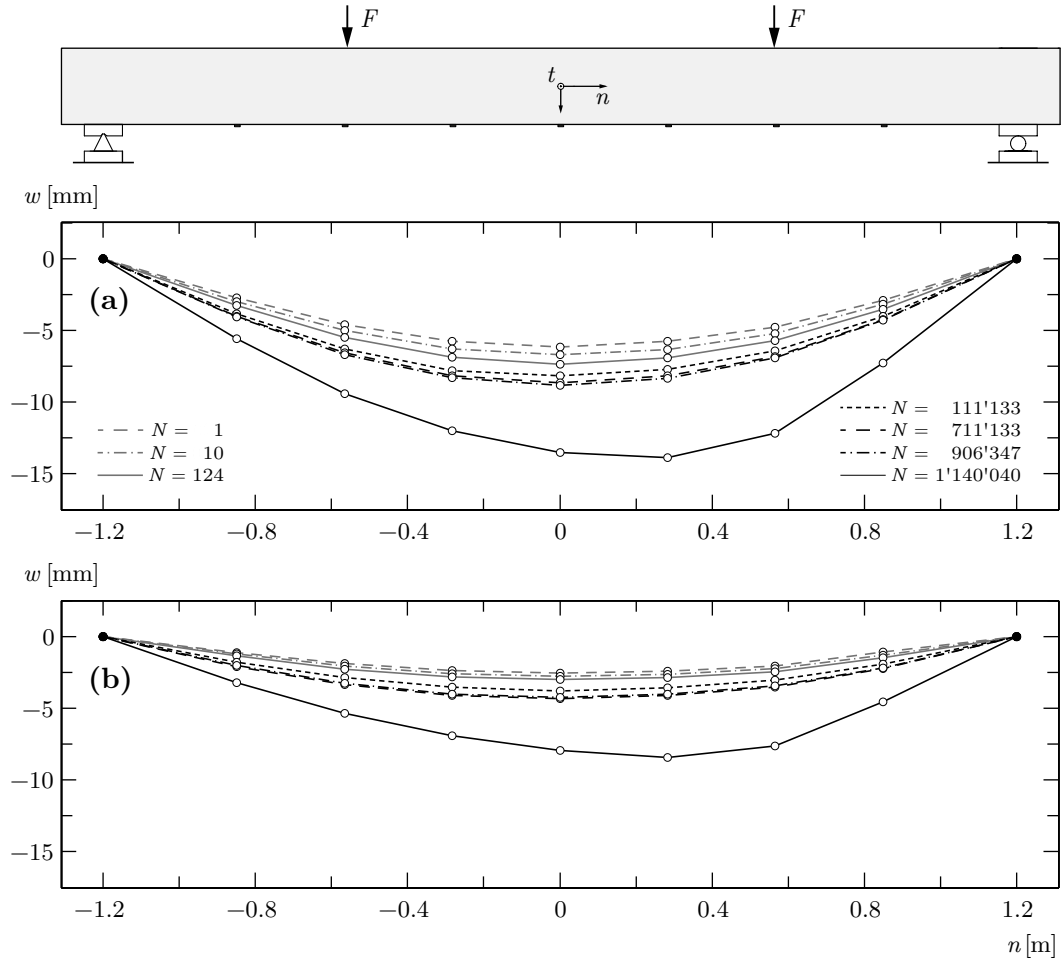


Fig. 6.15 – A1 – Deflection measurements: (a) maximum load; (b) minimum load.

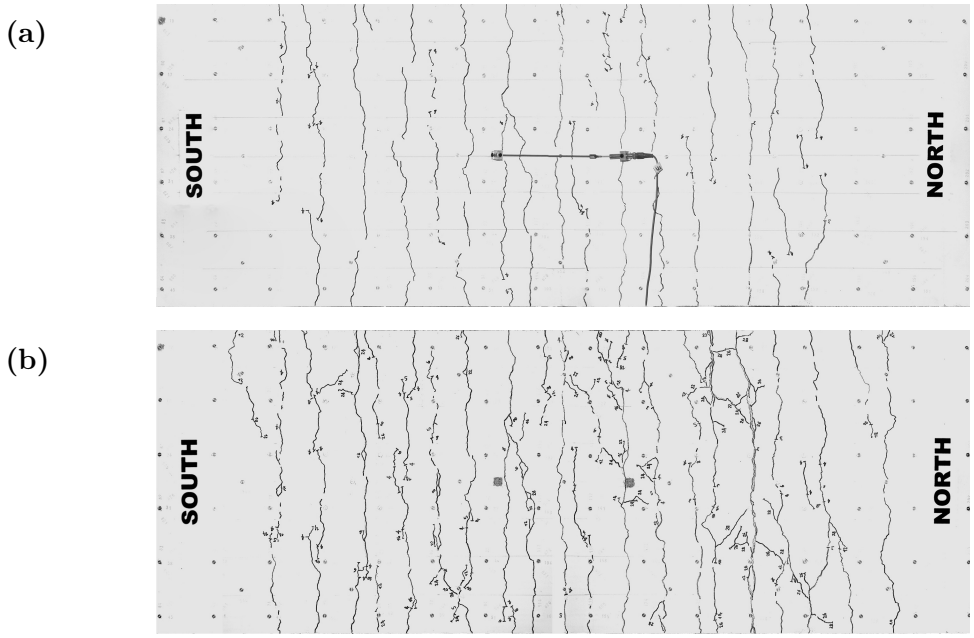


Fig. 6.16 – A1 – Soffit cracking pattern: (a) after initial loading; (b) at failure.

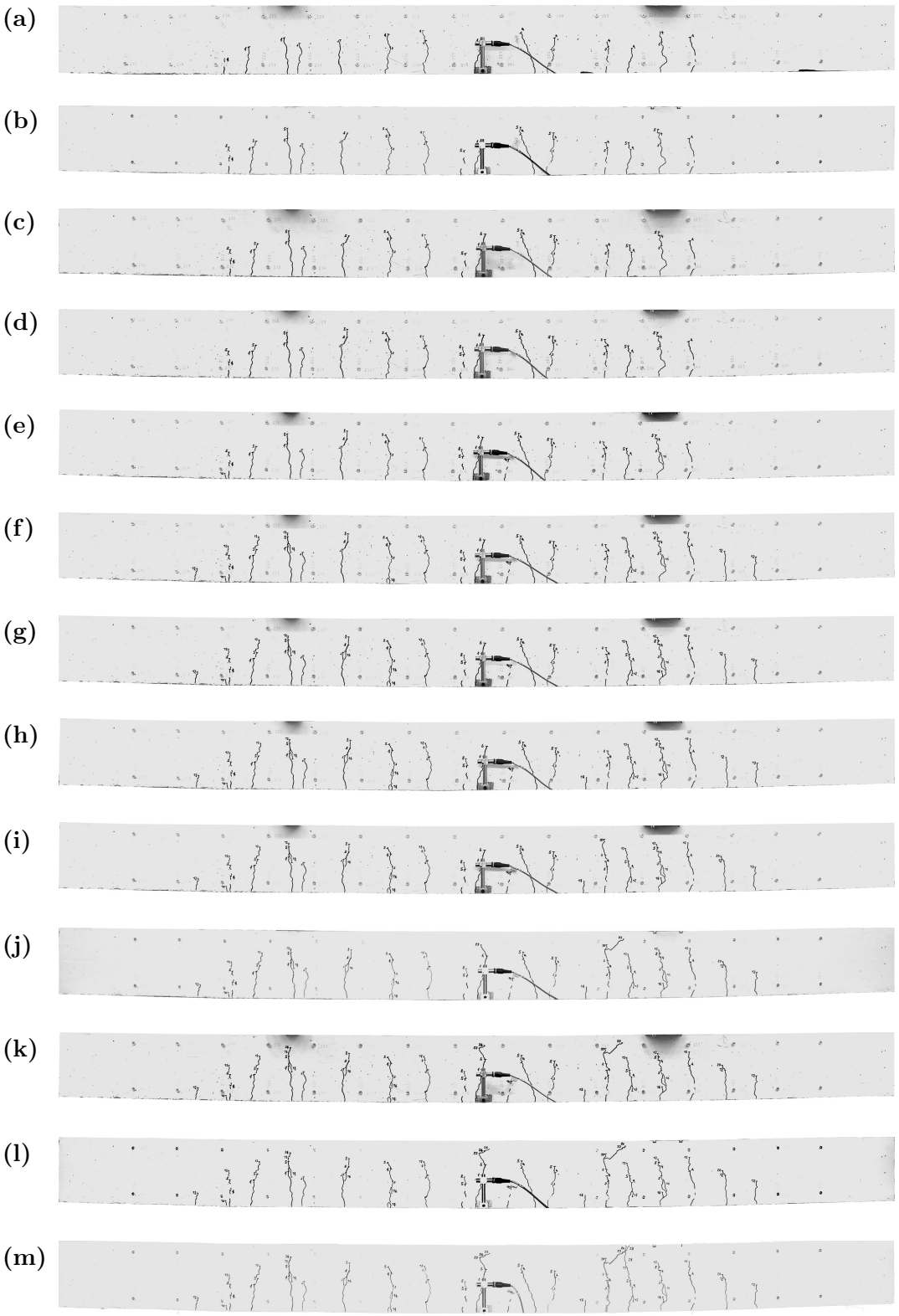


Fig. 6.17 – A1 – Front surface cracking pattern at maximum load in load cycle: (a) 1; (b) 10; (c) 124; (d) 1′124; (e) 111′133; (f) 311′133; (g) 511′133; (h) 711′133; (i) 906′347; (j) 961′995; (k) 1′035′895; (l) 1′131′296; (m) 1′140′040.

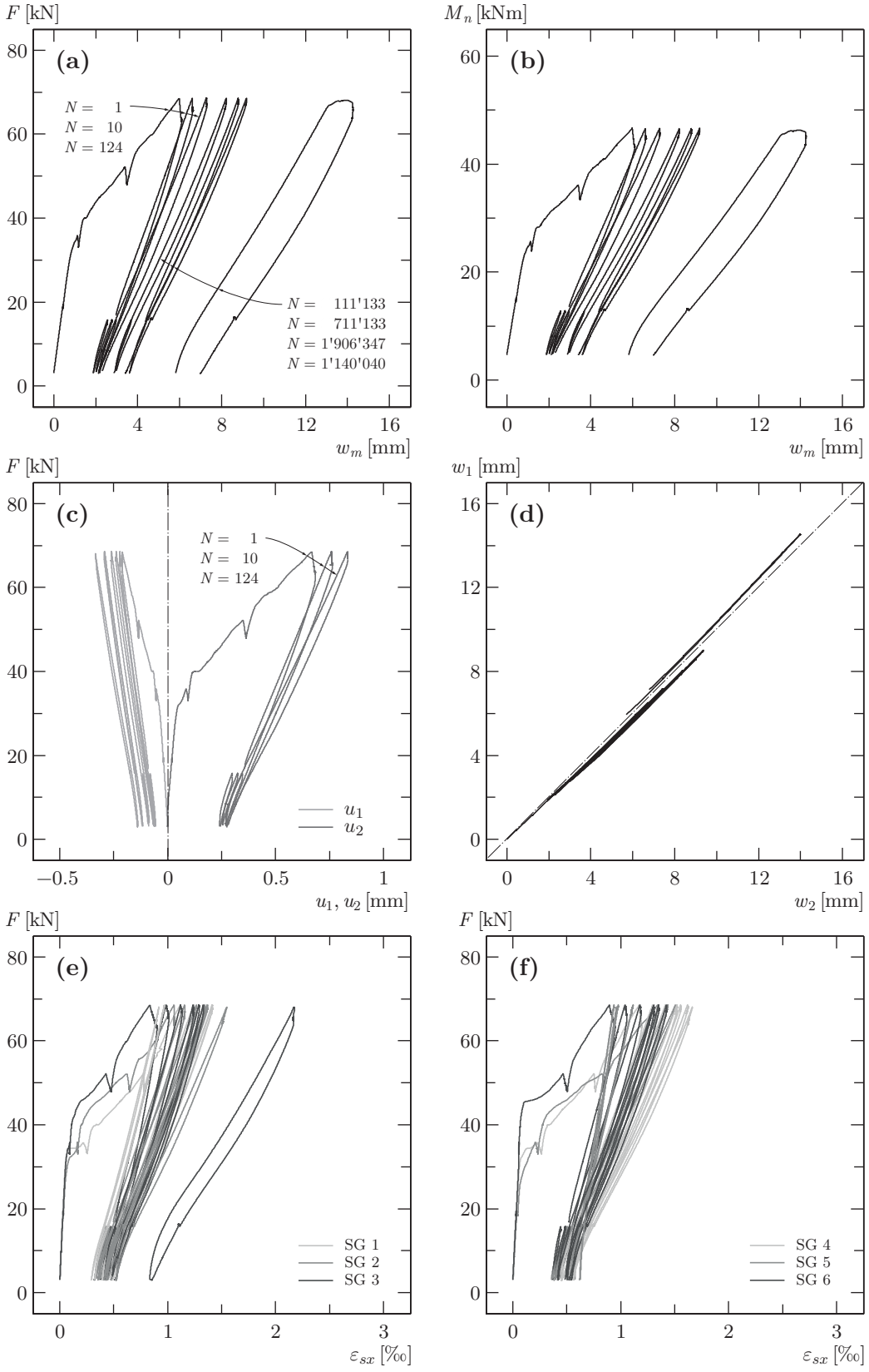


Fig. 6.18 – A1 – Quasi-static loading: (a) and (b) force and moment-deflection; (c) concrete deformations on top surface and soffit; (d) midspan deflection at front and back surface; (e) and (f) reinforcing steel strains in x-direction.

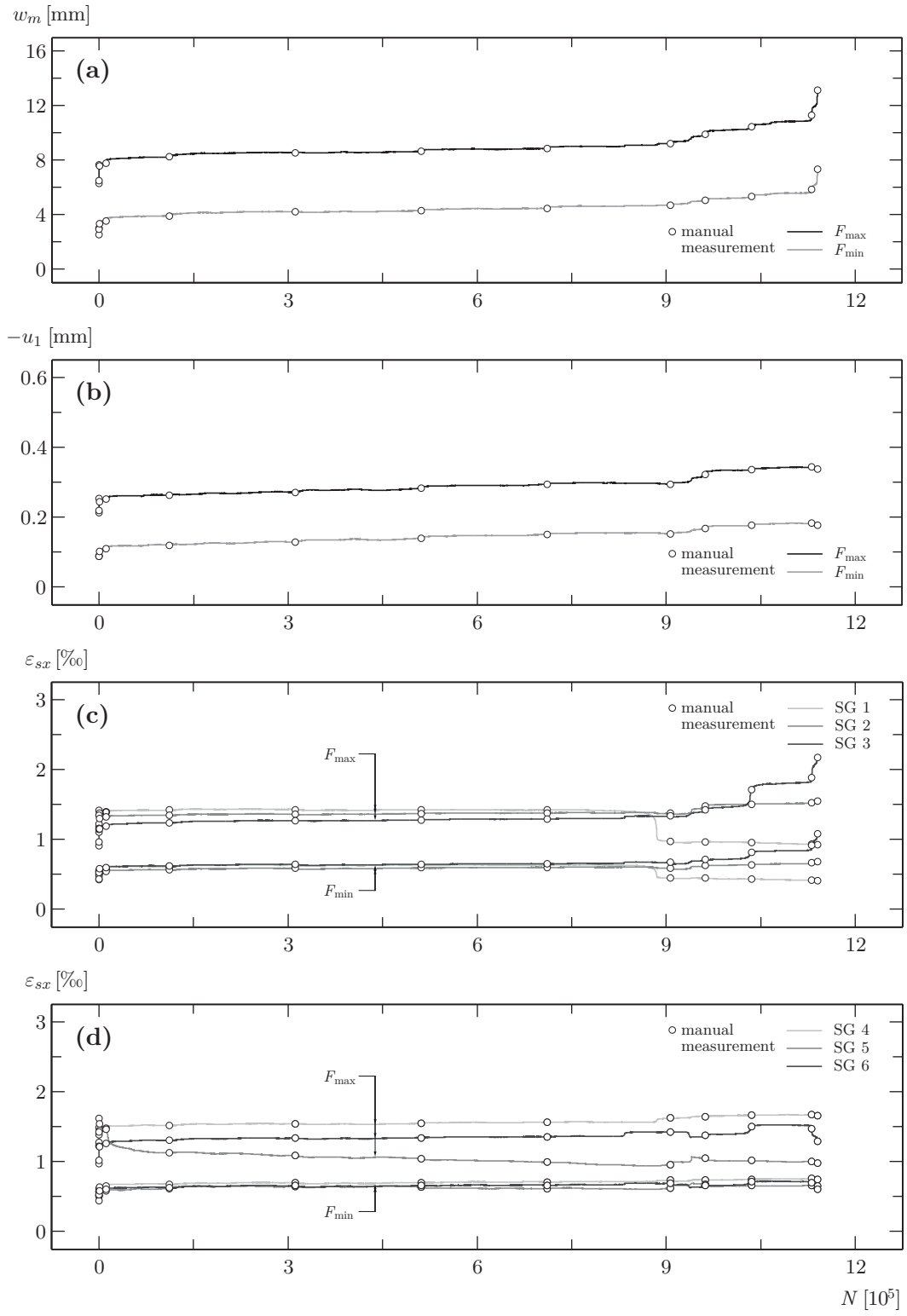


Fig. 6.19 – A1 – Development at maximum and minimum load: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x-direction.

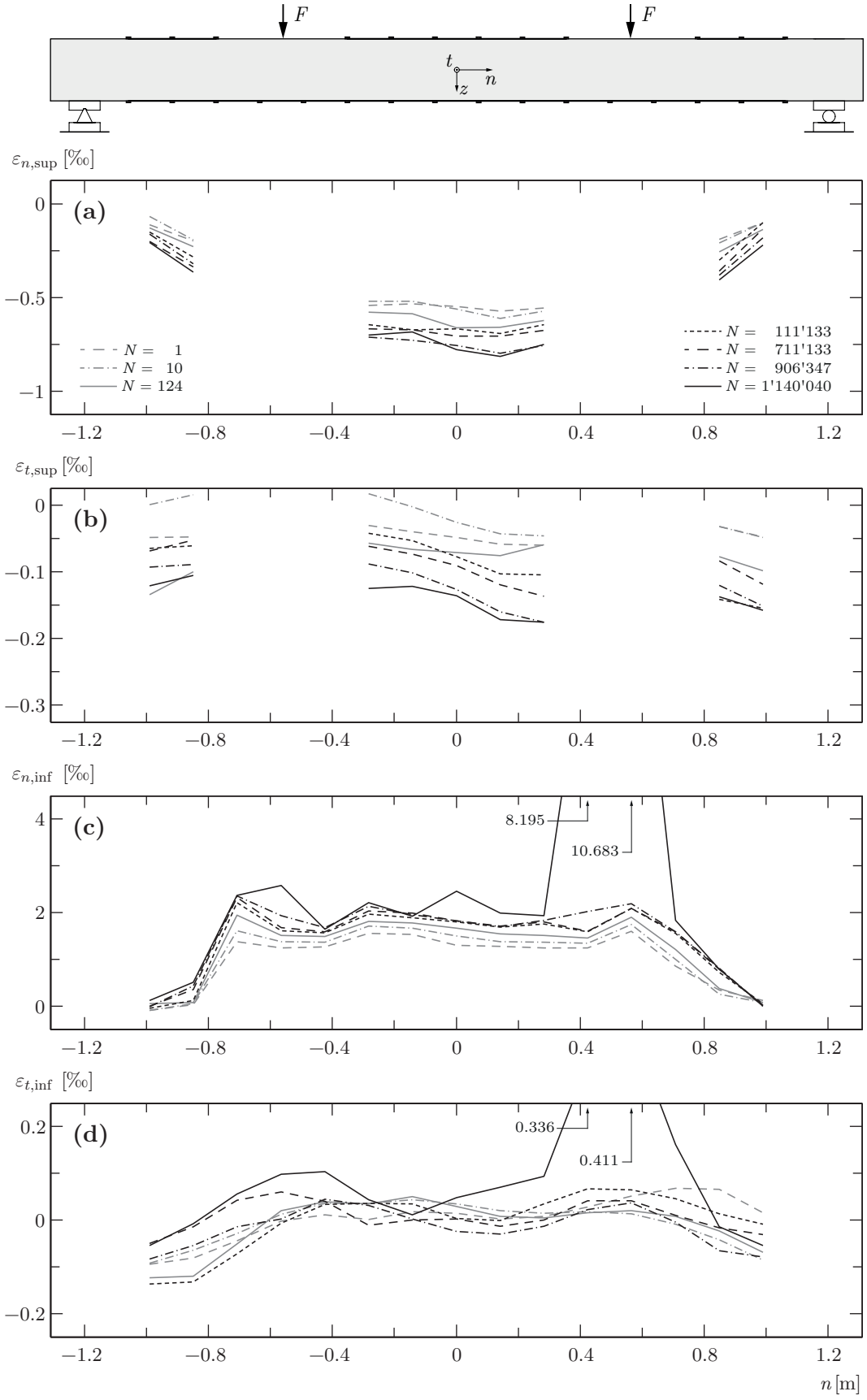


Fig. 6.20 – A1 – Average strains at maximum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

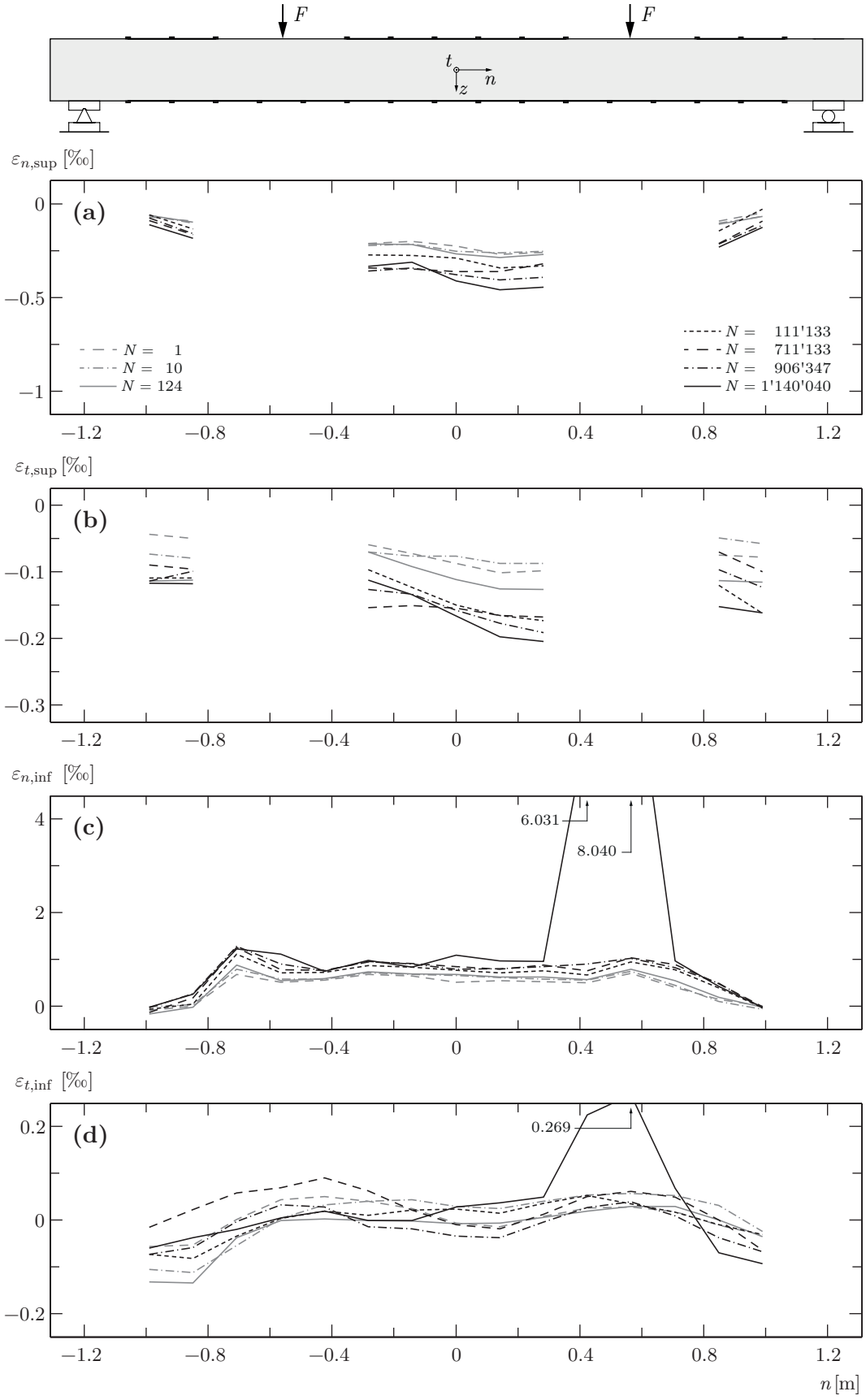


Fig. 6.21 – A1 – Average strains at minimum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

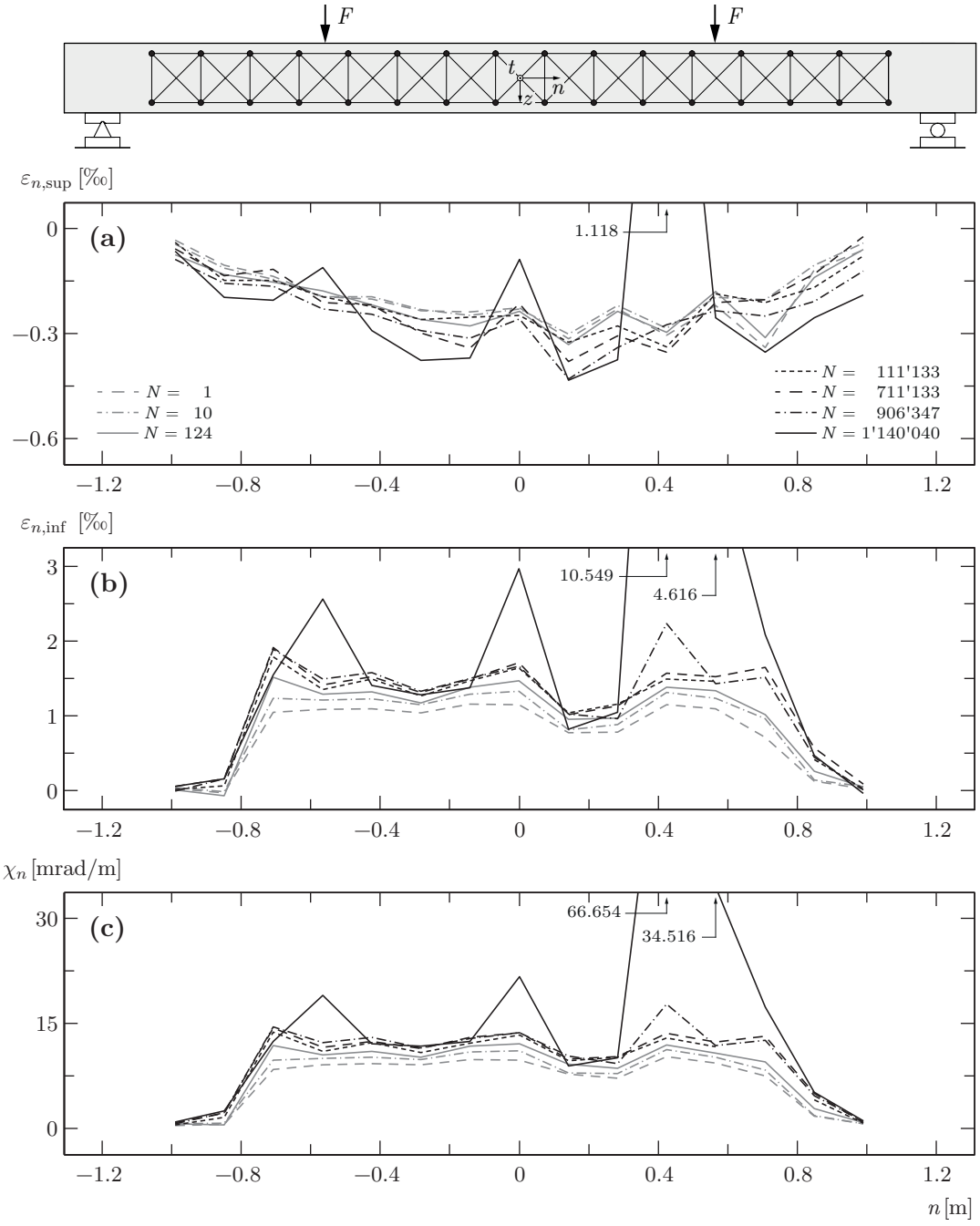


Fig. 6.22 – A1 – Average strains at maximum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

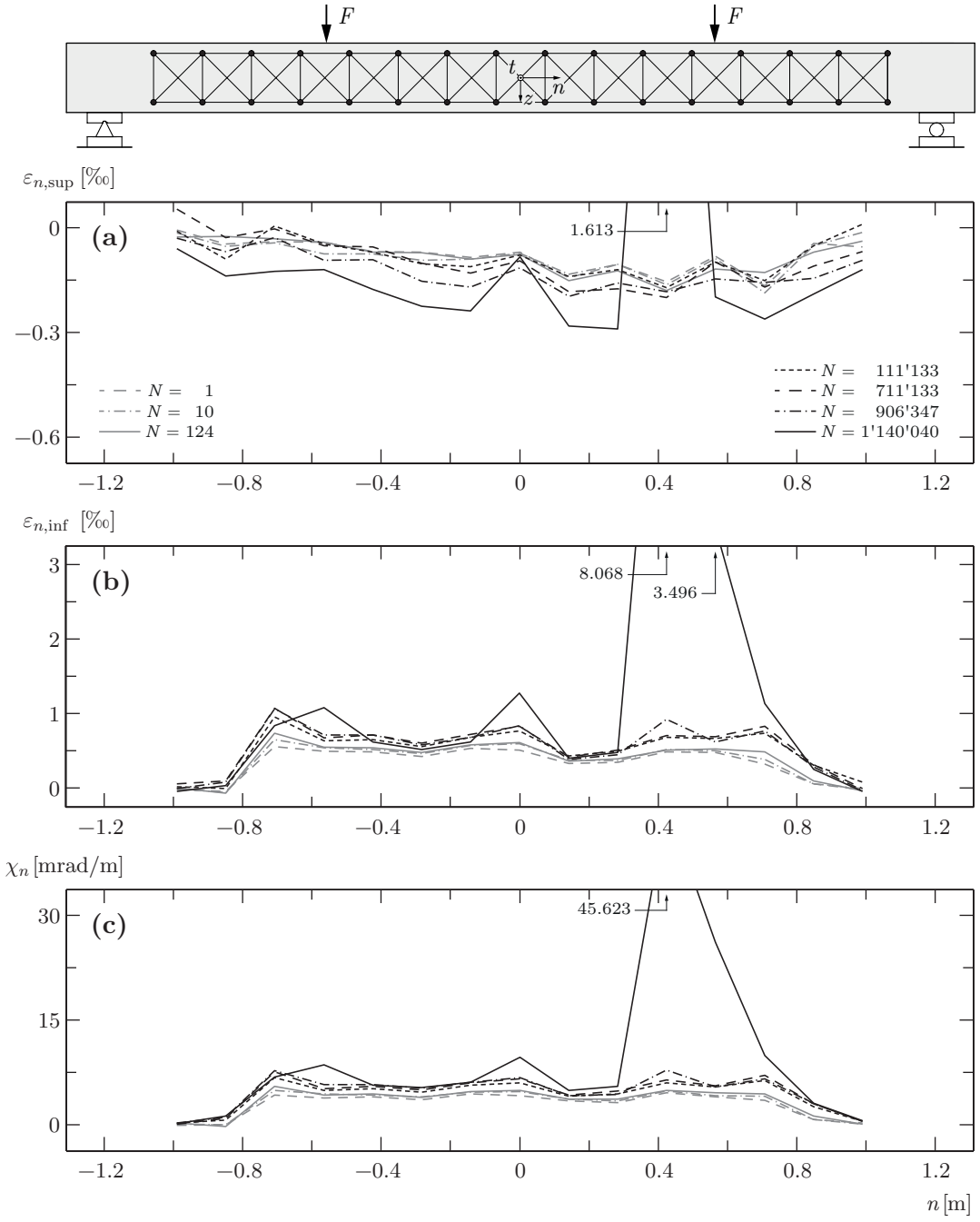


Fig. 6.23 – A1 – Average strains at minimum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

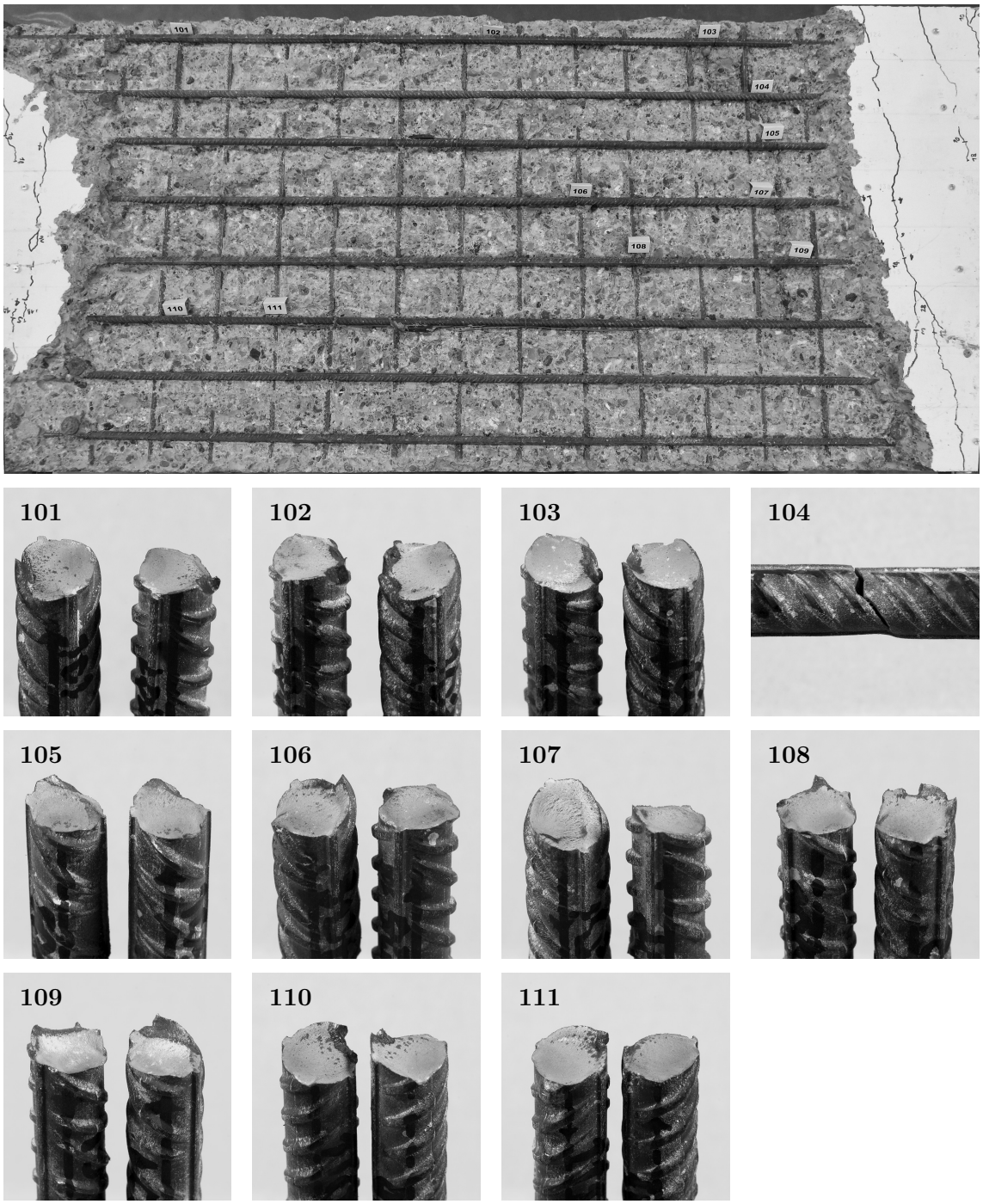


Fig. 6.24 – A1 – Overview and detail views of reinforcing bar fractures.

6.6.3 Test A2

Fig. 6.25 indicates the development of deflection in test A2 at maximum and minimum load for increasing load cycles. The midspan deflection more than doubled from 6.5 to 15.6 mm at maximum load between the initial loading and the final manual measurement shortly before failure. First cracks were recorded during the initial static loading at LS 3. The corresponding reduction in stiffness was observed at $M_n = 27.7$ kNm. The cracking pattern was practically complete after the initial loading, see Fig. 6.26 (a). Crack depths increased only marginally with further load cycles after $N = 10$, followed by another significant increase at $N = 755'347$, see Fig. 6.27. The crack width at maximum load within the cross-section at MC 5 doubled between $N = 755'347$ and $800'317$ from 0.25 to 0.5 mm. This crack extended vertically into the compression zone and developed another, almost horizontal, branch below the compression zone at $N = 809'232$. This development coincided with the detection of the first bar fractures. This critical crack measured 1.1 mm in width at maximum load in $N = 825'514$.

Similar to test A1, significant changes were observed in the measurements between the initial loading and $N = 110$. The relative change in subsequent load cycles remained small up to $N = 755'347$, see Fig. 6.28. Concrete strains in n-direction followed a similar trend to those in test A1. The large strains on the top surface and soffit at $N = 809'232$ shown in Fig. 6.30 were due to a localisation of curvature at the critical crack, see Fig. 6.32. Low strains were observed on the top surface and soffit in t-direction; these were of a similar magnitude to those measured in test A1. Strong localisations of strain only became evident in the final measurements at $N = 755'347$ and $809'232$ after failure of individual bars.

The first bar fracture occurred at fewer load cycles than in test A1. A total of 18 fractures were observed on five of the bars at failure, see Fig. 6.34. A number of disproportionally large strain increases were registered in all strain gauges between $N = 560'200$ and $748'600$. These indicate the rupture of individual reinforcing bars. A surprising number of 10 fractures were detected on three individual bars in $N = 755'347$ of which two bars ruptured at four positions each within the constant moment region. Three additional fractures were observed in $N = 800'317$ on a previously damaged bar. Three further fractures were detected in three previously intact bars at $N = 809'232$. Only three bars remained intact at this stage of cyclic loading. Two additional fractures were observed in the subsequent load cycles, resulting in failure of the specimen at $N = 825'784$. The last occurring fracture, No. 118, exhibited the characteristics of a plastic failure due to excessive deformation. A slightly diagonal crack was observed on the slab soffit between cross-sections MC 5 and MC 6. At failure, this crack extended through the compression zone and was visible across half of the specimen on the top surface.

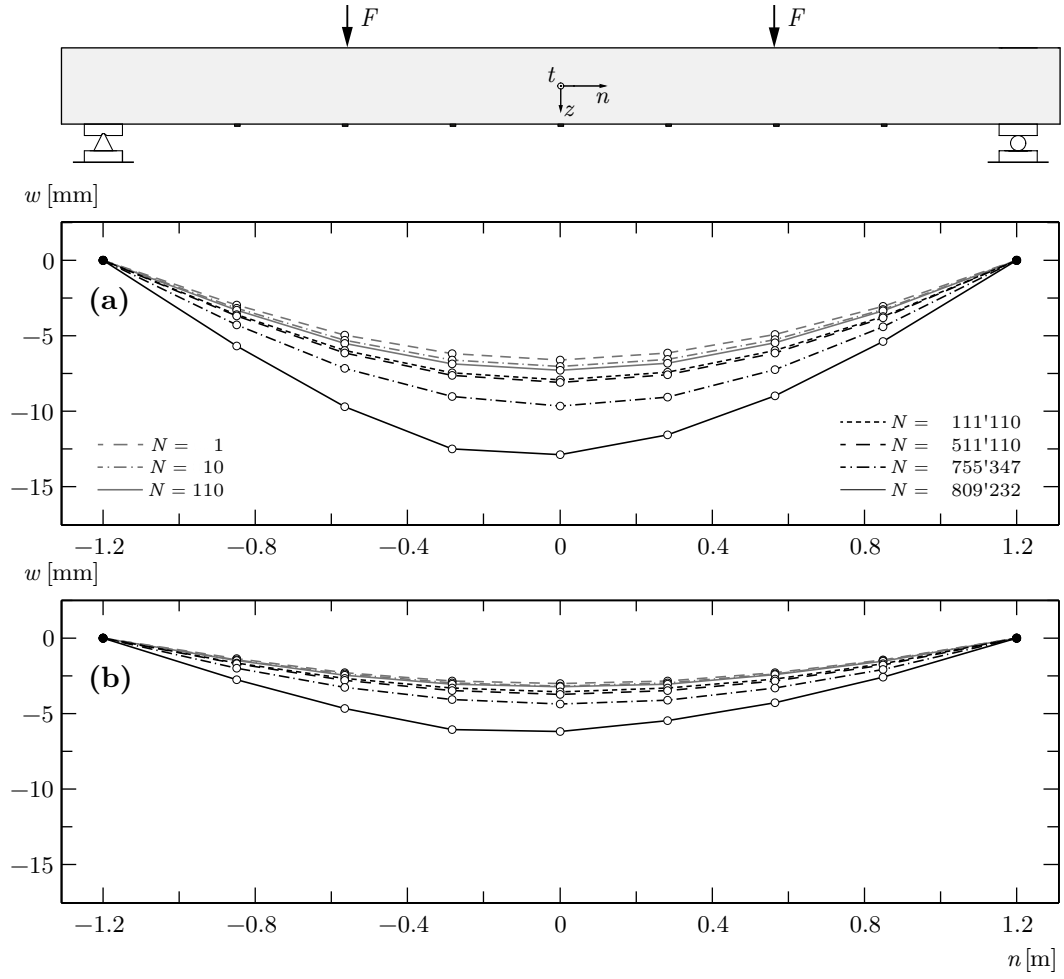


Fig. 6.25 – A2 – Deflection measurements: (a) maximum load; (b) minimum load.

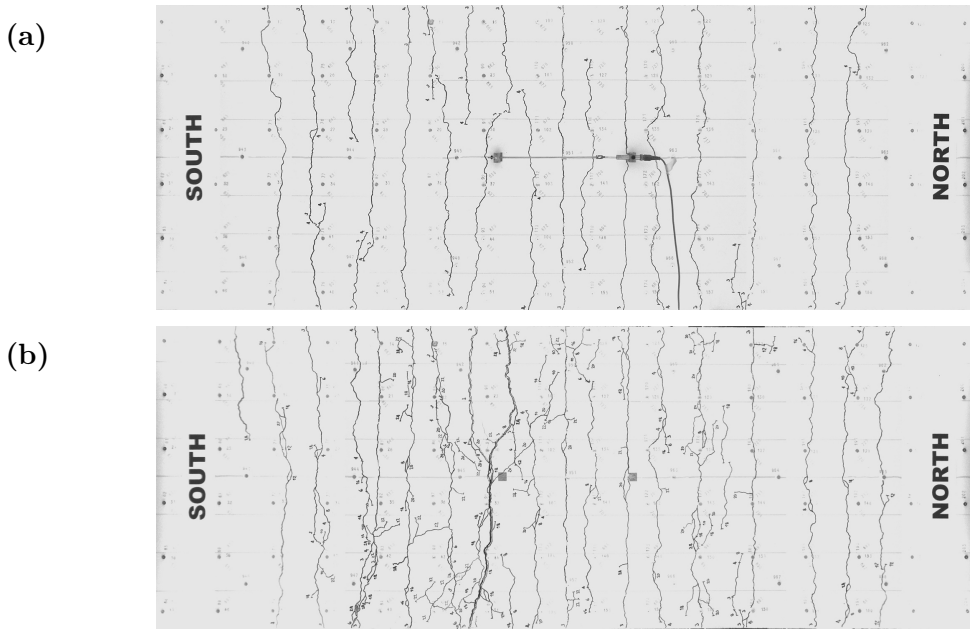


Fig. 6.26 – A2 – Soffit cracking pattern: (a) after initial loading; (b) at failure.

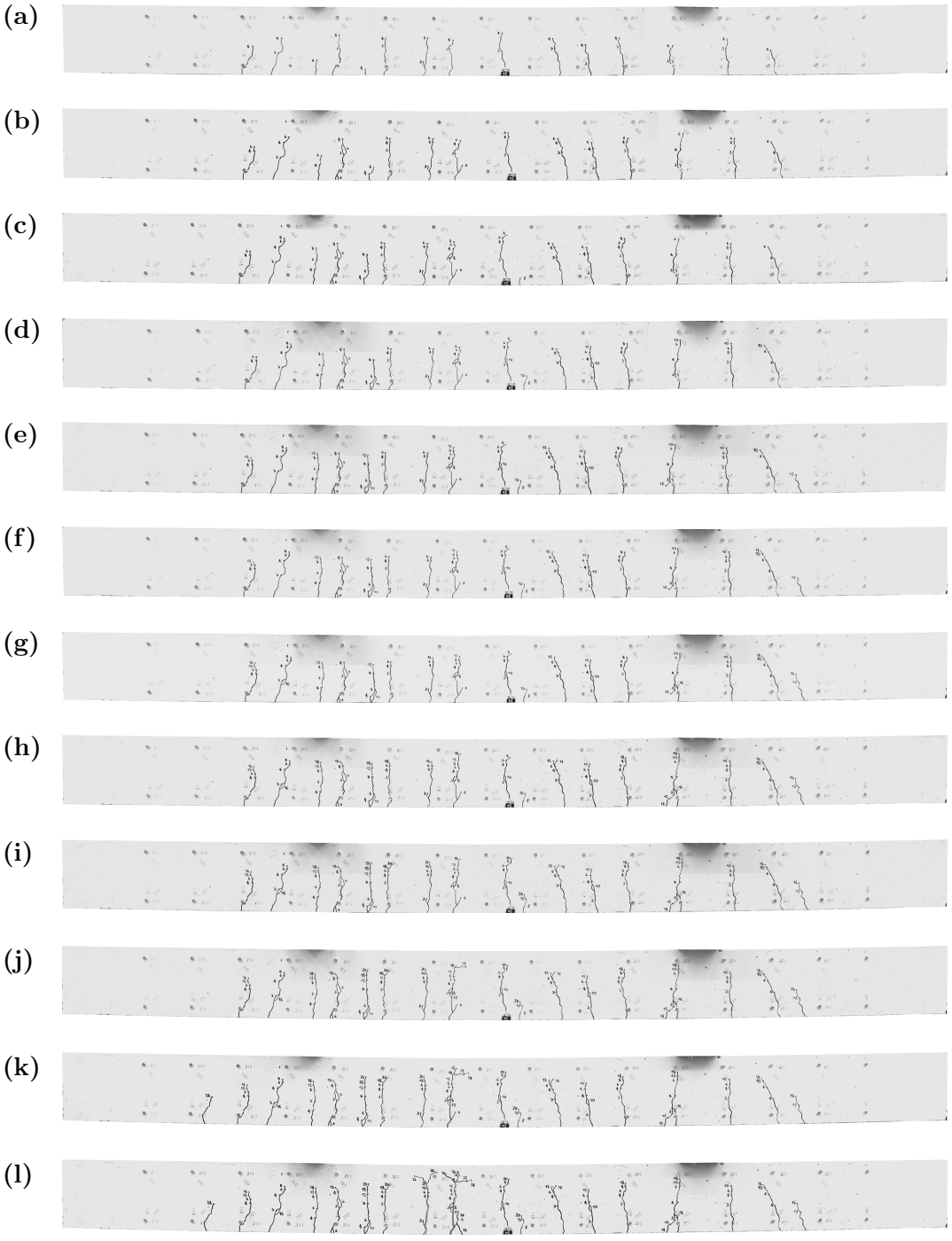


Fig. 6.27 – A2 – Front surface cracking pattern at maximum load in load cycle: (a) 1; (b) 10; (c) 110; (d) 1′110; (e) 11′110; (f) 111′110; (g) 511′110; (h) 755′347; (i) 800′317; (j) 809′232; (k) 825′514; (l) 825′784.

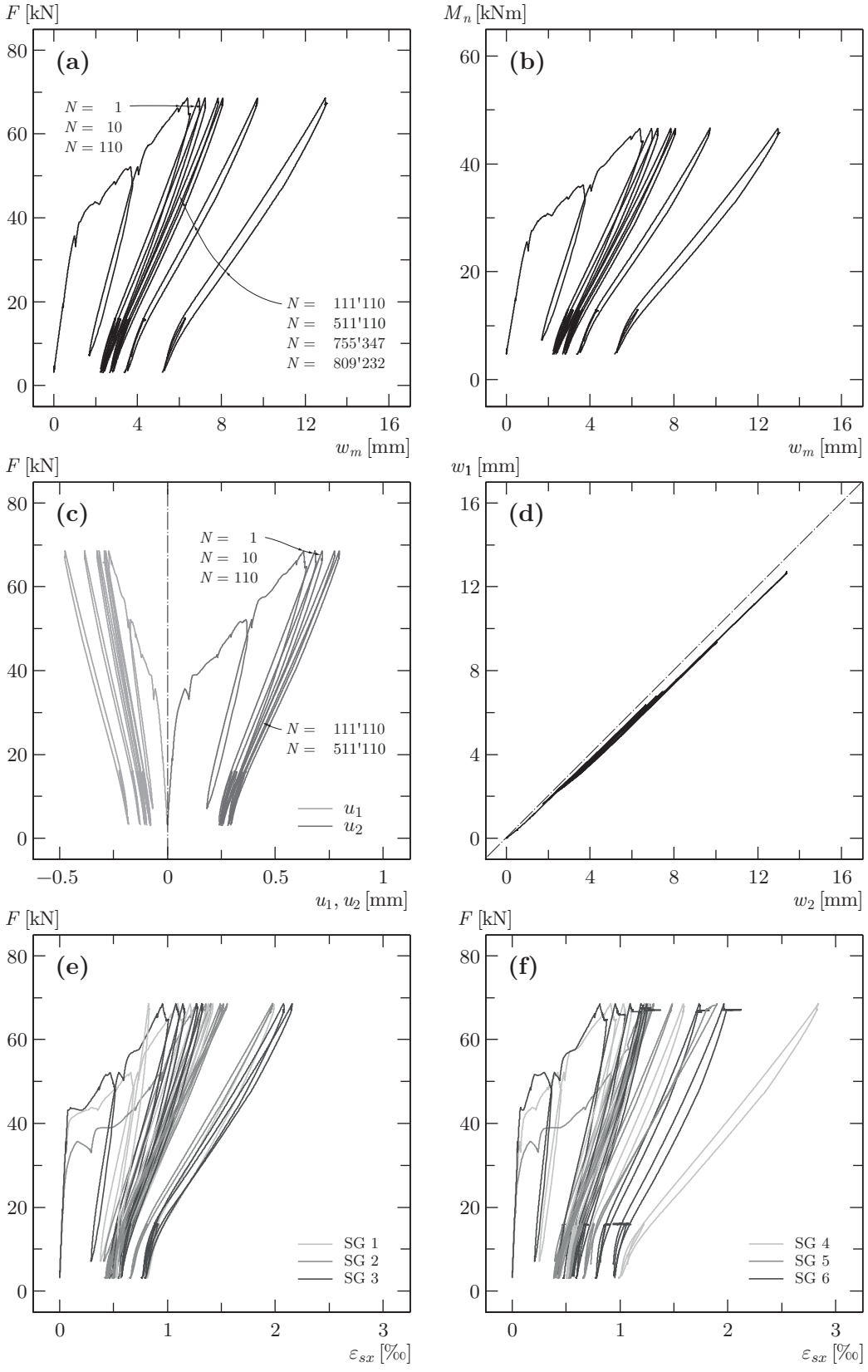


Fig. 6.28 – A2 – Quasi-static loading: (a) and (b) force and moment-deflection; (c) concrete deformations on top surface and soffit; (d) midspan deflection at front and back surface; (e) and (f) reinforcing steel strains in x-direction.

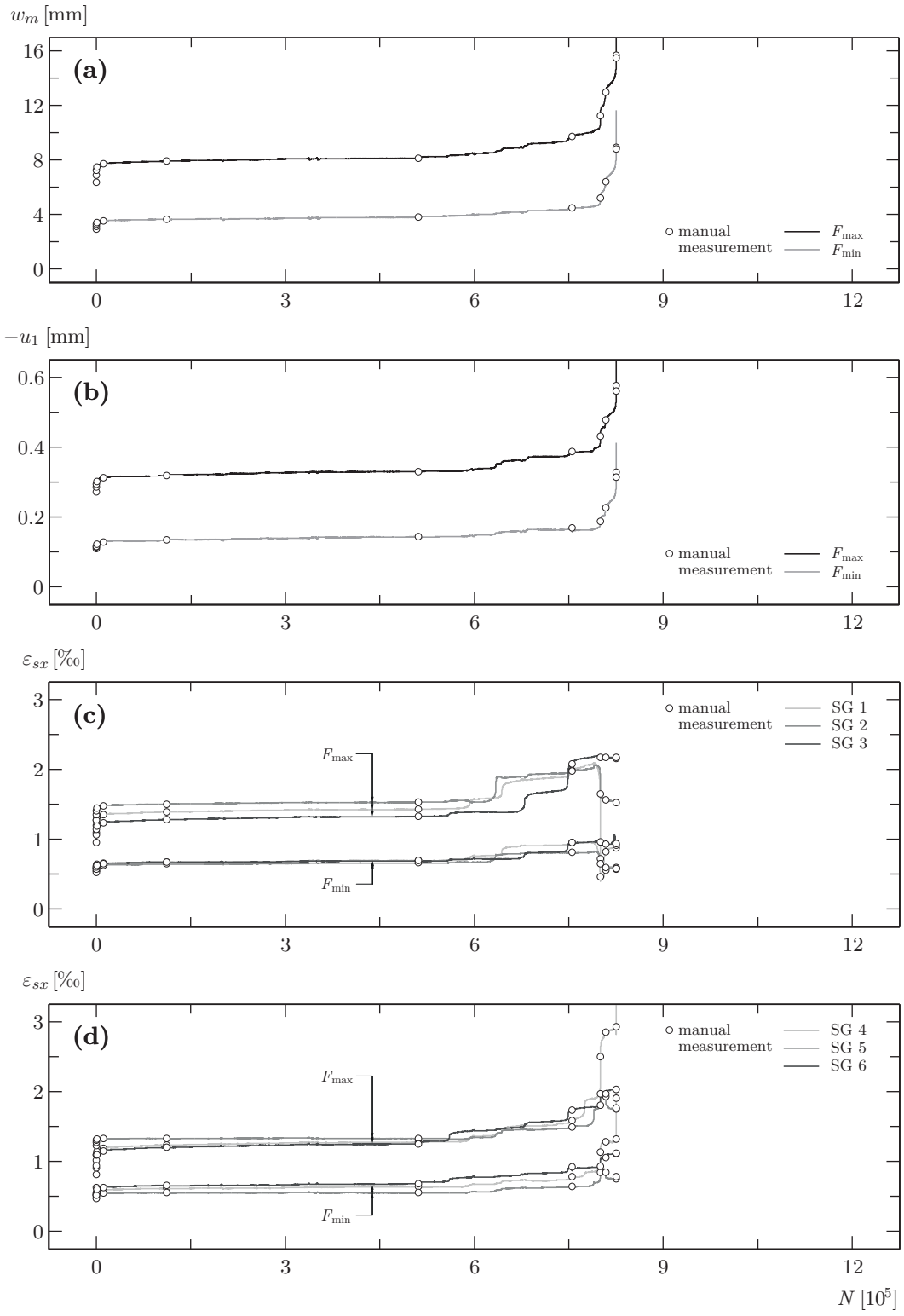


Fig. 6.29 – A2 – Development at maximum and minimum load: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x-direction.

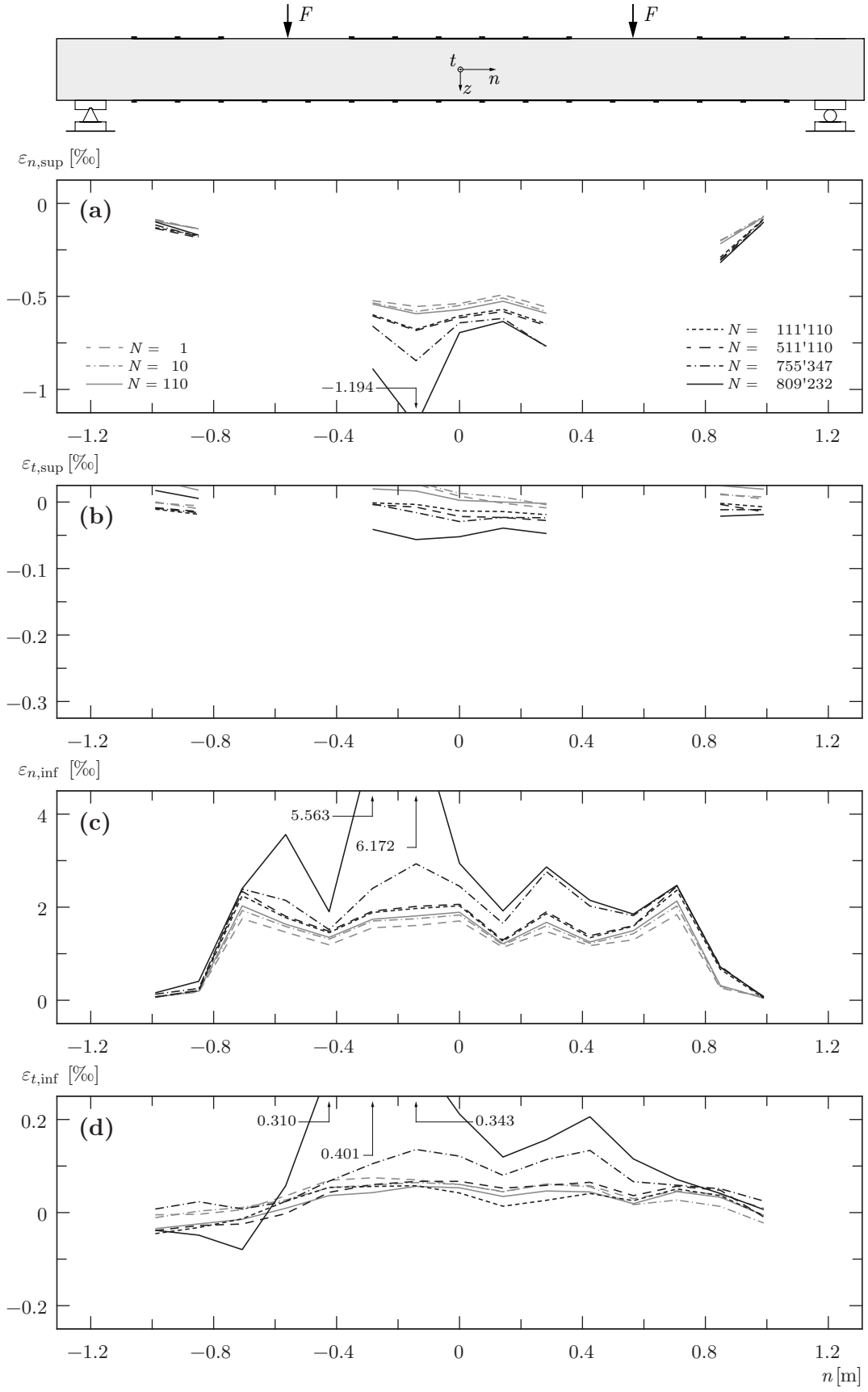


Fig. 6.30 – A2 – Average strains at maximum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

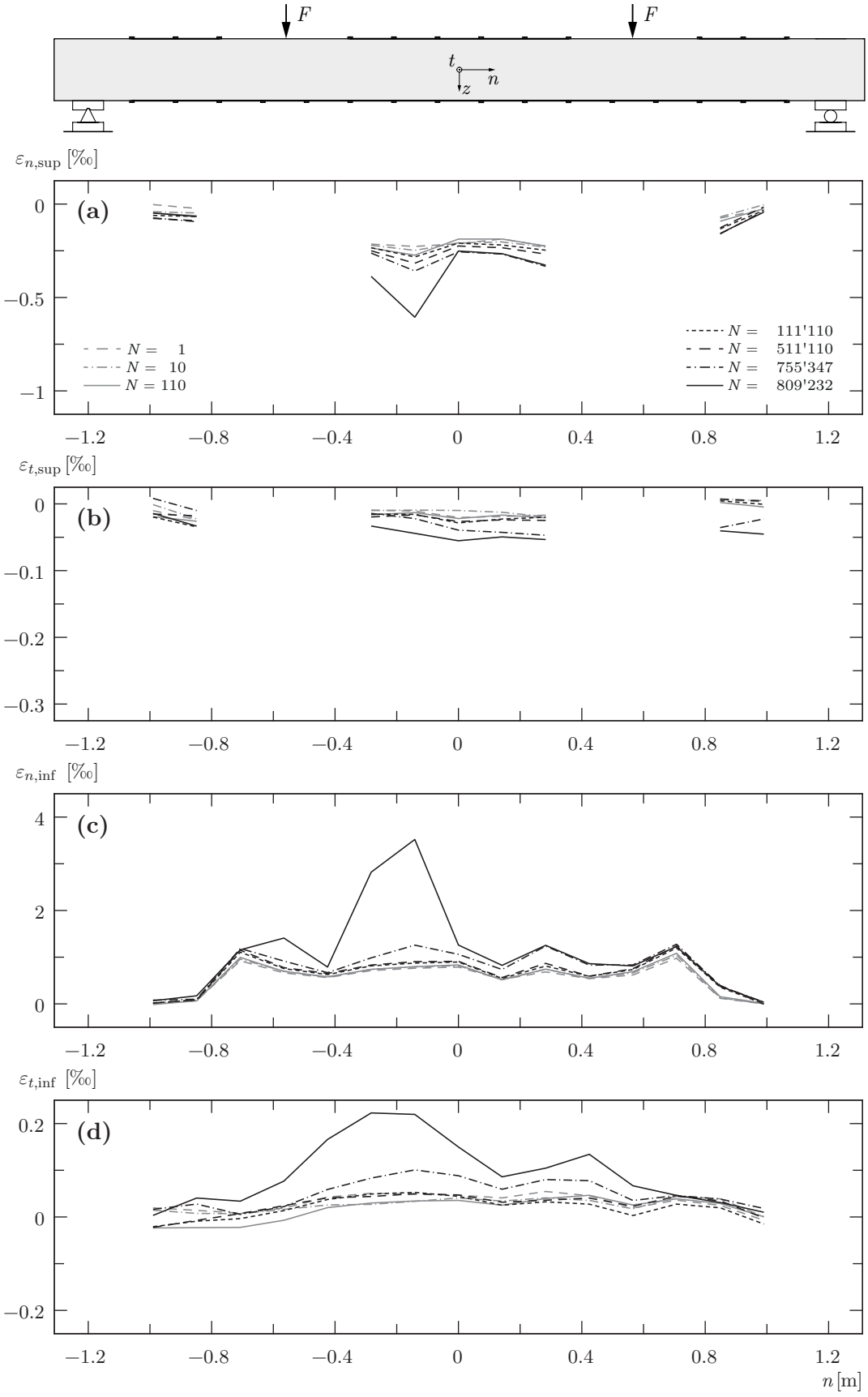


Fig. 6.31 – A2 – Average strains at minimum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

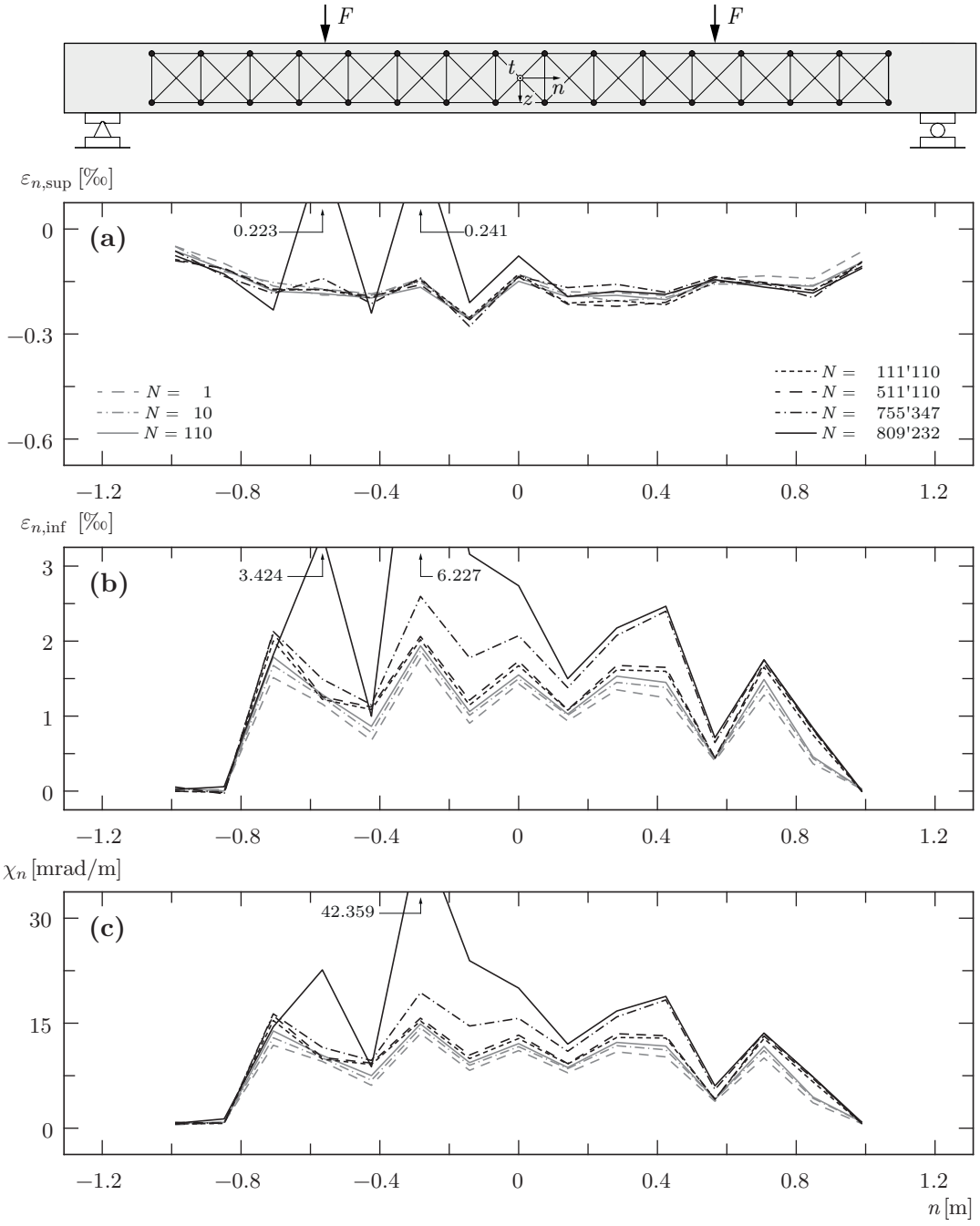


Fig. 6.32 – A2 – Average strains at maximum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

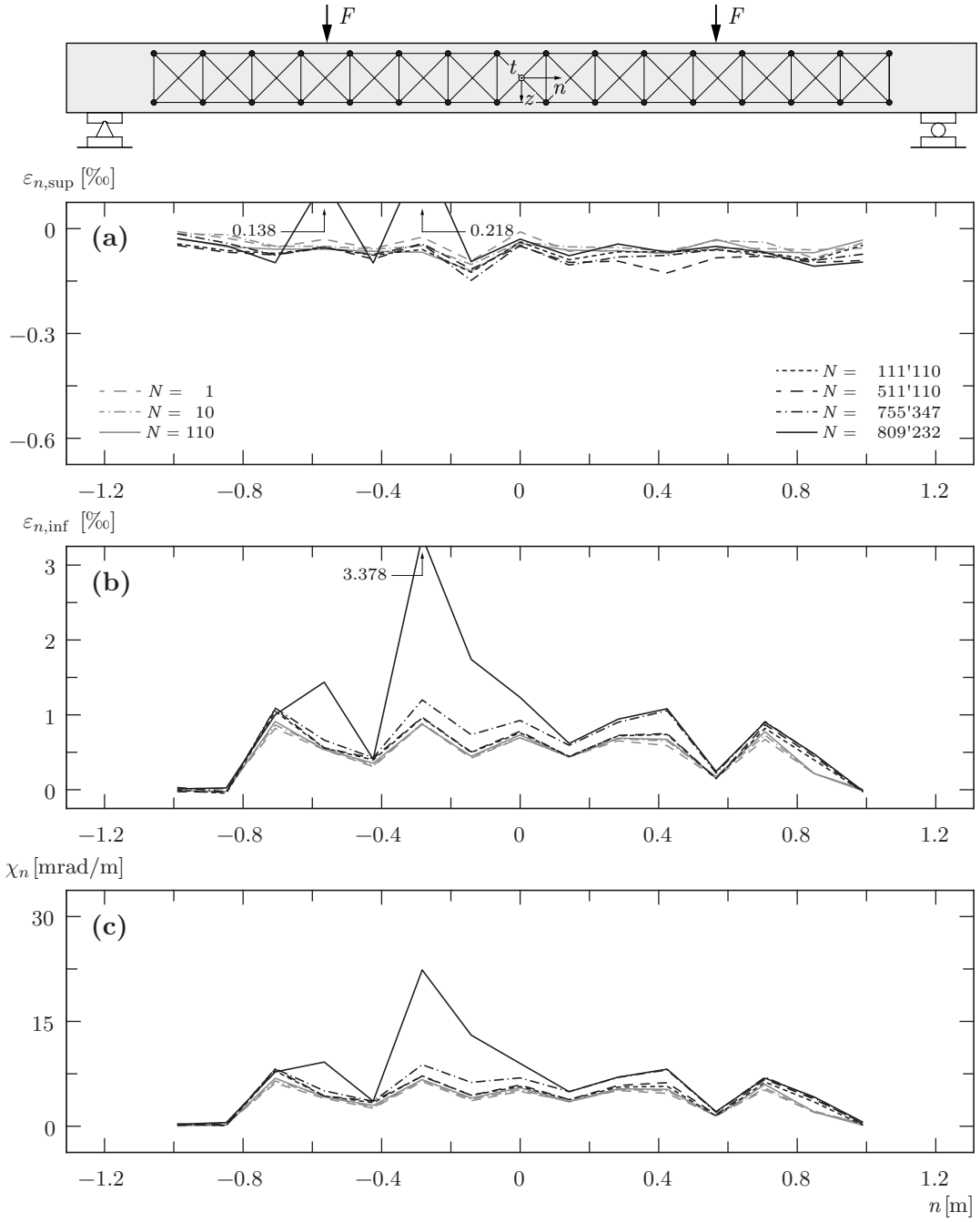


Fig. 6.33 – A2 – Average strains at minimum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

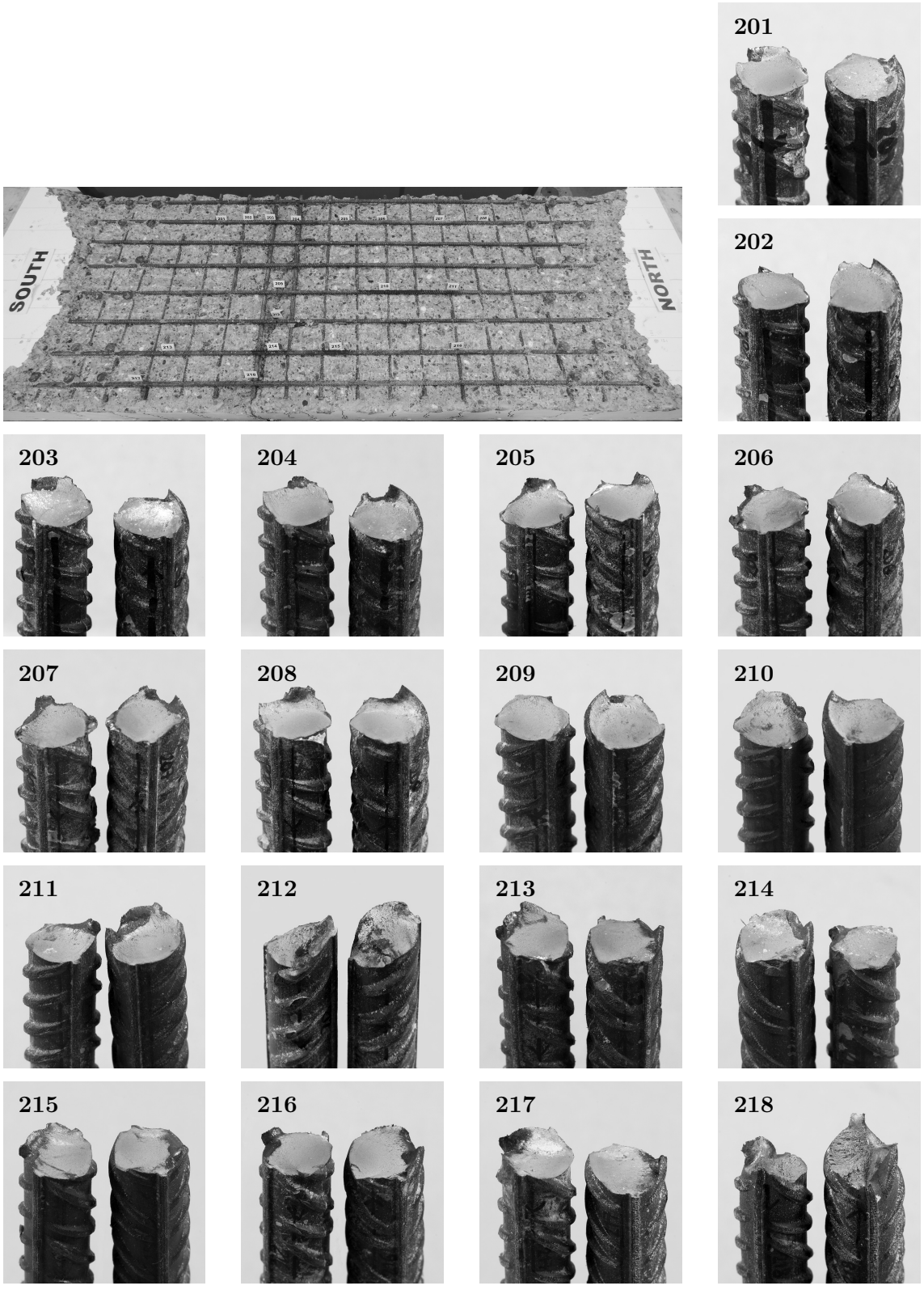


Fig. 6.34 – A2 – Overview and detail views of reinforcing bar fractures.

6.6.4 Test B1

The development of vertical deflection is indicated in Fig. 6.35 at maximum and minimum load. The midspan deflection at maximum load measured 15.9 mm in the initial loading and increased to 27.7 mm shortly before failure. First cracks were recorded during the initial loading at LS 2, whereas the associated reduction in stiffness was observed at $M_n = 21.1$ kNm. A practically complete pattern of primary flexural cracks were formed after the initial loading, see Fig. 6.36. Crack depths increased only marginally with further load cycles after $N = 10$. The formation of secondary diagonal cracks, joining up to the existing flexural cracks, was observed in $N = 269'807$, see Fig. 6.37. An almost horizontal crack formed at mid-height of the slab between two flexural cracks at the cross-section MC 6. The width of four other cracks within the test region also showed a significant increase. These changes were associated with the occurrence of the first bar fractures. The critical crack formed at a cross-section close to the load application. This developed from a secondary diagonal crack, which joined up to a vertical flexural crack at roughly mid-height of the slab. The crack extended through the compression zone at failure. Furthermore, some longitudinal cracks were observed on the top surface of the slab, i.e. perpendicular to the principal compressive direction.

Significant changes were observed in the measurements within the initial load cycles. Fig. 6.38 indicates an approximately equal increase in the measurements at upper load, albeit at a number of load cycles increasing by an order of magnitude. A disproportionally large increase was observed in the measurement at $N = 269'807$ when the first bar fractures were detected, see Fig. 6.39. Concrete strains in n-direction developed continuously with increasing cycles. Significant changes were observed within the first 10 load cycles and again after fatigue failure of the first bars, see Figs. 6.40 and 6.41. Transverse concrete strains in t-direction were clearly tensile on the top surface and compressive on the soffit. Large increases in strain on the soffit, as well as a localisation of curvature at the cracks, could only be observed after failure of individual bars, see Fig. 6.42.

A total of 13 bar fractures were observed on eleven bars at failure, see Fig. 6.44. Only two of these fractures occurred on bars in the inner layer of reinforcement. The first disproportionally large increases in reinforcing steel strain were registered at approximately $N = 214'400$. This most likely corresponded to the occurrence of fracture No. 303. A small crack initiated within the milled region of the bar at strain gauge SG 6. As a result, a continuous increase in strain was observed at both the maximum and minimum load, see fracture No. 306 in Fig. 6.44. Further, increases in reinforcing steel strain were observed at $N = 227'900$ and $256'100$. The former probably indicates the occurrence of fracture No. 307. Four additional fractures were observed with the magnetic flux leakage method at $N = 269'807$ and a further five at $271'345$ load cycles. Global failure of the specimen occurred shortly thereafter at $N = 271'524$. Two additional fractures were observed after removal of the concrete cover. Fracture No. 305 was a partial fatigue fracture with clearly visible crack initiation; fracture No. 302 was a complete break exhibiting a small fatigue and large plastic proportion on the fracture surface.

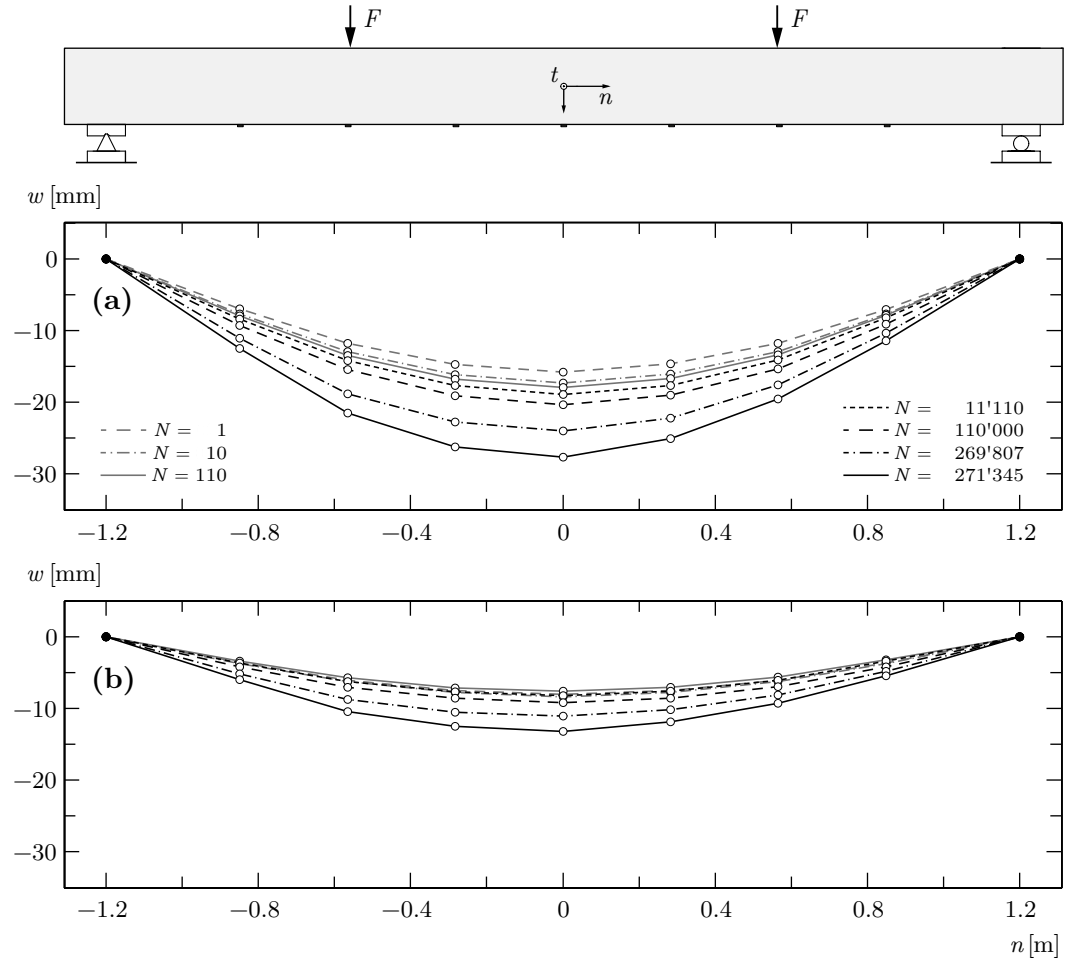


Fig. 6.35 – B1 – Deflection measurements: (a) maximum load; (b) minimum load.

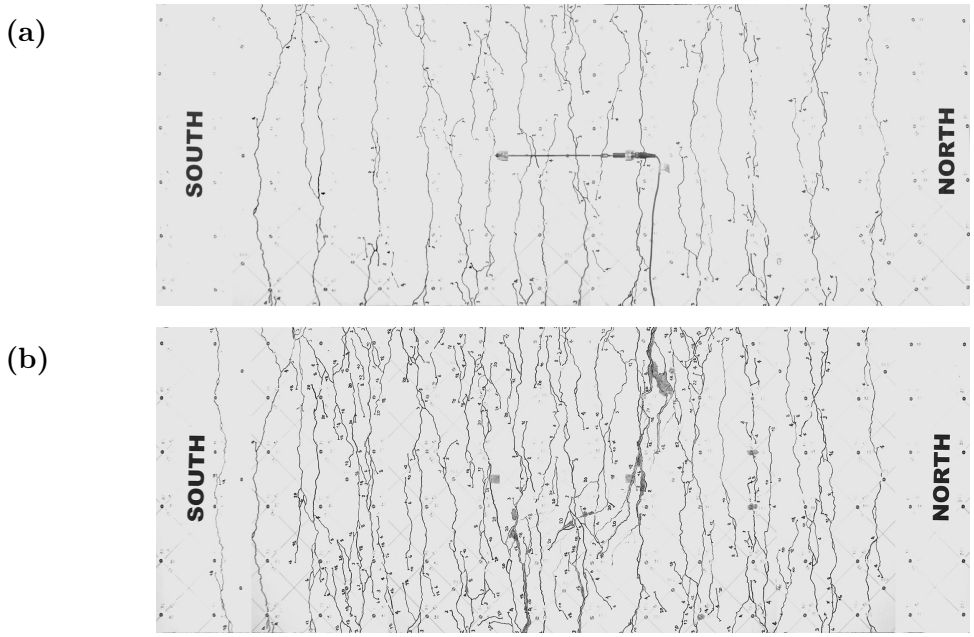


Fig. 6.36 – B1 – Soffit cracking pattern: (a) after initial loading; (b) at failure.

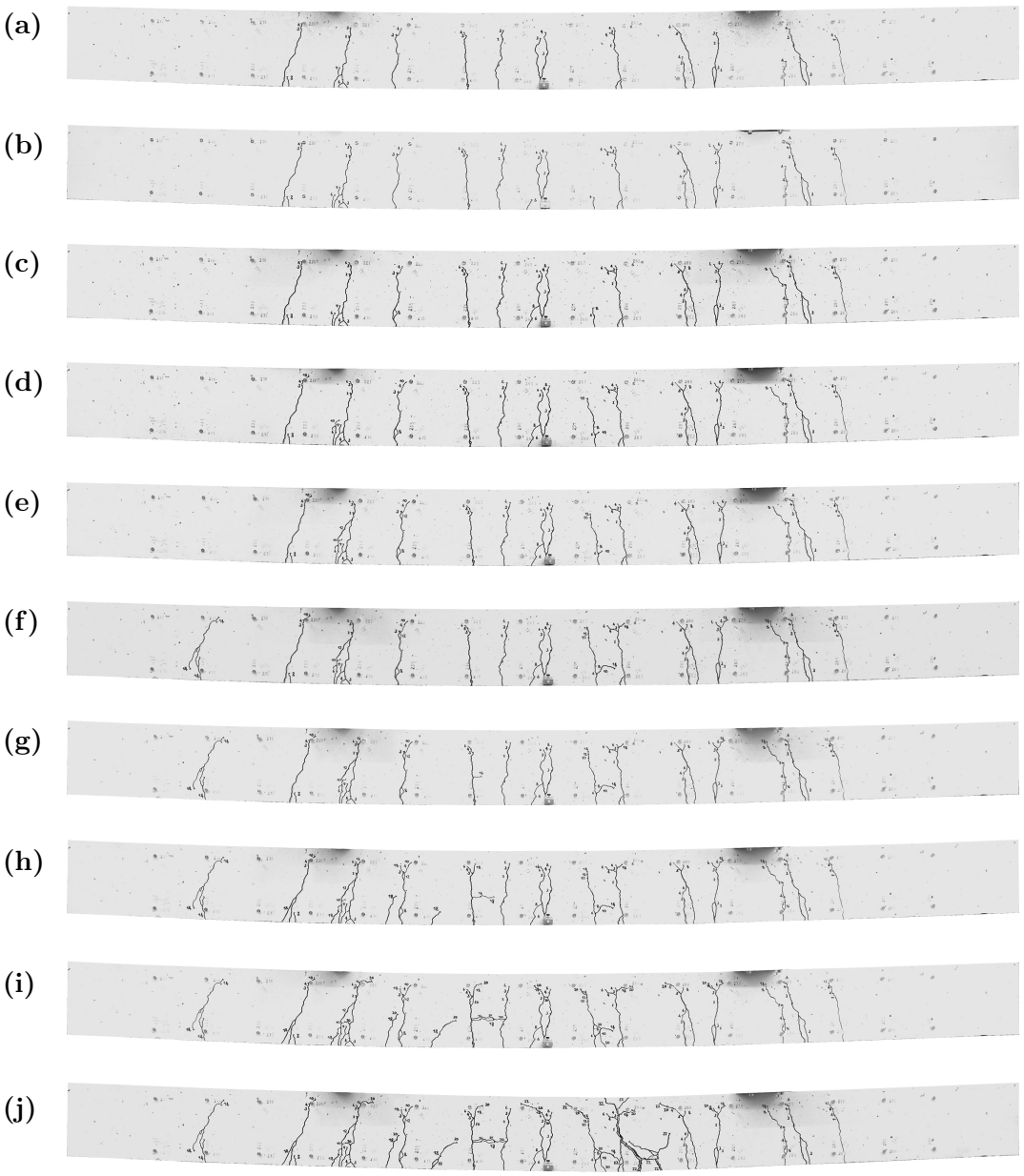


Fig. 6.37 – B1 – Front surface cracking pattern at maximum load in load cycle: (a) 1; (b) 10; (c) 110; (d) 1′110; (e) 11′110; (f) 60′278; (g) 110′000; (h) 269′807; (i) 271′345; (j) 271′524.

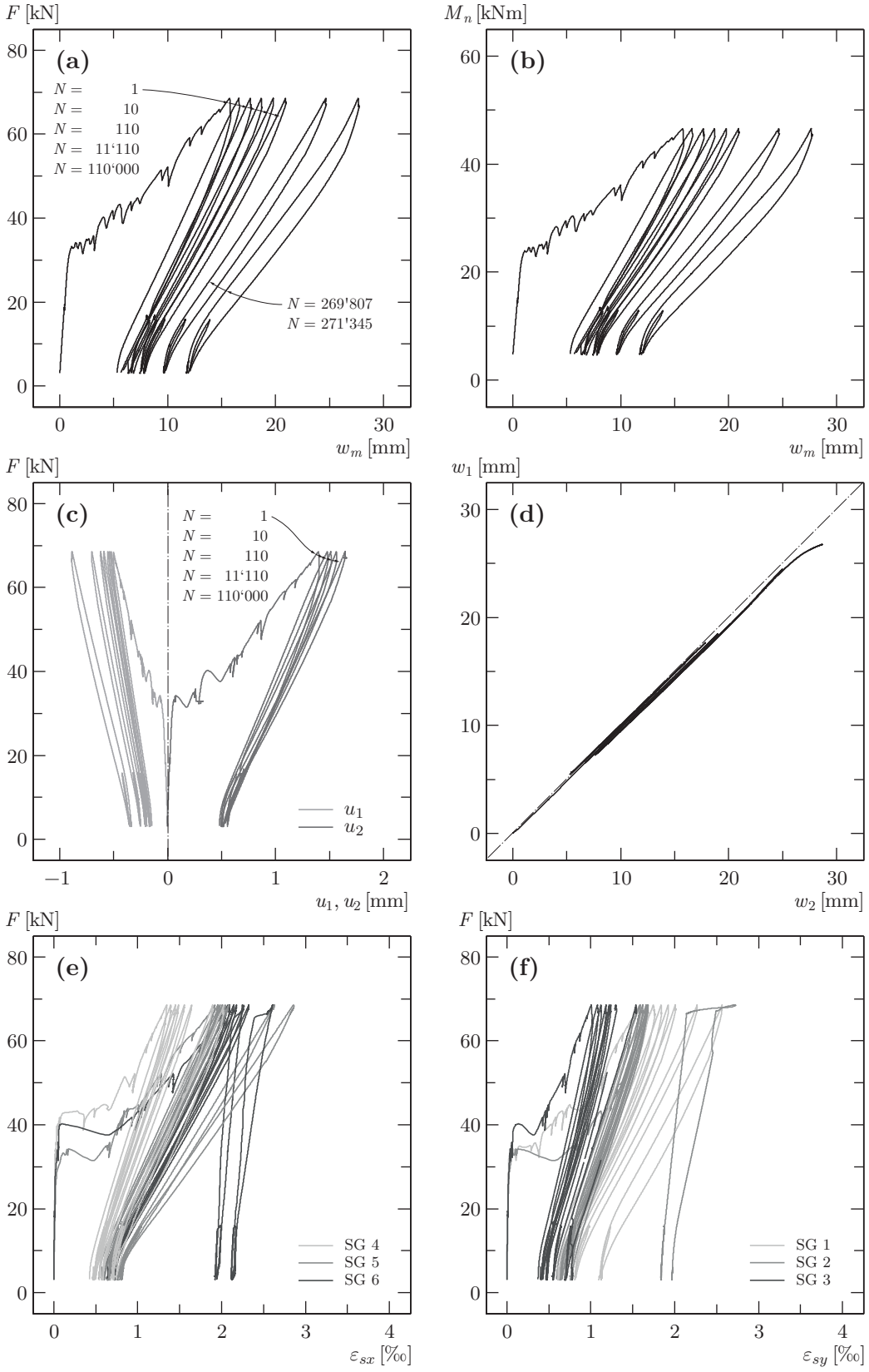


Fig. 6.38 – B1 – Quasi-static loading: (a) and (b) force and moment-deflection; (c) concrete deformations on top surface and soffit; (d) midspan deflection at front and back surface; (e) and (f) reinforcing steel strains in x- and y-direction.

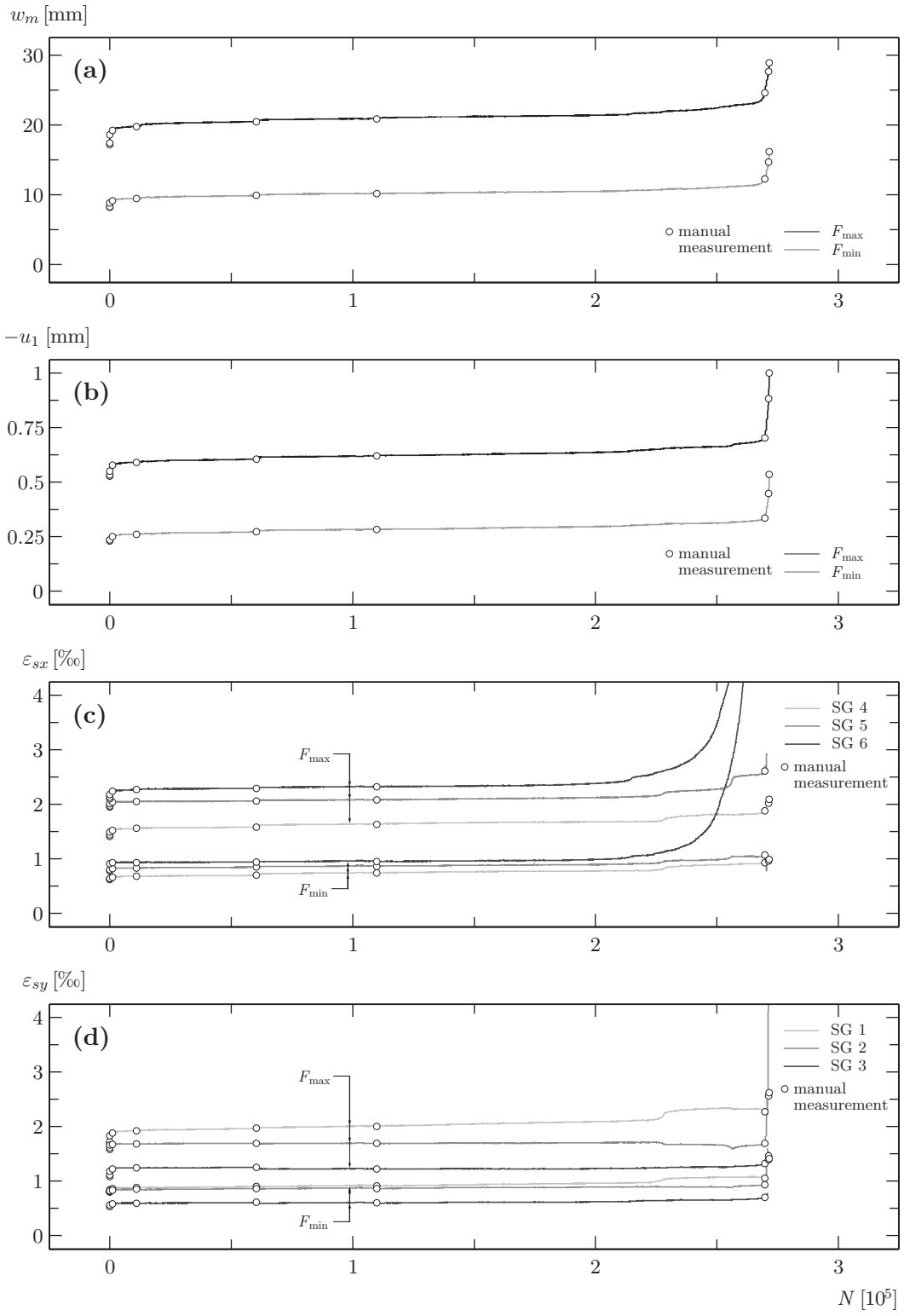


Fig. 6.39 – B1 – Development at maximum and minimum load: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x- and y-direction.

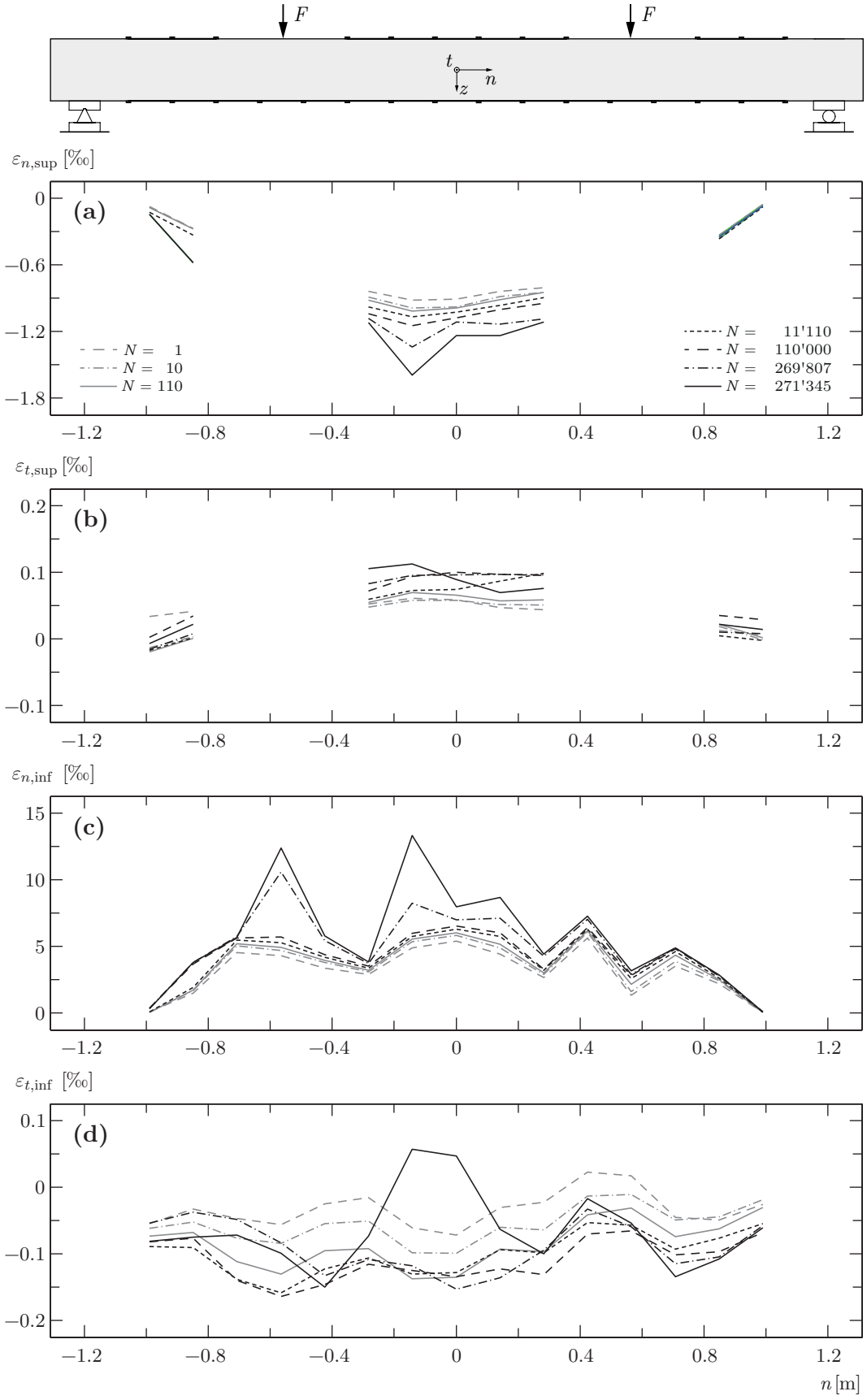


Fig. 6.40 – B1 – Average strains at maximum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

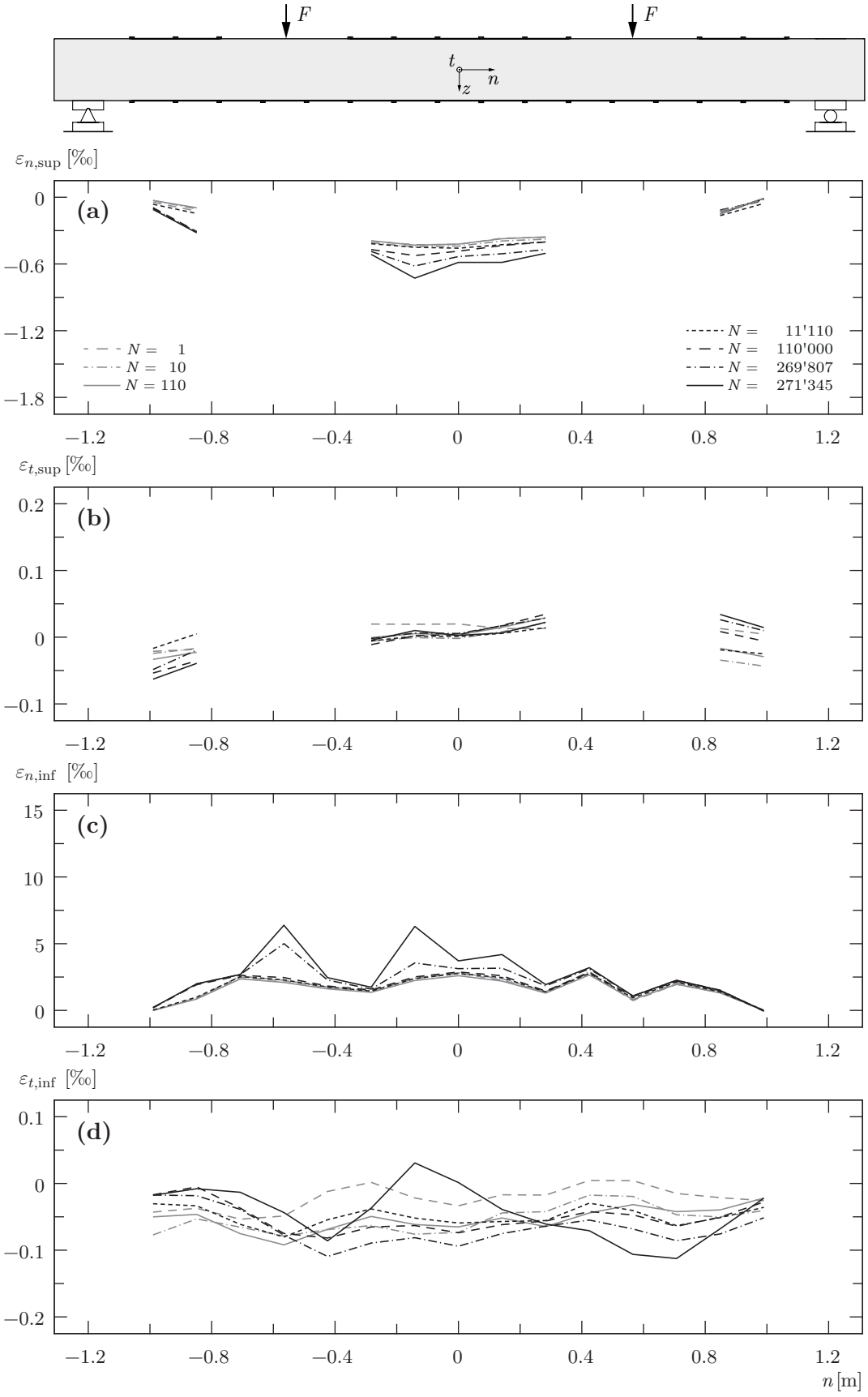


Fig. 6.41 – B1 – Average strains at minimum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

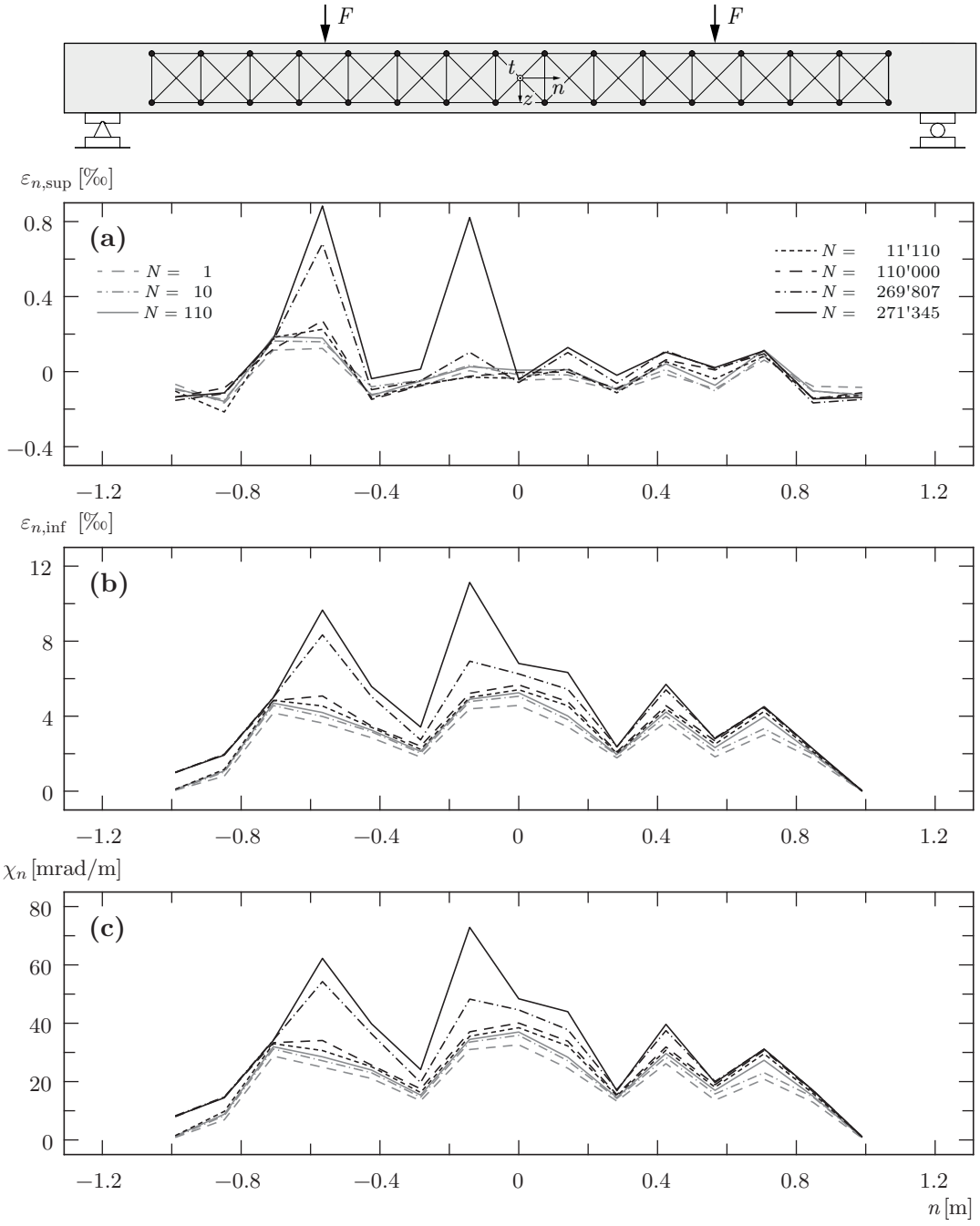


Fig. 6.42 – B1 – Average strains at maximum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

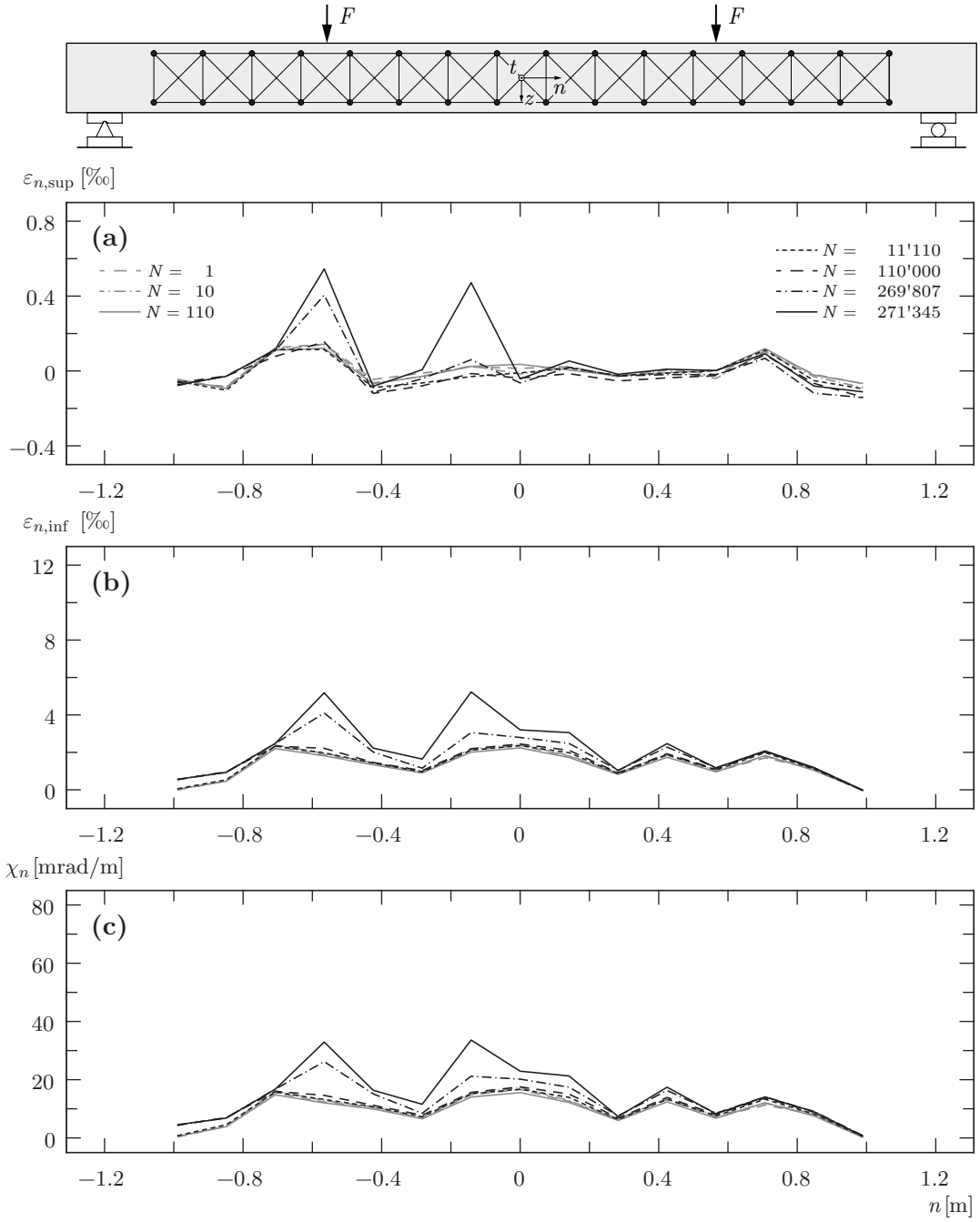


Fig. 6.43 – B1 – Average strains at minimum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

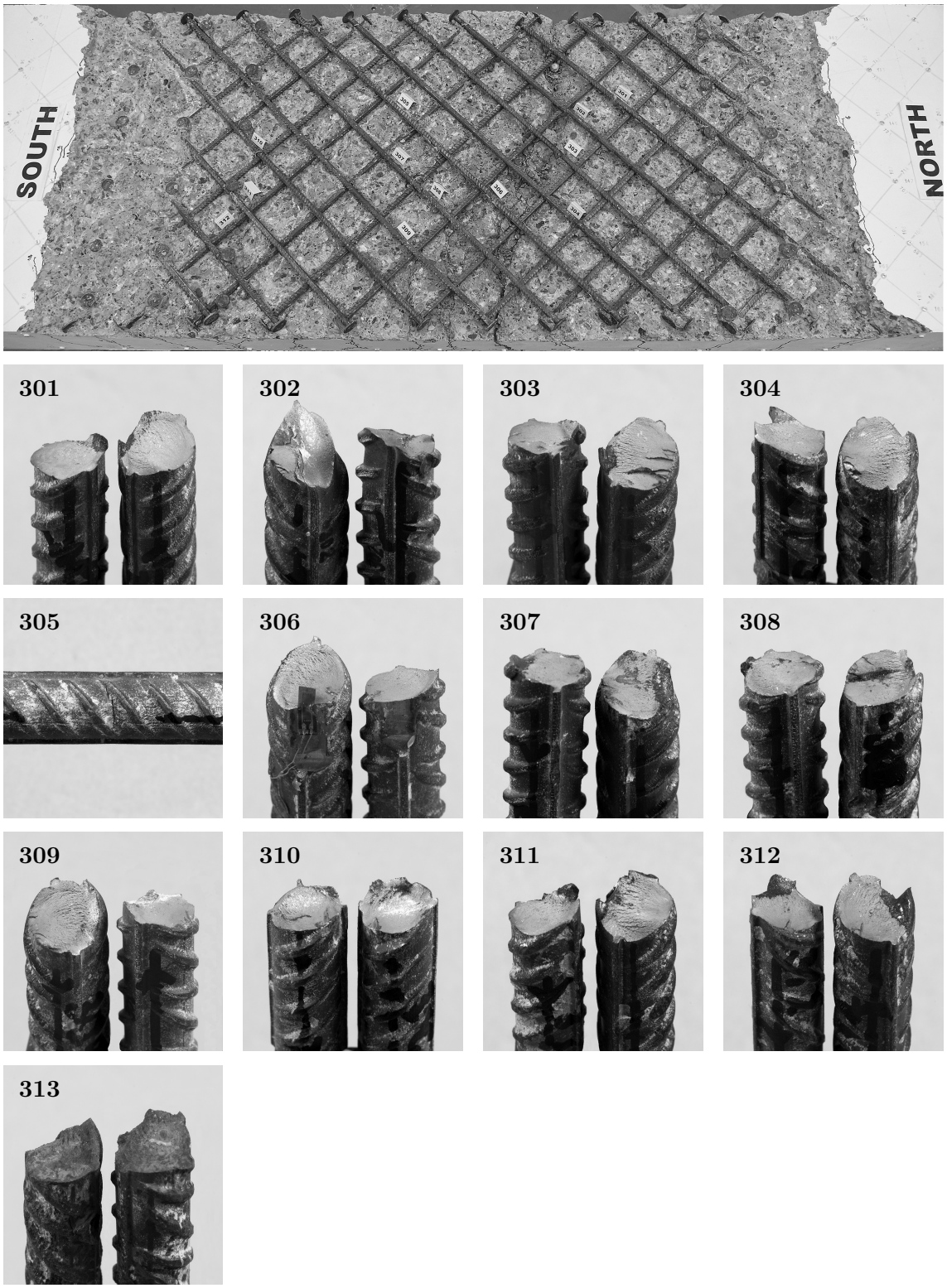


Fig. 6.44 – B1 – Overview and detail views of reinforcing bar fractures.

6.6.5 Test B2

Fig. 6.45 indicates the development of deflection at maximum and minimum load. A mid-span deflection of 16.3 mm was measured at maximum load after the initial loading which increased to 27.5 mm shortly before failure. Similar to test B1, first cracks were recorded at LS 2 during the initial loading. The reduction in stiffness associated with cracking was observed at $M_n = 22.7$ kNm. A practically complete cracking pattern was established in the initial loading, see Figs. 6.46 and 6.47. Crack depths increased only marginally with further load cycles up to $N = 10$ and even less to $N = 110$. Secondary diagonal cracks, similar to those observed in test B1, formed from $N = 217'555$ onwards. These extended with increasing load cycles up to their intersection with the existing flexural cracks, see Fig. 6.47. The crack widths at cross-sections corresponding to MC 2 and MC 11 almost doubled between the measurements at $N = 110'000$ and $217'555$. No fractures in the reinforcing steel were detected with the magnetic flux leakage method at this stage. A secondary diagonal crack was observed at $N = 219'079$, which joined up to a vertical flexural crack on the front surface at MC 6 and on the back surface at MC 5. This corresponded to the stage of loading when the first bar fractures were detected. The critical crack branched off below the compression zone and extended almost horizontally until failure of the specimen. The concrete on the slab soffit showed signs of spalling at the critical cross-section. This possibly indicated crushing of the concrete due to high transverse compressive strains perpendicular to the principal direction.

A similar development of the measurements under cyclic loading was observed as described for test B1. A significant change in response, associated with the formation of the first bar fractures, was observed in $N = 219'079$, see Fig. 6.49. Longitudinal concrete strains in n-direction developed continuously with increasing load cycles. Significant changes were first observed after failure of the first bars and then only highly localised around the ruptured bars, see Fig. 6.50. Transverse concrete strains in t-direction on the specimen top surface were clearly tensile. The transverse strains on the slab soffit were not as clearly compressive as observed in test B1. A large increase in curvature occurred after failure of the first bars. This was localised in the cross-section corresponding to the measurement grid at MC 6, see Fig. 6.52.

Eight reinforcing bar fractures were observed on seven of the bars at global failure. Half of these fractures occurred in the second layer of reinforcement, see Fig. 6.54. A large increase in reinforcing steel strain was registered at $N = 197'900$. This probably corresponds to the occurrence of fracture No. 406 or 407. These fractures were closely spaced and located directly adjacent to the critical crack. Fracture No. 405 occurred at the interface between the reinforcing bar and the anchor head. A portion of this fracture surface indicates fatigue striations, however, the final rupture might have occurred during the mechanical removal of the concrete. A small crack was observed within the milled region of the bar at SG 5, see fracture No. 402 in Fig. 6.54. Global failure of the slab occurred at $N = 219'741$ load cycles.

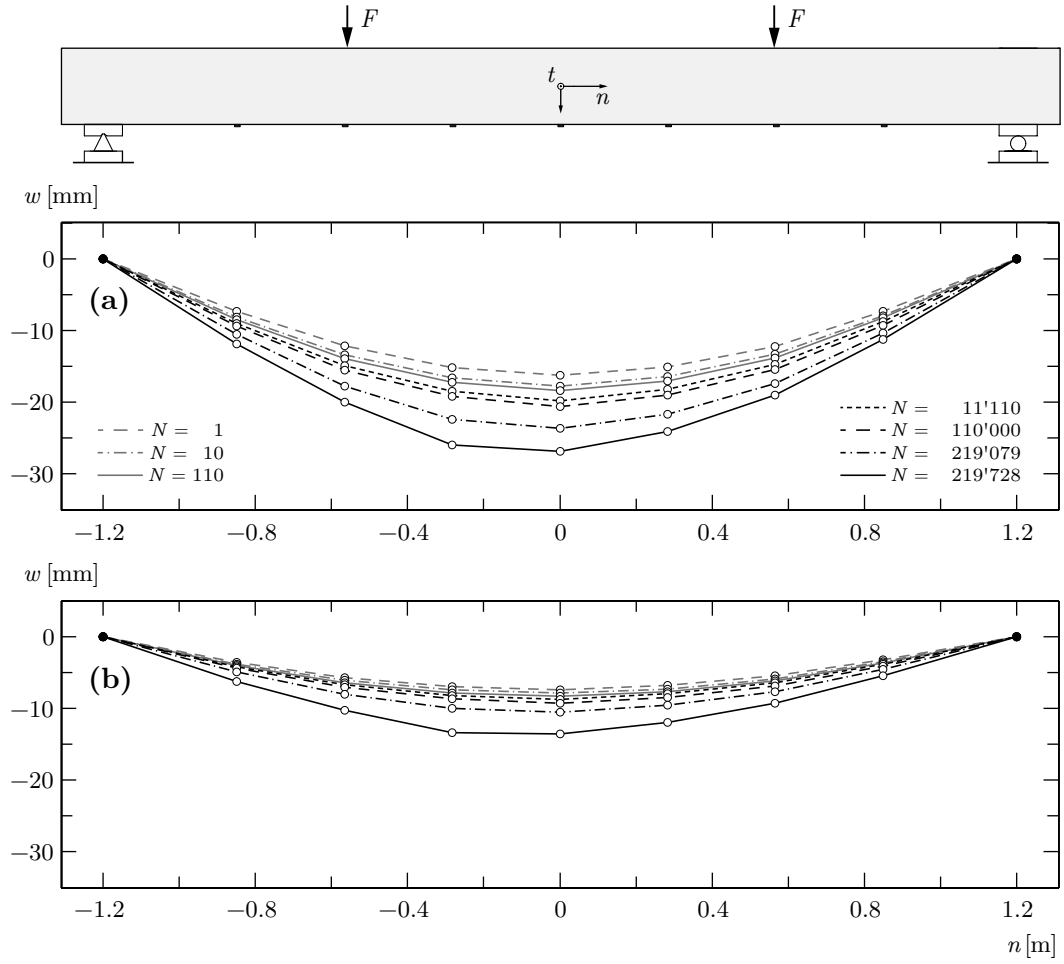


Fig. 6.45 – B2 – Deflection measurements: (a) maximum load; (b) minimum load.

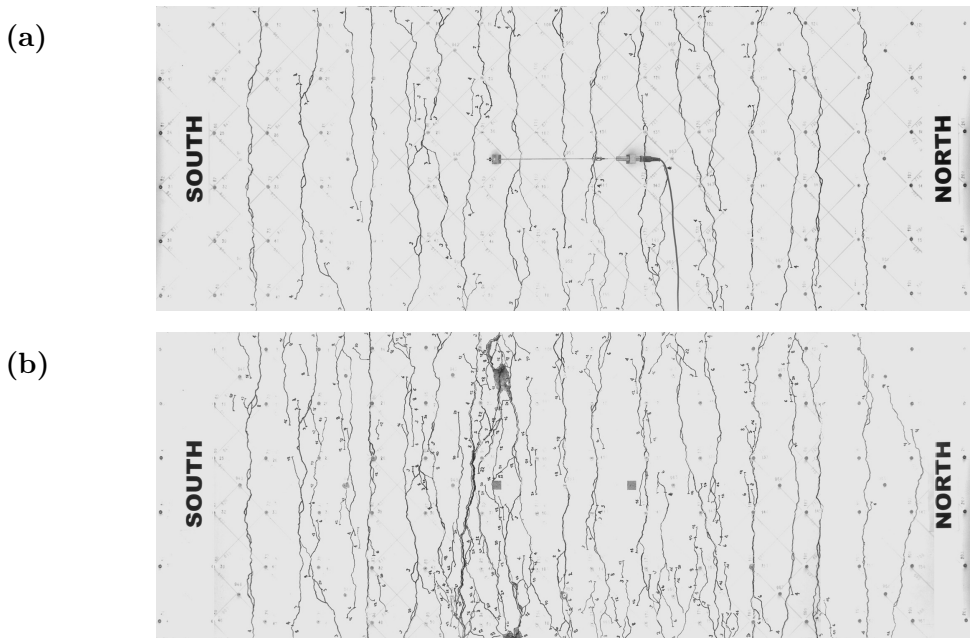


Fig. 6.46 – B2 – Soffit cracking pattern: (a) after initial loading; (b) at failure.

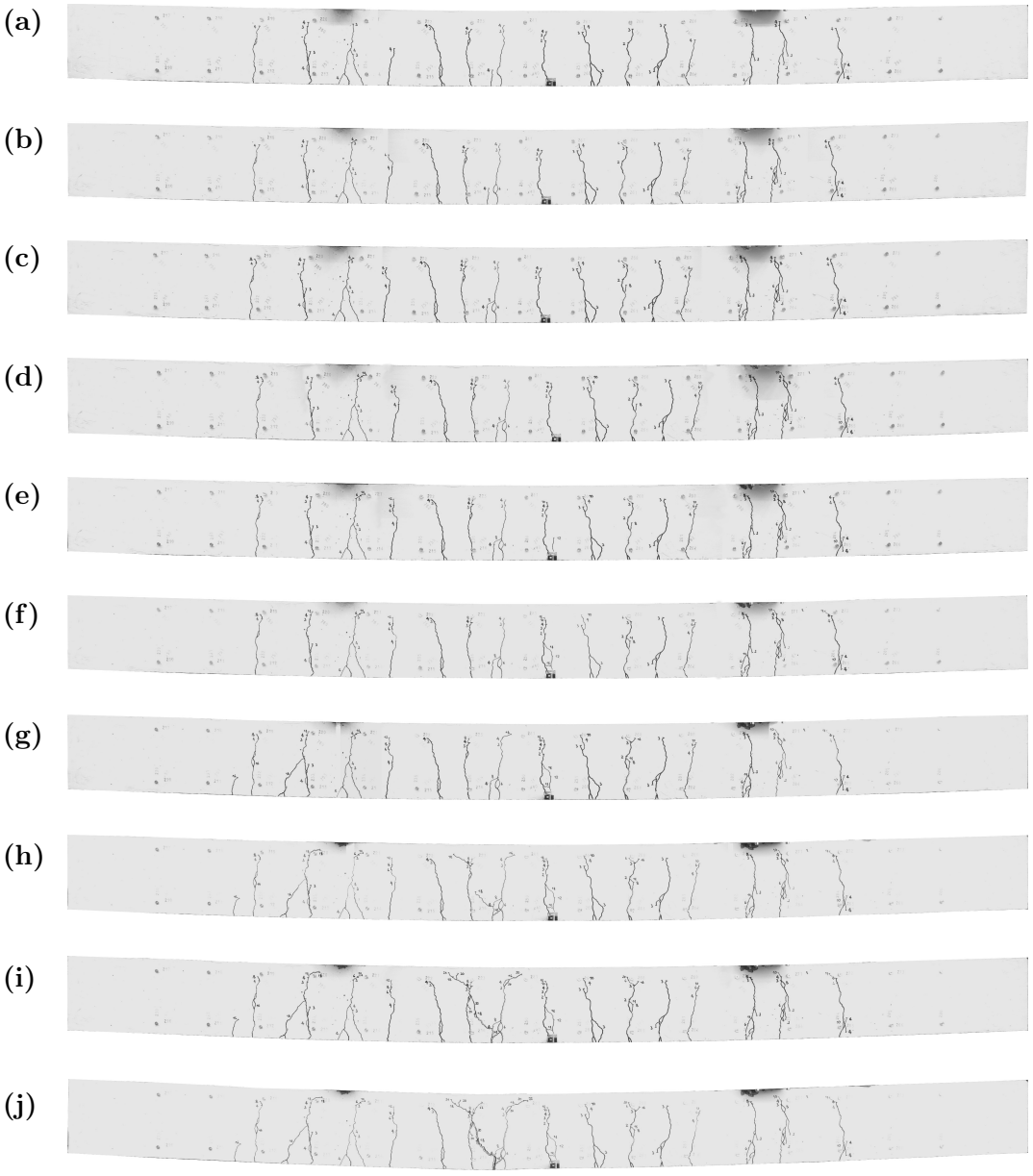


Fig. 6.47 – B2 – Front surface cracking pattern at maximum load in load cycle: (a) 1; (b) 10; (c) 110; (d) 1'110; (e) 11'110; (f) 110'000; (g) 217'555; (h) 219'079; (i) 219'728; (j) 219'741.

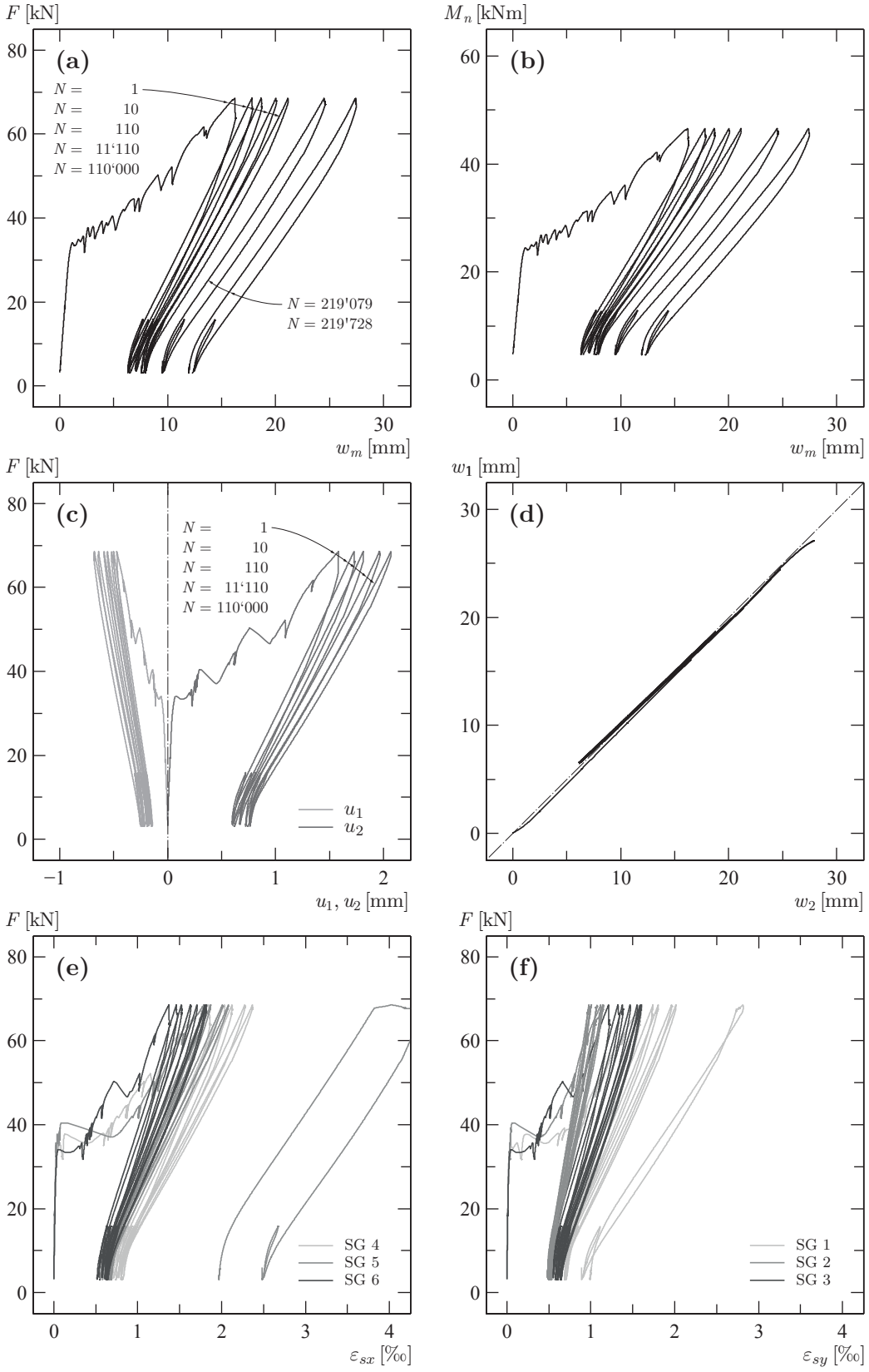


Fig. 6.48 – B2 – Quasi-static loading: (a) and (b) load and moment-deflection; (c) concrete deformations on top surface and soffit; (d) midspan deflection at front and back surface; (e) and (f) reinforcing steel strains in x- and y-direction.

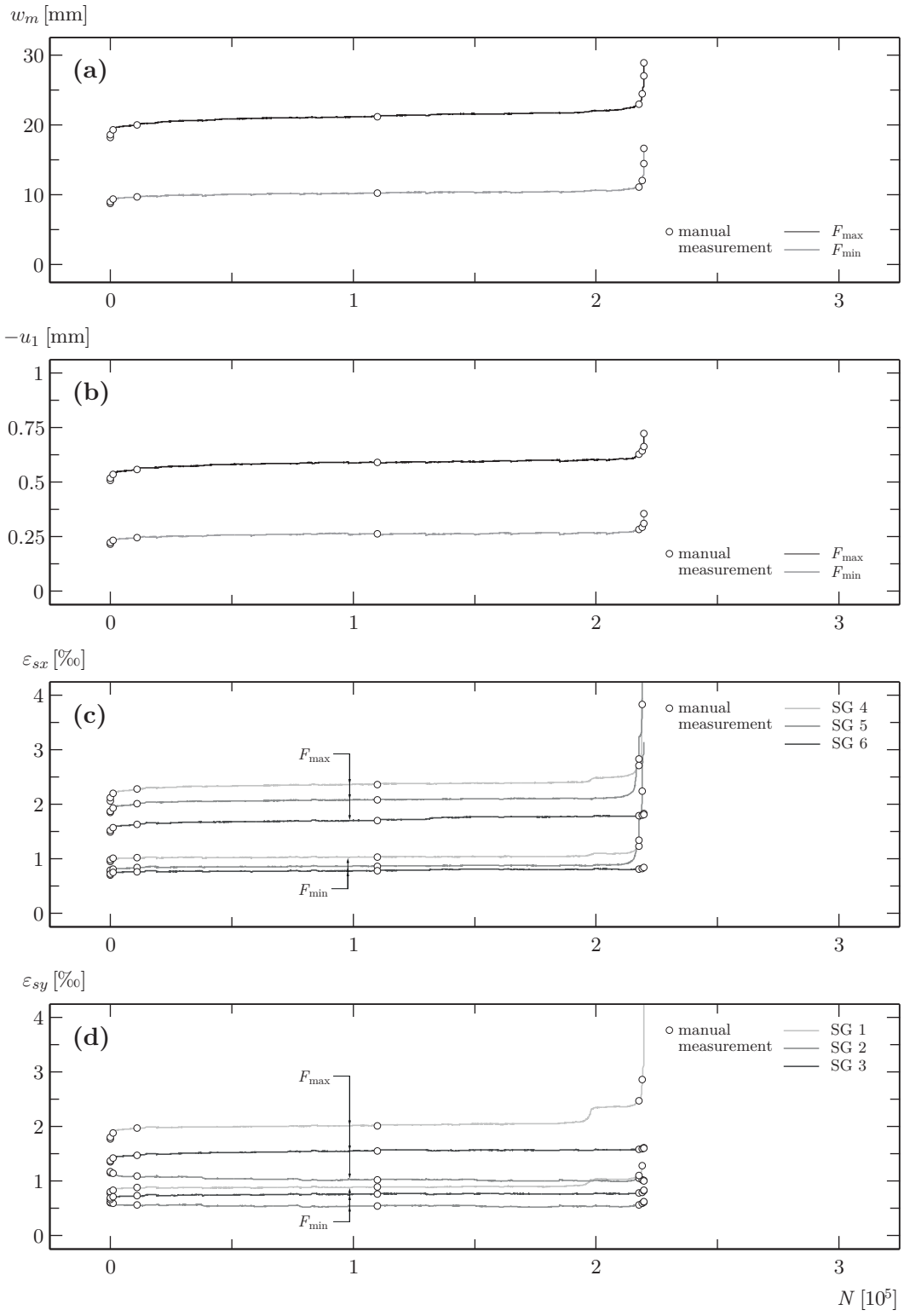


Fig. 6.49 – B2 – Development at maximum and minimum load: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x- and y-direction.

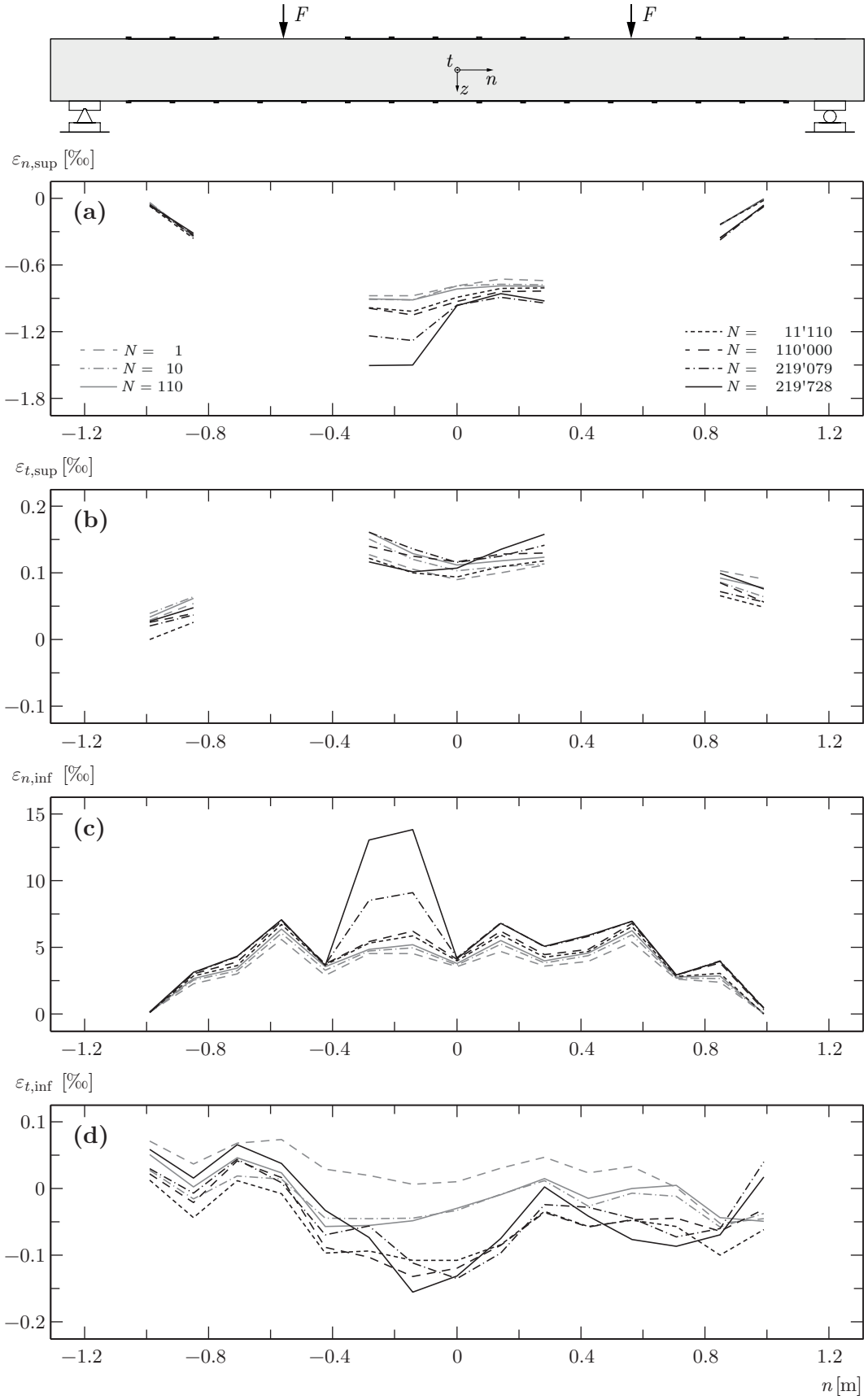


Fig. 6.50 – B2 – Average strains at maximum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

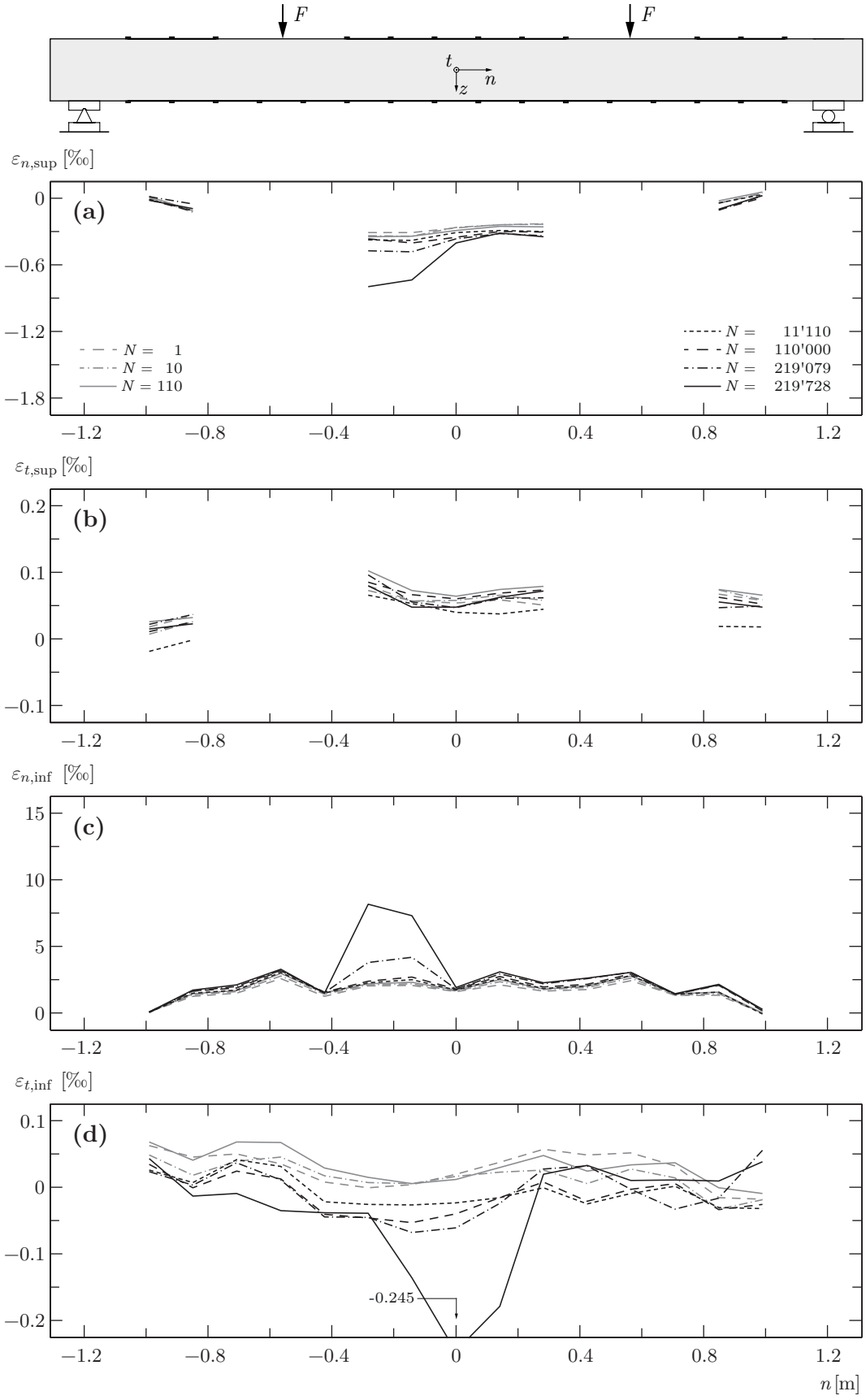


Fig. 6.51 – B2 – Average strains at minimum load: (a) longitudinal top; (b) transverse top; (c) longitudinal bottom; (d) transverse bottom.

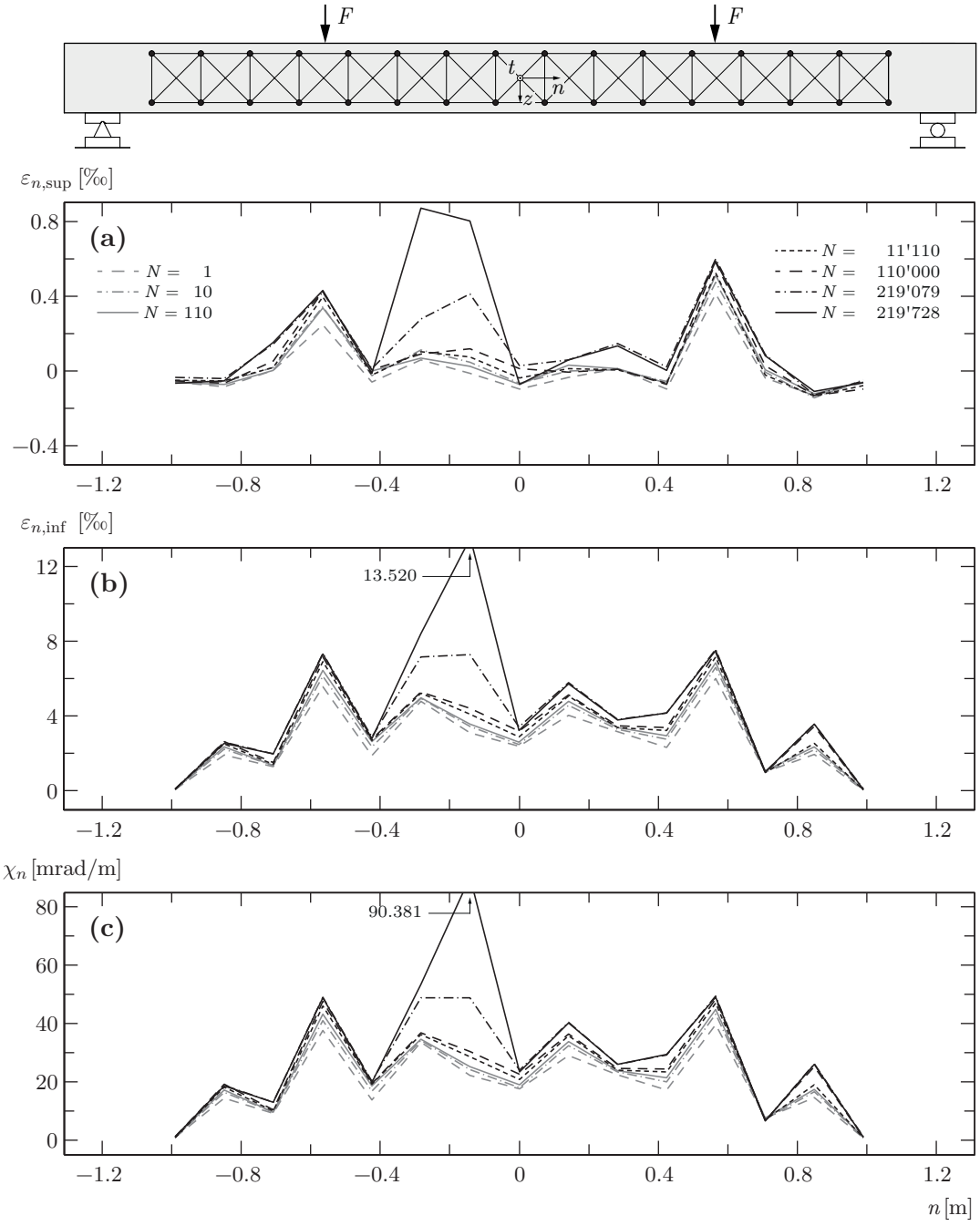


Fig. 6.52 – B2 – Average strains at maximum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

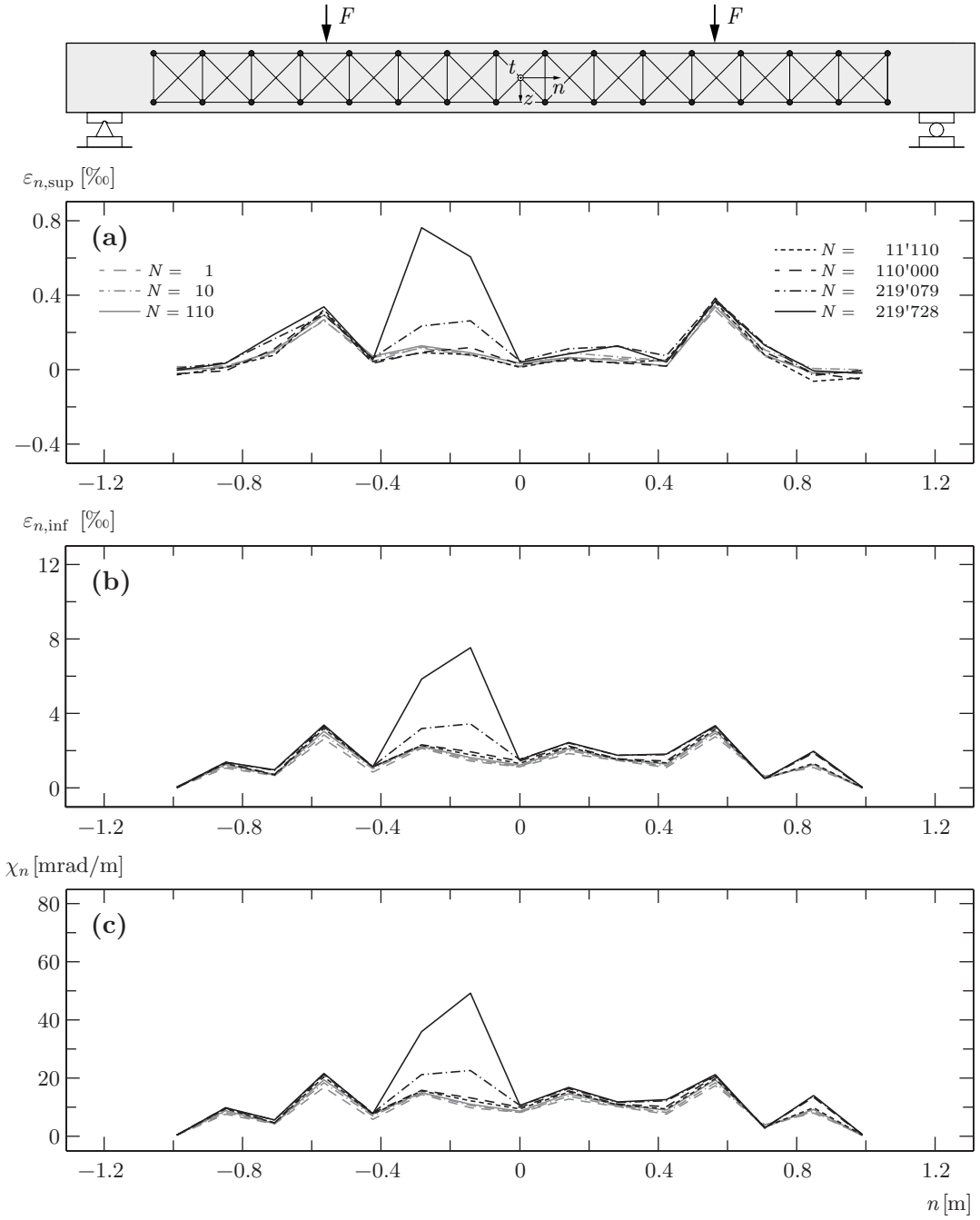


Fig. 6.53 – B2 – Average strains at minimum load: (a) longitudinal top; (b) longitudinal bottom; (c) curvature.

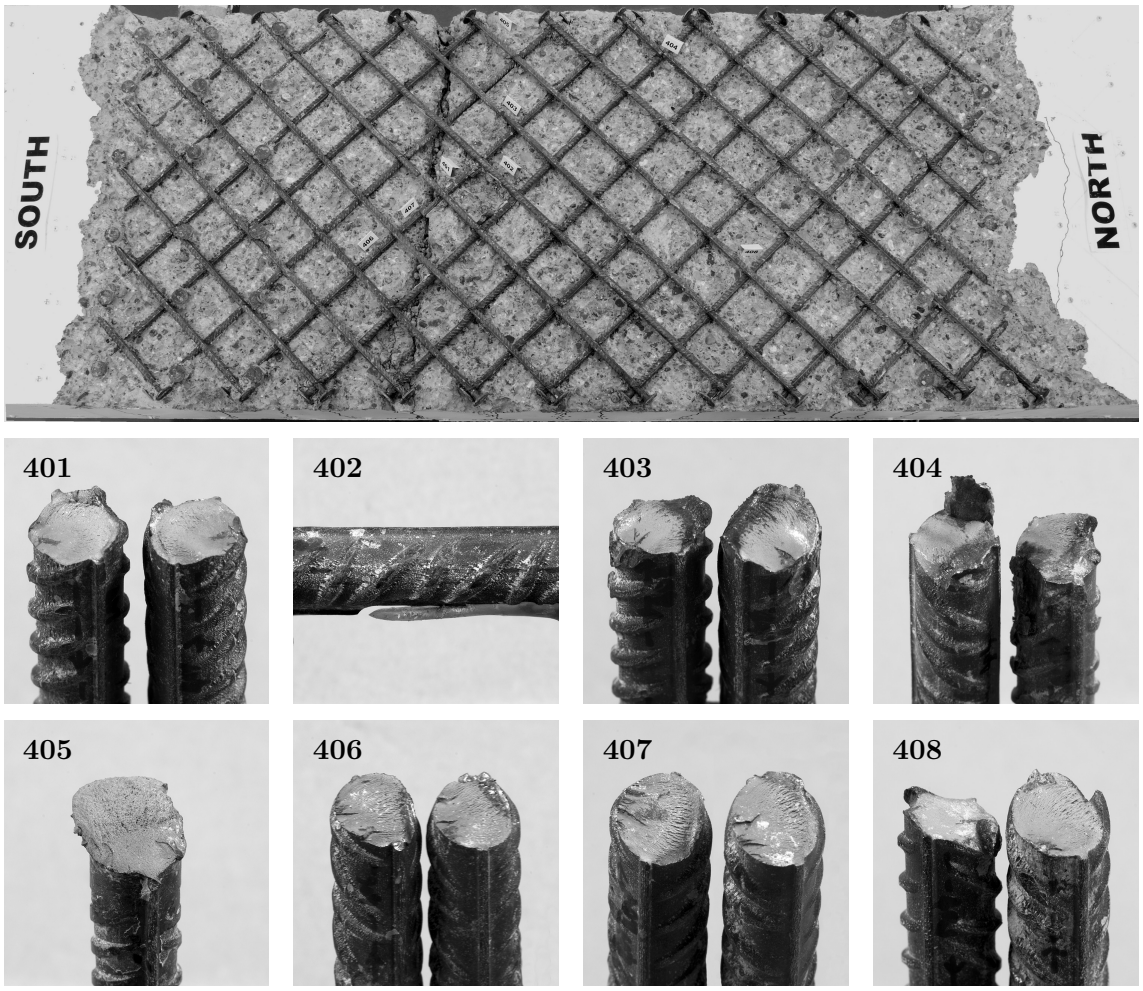


Fig. 6.54 – B2 – Overview and detail views of reinforcing bar fractures.

6.7 Comparison of results

6.7.1 Series A

Fig. 6.55 indicates the critical cross-sections in specimens A1 and A2 at failure. A comparison of continuous measurements in tests A1 and A2 is shown in Fig. 6.56 at maximum and minimum load.

The first reinforcing bar fracture occurred at a greater number of load cycles in test A1 than in A2. The remaining number of load cycles until global failure, however, was similar in both tests. The increase in midspan deflection from the initial loading to the final measurement shortly before failure amounted to 133% and 140% in tests A1 and A2, respectively. At maximum load, the measurements of concrete deformation on the slab top surface varied between the tests, see Fig. 6.56 (b). Larger concrete compressive deformations were measured in test A2 than in A1. The critical cross-section in test A2 developed close to the centre of the test region where the measurement was made, see Fig. 6.55 (b). In test A1, however, the critical crack formed close to the northern load application position, i.e. outside the measurement range, see Fig. 6.55 (a). Reinforcing steel strains were measured at three positions on two bars in x-direction in each specimen, see Figs. 6.56 (c) and (d). The strain measurements at different positions on the same bar varied markedly, particularly at maximum load. Such variations in strain are to be expected due to the relative position of the strain measurement to the concrete cracks. High steel strains are expected close to the cracks. The activation of bond stresses at the interface between the reinforcing bar and the surrounding concrete, however, results in lower steel strains between adjacent cracks. Large, nearly vertical flexural cracks developed in both tests at the critical cross-section after the rupture of a number of reinforcing bars. These cracks continued to extend into the flexural compression zone and became visible in both tests on the top surface of the specimen at failure.

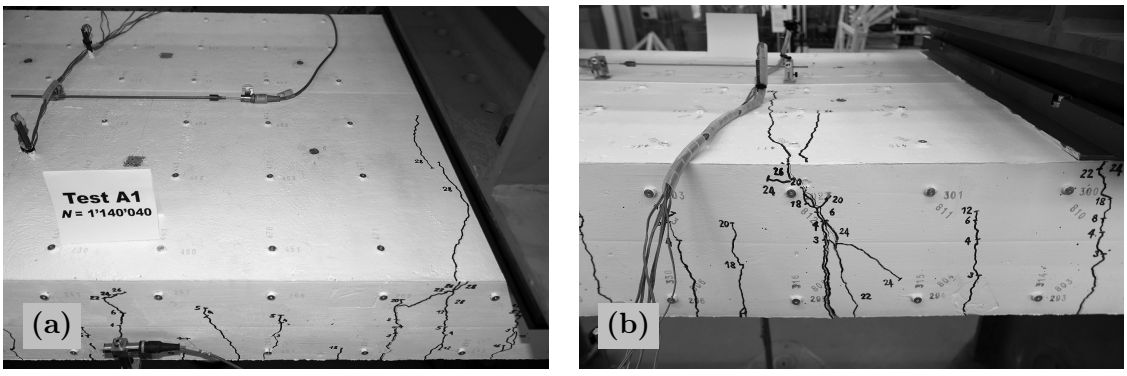


Fig. 6.55 – Critical cross-sections at failure: (a) test A1; (b) test A2.

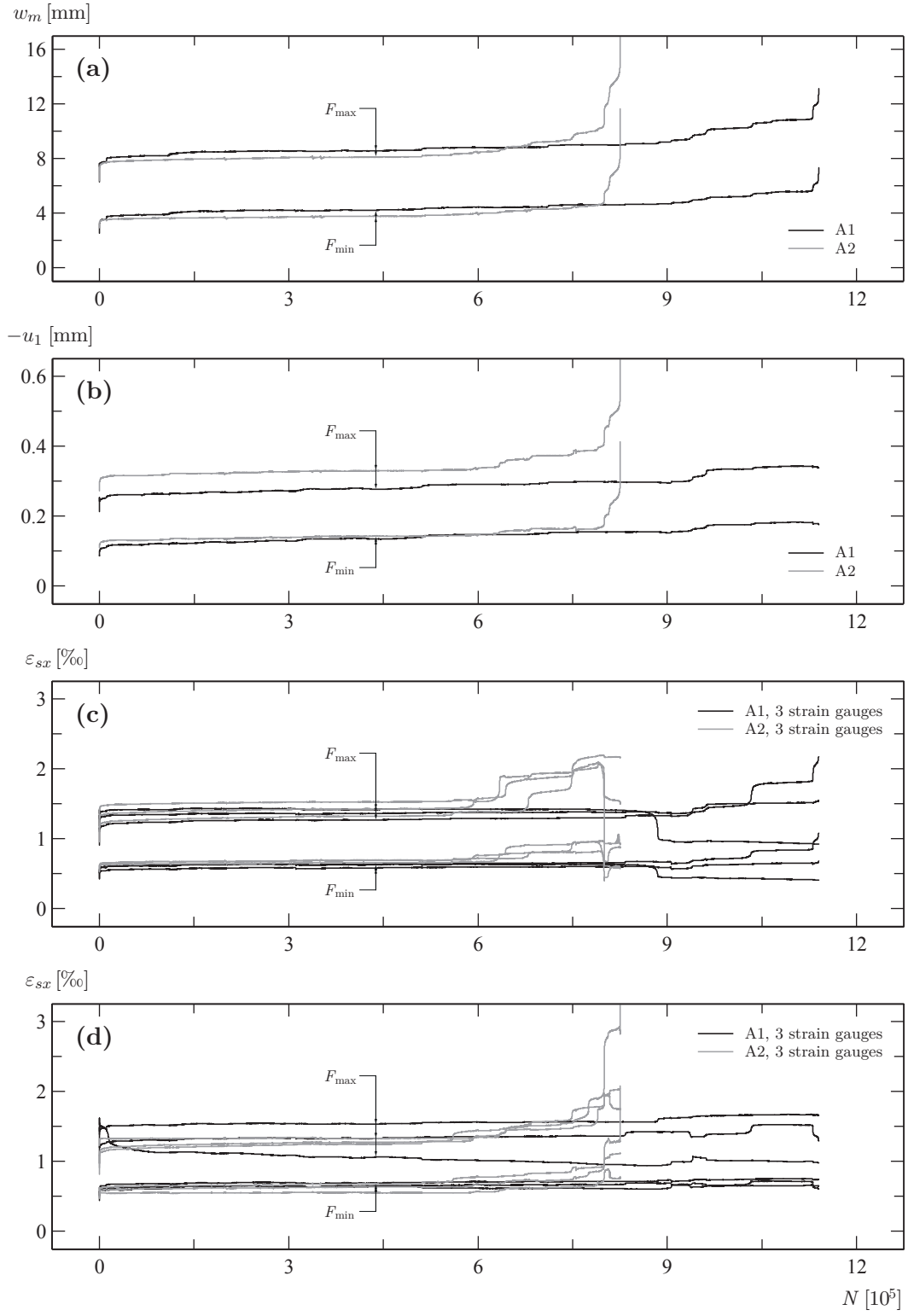


Fig. 6.56 – Comparison of tests A1 and A2: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x-direction.

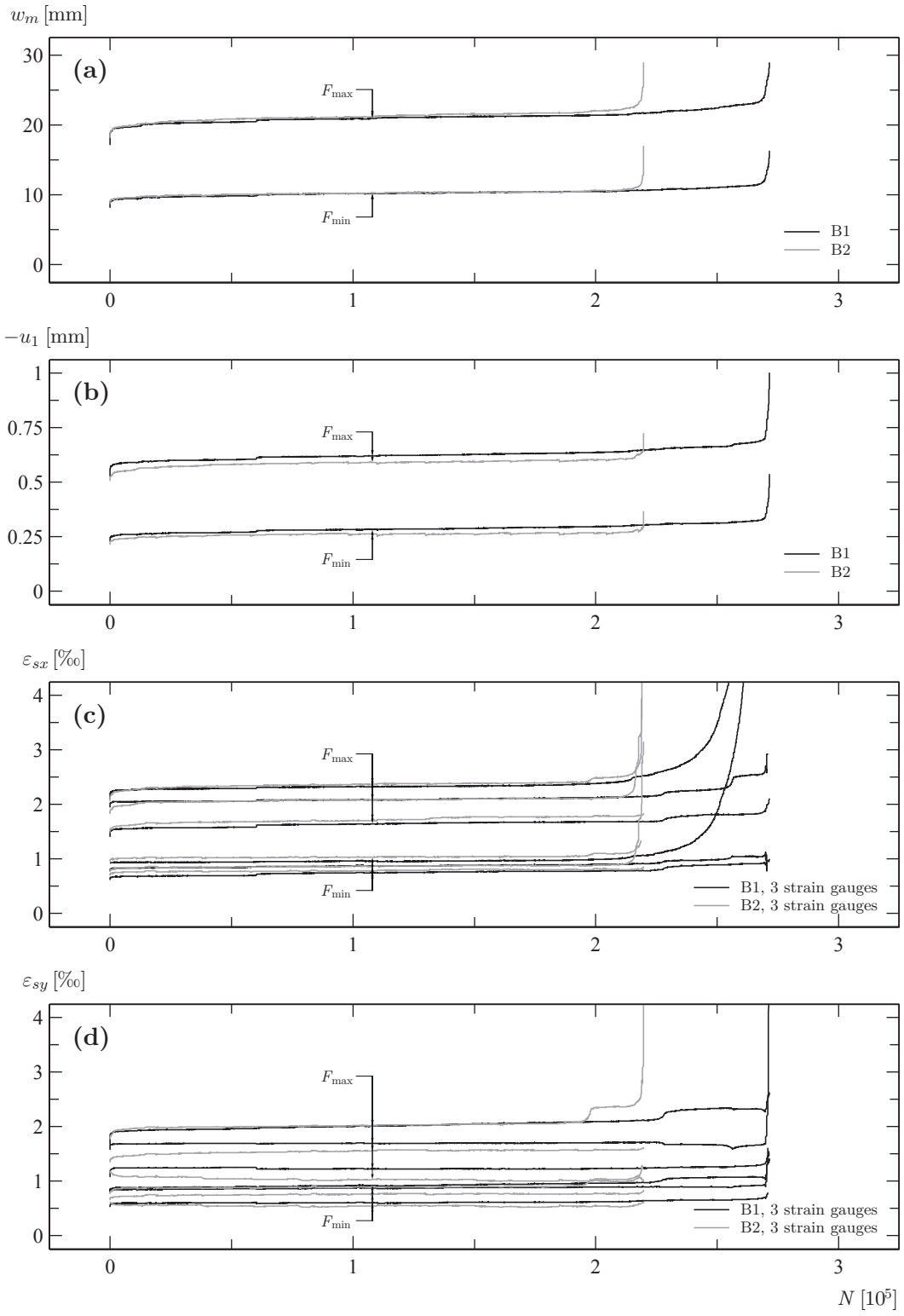


Fig. 6.57 – Comparison of tests B1 and B2: (a) midspan deflection; (b) concrete deformation on top surface; (c) and (d) reinforcing steel strains in x- and y-direction.

6.7.2 Series B

Fig. 6.57 indicates a comparison of the continuous measurements obtained in tests B1 and B2. The critical cross-sections of both tests at failure are shown in Fig. 6.58.

The midspan deflection of both tests in series B correlated closely at maximum and minimum load up until rupture of the first reinforcing bars. The ratio of uncracked to cracked stiffness was significantly lower than in the tests of series A. As a result, the midspan deflection was on average more than 2.5 times greater. The relative increase in midspan deflection between the initial loading and shortly before failure was lower in the tests of series B than in A. This amounted to only 74% and 69% in tests B1 and B2, respectively. The concrete deformation on the slab top surface correlated much more closely in the tests B1 and B2 than observed in the tests of series A. The average values of concrete deformation u_1 were approximately twice as high as the values observed in the tests of series A. This indicates that the top layer of concrete experienced high compressive strains in the principal compression direction. Fig. 6.57 (c) and (d) indicates the development of reinforcing steel strains. The measurements were obtained from strain gauges at three positions on a bar in the x-direction and another three on a bar in the y-direction. The variation in steel strains measured at different points on the same bar was larger than in the tests of series A. Moreover, the reinforcing steel strains on the outer layer of reinforcement in x-direction were higher than those on the inner layer in y-direction. The reinforcing steel strains shortly before failure were of a magnitude indicating the onset of yielding in some bars. In contrast to the transverse cracks observed on the top surfaces of specimens A1 and A2 at failure, cracks with an orientation approximately perpendicular to the direction of principal strain were observed in test B1, see Fig. 6.58 (a). Such cracks indicate high transverse tensile strains within the flexural compression zone of the specimen. A critical crack formed in both tests after the fracture of multiple bars. This crack developed as the combination of a vertical primary flexural crack and a diagonal secondary crack. Signs of spalling of the cover concrete to the reinforcement was observed in test B2 at failure. This indicates high compressive strains in the transverse direction on the slab soffit.

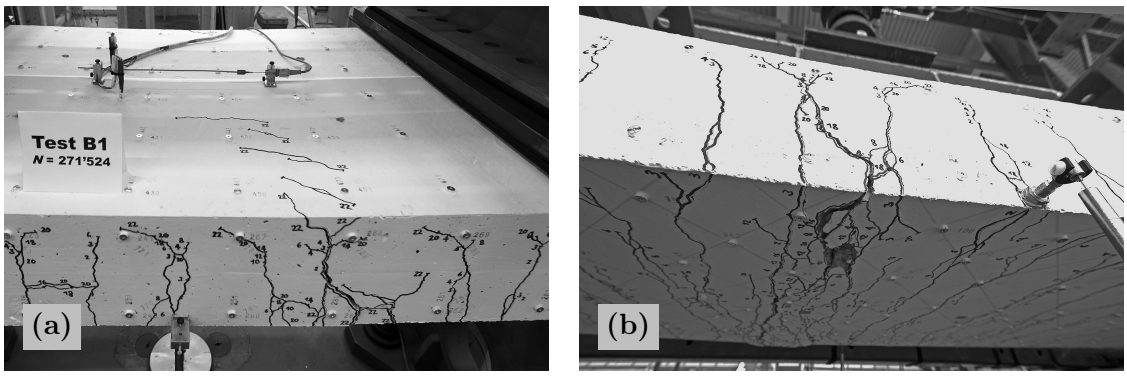


Fig. 6.58 – Critical cross-sections at failure: (a) test B1; (b) test B2.

6.8 Additional considerations

In the present section some additional considerations with regards to the experimental results are discussed. These include an evaluation of the stress range in the reinforcement as calculated from the measured reinforcing steel strains; an examination of the fatigue strength of individual reinforcing bars; and a discussion on the influence of reinforcing bars inclined to the crack direction with regards to their fatigue strength.

6.8.1 Stress range in the reinforcement

Reinforcing steel stresses are calculated from measurements of steel strains considering a bilinear stress-strain relationship. Average values of the static mechanical properties of the reinforcing steel, as summarised in Table 6.4, are considered. Subsequently, the stress range in the reinforcement is calculated as the difference between the maximum and minimum stress in a load cycle according to

$$\Delta\sigma_s = \sigma_{s,\max} - \sigma_{s,\min}. \tag{6.5}$$

Fig. 6.59 indicates the calculated stress ranges for tests A1 and A2 considering strain measurements on two reinforcing bars in x-direction for each test. A marked increase in stress range is observed within the initial roughly 1′000 load cycles. These stabilise thereafter at mean values of approximately $\Delta\sigma_{sx} = 112 \ldots 174\,\mathrm{N/mm^2}$ for the individual measurements. The variation in stress range from measurements at different points on the same bar amount to roughly $40\,\mathrm{N/mm^2}$.

For tests B1 and B2, the stress ranges were calculated from measurements on reinforcing bars in the x- and y-direction, see Fig. 6.60. A significant initial increase is observed within roughly the first 500 load cycles. The stress ranges stabilise thereafter at mean values corresponding to $\Delta\sigma_{sx} = 179 \ldots 273\,\mathrm{N/mm^2}$ and $\Delta\sigma_{sy} = 128 \ldots 226\,\mathrm{N/mm^2}$ for the bars in x- and y-directions, respectively. The variation in stress range at different points on the same bar is on average more than twice as high as observed in the tests of series A.

In both series of tests, sudden, disproportionally large increases in steel stress range are observed upon fracture of individual bars. This indicates a redistribution of internal forces within the remaining bars in the cross-section.

6.8.2 Fatigue strength of the reinforcement

Calculations with the model introduced in Chapter 5 enable a prediction of the expected stress range in the reinforcement due to cyclic loading. For the tests of series A, as will be presented in Section 7.3, this results in the stress range $\Delta\sigma_s = 223\,\mathrm{N/mm^2}$. In the tests of series B, however, the finite width of the slab strip specimens results in a variable reinforcement content within the test region, see discussion in Section 7.1. The effective reinforcement content is lower than in the tests of series A. As a result, higher stresses are expected in the reinforcement. The calculations indicate a stress range of $\Delta\sigma_s = 300.5\,\mathrm{N/mm^2}$ for both the x- and y-directions of reinforcement.

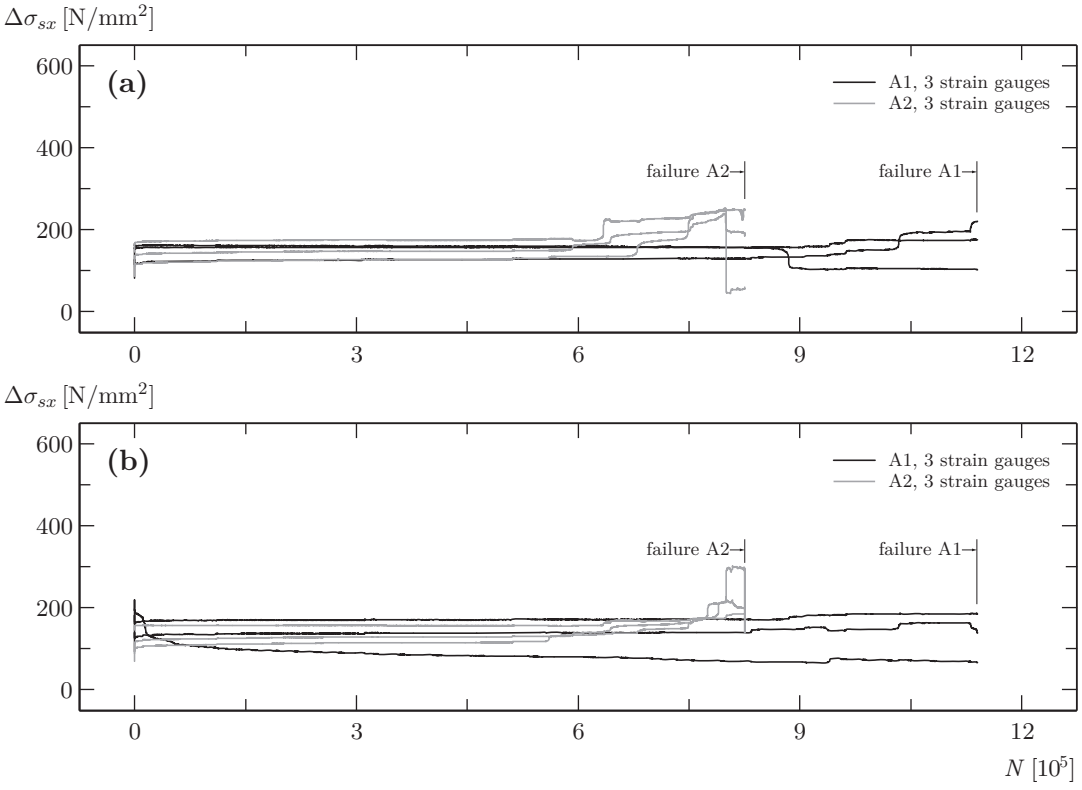


Fig. 6.59 – Reinforcement stress range in tests A1 and A2: (a) and (b) x-direction.

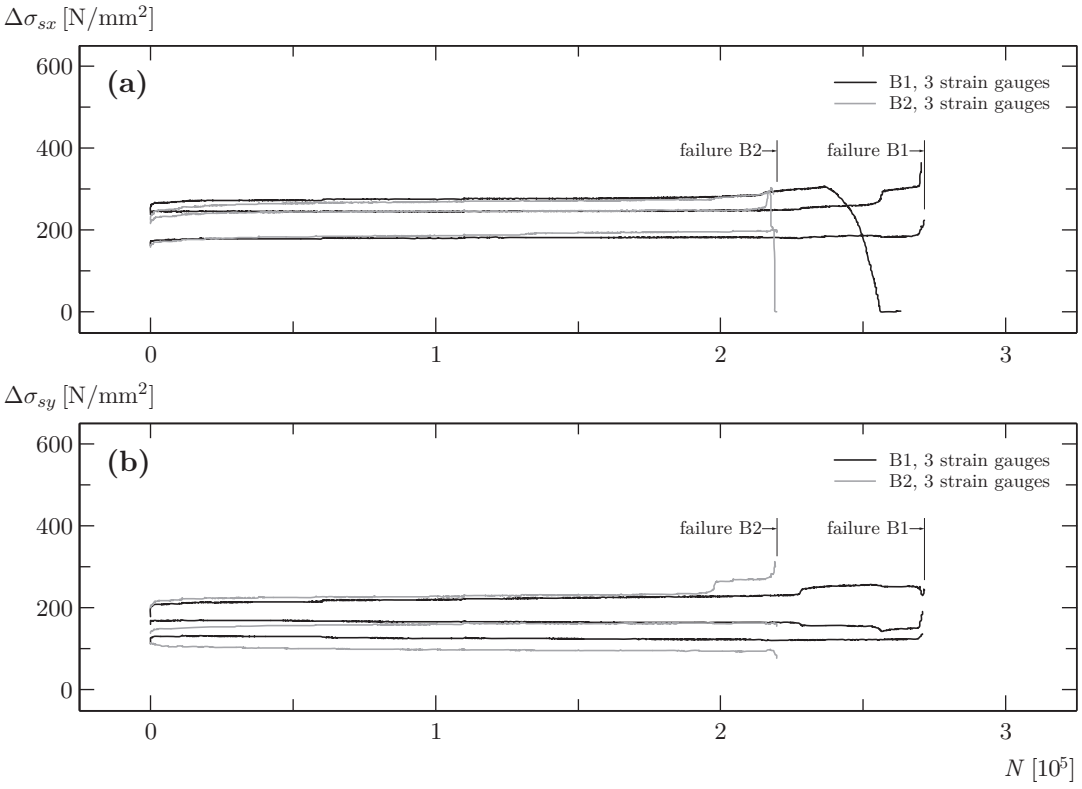


Fig. 6.60 – Reinforcement stress range in tests B1 and B2: (a) and (b) x- and y-direction.

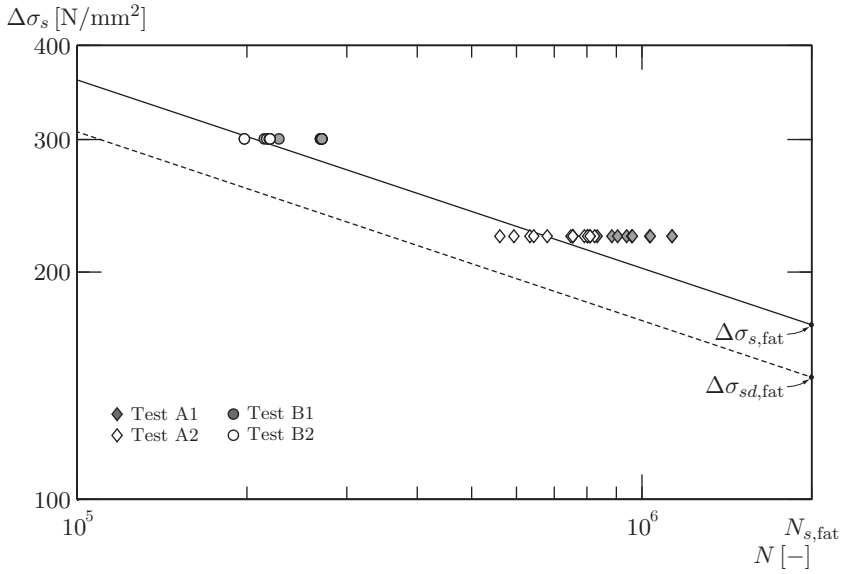


Fig. 6.61 – Fatigue fracture of reinforcing steel bars.

Fig. 6.61 represents the number of load cycles at which fractures were detected in the tests of series A and B at the corresponding expected stress range. In double logarithmic form, this represents an S-N diagram of the individual fractures in all specimens. The fatigue strength curve in the finite-life region is indicated for both the nominal fatigue strength to [84] and the design value to the Swiss standard [87]. The occurrence of reinforcing bar fractures in the tests of series A and B is summarised in Tables 6.9 and 6.10, respectively. Moreover, a characterisation of fracture type is indicated as either "fatigue" or "ductile" on the basis of an inspection of the fracture surface.

Between 8 and 18 fractures were found in each specimen at failure. Multiple fractures were observed in some bars within the test region. Some of these occurred at short intervals; the minimum distance approximately corresponding to the bar spacing. The sequence of occurrence could partially be derived from the manual and quasi-continuous measurements. All fractures occurred at a greater number of load cycles than represented by the fatigue strength curve for design. In both series of tests, however, some fractures occurred at less load cycles than indicated by the curve for nominal fatigue strength. The first fractures in tests A1 and A2 occurred at $N = 834'100$ and $560'200$, whereas global failure was defined at $N_u = 1'140'040$ and $825'784$ load cycles, respectively. In other words, the first fatigue fracture occurred at 73% and 68% of the total number of load cycles until failure in tests A1 and A2, respectively. In tests B1 and B2, first fractures were detected at approximately $N = 214'400$ and $197'900$ load cycles; global failure occurred at $N_u = 271'524$ and $219'741$ load cycles. Hence, the first fatigue fracture in tests B1 and B2 occurred at 79% and 90% of the total number of load cycles to global failure, respectively.

Test A1			Test A2		
Fracture No.	N_u [a,b]	Type	Fracture No.	N_u [a,b]	Type
103	(834′100)	Fatigue	207	(560′200)	Fatigue
111	(885′000)	Fatigue	209	(592′900)	Fatigue
110	906′347	Fatigue	215	(634′100)	Fatigue
102	(940′600)	Fatigue	214	(643′800)	Fatigue
106	(958′700)	Fatigue	216	(680′200)	Fatigue
105	961′995	Fatigue	211	(748′600)	Fatigue
108	(1′033′200)	Fatigue	204, 206, 208, 213	755′347	Fatigue
101	1′035′895	Fatigue	205	(790′100)	Fatigue
107, 109	1′131′296	Fatigue	201, 202	800′317	Fatigue
104	1′140′040	Fatigue-ductile	212	(800′500)	Fatigue
			210, 217	809′232	Fatigue
			203	825′514	Fatigue
			218	825′784	Ductile

[a] the load cycle at failure is based on the detection with the magnetic flux leakage method.
[b] values in brackets indicate an estimation from strain measurements.

Tab. 6.9 – Sequence of reinforcing bar fractures for tests A1 and A2.

Test B1			Test B2		
Fracture No.	N_u [a,b]	Type	Fracture No.	N_u [a,b]	Type
303	(214′400)	Fatigue	406, 407	(197′900)	Fatigue
307	(227′900)	Fatigue	402	(216′500)	Fatigue (SG 5)
301, 308, 310, 311	269′807	Fatigue	401, 403	219′079	Fatigue
304, 309, 312, 313	271′345	Fatigue	404, 408	219′741	Fatigue
306	271′345	Fatigue (SG 6)	405	219′741	Fatigue
305	271′524	Fatigue			
302	271′524	Fatigue-ductile			

[a] the load cycle at failure is based on the detection with the magnetic flux leakage method.
[b] values in brackets indicate an estimation from strain measurements.

Tab. 6.10 – Sequence of reinforcing bar fractures for tests B1 and B2.

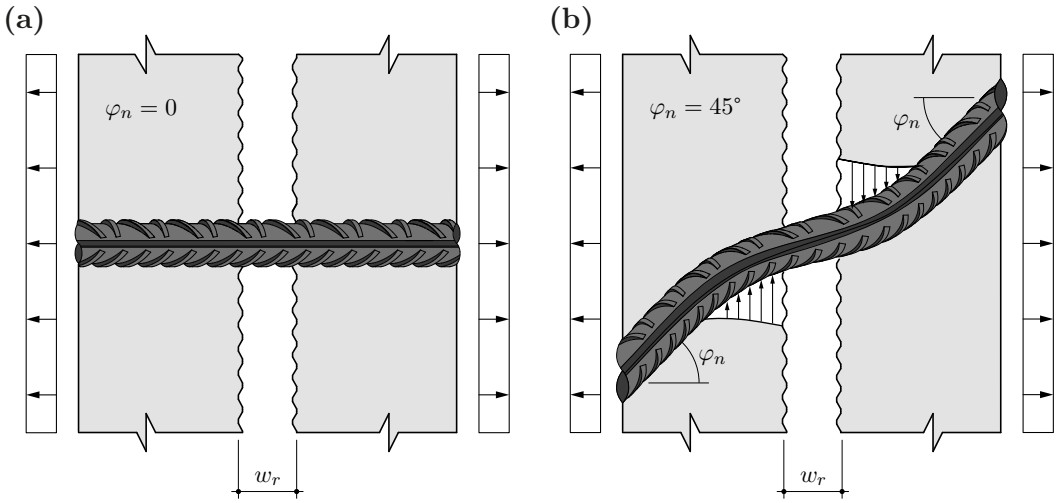


Fig. 6.62 – Reinforcing bar orientation: (a) perpendicular to cracks; (b) inclined with respect to cracks, adapted from [54]. Note: the crack width is exaggerated.

6.8.3 Reinforcing bars inclined to crack direction

Flexural cracks in reinforced concrete slabs initially form perpendicular to the principal moment direction. In cases where the principal moment and reinforcement directions coincide, i.e. uniaxial or biaxial bending, the cracks form perpendicular to the reinforcement direction, see Fig. 6.62 (a). Such bars may be assumed to be subjected to purely uniaxial loading. In slab regions subjected to high torsional moments the direction of principal moment deviates from the reinforcement direction. The crack orientation is no longer perpendicular to the reinforcement direction. Considering the geometric compatibility of crack opening requires such bars to undergo a reorientation in a direction normal to the crack [54], see Fig. 6.62 (b). High loads will result in local crushing of the concrete at the inner surface of the bar; thus, allowing only a partial reorientation. The reinforcing bars in such regions experience, in addition to axial tensile load, a local bending and/or possibly shear action. Such a reorientation of bars at the cracks has little influence on the ultimate resistance as verified experimentally in [54]. Fatigue, however, occurs under service conditions; typically at significantly lower stress levels than the ultimate strength.

The fracture pattern of reinforcing bars adjacent to the critical crack in test B2 suggests some degree of reorientation, see Fig. 6.63. Bar fractures were observed within both the outer and inner layer of reinforcement at failure. The fractures denoted by No. 406 and 407 occurred in the same bar within the inner layer of reinforcement at a distance roughly corresponding to the bar spacing. Furthermore, the fracture surfaces on both sides of the critical crack indicate some misalignment due to the opening of the crack at an inclination to the bar direction, see Fig. 6.63 (b).

All fractures in the tests of series B occurred at significantly fewer load cycles than those in the tests of series A. Furthermore, global failure in tests B1 and B2 occurred on average at only 25% of the number of load cycles sustained in tests A1 and A2. The

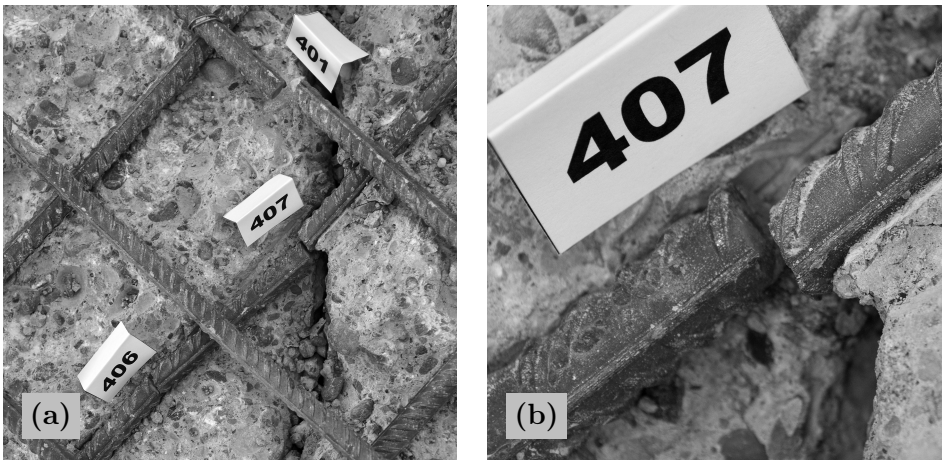


Fig. 6.63 – Exposed bar fractures in test B2: (a) overview at critical cross-section; (b) detail of fracture No. 407.

lower fatigue lives, however, can be attributed to the higher reinforcing steel stress range observed. Considering the S-N diagram of bar fractures shown in Fig. 6.61, the reduced number of load cycles to failure observed in the tests of series B is appropriately reflected in the fatigue strength curve. Hence, the tests indicate that an inclination of reinforcing bars with respect to the direction of principal stress does not detrimentally affect their fatigue strength.

6.9 Conclusions

The present experimental campaign was carried out to investigate the fatigue performance of orthogonally reinforced concrete slabs wherein the principal moment and reinforcement direction do not coincide. On the basis of the experimental results, a number of conclusions can be drawn with regards to the load-carrying behaviour of slabs under cyclic loading. The main conclusions are summarised as follows:

General Fatigue represents a realistic hazard scenario for reinforced concrete slabs with a typical reinforcement content. For predominantly bending action, a relatively gradual but continuous deterioration of the load-deformation behaviour is observed. Measurable changes in structural response occur upon failure of individual bars; especially pronounced were the increases in reinforcing steel strains. Such changes could also be identified in measurements of concrete deformations on the slab top surface and soffit. The occurrence of bar failures was characterised by localisations of concrete strains and curvatures in the critical cross-sections. Furthermore, local increases in crack widths were observed in the proximity of bar fractures as well as the formation of secondary cracks in some cases. Global failure in bending occurs when the residual flexural strength, following rupture of individual reinforcing bars, is lower than the maximal bending demand. Such a mode of failure was observed in all four tests in the present investigation.

Reinforcing bars inclined to crack direction An evaluation of reinforcing steel strain measurements indicates higher stress ranges in the tests of series B than in series A. As a result, tests B1 and B2 failed on average at only 25% of the number of load cycles observed for tests A1 and A2 at failure. Moreover, the number of load cycles corresponding to bar fractures at the expected stress ranges in the tests of both series correlate well with the fatigue strength curve in an S-N diagram representation. The fatigue strength of the tests in series B is therefore similar to the tests in series A. Hence, the tests indicate that the fatigue strength of bars inclined with respect to the direction of principal stress is comparable to that of bars orientated in the principal stress direction.

Redistribution of internal forces Measurements of reinforcing steel strains during cyclic loading indicated sudden, disproportionally large increases, which corresponded to the fracture of individual bars. Moreover, the failure of a bar in one layer of reinforcement in the tests of series B was observed as an increase in steel strains on the other layer activated in bending. This confirms that a redistribution of internal forces occurs under cyclic loading upon rupture of individual bars. Such a redistribution may even extend to other layers of reinforcement activated in bending.

Remaining service life after first fatigue damage Failure of a single reinforcing bar in members reinforced with multiple bars generally does not signify immediate collapse. Slabs are typically reinforced with layers of closely spaced bars and, therefore, relatively tolerant to the failure of single bars. The *remaining fatigue life*, defined as the phase after failure of the first bar until global failure of the structural member [23], amounted on average to 29% and 15% of the total fatigue life for the tests in series A and B, respectively. The progressive deterioration of load-carrying capacity associated with fatigue fracture of individual bars is characterised by sudden, disproportionally large increases in deflection as well as existing crack widths, possibly localised formation of secondary cracks and/or spalling of the cover concrete to the reinforcement. The identification of these characteristics in existing concrete structures could serve as warning signals for possible fatigue damage.

Chapter 7

Model validation with experimental results

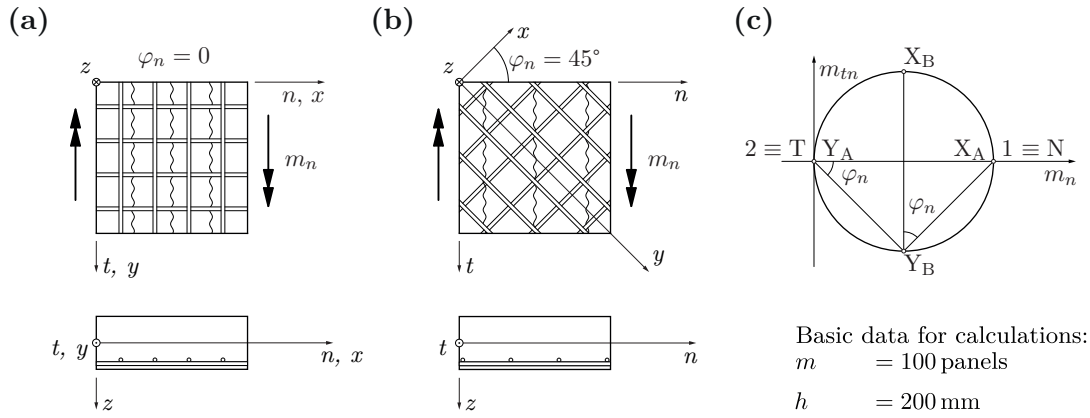
7.1 General

In the present chapter, the results obtained from calculations with the model presented in Chapter 5 will be compared to the experimental results of fatigue tests on orthogonally reinforced concrete slabs presented in Chapter 6.

In the tests, the longitudinal or n-axis of the slab strip specimens corresponded to the direction of applied moment such that $m_n = m_1$. A four-point bending test setup was used to produce a constant moment region in the specimens between the applied loads. In the tests of series A, the x-direction of reinforcement was placed to coincide with the direction of applied moment m_n, resulting in purely uniaxial bending $m_i = m_n\,[1,\, 0,\, 0]$ with respect to the reinforcement directions. The reinforcement content in the tests of series B was the same as in series A, however, the orthogonal layout was rotated by an angle of $-45°$ with respect to the n-axis of the specimen. A combination of bending and torsional moments of $m_i = m_n\,[0.5,\, 0.5,\, 0.5]$ resulted in the test region of the specimens with respect to the reinforcement directions. An overview of the test concept as well as basic parameters underlying the calculations of the tests of both series is shown in Fig. 7.1.

In order to facilitate a comparison between the calculations and experimental results, the finite width of the slab strip specimens of 0.8 m had to be considered. Thus, the applied bending moment per unit length m_n [kNm/m] will be presented as the bending moment M_n [kNm] corresponding to the specimen width. The applied bending moment at maximum and minimum load during constant amplitude cyclic loading amounted to 46.64 and 12.80 kNm, respectively.

The calculations are based on the assumption of a uniformly distributed geometric reinforcement content ρ within the element. In the tests of series A, the reinforcement content ρ_x remains constant in every cross-section perpendicular to the x-axis and all bars are fully anchored within the considered region, see Fig. 7.2 (a). The results from this series of tests could, therefore, be considered independent of the specimen width. The specimens



$m_x = m_n$
$m_y = m_{xy} = 0$

$f_{cc} = 51.61\,\mathrm{N/mm^2}$
$f_{ct} = 3.77\,\mathrm{N/mm^2}$
$\varepsilon_{c0} = 2.427‰$
$E_c = 34.79\,\mathrm{kN/mm^2}$

$m_x = m_y = m_{xy} = 0.5 m_n$

$f_{cc} = 52.01\,\mathrm{N/mm^2}$
$f_{ct} = 3.63\,\mathrm{N/mm^2}$
$\varepsilon_{c0} = 2.329‰$
$E_c = 34.31\,\mathrm{kN/mm^2}$

Basic data for calculations:

m	$= 100$ panels
h	$= 200\,\mathrm{mm}$
c_{nom}	$= 20\,\mathrm{mm}$
$\varnothing_{x,\mathrm{B}}$	$= \varnothing_{y,\mathrm{B}} = 12\,\mathrm{mm}$
$s_{x,\mathrm{B}}$	$= s_{y,\mathrm{B}} = 100\,\mathrm{mm}$
f_{sy}	$= 554.6\,\mathrm{N/mm^2}$
f_{su}	$= 612.5\,\mathrm{N/mm^2}$
E_s	$= 201.3\,\mathrm{kN/mm^2}$
ε_{su}	$= 91.6‰$

Fig. 7.1 – Test concept and basic data for calculations: (a) series A; (b) series B; (c) *Mohr's* circle for bending moments.

of series B, however, exhibit a variable reinforcement content for the layers orientated in the x- and y-directions. Within a central region of the specimen, corresponding to approximately one bar spacing, all reinforcing bars in a strip of 1 m width can be considered to be fully anchored for both reinforcement directions. The reinforcement content ρ can be assumed to be uniform. However, an approximately linear reduction in reinforcement content results along the considered strip; reaching a value of $\rho/2$ at the cross-sections where both the x- and y-coordinate reach values of -0.5 and 0.5 m, see Fig. 7.2 (b). The effective reinforcement content considered in the calculations for the tests of series B amounts to $\rho_{\mathrm{eff}} = 0.78\rho$.

7.2 Comparisons of initial loading response

A comparison of the experimental results obtained from the initial loading in the tests of series A and B and the calculated load-deformation response is indicated in Figs. 7.3 and 7.4, respectively. Both analyses in which tension stiffening effects are considered with $\lambda = 1$ and without tension stiffening are indicated. Markers on the experimental results indicate the end of manual measurements in each load step. In the displacement controlled tests, this corresponded to a slight decrease in the applied load. The measurements presented for comparison include midspan deflection, curvature from concrete deformations on the slab soffit and top surface within the test region and reinforcing steel strains measured in three strain gauges on two bars in each specimen. Furthermore, deformeter measurements on the slab soffit and top surface within the constant moment region of the slab at maximum load are presented as the concrete strains $\varepsilon_{n,\mathrm{inf}}$ and $\varepsilon_{n,\mathrm{sup}}$, respectively.

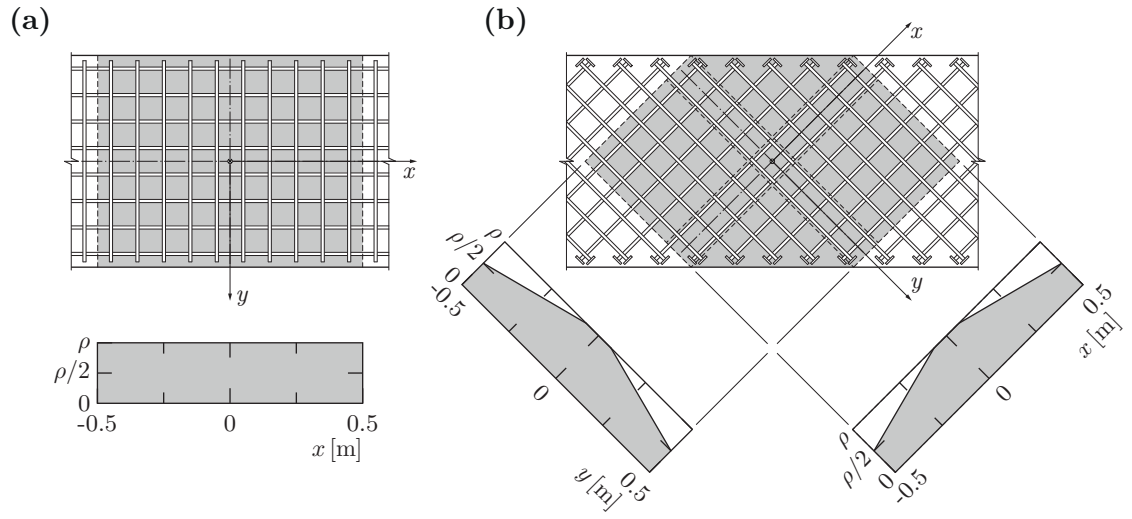


Fig. 7.2 – Effective reinforcement content considered in calculations: (a) series A; (b) series B.

7.2.1 Tests of series A

In general, the calculations show a good correlation with the measurements in tests A1 and A2 made during both the uncracked and the cracked phases. The calculations considering tension stiffening effects indicate a better prediction of the response than those without tension stiffening. Measured concrete strains on the slab soffit $\varepsilon_{n,\mathrm{inf}}$ are partially lower than the calculated strain considering tension stiffening effects, see Fig. 7.3 (b). These strain measurements within the test region, however, also indicate some variation due to the irregular spacing and position of cracks. All deformeter measurements of concrete strain on the slab top surface correlate well with the calculations in both series of tests, see Fig. 7.3 (c). Only the x-direction of reinforcement is activated in the tests of series A. A comparison of the reinforcing steel strains, measured on two bars in each specimen to the calculations is shown in Fig 7.3 (e) and (f). Strain gauge measurements at different positions on the same bar indicate a variation of roughly 7 to 12% at the maximum load. This is primarily due to the position of the strain gauges relative to the cracks. On average, the measured reinforcing steel strains correspond well with the calculations considering tension stiffening effects.

7.2.2 Tests of series B

A deviation between the reinforcement directions and the direction of applied moment in tests B1 and B2 results in the activation of both layers of reinforcement. A significantly lower stiffness in bending is observed than in the tests of series A. This is due to the reduced efficiency of the inclined reinforcement with respect to the direction of applied load. The calculations are able to account for this reduction in stiffness and are generally in good agreement with the measurements. Similar to the tests of series A, the calculations considering tension stiffening effects correlate more closely with the measurements than

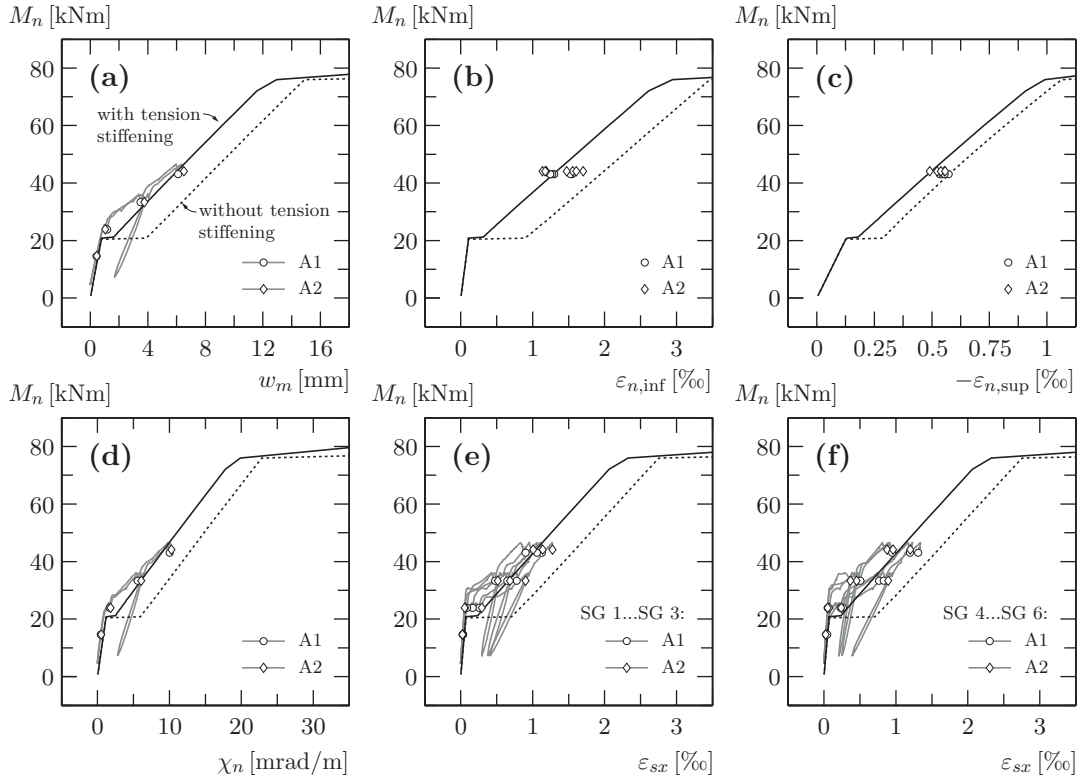


Fig. 7.3 – Comparison to tests of series A during initial loading: (a) midspan deflection; (b) and (c) principal concrete strains on soffit and top surface; (d) curvature; (e) and (f) reinforcing steel strains in x-direction.

the calculations where tension stiffening is neglected. Deformeter measurements on the slab soffit within the test region exhibited variations of 28% and 18% for tests B1 and B2, respectively, see Fig. 7.4 (b). This resulted from crack spacings, which were generally larger than the base length of the measurements. The average of all measurements, i.e. "smeared" over many cracks, lies within the envelope of the two sets of calculations. The calculations of concrete compressive strains on the slab top surface show a better correlation to the deformeter measurements, see Fig. 7.4 (c). The variation in strain gauge measurements at different positions on the same bar was greater than observed in tests A1 and A2. The variation in measurements ranged between 16 and 22% at the maximum load. In general, the calculations predicted lower steel strains than the measurements for the x-direction of reinforcement and a higher steel strain than those measured in the y-direction, see Fig. 7.4 (f). These discrepancies can be attributed to the simplified model formulation considering a single cover element on the slab soffit containing both layers of reinforcement. As a result, the same strains are considered in both layers of reinforcement. In reality, however, the x-direction of reinforcement experienced higher strains than those in the y-direction due to the slightly larger distance from the neutral axis. Considering the inherent variation in the measurements of reinforcing steel strains, the calculations represent a satisfactory prediction.

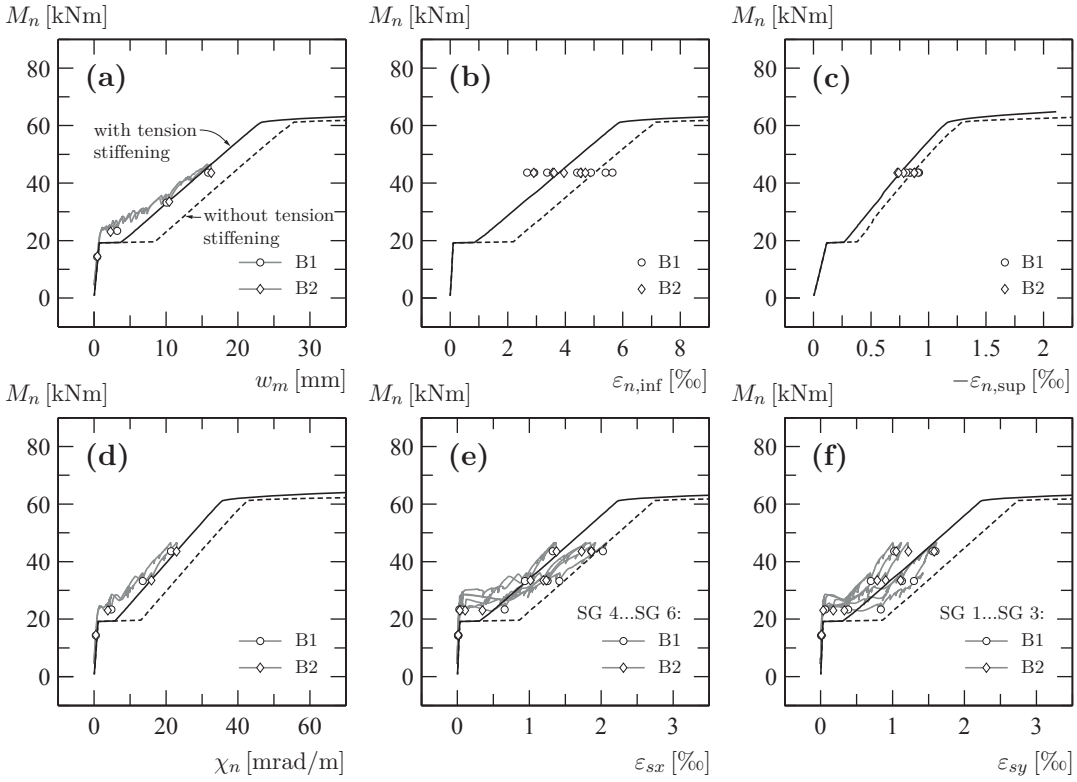


Fig. 7.4 – Comparison to tests of series B during initial loading: (a) midspan deflection; (b) and (c) principal concrete strains on soffit and top surface; (d) curvature; (e) and (f) reinforcing steel strains in x- and y-direction.

7.3 Comparisons of cyclic loading response

Calculations considering cyclic loading are compared to the experimental results in the present section. First, the predicted response in the initial loading and the final load cycle shortly before global failure is indicated on a cross-sectional level. Subsequently, representative calculations of the development of response under cyclic loading is compared to continuous measurements in both series of tests. All calculations are performed considering tension stiffening effects with $\lambda = 1$. Furthermore, the fatigue strengths of individual reinforcing bars are assigned according to the normal distribution of fatigue strengths considering only values between the 5^{th} and 95^{th} percentiles, see Section 5.6.2.

7.3.1 Cross-sectional analyses

Fig. 7.5 illustrates the calculated distribution of principal strains and concrete stresses over the slab thickness as well as reinforcing steel stresses for the tests of series A. A summary of the results at maximum and minimum load level as well as the predicted ranges is provided in Table 7.1. The larger principal strain ε_1 on the slab soffit at maximum load is predicted to increase by more than a factor of two from the initial load cycle to shortly before global failure occurs. An upwards shift of the neutral axis and corresponding increase in the

depth of flexural cracks is predicted. The major changes in response are primarily due a redistribution of internal forces after failure of individual reinforcing bars. Flexural cracks on the slab tensile surface are orientated at $\theta_r = 90°$, i.e. perpendicular to the direction of applied moment. The principal concrete compressive stress profile remains approximately linear during the initial load cycle and shortly before failure. Furthermore, the maximum principal concrete compressive stress on the slab top surface remains well below the peak compressive strength f_c. The reinforcement stress in the x-direction in the initial load cycle results to 340.8 and 117.8 N/mm^2 at maximum and minimum load, respectively. An initial stress range of $\Delta\sigma_{sxr} = 223.0\,\mathrm{N/mm^2}$ is predicted. In the final load cycle before failure, the reinforcing steel stress at maximum load increases to marginally below the yield strength. The calculations predict global failure in the tests of series A due to yielding of the reinforcement after the fracture of individual reinforcing bars.

The cross-sectional analyses for the tests of series B at maximum and minimum load are shown in Fig. 7.6 during the initial load cycle and in the load cycle prior to global failure. Table 7.2 provides a summary of the calculated results. The larger principal tensile strain on the slab soffit is predicted to increase by more than a factor of four from the initial to the final load cycle; reaching a final value of 18.11‰. Cracks on the slab soffit form at $\theta_r = 45°$ with respect to the x-axis, hence perpendicular to the direction of applied moment. Principal concrete compressive stresses result at both the slab soffit and top surface; offset by an angle of 90° as shown in the distribution of θ_r. The larger principal compressive stress on the slab top surface acts parallel to the direction of applied moment. This approaches the peak compressive strength $f_c = 33.1\,\mathrm{N/mm^2}$ in the final load cycle before failure. The principal concrete stress on the slab soffit acts parallel to the direction of the flexural cracks and results from the activation of both layers of reinforcement. The reinforcing steel stresses are predicted to be approximately 24% higher than in the tests of series A. The same reinforcement stresses are predicted in the x- and y-directions as they are considered to act within the same fictitious cover element. Reinforcing steel stresses of $\sigma_{sr} = 423.3\,\mathrm{N/mm^2}$ are predicted at the cracks in the initial load cycle at maximum load. The corresponding stress range in the reinforcement amounts to $\Delta\sigma_s = 300.5\,\mathrm{N/mm^2}$. Failure of individual bars in the calculations results in yielding of the remaining bars. Steel stresses approaching the tensile strength of the reinforcement are predicted in the final load cycle. Thus, global failure of the tests in Series B is predicted as an almost simultaneous crushing of the concrete in the flexural compression zone and rupture of the remaining reinforcing bars within the cross-section.

7.3.2 Tests of series A

A comparison of calculations representing the tests of series A and the experimental results for tests A1 and A2 is shown in Fig. 7.7. Table 7.3 provides a summary of calculations and experimental results for selected load cycles at maximum load. Experimental results are presented as average values of the measurements in tests A1 and A2. The calculations show a good correlation to the measurements at maximum load. The failure of individual

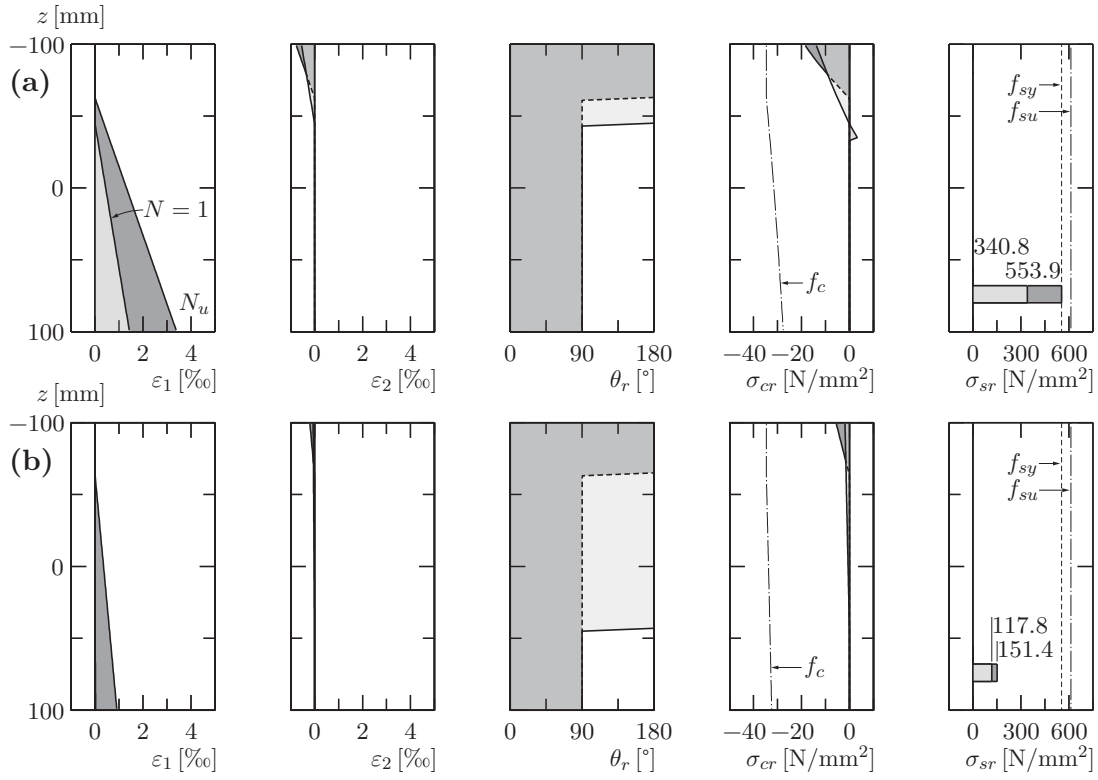


Fig. 7.5 – Cross-sectional analysis for tests of series A: (a) maximum load; (b) minimum load.

N	χ_n	$\varepsilon_{1,\text{inf}}$	$\varepsilon_{1,\text{sup}}$	$\varepsilon_{2,\text{inf}}$	$\varepsilon_{2,\text{sup}}$	$\sigma_{c2r,\text{inf}}$	$\sigma_{c2r,\text{sup}}$	σ_{sxr}	σ_{syr}	Load level
[-]	[mrad/m]	[‰]	[‰]	[‰]	[‰]	[N/mm^2]	[N/mm^2]	[N/mm^2]	[N/mm^2]	
	9.95	1.427	0	0	-0.542	0	-13.8	340.8	0	$F_{\max}$
1	0.46	0.025	0	0	-0.065	0	-1.8	117.8	0	$F_{\min}$
	9.49	1.402	0	0	-0.477	0	-11.9	223.0	0	Δ
	20.95	3.382	0	0	-0.765	0	-18.4	553.9	0	$F_{\max}$
N_u	5.57	0.906	0	0	-0.198	0	-5.4	151.4	0	$F_{\min}$
	15.37	2.476	0	0	-0.568	0	-13.0	402.5	0	Δ

Tab. 7.1 – Calculations for tests of series A at maximum and minimum load.

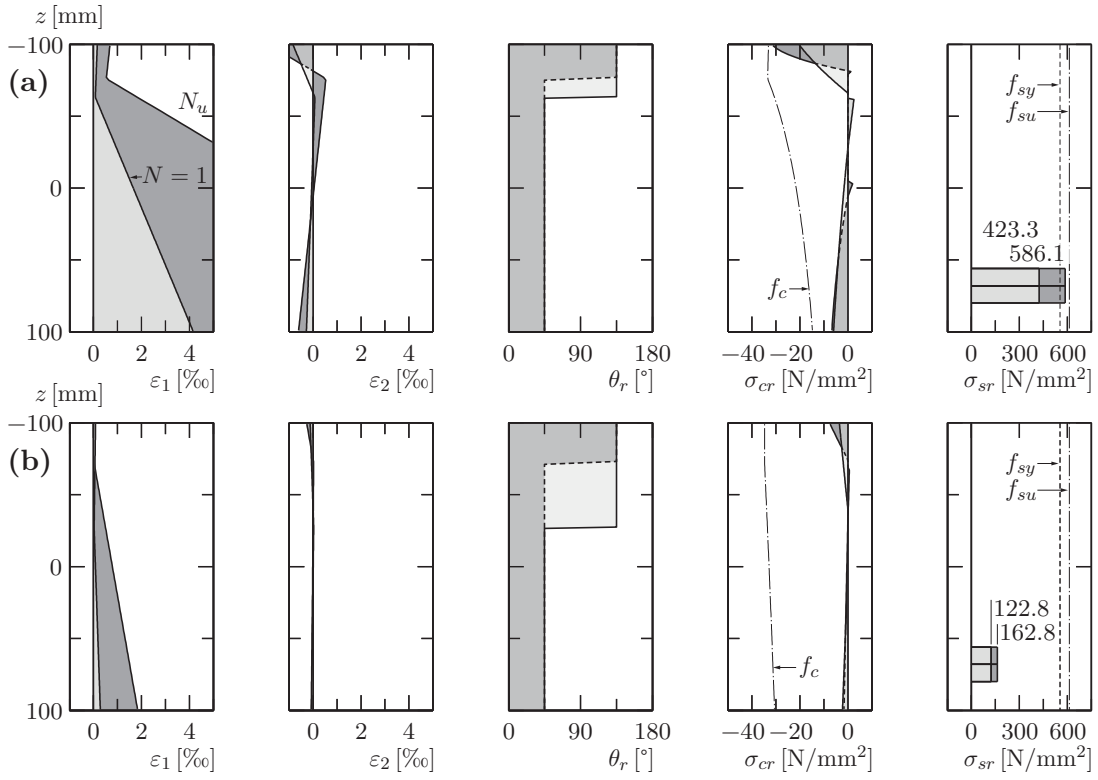


Fig. 7.6 – Cross-sectional analysis for tests of series B: (a) maximum load; (b) minimum load.

N [-]	χ_n [mrad/m]	$\varepsilon_{1,\text{inf}}$ [‰]	$\varepsilon_{1,\text{sup}}$ [‰]	$\varepsilon_{2,\text{inf}}$ [‰]	$\varepsilon_{2,\text{sup}}$ [‰]	$\sigma_{c2r,\text{inf}}$ [N/mm^2]	$\sigma_{c2r,\text{sup}}$ [N/mm^2]	σ_{sxr} [N/mm^2]	σ_{syr} [N/mm^2]	Load level
	25.07	4.140	0.156	-0.277	-0.823	-5.9	-20.0	423.3	423.3	$F_{\max}$
1	2.04	0.284	0.084	-0.077	-0.121	-2.2	-3.5	122.8	122.8	$F_{\min}$
	23.02	3.856	0.071	-0.200	-0.702	-3.7	-16.5	300.5	300.5	Δ
	100.43	18.110	0.681	-0.605	-1.771	-6.7	-31.2	586.1	586.1	$F_{\max}$
N_u	10.58	1.833	0.033	-0.063	-0.262	-1.6	-7.4	162.8	162.8	$F_{\min}$
	89.85	16.277	0.649	-0.542	-1.509	-5.0	-23.9	423.3	423.3	Δ

Tab. 7.2 – Calculations for tests of series B at maximum and minimum load.

	Series A						Calculation					
N	w_m	χ_n	$\varepsilon_{n,\text{inf}}$	$\varepsilon_{n,\text{sup}}$	ε_{sx}	ε_{sy}	w_m	χ_n	$\varepsilon_{n,\text{inf}}$	$\varepsilon_{n,\text{sup}}$	ε_{sx}	ε_{sy}
[-]	[mm]	[mrad/m]	[‰]	[‰]	[‰]	[‰]	[mm]	[mrad/m]	[‰]	[‰]	[‰]	[‰]
1	6.31	10.16	1.375	-0.541	1.083	-	6.48	9.95	1.427	-0.542	1.119	0
10	6.79	10.98	1.496	-0.554	1.180	-	6.91	10.61	1.544	-0.556	1.215	0
100	7.26	11.68	1.585	-0.593	1.266	-	7.34	11.26	1.660	-0.569	1.311	0
1′000	7.53	12.06	1.628	-0.600	1.289	-	7.76	11.90	1.777	-0.581	1.407	0
10′000	7.75	12.42	1.670	-0.613	1.307	-	8.17	12.54	1.888	-0.595	1.499	0
100′000	8.01	11.94	1.719	-0.642	1.332	-	8.58	13.17	1.999	-0.609	1.590	0
f1[a]	9.39	-	2.063	-0.729	1.648	-	8.89	13.63	2.080	-0.619	1.658	0
u[b]	14.91	-	3.133	-0.791	2.029	-	13.61	20.89	3.372	-0.764	2.724	0

[a] measurement after detection of first bar fracture.

[b] final measurement corresponding to global failure.

Tab. 7.3 – Comparison of experimental results and calculations at maximum load for series A.

reinforcing bars is observed as abrupt, disproportionally large changes in the measurements. The response stabilises thereafter at a new level of equilibrium until failure of the next bar within the cross-section. This effect is predicted satisfactorily in the calculations. The response at minimum load, however is invariably underestimated due to the simplified consideration of unloading, see Section 5.8.2. The curvature is calculated from continuous measurements of concrete deformations on the slab soffit and top surface. However, in order to facilitate the detection of reinforcing bar fractures, the sensor on the slab soffit was removed before the first bar fractures were expected. The continuous recording of curvature was not possible thereafter. Concrete strains on the slab soffit and top surface represent average values of deformeter measurements within the test region. The calculated concrete strains show a good correlation with these measurements on both the tensile and compressive surfaces. Reinforcing steel strains at maximum and minimum load, recorded with six strain gauges in tests A1 and A2, are shown in Fig. 7.7 (b). Some variation in measured steel strains is observed for the x-direction of reinforcement; this is most pronounced at maximum load level. However, the calculated maximum steel strain at the cracks correlates well to highest strain measurements under maximum load. Reinforcing steel strains at minimum load are underestimated on average by 34%. As a result, a higher stress range is predicted for the reinforcement. This would result in a conservative evaluation of fatigue life.

7.3.3 Tests of series B

Fig. 7.8 indicates a comparison of the calculations with experimental results for tests B1 and B2. Table 7.4 summarises the calculations and experimental results for selected load cycles at maximum load. Again, the presented experimental results refer to average values of the measurements in tests B1 and B2. Similar to the comparison for the tests of series A, the

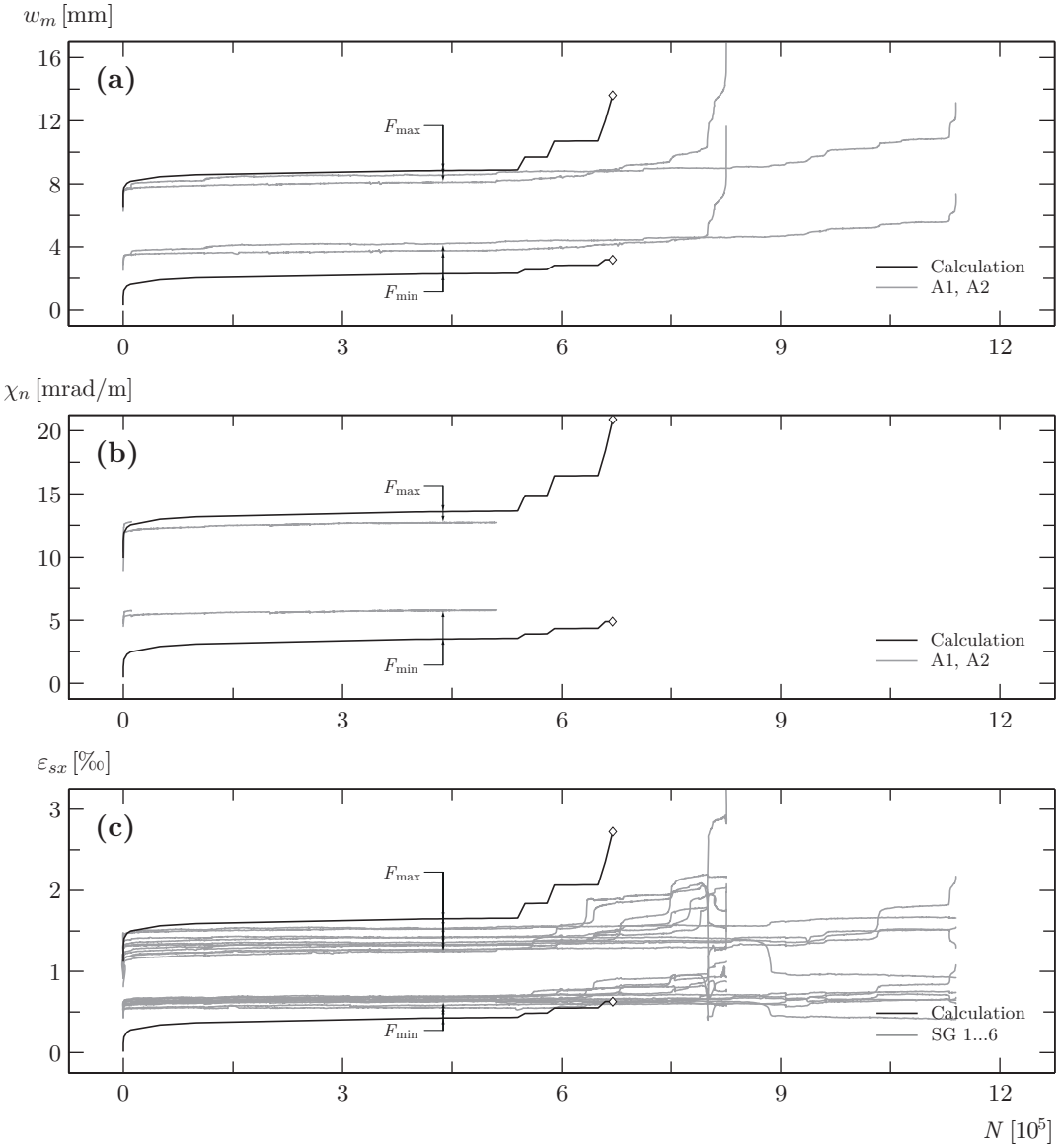


Fig. 7.7 – Comparison to tests of series A at maximum and minimum load: (a) midspan deflection; (b) curvature; (c) reinforcing steel strains in x-direction.

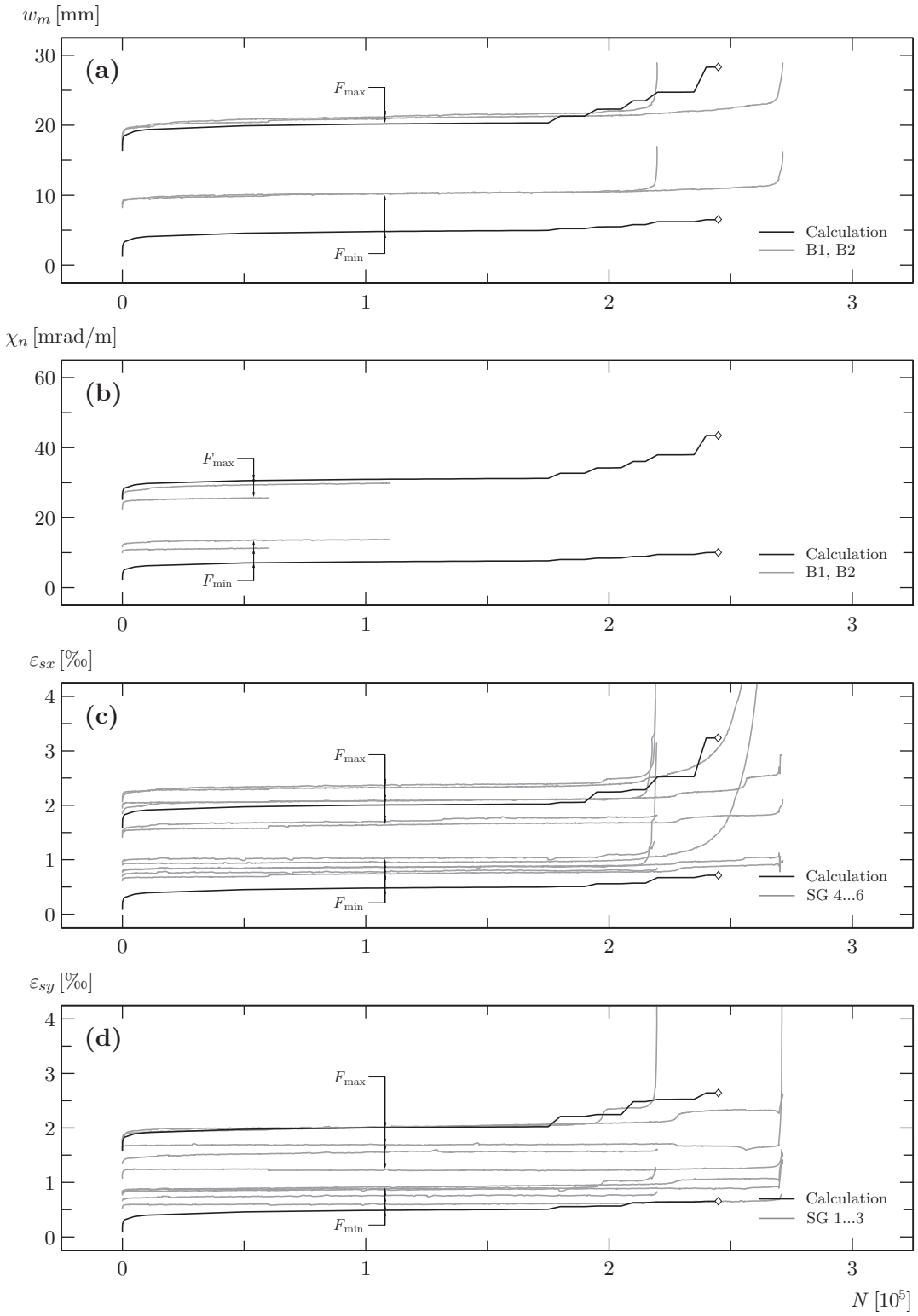


Fig. 7.8 – Comparison to tests of series B at maximum and minimum load: (a) midspan deflection; (b) curvature; (c) and (d) reinforcing steel strains in x- and y-direction.

	Series B						Calculation					
N	w_m	χ_n	$\varepsilon_{n,\text{inf}}$	$\varepsilon_{n,\text{sup}}$	ε_{sx}	ε_{sy}	w_m	χ_n	$\varepsilon_{n,\text{inf}}$	$\varepsilon_{n,\text{sup}}$	ε_{sx}	ε_{sy}
[-]	[mm]	[mrad/m]	[‰]	[‰]	[‰]	[‰]	[mm]	[mrad/m]	[‰]	[‰]	[‰]	[‰]
1	16.07	22.16	4.080	-0.832	1.697	1.338	16.34	25.07	4.140	-0.822	1.577	1.577
10	17.83	24.05	4.436	-0.875	1.805	1.447	17.07	26.19	4.345	-0.841	1.661	1.661
100	18.74	24.96	4.612	-0.890	1.850	1.496	17.81	27.33	4.551	-0.859	1.746	1.746
1′000	19.36	25.81	4.745	-0.925	1.903	1.612	18.54	28.45	4.757	-0.877	1.831	1.831
10′000	19.94	26.70	4.898	-0.944	1.956	1.654	19.38	29.74	4.984	-0.905	1.915	1.915
100′000	21.07	29.98	5.086	-0.985	2.029	1.697	20.16	30.93	5.202	-0.922	2.002	2.002
f1[a]	26.09	-	6.145	-1.107	2.134	1.844	20.28	31.12	5.237	-0.926	2.016	2.016
u[b]	30.02	-	7.405	-1.205	2.460	2.349	24.73	37.94	6.484	-1.029	2.526	2.526

[a] measurement after detection of first bar fracture.
[b] final measurement corresponding to global failure.

Tab. 7.4 – Comparison of experimental results and calculations at maximum load for series B.

calculations generally show a good correlation with the experimental results at maximum load. A similar underestimation of the response at minimum load is observed. The failure of individual bars in the tests of series B resulted in the same abrupt changes in response. Due to the significantly less stiff response in comparison to the tests of series A, these relative changes were less distinct. This characteristic behaviour is also reflected in the calculations. The detection of reinforcing bar fractures during manual measurements again necessitated the removal of the sensor on the slab soffit recording concrete deformations. As a result, the curvature is only indicated up to the load cycle in which the first bar fractures were suspected, see Fig. 7.8 (b). Reinforcing steel strains at maximum load exhibit a variation of roughly 20% between measurements at different positions on the same bar. Fig. 7.8 (c) and (d) indicates that the calculations provide a somewhat lower prediction of steel strain in the x-direction (outer layer) at maximum load, whereas the correlation with strains in the y-direction (inner layer) is much closer. Reinforcement strains in the outer layer are expected to be higher than in the inner layer due to the larger distance from the neutral axis. Such a differentiation is not considered in the model formulation. The calculations consistently underestimate the measured reinforcing steel strains in both the x- and y-directions of reinforcement at minimum load. Similar to the tests of series A, this leads to a conservative prediction of the stress range.

7.4 Comparisons of fatigue life prediction

The Monte Carlo method is used to generate fatigue life predictions with calculations considering stochastically assigned fatigue strengths for the reinforcement. The results of 150 calculations each are considered in the evaluation of both series of tests. Moreover, the mean number of load cycles until global failure as well as the variation between simulations is evaluated and compared to the fatigue lives observed in the tests. Additional calculations

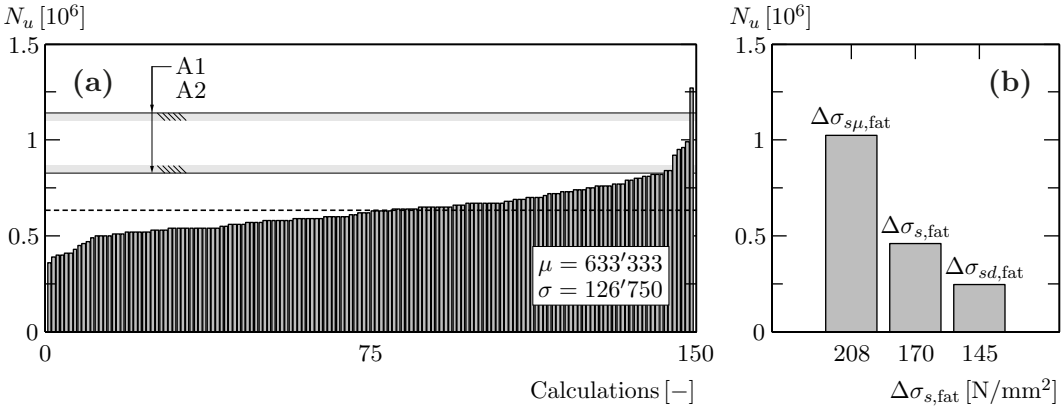


Fig. 7.9 – Fatigue life predictions for tests of series A: (a) 150 calculations with stochastically assigned reinforcing bar fatigue strengths; (b) mean fatigue strength of considered database, nominal fatigue strength to [84] and design value to [87].

are performed for each series of tests considering the following deterministic values of fatigue strength for all reinforcing bars:

- the mean value from the database of considered fatigue tests, $\Delta\sigma_{s\mu,\mathrm{fat}} = 208\,\mathrm{N/mm^2}$
- the nominal fatigue strength $\Delta\sigma_{s,\mathrm{fat}} = 170\,\mathrm{N/mm^2}$ recommended in [84]
- the design value of the fatigue strength $\Delta\sigma_{sd,\mathrm{fat}} = 145\,\mathrm{N/mm^2}$ prescribed in [87].

Global failure of tests A1 and A2 was defined at $N_u = 1'140'040$ and $825'784$, respectively. The corresponding set of 150 generated fatigue life predictions indicates a mean number of load cycles to global failure of $633'333$ with a standard deviation of $126'750$ load cycles between simulations, see Fig. 7.9 (a). The mean number of load cycles to failure represents a conservative prediction for both tests. This result, however is satisfactory considering the typical variation in fatigue test results. Calculations in which the mean fatigue strength of the considered database is assigned to all bars result in a prediction of $N_u = 1'025'000$, see Fig. 7.9 (b). This value correlates well with the highest fatigue life predictions considering a stochastic distribution of fatigue strength and also with the mean of the tests. Furthermore, calculations considering the nominal fatigue strength for straight, small diameter bars as well as the design value according to the Swiss standard result in predictions of $N_u = 460'000$ and $245'000$, respectively. The former result is in good agreement with the mean predicted fatigue life from the generated set; the latter result corresponds to the lowest predictions of fatigue life from the set.

The mean predicted fatigue life of a set of 150 calculations for the tests of series B amounts to $244'400$ load cycles; the standard deviation is $45'390$ load cycles as shown in Fig. 7.10 (a). This correlates exceptionally well with the number of load cycles to failure of tests B1 and B2 at $N_u = 271'635$ and $219'741$ load cycles, respectively. Using the mean prediction of fatigue life, however, would result in an unsafe evaluation for test B2. The calculations considering the mean value of fatigue strength from the database results in a

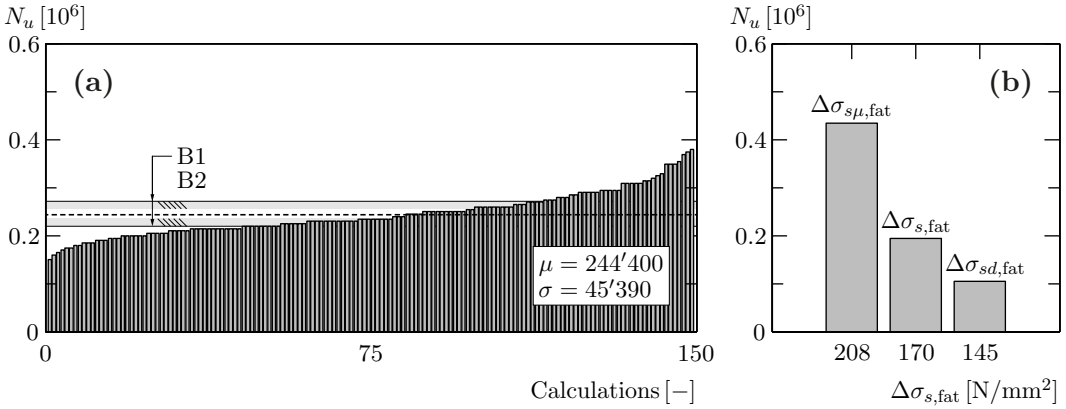


Fig. 7.10 – Fatigue life predictions for tests of series B: (a) 150 calculations with stochastically assigned reinforcing bar fatigue strengths; (b) mean fatigue strength of considered database, nominal fatigue strength to [84] and design value to [87].

prediction of $N_u = 435'000$. This again corresponds to the highest predictions considering the distribution of fatigue strength. However, this prediction significantly overestimates the number of load cycles to failure observed in the tests. The calculations considering the nominal and design values of fatigue strength result in $N_u = 195'000$ and $105'000$, respectively. As observed for the tests of series A, the former corresponds to the mean of the set of generated fatigue life predictions and the latter to the lowest fatigue life predictions.

7.5 Summary of results

The validation of calculations using the model proposed in Chapter 5 with experimental results from two series of four fatigue tests on orthogonally reinforced concrete slabs is presented. The tests of series A represent the case of pure uniaxial bending with the x-direction of reinforcement coinciding with the principal moment direction. A rotation of the reinforcement layout in the tests of series B results in a combination of bending and torsional moment with respect to the reinforcement directions.

The load-deformation response of both series of tests is predicted satisfactorily with the proposed model. The calculations of initial loading generally correlate well with the experimental results. Furthermore, considering tension stiffening effects with $\lambda = 1$ provides a more realistic prediction of the response than the calculations assuming "unbonded" bars. Calculations representing cyclic loading also indicate a good correlation with the experimental results, albeit only at maximum load. At minimum load, the response is invariably underestimated. For this reason, the calculated reinforcing steel stresses at minimum load are lower than expected from the measurements. A conservative (higher) prediction of the expected stress range in the reinforcement results.

The distance between the neutral axis and individual layers of reinforcement is not considered in the model formulation. As a result, a constant strain state is assumed for both

the outer and inner layer of reinforcement in each cover element. In the tests of series B, this results in a slight underestimation of steel strain for the x-direction of reinforcement and an overestimation for the y-direction. However, due to the large degree of variation in measured reinforcing steel strains a conservative calculation of the corresponding stress range might be sufficiently accurate.

Finally, fatigue life predictions for both series of tests are generated with the Monte Carlo method. The mean number of load cycles to failure of 150 calculations for series A represents a conservative fatigue life prediction compared to test results. In the evaluation of the tests of series B, the mean value from all 150 calculations corresponded surprisingly well to the average number of load cycles to failure observed in the tests B1 and B2. An unsafe prediction, however, would result for test B2. This indicates, that some degree of safety has to be included for this approach to be used in the evaluation of structures with regards to safety against fatigue.

In summary, the proposed approach provides a realistic prediction of fatigue life for both combinations of bending and torsional moments represented by the series of tests. In addition, the calculations provide an indication of the expected variation in the number of load cycles to failure for realistic values of reinforcement fatigue strength.

Chapter 8

Summary and conclusions

8.1 Summary

This thesis aims at contributing to a better understanding of the fatigue performance of orthogonally reinforced concrete slabs; the influence of a deviation between principal moment and reinforcement direction with regards to the fatigue strength is of particular interest. A new physical-mechanical model is presented that allows (i) a realistic assessment of the load-deformation behaviour of cyclically loaded slabs under various combinations of bending and torsional moments, (ii) the characterisation of changes in the response upon rupture of individual reinforcing bars, and (iii) a realistic prediction of fatigue life to supplement the current provisions for the design and evaluation of concrete structures.

In the first part of the thesis, Chapters 2 and 3, provide a basic outline of the fatigue phenomenon and examines the material properties of concrete, reinforcing steel and their interaction through bond.

In Chapter 2, the terminology related to work on fatigue is introduced and the characteristics of fractures as well as the interpretation of fatigue test results is described. The theory of damage accumulation is examined with emphasis on the linear rule that is considered in the model formulation. An overview of previous work on fatigue of reinforced concrete concludes the chapter.

In Chapter 3, the characteristics of concrete and reinforcing steel as well as their interaction through bond are examined. In each case, the material behaviour under static loads and the typical stress-strain relationships are presented. The behaviour of the two materials and their interaction under cyclic loading is introduced with reference to the deterioration of the load-deformation response and experimentally determined fatigue strengths as described in literature. The Tension Chord Model is introduced, facilitating a theoretical description of the interaction of concrete and reinforcement through bond. Furthermore, the consideration of tension stiffening effects in the analysis of cracked reinforced concrete is presented in the form of a modified stress-strain relationship for the reinforcing steel.

The second part of this thesis, Chapters 4 and 5, relates to the theoretical modelling of the response of reinforced concrete slabs to cyclic loading.

In Chapter 4, a general overview of the static and kinematic relationships governing the load-deformation behaviour of membrane and plate elements is presented. A compression field approach, namely, the Cracked Membrane Model is introduced, which allows the analysis of reinforced concrete membranes in a homogeneous state of plane stress. Subsequently, the analysis of reinforced concrete plates using a layered model formulation is discussed. The system of equations relating bending and torsional moments as well as membrane forces to the element mid-plane strains and curvatures is introduced.

Chapter 5 begins with an overview of the basic assumptions and idealisations underlying to the model formulation as well as a description of the implemented constitutive relationships. The equilibrium conditions in the analysis of reinforced concrete membranes for the uncracked and cracked response are classified into six distinct states; the corresponding constitutive relationships are summarised in each case. The geometry of the element and the notation used in the formulation is introduced. Furthermore, the equilibrium conditions governing the response of reinforced concrete plates under various combinations of bending and torsional moments are described for the uncracked and cracked states. The numerical method for the analysis of orthogonally reinforced concrete slabs under cyclic loading is outlined. This includes the presentation of a general solution algorithm which allows the determination of complete stress and strain states for a given spectrum of loads. Example calculations illustrate the prediction of failure through either crushing of the concrete or yielding of the reinforcement. The calculations of load-deformation response under monotonically increasing loads are validated with experimental results reported in literature. Thereafter, the consideration of cyclic loading is examined with reference to: the accumulation of damage in individual bars; the shortcomings of the model formulation in the prediction of the unloading response; and the influence of the level of stress range in the reinforcement on the fatigue performance. Finally, the sensitivity of the fatigue life prediction to the probabilistically assigned fatigue strengths of the individual bars is examined.

The third part of this thesis, Chapter 6 provides details of an experimental campaign in which the fatigue performance of orthogonally reinforced concrete slabs was investigated. The testing program constituted two series of tests exhibiting differently orientated reinforcement layouts. Two identical slab strip specimens in each of the two series were tested under constant force-amplitude loading until failure. In the tests of series A, the principal moment and reinforcement direction coincided; whereas, a deviation of 45° between the principal moment and reinforcement directions was introduced in the tests of series B. Comparison of the results indicates that the latter tests exhibit a significantly lower stiffness than those of series A. Furthermore, generally higher strains were measured on individual reinforcing bars in the tests of series B; thus, indicating higher steel stresses. The rupture of individual reinforcing bars within the test region of the slabs resulted in measurable changes in the load-deformation response. These were most clearly visible in the reinforcing steel strain measurements. The remaining fatigue life, defined as the phase between failure of the first reinforcing bar and global failure of the specimen, amounted on average

to 29% and 15% for the tests of series A and B, respectively. Global failure in all four tests resulted due to rupture of multiple reinforcing bars within a critical cross-section. Some additional considerations regarding the expected stress range in the reinforcement, the observed fatigue strength of the reinforcement and the influence of reinforcing bars inclined with respect to the crack direction are discussed. Finally, the main conclusions from the experimental work are presented.

The fourth part of this thesis, Chapter 7, relates to the validation of the proposed new model through a comparison of the calculations to own experimental results as described in Chapter 6. The comparison starts with the initial loading response, proceeds to the response under cyclic loading and concludes with a comparison of the maximum number of load cycles observed in the tests to fatigue life calculations. A good agreement between the calculations and experimental results is emphasised and discrepancies are discussed. Finally, a summary of results concludes the chapter.

8.2 Conclusions

8.2.1 Assessment of load-deformation behaviour

A realistic assessment of the load-deformation behaviour of slabs under cyclic loading, i.e. the development of deflections as well as concrete and reinforcing steel strains, would contribute significantly to the evaluation of remaining fatigue life for existing structures.

Generally, the calculations show a good correlation with experimental results regarding the load-deformation behaviour under monotonically increasing load. This applies for the comparison to tests reported in literature, see Section 5.7.3, as well as to measurements made during the initial loading in own tests, see Section 7.2. Herein, various combinations of bending and torsional moments with respect to the reinforcement directions are represented in addition to a wide range of geometric reinforcement contents in isotropic and severely orthotropic layouts. The calculations result in a realistic prediction of the moment-curvature response as well as concrete strains in the principal directions and reinforcing steel strains. The comparison indicates that calculations considering tension stiffening effects with crack spacing coefficient $\lambda = 1$ generally correlate more closely with the experimental results than calculations where tension stiffening is not considered.

Calculations considering cyclic loading similarly show a good correlation with own experimental results at maximum load. The load-deformation behaviour can be predicted for tests in which the principal moment and reinforcement direction coincides and for tests exhibiting a deviation between principal moment and the reinforcement directions.

The calculations at minimum load consistently underestimate the response. Hence, the simplified consideration of loading history through a recording of the extent of cracking sustained in each load cycle is not suitable for the prediction of response in the unloaded state. Rather, as highlighted in Section 5.8.2, path-dependant constitutive relationships are recommended to consider the influence of residual deformations on the unloading path.

8.2.2 Changes in response upon reinforcing bar rupture

The rupture of individual reinforcing bars during cyclic loading was identified in the continuous measurements of both series of tests as abrupt, disproportionally large changes in the response. This was typically followed by a stabilisation of the measurements at a new level of equilibrium until further load repetitions led to failure of the next bar within the cross-section. The magnitude of these changes corresponding to reinforcing bar ruptures is predicted realistically in both the calculations at maximum and at minimum load. The residual load-carrying capacity of the specimens after rupture of the first reinforcing bar was sufficient to resist the maximum applied bending moment; with global failure only occurring after fatigue failure of multiple bars within a critical cross-section. A unique capability of the proposed model is the characterisation of the subsequent development in load-deformation behaviour between fatigue failure of the "weakest" bar and global failure of the slab. This is especially relevant as the large-scale tests indicated that the remaining life phase constitutes a significant proportion of the total fatigue life.

8.2.3 Prediction of fatigue life

Stochastically distributed fatigue strengths assigned to the individual bars in the model lead to different rates of damage accumulation under cyclic loading; and consequently different fatigue lives for each set of calculations. For this reason, the Monte Carlo method is used to generate a sample of fatigue life predictions that allows a statistical evaluation.

It is shown in Section 7.4 that the mean fatigue life for a large number of calculations represents a realistic prediction of the maximum number of load cycles to global failure in comparison to own tests. Furthermore, this approach additionally provides an indication of the measure of variability in the expected fatigue life. The mean predicted fatigue life partially results in an unsafe evaluation for the tests of series B. Hence, some margin of safety is recommended in the interpretation of the generated fatigue life predictions such as could be evaluated in terms of confidence intervals.

8.2.4 Reinforcing bars inclined to crack direction

The higher reinforcing steel stress range in the tests of series B resulted in the significant reduction in the number of load cycles to failure compared to the tests of series A, see discussion in Section 6.8.3. This observation is most clearly illustrated on the S-N diagram of detected bar fractures shown in Fig. 6.61. Herein, the general trend of the bar fractures in both series of tests at their respective stress ranges is appropriately represented by the fatigue strength curve. The fatigue strength of the bars in the tests of series B, in which these were inclined to the flexural cracks at roughly 45°, are therefore similar to the fatigue strengths observed in the tests of series A.

The tests indicate that the fatigue strength of reinforcing bars inclined with respect to the direction of principal stress (and the direction of cracks) are comparable to that of bars perpendicular to the cracks.

8.3 Recommendations for future research

The physical-mechanical model developed in this work allows for a realistic assessment of the fatigue performance of orthogonally reinforced concrete slabs. This approach shows promise in the calculation of load-deformation behaviour under cyclic loading as well as in the prediction of fatigue life. Subsequently, several recommendations for future work have been compiled which may proceed along the following lines:

The proposed model is based on simple, consistent mechanical principles, thus, allowing for a straightforward extension of the formulation. The assessment of slabs containing skew layers of reinforcement, typically observed in skew-supported bridge decks, would require some modifications on the current formulation; the theoretical principles of which are presented in [88]. A refinement of the element discretisation to consider individual layers of reinforcement to act within separate fictitious cover elements would allow a more realistic treatment of the distance between the centre of each layer and the neutral axis.

Consideration of path-dependant constitutive relationships would greatly improve the response prediction at the minimum load level of cyclic loading; thus, enabling a more realistic calculation of the fatigue-inducing stress range in reinforcement and concrete.

The activation of bond stresses causes a reduction in reinforcing steel strains along the bars at a distance away from the cracks. Fatigue relevant stress ranges in the reinforcement therefore occur at the cracks and the probability of these coinciding with critical surface defects on the bars are reduced; possibly resulting in higher fatigue strengths. However, a deterioration of the bond stress-slip relationship between reinforcing bars and the surrounding concrete has been observed under repeated loading. This results in higher average reinforcing steel strains within the tensile zone and a corresponding decrease in tension stiffening. This complex process is considered in a simplified manner in the present formulation through a reduction of the tension stiffening effect as a logarithmic function of the number of load cycles, see Eq. (5.23). The rate and extent of deterioration of bond under cyclic loading needs to be verified experimentally through strain measurements along the length of reinforcing bars embedded in concrete. The use of fibre optic sensors to obtain continuous strain measurements represents a promising new approach.

The current model represents a single plate or slab element subjected to a constant set of loads. Implementation of this element formulation in a nonlinear finite element software would enable an investigation of the redistribution of moments corresponding to cracking and the rupture of individual reinforcing bars. Moreover, the redistribution of steel and concrete stresses in statically indeterminate structures could be evaluated. For static loads, *Thoma et al.* [95] compared calculations of a similar element formulation implemented in different FE software packages with experimental tests and obtained promising results.

Nomenclature

Abbreviations

ASTM	American Society for Testing and Materials
ASTRA	Bundesamt für Strassen (Swiss Federal Roads Office)
B	Bottom
CMM	Cracked Membrane Model
EOL	End of Loading
EOM	End of Measurement
FEM	Finite Element Method
it	iteration
LC	Load Cycle
LED	Light-Emitting Diode
LS	Load Stage
LVDT	Linear Variable Differential Transformer
SG	Strain Gauge
SIA	Schweizerischer Ingenieur- und Architektenverein
SRSS	Square Root of the Sum of Squares
T	Top
TCM	Tension Chord Model
TS	Tension Stiffening

Upper-case roman letters

A	cross-sectional area
A_{gt}	reinforcing steel strain at peak load

C	regression coefficient for fatigue strength curve
D	total accumulated damage
E_c	modulus of elasticity of concrete
E_{c1}, E_{c1}	modulus of elasticity of concrete in principal directions
E_s	modulus of elasticity of reinforcing steel
E_{sh}	modulus of strain hardening of reinforcing steel
E_{sx}, E_{sy}	modulus of elasticity of reinforcing steel in x- and y-directions
F	force or load
F_{act}	actuator force
$F_{\max}$, $F_{\min}$	maximum and minimum applied load during cyclic loading
F_u	peak load
G	own weight of loading yoke and rocker bearings in test setup
M	bending moment
M_n	bending moment in n-direction
$M_{n,\max}$, $M_{n,\min}$	maximum and minimum bending moment in n-direction in load cycle
N	number of load cycles
N^*	reference number of load cycles
$N_{s,\text{D}}$	number of load cycles corresponding to fatigue endurance limit $\Delta\sigma_{s,\text{D}}$
$N_{s,\text{fat}}$	number of load cycles corresponding to nominal fatigue strength $\Delta\sigma_{s,\text{fat}}$
N_u	maximum number of load cycles to failure
R	stress ratio; number of load cycles in *Basquin* equation; reactions
S	maximum applied stress in *Basquin* equation
T_{cij}	transformation matrix, see Eq. (5.19)
V	shear force

Lower-case roman letters

a_s	cross-sectional area of reinforcement per unit length
c	proportionality constant
c_{1ij}	extensional stiffness matrix
c_{12ij}, c_{21ij}	extension-bending coupling matrix

c_{2ij}	bending stiffness matrix
$c_{\mathrm{I}1ij}$, $c_{\mathrm{II}1ij}$	uncracked and cracked extensional stiffness matrices
$c_{\mathrm{I}12ij}$, $c_{\mathrm{II}12ij}$	uncracked and cracked extension-bending coupling matrices
$c_{\mathrm{I}2ij}$, $c_{\mathrm{II}2ij}$	uncracked and cracked bending stiffness matrices
$c_{\mathrm{I}ij}$, $c_{\mathrm{II}ij}$	uncracked and cracked global stiffness matrices
$c_{\mathrm{cI}ij}$, $c_{\mathrm{cII}ij}$	uncracked and cracked concrete contribution to stiffness matrix
$c_{c1,2}$	concrete stiffness matrix in principal directions
$c_{c\theta}$	rotational stiffness matrix, see Eq. (5.20)
c_{nom}	nominal concrete cover to the reinforcement
c_{sk}	reinforcement contribution to stiffness matrix
d_i	damage increment or cycle ratio
e	eccentricity
f_c	peak concrete compressive strength
$f_{c,\mathrm{cube}}$	concrete cube compressive strength
$f_{c,\mathrm{fat}}$	concrete fatigue strength in compression
$f_{c,\mathrm{stat}}$	concrete static compressive strength
f_{cc}	concrete cylinder compressive strength
f_{ct}	concrete tensile strength
f_{ct0}	concrete tensile strength before onset of cyclic loading
$f_{s,0.2}$	equivalent yield strength of reinforcing steel at 2‰ strain
f_{sy}	yield strength of reinforcing steel
$f_{sy,\mathrm{dyn}}$, $f_{sy,\mathrm{stat}}$	dynamic and static yield strength of reinforcing steel
f_{su}	tensile strength of reinforcing steel
$f_{su,\mathrm{dyn}}$, $f_{su,\mathrm{stat}}$	dynamic and static tensile strength of reinforcing steel
g_{c0}, g_{m0}	own weight of specimens per unit length
h	height
i, j	subscript indices
k	subscript index for layers of reinforcement; exponent in Eq. (3.14)
k_E	coefficient to describe modulus of elasticity of concrete
k_{corr}	calibration coefficient for strain gauge measurements

k_σ	coefficient to describe concrete stress-strain relationship
l_0	free length of uniaxial tension test
l_b	embedment length considered in bond stress calculations
m	number of plane stress concrete layers; slope of fatigue strength curve
m_1, m_2	principal moments per unit length
m_i	vector of bending and torsional moments per unit length
$m_{i,\mathrm{it}}$	vector of calculated bending and torsional moments in an iteration
$m_{i,\mathrm{norm}}$	vector of normalised bending and torsional moments
m_n, m_t	bending moments per unit length in the n- and t-directions
m_{nt}	torsional moment per unit length in relation to the n- and t-directions
m_x, m_y	bending moments per unit length in the x- and y-directions
m_{xy}	torsional moment per unit length in relation to the x- and y-directions
n	generalised coordinate; number of data points in sample; slope
n_1, n_2	principal membrane forces per unit length
n_i	vector of membrane forces per unit length
$n_{i,\mathrm{it}}$	vector of calculated membrane forces in an iteration
$n_{i,\mathrm{norm}}$	vector of normalised membrane forces
n_x, n_y	membrane forces per unit length in the x- and y-directions
n_{xy}	shear force per unit length in relation to the x- and y-directions
r	index for number of reinforcement layers
$r_{n,\mathrm{it}}$, $r_{m,\mathrm{it}}$	residuals on membrane forces and bending moments in an iteration
s_r	crack spacing
s_{r0}	maximum crack spacing, see Eq. (3.17)
s_{rx0}, s_{ry0}	maximum crack spacing in the x- and y-directions
s_x, s_y	spacing of reinforcing bars in the x- and y-directions
t	generalised coordinate; time
t_i	angle dependant vector, see Eq. (5.1)
u_1, u_2	concrete deformations on top surface and soffit in n-direction
u_c	deformation of concrete in the x-direction
u_x, u_y	displacements in the x- and y-directions

v_x, v_y	shear forces per unit length in relation to the x- and y-directions
w	deflection in the z-direction
w_1, w_2	deflection measured at front and back surface of specimen
w_{act}	actuator stroke
w_m	midspan deflection
w_r	crack width
x, y	Cartesian coordinates (horizontal)
z	Cartesian coordinate (vertical)
$z_{c,\text{B}}$, $z_{c,\text{T}}$	z-coordinate to the centre of cover elements
z_j	z-coordinate to the centre of j^{th} layer

Upper-case greek letters

Δz	concrete layer thickness
$\Delta z_{c,\text{B}}$, $\Delta z_{c,\text{T}}$	thickness of bottom and top cover elements
$\Delta\varepsilon_{0j,\text{it}}$	vector of mid-plane strain increments in an iteration
$\Delta\sigma$	stress range
$\Delta\sigma_0$	stress range at onset of cyclic loading
$\Delta\sigma_{s,\text{D}}$	fatigue endurance limit of reinforcing steel
$\Delta\sigma_{s,\text{fat}}$, $\Delta\sigma_{sd,\text{fat}}$	nominal and design fatigue strength of reinforcing steel
$\Delta\sigma_{s,N^*}$	fatigue strength of reinforcing steel at N^*
$\Delta\sigma_{sx}$, $\Delta\sigma_{sy}$	reinforcing steel stress range in the x- and y-directions
$\Delta\sigma_{s\mu,\text{fat}}$	mean value of fatigue strength from database of tests
$\Delta\chi_{j,\text{it}}$	vector of curvature increments in an iteration
$\sum D_{sk}$	sum of accumulated damage in a reinforcement layer

Lower-case greek letters

β	coefficient; angle defining direction of minimum principal strain
γ_{nt}	shear strain in relation to the n- and t-directions
γ_{xy}	shear strain in relation to the x- and y-directions
γ_{xy0}	element mid-plane shear strain in relation to the x- and y-directions

δ	relative displacement
δ_b	bond slip
δ_{b0}	bond slip at onset of cyclic loading
δ_{b1}	bond slip corresponding to onset of yielding of reinforcement
ε_{0j}	vector of strains at element mid-plane $\varepsilon_{0j} = [\varepsilon_{x0}, \varepsilon_{y0}, \gamma_{xy0}]$
ε_1, ε_2, ε_3	principal strains
ε_c	concrete strain
ε_{c0}, ε_{cu}	concrete strain at peak load
ε_{ct}	strain corresponding to concrete tensile strength $\varepsilon_{ct} = f_{ct} \,/\, E_c$
ε_j	vector of strains $\varepsilon_j = [\varepsilon_x, \varepsilon_y, \gamma_{xy}]$
ε_m	average strain
ε_n, ε_t	normal strains in the n- and t-directions
$\varepsilon_{n,\mathrm{sup}}$, $\varepsilon_{n,\mathrm{inf}}$	strains in n-direction on top slab surface and soffit
ε_s	reinforcing steel strain
$\varepsilon_{s(300)}$, $\varepsilon_{s(750)}$	reinforcing steel strain measurements indicating base lengths
ε_{sr}	reinforcing steel strain at cracks
ε_{su}	rupture strain of reinforcing steel
ε_{sv}	reinforcing steel strain at onset of strain-hardening
ε_{sx}, ε_{sy}	reinforcing steel strains in the x- and y-directions
ε_{sy}	yield strain of reinforcing steel
$\varepsilon_{t,\mathrm{sup}}$, $\varepsilon_{t,\mathrm{inf}}$	strains in t-direction on top slab surface and soffit
ε_x, ε_y	normal strains in the x- and y-directions
ε_{x0}, ε_{y0}	element mid-plane strains in the x- and y-directions
ζ	strain ratio for concrete; under-relaxation factor
η_{TS}	coefficient describing the deterioration of bond under cyclic loading
θ_r	crack orientation
κ	scaling factor
$\kappa_{\max}$, $\kappa_{\min}$	scaling factor for maximum and minimum load level in a load cycle
κ_u	scaling factor corresponding to ultimate strength
λ	coefficient regulating crack spacing $[0.5 \leq \lambda \leq 1.0]$

μ	mean of sample
ν_c	*Poisson's* ratio for concrete
ρ	geometric reinforcement content
ρ_c	density of concrete
ρ_{eff}	effective geometric reinforcement content in slab strip specimen
ρ_x, ρ_y	geometric reinforcement content in the x- and y-directions
σ	normal stress; standard deviation of sample
σ_1, σ_2, σ_3	principal stresses
σ_a	stress amplitude
σ_c	concrete stress
σ_{c1}, σ_{c2}, σ_{c3}	principal concrete stresses
σ_{c1r}, σ_{c2r}	principal concrete stresses at the cracks
$\sigma_{c,m}$	mean concrete stress in load cycle
$\sigma_{c,\max}$, $\sigma_{c,\min}$	maximum and minimum concrete compressive stress in load cycle
σ_{ci}	vector of concrete stresses $\sigma_{ci} = [\sigma_{cx}, \sigma_{cy}, \tau_{cxy}]$
σ_{cij}	matrix of concrete stresses in a plane stress layer
$\sigma_{ct,\max}$, $\sigma_{ct,\min}$	maximum and minimum concrete tensile stress in load cycle
σ_{cx}, σ_{cy}	concrete stresses in the x- and y-directions
σ_{cnr}	concrete stress at the cracks in the n-direction
σ_{ctr}	concrete stress at the cracks in the t-direction
σ_m	mean stress
$\sigma_{\max}$, $\sigma_{\min}$	maximum and minimum stress in load cycle
σ_n, σ_t	normal stresses in the n- and t-directions
σ_s	reinforcing steel stress
$\sigma_{s,\max}$, $\sigma_{s,\min}$	maximum and minimum reinforcing steel stress in load cycle
σ_{si}	vector of reinforcing steel stress
σ_{sr}	reinforcing steel stress at cracks
σ_{sxr}, σ_{syr}	reinforcing steel stress at cracks in the x- and y-directions
σ_x, σ_y, σ_z	normal stresses in the x-, y- and z-directions
τ_b	bond stress

$\tau_{b,\max}$, $\tau_{b,\min}$	maximum and minimum bond stress during cyclic loading
τ_{b0}, τ_{b1}	levels of bond stress describing bond stress-slip relationship
τ_{bu}	bond strength
τ_{cntr}	concrete shear stress at cracks in relation to the n- and t-directions
τ_{cxy}	concrete shear stress in relation to the x- and y-directions
τ_{nt}	shear stress in relation to the n- and t-directions
τ_{xy}	shear stress in relation to the x- and y-directions
φ	angle defining the generalised coordinate system
φ_1	angle defining the principal direction
φ_n	angle defining reinforcement direction
χ_j	vector of curvatures $\chi_j = [\chi_x, \chi_y, \chi_{xy}]$
χ_n	curvature in the n-direction
χ_x, χ_y	curvatures in the x- and y-directions
χ_{xy}	curvature in relation to the x- and y-directions
ψ_k	angle defining reinforcement direction for skew layers
ω	circular frequency

Other symbols

1, 2, 3	principal directions
$\varnothing$	diameter
$\varnothing_{\text{eff}}$	effective diameter of reinforcing bar
$\varnothing_s$	diameter of reinforcing bar
$\varnothing_x$, $\varnothing_y$	diameter of reinforcing bar in the x- and y-directions

Appendix A

Details on measurements

A.1 Strain gauges

Strain gauges applied to individual reinforcing bars in each of the four slab strip specimens described in Chapter 6 were calibrated prior to placing the instrumented bars in the formwork. Each bar was loaded under deformation control in a servo-hydraulic testing machine to a tensile force resulting in a nominal stress of roughly $350\,\mathrm{N/mm^2}$. The maximal reinforcement stress corresponded to approximately 60% of the static yield strength. The stress-strain characteristic was determined for each strain gauge and compared to that of an extensometer (base length 300 mm) placed at the centre of the bar. In each case the secant modulus of elasticity was determined between a lower and upper stress level corresponding to 90 and $330\,\mathrm{N/mm^2}$, respectively. The correction factor on the strain measurements of individual strain gauges result from

$$k_{\mathrm{corr}} = \frac{\Delta\varepsilon_{\mathrm{SG}}}{\Delta\varepsilon_{s(300)}} \tag{A.1}$$

where $\Delta\varepsilon_{\mathrm{SG}}$ denotes the difference in strain gauge measurements and $\Delta\varepsilon_{s(300)}$ the corresponding difference measured with an extensometer. Figs. A.1 and A.2 indicate the stress-strain relationships of the uncorrected strain gauges compared to the extensometer measurement and the corresponding correction factors.

A.2 Deformeters

The following demountable deformeters were used in the experimental campaign:

- deformeter with a base length of 141 mm, measurement range ±20 mm;
- deformeter with a base length of 200 mm, measurement range ±10 mm;
- deformeter with adjusted base length of 1530 mm, measurement range ±100 mm.

Table A.1 provides a summary of deformeter readings taken during periods of manual measurement at maximum and minimum load. The grid numbering of measured distances on the specimen top surface, sides and soffit is indicated in Figs. 6.9 and 6.10.

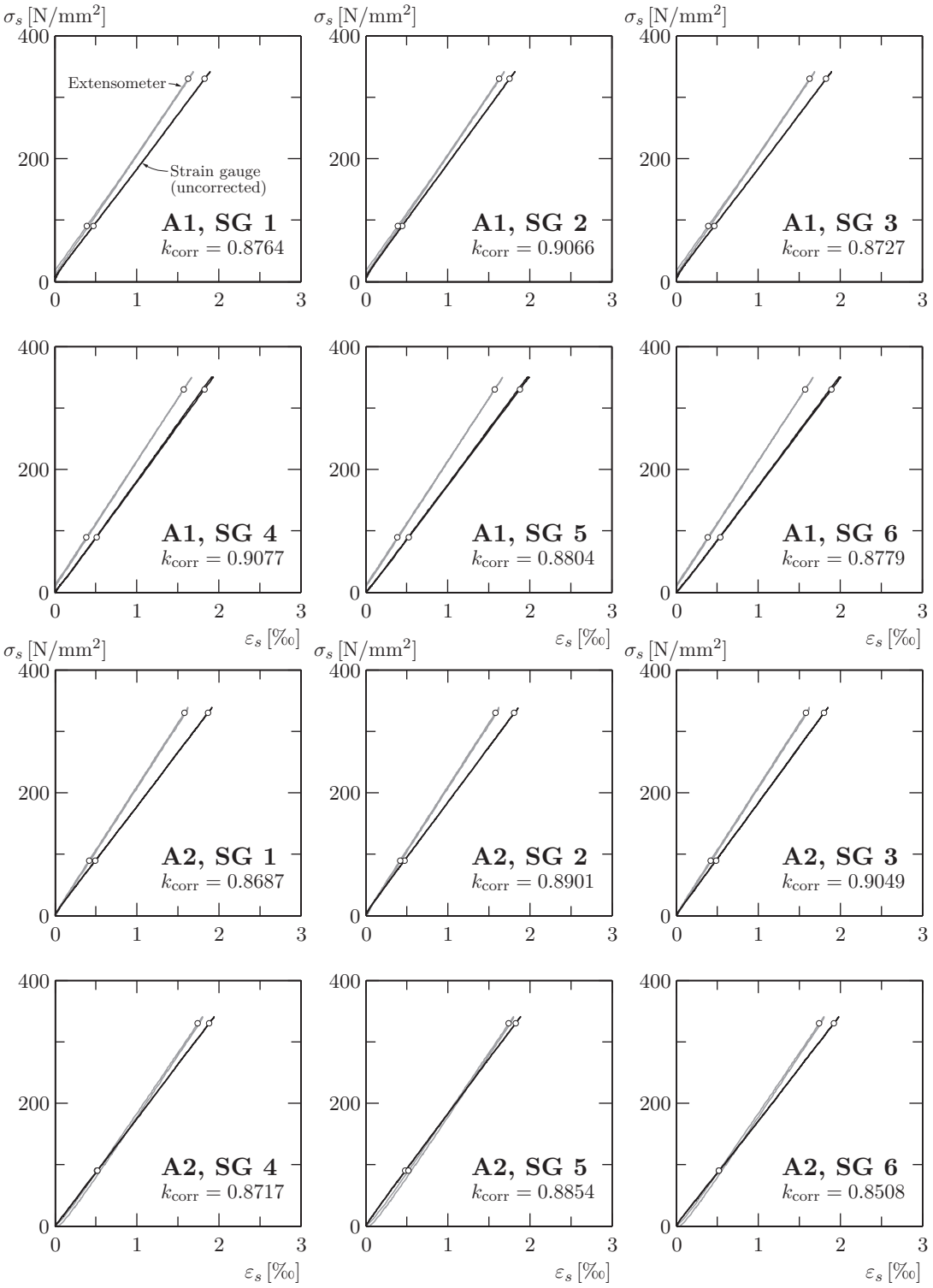


Fig. A.1 – Calibration of stress-strain characteristic for strain gauges used in tests of series A. Note: uncorrected strain gauge characteristic indicated.

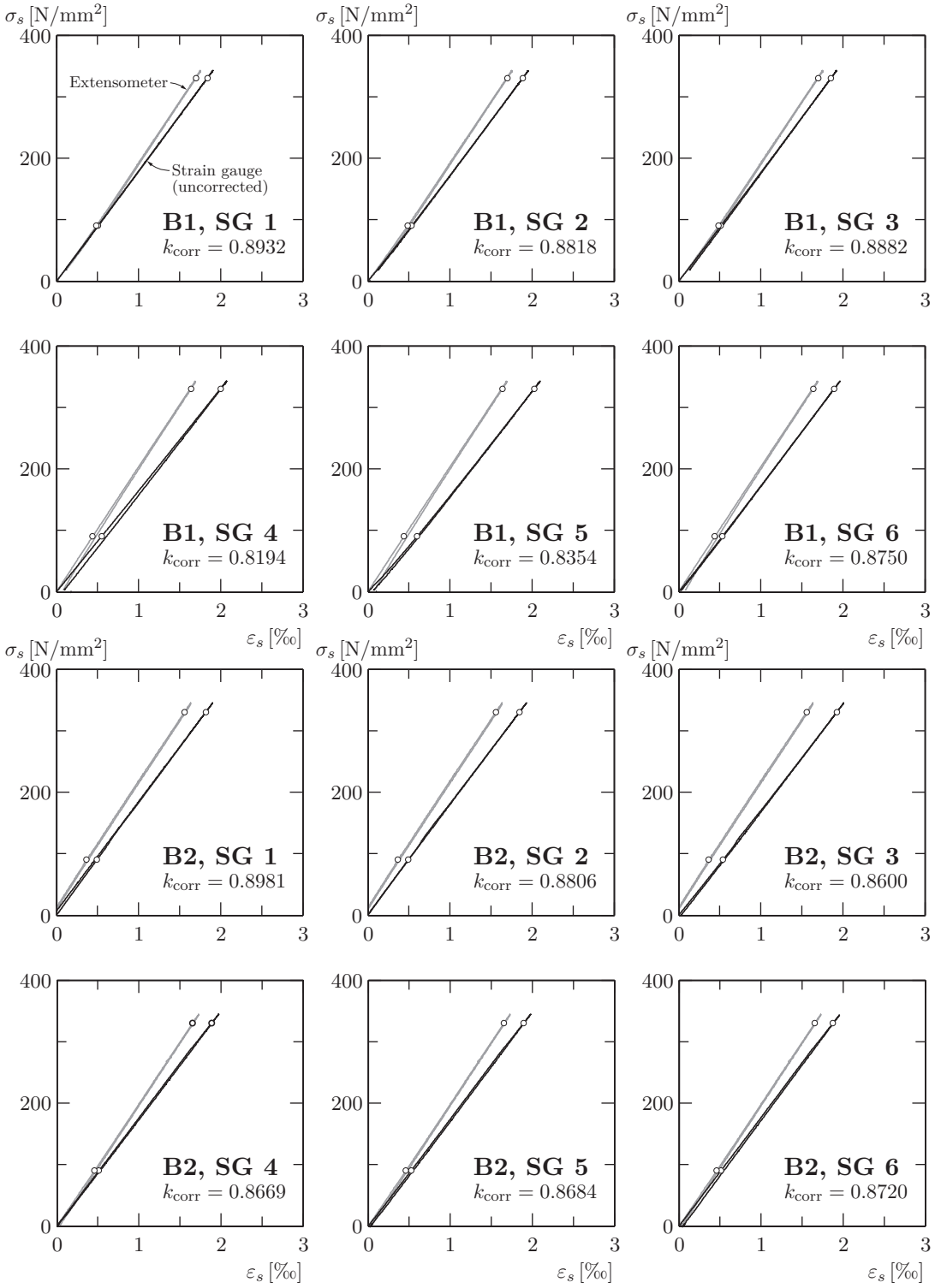


Fig. A.2 – Calibration of stress-strain characteristic for strain gauges used in tests of series B. Note: uncorrected strain gauge characteristic indicated.

Measurements	Base length [mm]	Position	Direction	No. of measurements
10...30	141	Soffit	Longitudinal	21
31...51				21
100...105				6
120...140				21
141...161				21
60...79	141	Soffit	Transverse	20
80...94				15
110...119				10
170...184				15
185...204				20
850...861	200	Top	Diagonal	12
870...889				20
890...899				10
900...919				20
920...931				12
210...223	141	Front	Horizontal	14
240...241				2
260...273				14
230...236	141	Front	Vertical	7
250...251				2
280...286				7
750...763	200	Front	Diagonal	14
770...771				2
780...793				14
290...303	141	Back	Horizontal	14
320...321				2
340...353				14
310...316	141	Back	Vertical	7
330...331				2
360...366				7
800...813	200	Back	Diagonal	14
820...821				2
830...843				14
370...381	141	Top	Longitudinal	12
400...411				12
430...435				6
450...461				12
480...491				12
390...398	141	Top	Transverse	9
420...429				10
440...449				10
470...479				10
500...508				9
600...627	200	Soffit	Diagonal	28
628...641				14
642...669				28
670...679				10
680...707				28
708...721				14
722...749				28
940...948	1530	Soffit, deflection	Vertical	9
950...952				3
960...968				9

Tab. A.1 – Overview of deformeter measurements.

Appendix B

Summary of test results

A summary of selected experimental results from the tests on orthogonally reinforced concrete slabs presented in Chapter 6 is provided. The values of continuous measurements made during the initial loading are listed at each load step immediately after ending the loading as well as after completing all manual measurements, see Tables B.1 and B.3 for tests A1 and A2 and Tables B.5 and B.7 for tests B1 and B2, respectively. Since the actuator stroke was kept constant during taking measurements, a slight decrease in applied load is observed.

A summary of continuous measurements recorded during phases of quasi-static loading at maximum and minimum load is provided in Tables B.2 and B.4 for tests A1 and A2 and in Tables B.6 and B.8 for tests B1 and B2, respectively.

Load Step [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level[a]
1	19.45	15.17	0.43	0.45	-0.02	0.02	0.028	0.032	0.029	0.033	0.040	0.031	EOL
	18.52	14.58	0.43	0.45	-0.02	0.02	0.029	0.033	0.030	0.034	0.045	0.031	EOM
2	35.80	25.63	1.10	1.16	-0.05	0.09	0.217	0.163	0.092	0.222	0.208	0.056	EOL
	33.03	23.86	1.14	1.22	-0.05	0.10	0.256	0.165	0.090	0.266	0.231	0.053	EOM
3	52.18	36.12	3.32	3.48	-0.14	0.35	0.779	0.621	0.429	0.752	0.828	0.466	EOL
	47.86	33.35	3.40	3.58	-0.14	0.36	0.779	0.647	0.477	0.766	0.832	0.505	EOM
4, F_{max}	68.56	46.60	5.87	6.08	-0.21	0.67	1.118	1.059	0.834	1.172	1.342	0.894	EOL
	63.03	43.06	5.99	6.21	-0.21	0.68	1.127	1.061	0.903	1.194	1.310	0.948	EOM

[a] EOL End of Loading
EOM End of Measurement

Tab. B.1 – Continuous measurements of test A1 during initial loading.

N [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level
0	63.03	43.06	5.99	6.21	-0.21	0.68	1.127	1.061	0.903	1.194	1.310	0.948	F_{max}
1	14.48	11.99	2.44	2.57	-0.08	0.30	0.463	0.424	0.436	0.515	0.493	0.437	F_{min}
10	65.01	44.33	6.49	6.73	-0.22	0.76	1.232	1.149	1.011	1.295	1.401	1.056	F_{max}
10	15.04	12.35	2.69	2.81	-0.09	0.32	0.502	0.460	0.477	0.554	0.525	0.484	F_{min}
124	66.41	45.22	7.15	7.42	-0.24	0.83	1.336	1.261	1.136	1.413	1.485	1.189	F_{max}
124	14.91	12.27	2.84	3.05	-0.09	0.35	0.530	0.485	0.513	0.576	0.541	0.522	F_{min}
1′124	66.16	45.07	7.39	7.71	-0.25	0.86	1.353	1.281	1.165	1.443	1.467	1.224	F_{max}
1′124	14.72	12.14	2.97	3.29	-0.10	0.36	0.542	0.498	0.533	0.591	0.545	0.544	F_{min}
11′124	67.98	46.23	7.58	7.99	-0.25	0.89	1.387	1.312	1.194	1.473	1.440	1.260	F_{max}
11′124	14.74	12.16	3.24	3.51	-0.10	-	0.562	0.512	0.549	0.607	0.568	0.564	F_{min}
111′133	65.98	44.95	7.98	8.35	-0.26	-	1.399	1.328	1.240	1.508	1.091	1.304	F_{max}
111′133	15.34	12.54	3.52	3.84	-0.12	-	0.591	0.541	0.581	0.643	0.611	0.598	F_{min}
311′133	66.52	45.30	8.23	8.62	-0.27	-	1.404	1.338	1.257	1.518	1.054	1.318	F_{max}
311′133	16.61	13.35	3.96	4.32	-0.13	-	0.630	0.582	0.631	0.690	0.656	0.643	F_{min}
511′133	67.23	45.75	8.42	8.82	-0.28	-	1.412	1.358	1.280	1.544	1.009	1.342	F_{max}
511′133	16.31	13.16	4.08	4.47	-0.14	-	0.634	0.594	0.644	0.704	0.641	0.658	F_{min}
711′133	66.94	45.57	8.58	9.01	-0.29	-	1.407	1.363	1.291	1.552	0.963	1.352	F_{max}
711′133	16.28	13.14	4.25	4.68	-0.15	-	0.641	0.608	0.656	0.719	0.635	0.672	F_{min}
906′347	66.50	45.28	8.94	9.33	-0.29	-	0.954	1.352	1.331	1.611	0.924	1.420	F_{max}
906′347	16.03	12.98	4.43	4.82	-0.15	-	0.450	0.586	0.663	0.730	0.631	0.681	F_{min}
961′995	66.96	45.58	9.74	9.93	-0.32	-	0.942	1.460	1.425	1.631	0.989	1.370	F_{max}
961′995	16.48	13.27	4.89	5.17	-0.17	-	0.445	0.632	0.707	0.740	0.669	0.657	F_{min}
1′035′895	66.92	45.55	10.37	10.49	-0.33	-	0.942	1.486	1.723	1.655	0.988	1.505	F_{max}
1′035′895	16.29	13.15	5.25	5.47	-0.18	-	0.436	0.647	0.817	0.754	0.676	0.721	F_{min}
1′131′296	66.76	45.45	11.50	11.33	-0.34	-	0.926	1.505	1.929	1.656	0.965	1.420	F_{max}
1′131′296	16.48	13.27	5.85	5.92	-0.18	-	0.431	0.664	0.938	0.755	0.676	0.659	F_{min}
1′140′040	63.84	43.58	14.52	13.97	-0.33	-	0.886	1.504	2.167	1.619	0.932	1.260	F_{max}
1′140′040	16.25	13.12	8.75	8.45	-0.18	-	0.418	0.690	1.100	0.748	0.664	0.610	F_{min}

Tab. B.2 – Development of continuous measurements of test A1 at maximum and minimum load.

Load Step [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level[a]
1	19.39	15.13	0.39	0.50	-0.02	0.02	0.025	0.029	0.026	0.028	0.026	0.025	EOL
	18.56	14.60	0.39	0.50	-0.02	0.02	0.024	0.030	0.025	0.028	0.029	0.024	EOM
2	35.71	25.58	0.90	1.07	-0.06	0.08	0.059	0.163	0.058	0.070	0.167	0.050	EOL
	33.11	23.91	0.95	1.12	-0.06	0.10	0.063	0.294	0.055	0.078	0.241	0.050	EOM
3	52.16	36.10	3.56	3.78	-0.17	0.36	0.660	0.938	0.469	0.422	0.919	0.296	EOL
	47.86	33.35	3.65	3.87	-0.17	0.37	0.687	0.901	0.516	0.454	0.892	0.366	EOM
4, F_{max}	68.48	46.55	6.22	6.53	-0.27	0.63	1.096	1.273	0.953	0.911	1.230	0.811	EOL
	64.71	44.14	6.35	6.66	-0.27	0.64	1.139	1.276	1.008	0.960	1.197	0.876	EOM

[a] EOL End of Loading
EOM End of Measurement

Tab. B.3 – Continuous measurements of test A2 during initial loading.

N [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level
0	64.71	44.14	6.35	6.66	-0.27	0.64	1.139	1.276	1.008	0.960	1.197	0.876	F_{max}
1	16.03	12.98	2.84	3.04	-0.11	0.30	0.553	0.554	0.528	0.478	0.515	0.481	F_{min}
10	65.77	44.82	6.77	7.18	-0.29	0.69	1.226	1.350	1.106	1.053	1.234	1.046	F_{max}
10	15.98	12.95	2.96	3.22	-0.11	0.31	0.577	0.574	0.557	0.506	0.520	0.575	F_{min}
110	66.34	45.18	7.01	7.47	-0.29	0.72	1.277	1.395	1.158	1.105	1.266	1.174	F_{max}
110	15.86	12.87	3.00	3.36	-0.12	0.32	0.593	0.589	0.575	0.523	0.530	0.550	F_{min}
1′110	67.28	45.78	7.24	7.77	-0.30	0.75	1.321	1.439	1.204	1.155	1.292	1.126	F_{max}
1′110	15.97	12.94	3.09	3.49	-0.12	0.33	0.613	0.606	0.595	0.544	0.540	0.588	F_{min}
11′110	67.93	46.19	7.44	8.01	-0.31	0.77	1.358	1.474	1.246	1.198	1.307	1.166	F_{max}
11′110	15.62	12.72	3.16	3.58	-0.12	0.34	0.627	0.616	0.611	0.561	0.540	0.594	F_{min}
111′110	67.29	45.78	7.56	8.14	-0.32	0.78	1.383	1.487	1.277	1.228	1.298	1.204	F_{max}
111′110	15.70	12.77	3.26	3.70	-0.13	0.35	0.644	0.629	0.630	0.580	0.540	0.616	F_{min}
511′110	66.88	45.52	7.74	8.36	-0.33	0.79	1.417	1.508	1.321	1.268	1.293	1.250	F_{max}
511′110	15.99	12.96	3.41	3.86	-0.14	0.36	0.666	0.648	0.653	0.604	0.546	0.682	F_{min}
755′347	66.31	45.16	9.30	9.99	-0.38	-	1.979	1.947	2.073	1.585	1.461	1.717	F_{max}
755′347	15.88	12.88	4.06	4.51	-0.16	-	0.924	0.808	0.908	0.757	0.639	0.868	F_{min}
800′317	66.21	45.10	10.89	11.64	-0.43	-	0.701	1.584	2.145	2.539	1.914	1.836	F_{max}
800′317	15.59	12.70	4.79	5.32	-0.18	-	0.452	0.609	0.891	1.109	0.790	0.993	F_{min}
809′232	67.13	45.68	12.63	13.36	-0.48	-	0.829	1.537	2.152	2.826	1.799	1.987	F_{max}
809′232	16.14	13.05	6.02	6.54	-0.22	-	0.546	0.588	0.885	1.221	0.756	1.059	F_{min}
825′514	66.28	45.14	15.24	15.92	-0.56	-	0.883	1.489	2.332	2.857	1.700	2.025	F_{max}
825′514	15.77	12.81	8.45	8.88	-0.31	-	0.584	0.572	2.379	1.253	0.735	1.084	F_{min}

Tab. B.4 – Development of continuous measurements of test A2 at maximum and minimum load.

Load Step [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level[a]
1	19.42	15.15	0.38	0.48	-0.02	0.02	0.009	0.012	0.010	0.008	0.014	0.008	EOL
	18.12	14.32	0.38	0.49	-0.02	0.02	0.010	0.013	0.011	0.009	0.015	0.010	EOM
2	35.78	25.62	3.04	3.22	-0.14	0.26	0.337	0.840	0.057	0.027	0.666	0.052	EOL
	32.23	23.35	3.12	3.29	-0.14	0.29	0.383	0.838	0.064	0.028	0.657	0.053	EOM
3	52.18	36.12	9.92	10.09	-0.34	0.86	1.126	1.423	0.691	0.963	1.282	1.425	EOL
	47.67	33.23	10.00	10.16	-0.34	0.87	1.130	1.299	0.699	0.940	1.254	1.418	EOM
4, F_{max}	68.54	46.59	15.66	15.83	-0.50	1.40	1.586	1.586	1.004	1.345	1.920	2.074	EOL
	63.92	43.63	15.77	15.93	-0.50	1.41	1.598	1.556	1.022	1.324	1.864	2.025	EOM

[a] EOL End of Loading
EOM End of Measurement

Tab. B.5 – Continuous measurements of test B1 during initial loading.

N [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level
0	63.92	43.63	15.77	15.93	-0.50	1.41	1.598	1.556	1.022	1.324	1.864	2.025	F_{max}
1	16.65	13.38	8.12	8.01	-0.24	0.72	0.824	0.819	0.555	0.630	0.813	0.926	F_{min}
10	66.20	45.09	17.68	17.84	-0.54	1.52	1.764	1.614	1.126	1.438	1.953	2.122	F_{max}
10	16.93	13.56	8.81	8.73	-0.24	0.70	0.845	0.829	0.555	0.650	0.815	0.931	F_{min}
110	66.37	45.20	18.47	19.00	-0.56	1.57	1.844	1.654	1.182	1.475	1.981	2.161	F_{max}
110	15.17	12.43	8.23	8.43	-0.22	0.64	0.775	0.771	0.515	0.600	0.760	0.854	F_{min}
1′110	67.08	45.65	19.09	19.62	-0.58	1.61	1.892	1.667	1.224	1.517	2.015	2.223	F_{max}
1′110	15.52	12.66	8.59	8.85	-	0.65	0.807	0.790	0.543	0.628	0.788	0.881	F_{min}
11′110	66.95	45.57	19.41	20.16	-0.59	1.64	1.924	1.674	1.237	1.545	2.025	2.249	F_{max}
11′110	15.60	12.71	8.77	9.24	-0.25	0.68	0.826	0.800	0.552	0.646	0.804	0.889	F_{min}
60′278	67.06	45.64	20.12	21.00	-0.61	1.69	1.973	1.686	1.245	1.574	2.041	2.280	F_{max}
60′278	15.12	12.40	8.98	9.60	-0.26	0.69	0.835	0.803	0.553	0.655	0.807	0.885	F_{min}
110′000	67.63	46.00	20.60	21.37	-0.62	-	2.012	1.683	1.218	1.639	2.075	2.316	F_{max}
110′000	15.98	12.95	9.49	9.98	-0.27	-	0.863	0.822	0.557	0.714	0.847	0.914	F_{min}
269′807	67.15	45.70	24.42	24.97	-0.71	-	2.263	1.675	-	1.889	2.593	2.242	F_{max}
269′807	15.88	12.88	11.36	11.94	-0.32	-	0.975	0.863	0.639	0.828	1.030	1.983	F_{min}
271′345	66.46	45.26	26.78	28.69	-0.89	-	2.558	2.140	-	2.023	2.816	2.542	F_{max}
271′345	15.76	12.81	13.55	14.26	-0.42	-	1.234	1.878	0.758	0.814	1.040	2.164	F_{min}

Tab. B.6 – Development of continuous measurements of test B1 at maximum and minimum load.

Load Step [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level[a]
1	19.38	15.12	0.36	0.59	-0.02	0.02	0.011	0.009	0.011	0.010	0.011	0.013	EOL
	18.33	14.45	0.35	0.58	-0.02	0.02	0.011	0.009	0.011	0.010	0.010	0.013	EOM
2	35.75	25.60	1.96	2.42	-0.11	0.22	0.123	0.034	0.308	0.074	0.026	0.335	EOL
	31.79	23.07	2.06	2.52	-0.11	0.23	0.174	0.036	0.329	0.108	0.033	0.349	EOM
3	52.17	36.11	10.17	10.59	-0.34	1.09	1.136	0.812	0.905	1.149	1.269	1.021	EOL
	48.06	33.48	10.23	10.65	-0.33	1.09	1.111	0.787	0.905	1.194	1.237	1.017	EOM
4, F_{max}	68.56	46.60	16.01	16.44	-0.47	1.58	1.612	1.121	1.213	1.846	1.787	1.373	EOL
	63.78	43.54	16.08	16.51	-0.47	1.58	1.579	1.049	1.221	1.862	1.727	1.379	EOM

[a] EOL End of Loading
EOM End of Measurement

Tab. B.7 – Continuous measurements of test B2 during initial loading.

N [-]	F [kN]	M_n [kNm]	w_1 [mm]	w_2 [mm]	u_1 [mm]	u_2 [mm]	ε_{SG1} [‰]	ε_{SG2} [‰]	ε_{SG3} [‰]	ε_{SG4} [‰]	ε_{SG5} [‰]	ε_{SG6} [‰]	Level
0	63.78	43.54	16.08	16.51	-0.47	1.58	1.579	1.049	1.221	1.862	1.727	1.379	F_{max}
1	15.02	12.33	7.63	7.41	-0.19	0.71	0.677	0.555	0.568	0.837	0.710	0.603	F_{min}
10	65.99	44.95	17.86	17.93	-0.50	1.73	1.732	1.106	1.342	2.044	1.800	1.475	F_{max}
10	15.39	12.57	8.36	8.04	-0.20	0.78	0.736	0.588	0.624	0.901	0.745	0.656	F_{min}
110	66.32	45.17	18.67	18.81	-0.52	1.81	1.785	1.131	1.381	2.118	1.844	1.523	F_{max}
110	15.80	12.84	8.78	8.52	-0.21	0.81	0.757	0.598	0.642	0.930	0.760	0.680	F_{min}
1′110	66.64	45.37	19.25	19.47	-0.54	1.88	1.857	1.088	1.421	2.184	1.907	1.569	F_{max}
1′110	15.51	12.65	8.97	8.75	-0.22	0.83	0.779	0.572	0.653	0.938	0.773	0.690	F_{min}
11′110	67.75	46.08	19.93	20.24	-0.56	1.97	1.963	1.062	1.474	2.278	2.002	1.635	F_{max}
11′110	15.43	12.60	9.13	8.95	-0.23	0.86	0.819	0.565	0.681	0.962	0.809	0.717	F_{min}
110′000	67.53	45.94	20.91	21.41	-0.59	2.07	2.012	0.993	1.557	2.364	2.072	1.709	F_{max}
110′000	15.81	12.84	9.74	9.69	-0.25	0.92	0.846	0.559	0.722	0.994	0.846	0.744	F_{min}
217′555	67.80	46.11	22.86	23.30	-0.63	-	2.482	1.006	1.585	2.726	3.200	1.804	F_{max}
217′555	15.50	12.64	10.59	-	-0.27	-	1.025	0.563	0.733	1.171	1.659	0.768	F_{min}
219′079	67.06	45.64	24.43	24.77	-0.65	-	2.730	0.985	1.584	-	4.369	1.796	F_{max}
219′079	15.21	12.46	11.25	11.23	-0.27	-	1.088	0.569	0.732	1.149	2.645	0.756	F_{min}
219′728	66.42	45.23	27.10	27.87	-0.68	-	-	0.961	1.594	-	-	1.811	F_{max}
219′728	15.56	12.68	14.29	14.43	-0.31	-	-	0.559	0.756	-	-	0.769	F_{min}
219′741	53.23	36.79	29.20	35.42	-0.66	-	1.660	0.924	1.431	1.783	1.478	1.597	F_{max}
219′741	15.54	12.67	25.07	24.67	-0.37	-	1.532	0.597	0.773	1.487	1.418	0.770	F_{min}

Tab. B.8 – Development of continuous measurements of test B2 at maximum and minimum load.

Appendix C

Preliminary tests

Preliminary tests were performed with reinforced concrete tension tie elements to investigate the fatigue performance of reinforcing steel bars embedded in concrete. Of particular importance for the subsequent tests on slabs was to establish a reliable means of measuring reinforcing steel strains directly under cyclic loading. For this purpose, foil strain gauges of type FLA-5-11-3L (gauge length 5 mm) from Tokyo Sokki Kenkyujo Co. were applied directly to prepared surfaces on the bar before casting the concrete. Two series of 10 specimens were tested at the structural laboratory of ETH Zurich, Hönggerberg; the first series of tests were conducted by the author and the second by *Grube and Lötscher* during their project work in the autumn semester 2015 [30]. Specimens were subjected to constant force-amplitude cyclic loading in a servo-hydraulic testing machine. The tests parameters are summarised in Table C.1.

The tension tie specimens comprised a concrete cylinder with a diameter of 150 mm, a height of 800 mm, and were concentrically reinforced with a single ⌀16 mm profiled steel bar. The specimen geometry and test setup is illustrated in Fig. C.1. Mechanical couplers were used to attach ⌀20 mm reinforcing bars at both ends of the embedded bar in order to prevent premature failure occurring at the grips of the testing machine. For this purpose, threaded couplers of the type BARTEC® DYN, see product documentation in [17], proved to be a suitable solution. No fatigue failures occurred at the connections or in the larger diameter bars. All specimens were constructed with a self-compacting concrete containing aggregate with a maximum diameter of 16 mm. Specimens were cast

Series			Series S1	Series S2
Nominal bar diameter	$\varnothing_{\text{nom}}$	[mm]	16	16
Maximum tensile force	$F_{\max}$	[kN]	83.00	83.00
Minimum tensile force	$F_{\min}$	[kN]	16.60	35.75
Stress range in reinforcement[a]	$\Delta\sigma_s$	[N/mm^2]	330	235

[a] based on nominal cross-sectional area of reinforcing bar.

Tab. C.1 – Test parameters for preliminary tests.

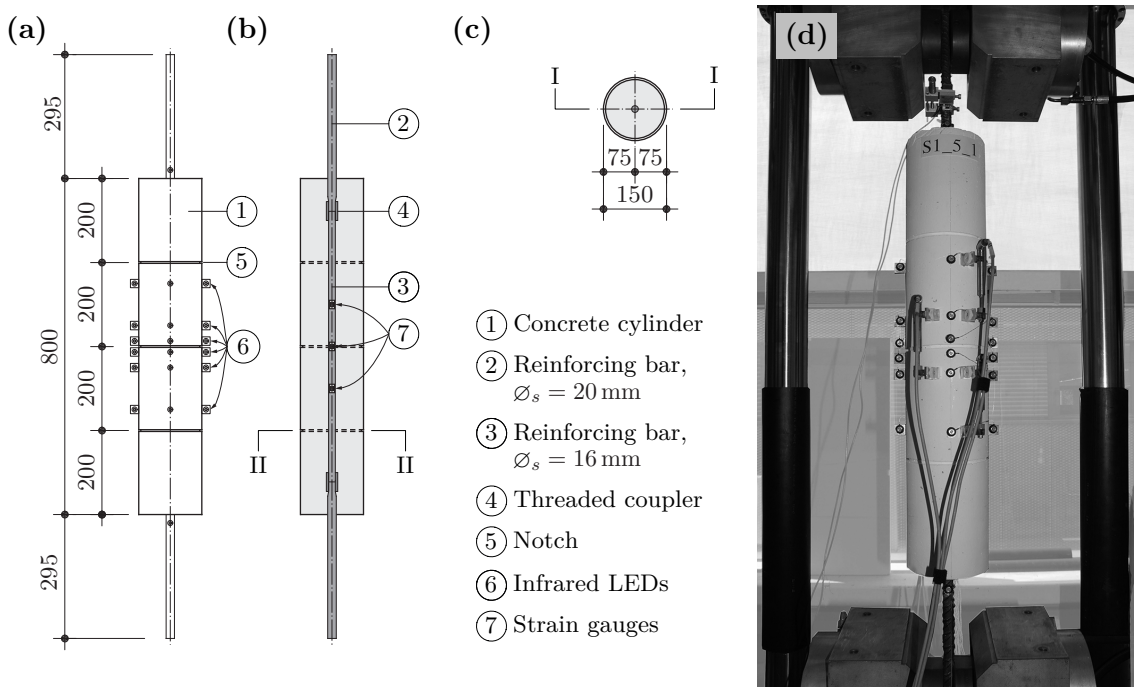


Fig. C.1 – Reinforced concrete tension tie: (a) front view; (b) section I-I; (c) section II-II; (d) setup. Dimensions in mm.

in the vertical position. After removal of the formwork three approximately 5 mm wide notches were sawed into the concrete surface at quarter-points to a depth of approximately 7 mm. These were introduced in an attempt to localise cracking of the concrete at specific locations. Profiled reinforcing steel bars of the type BSW Tempcore were used in all specimens. The mechanical properties of the reinforcing steel were determined in a servo-hydraulic testing machine. The static yield and ultimate strengths amounted on average to 496.6 N/mm^2 and 621.2 N/mm^2, respectively

Results of both series of fatigue tests are indicated in the S-N diagram in Fig. C.2; fatigue test results reported in literature with bars tested in air are indicated for comparison. Table C.2 provides a summary of the number of load cycles sustained to failure. The majority of specimens indicated concrete cracks which did not extend around the full circumference of the cylinder – even shortly before fatigue failure. Furthermore, crack widths were typically greater on one side of the specimen. In some cases, the cracks on the opposing side of the specimen remained practically closed at both maximum and minimum load level. All fatigue fractures occurred at cracks that formed either at the top or bottom quarter-point of the specimen. These observations indicate an eccentricity of the applied load, possibly introduced due to imperfections in the specimen geometry and/or due to the rotationally fixed boundary conditions at the grips of the testing machine.

The bars embedded in concrete failed on average at less load cycles than those tested in air. One specimen in each series failed at fewer load cycles than indicated by the fatigue strength curve based on the design fatigue strength according to SIA 262(2013) [87]. The low fatigue strengths observed in all tests also signify that an additional loading occurred.

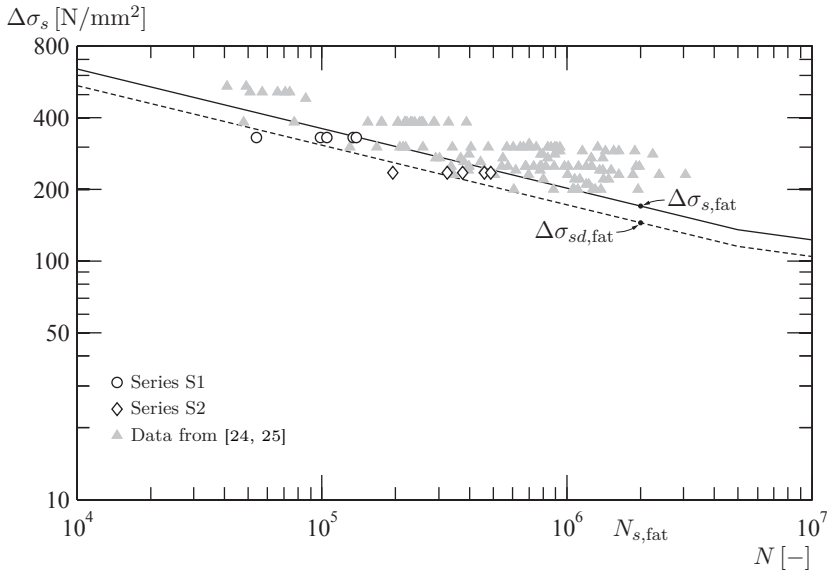


Fig. C.2 – Fatigue failure of reinforcing steel bars embedded in concrete.

Eccentricity of the applied loading would result in an additional, localised bending of the bar at the cracks; thus, increasing the stress range on the bar outer surface.

In general, the strain gauges embedded in concrete performed well under cyclic loading and provided satisfactory readings until fatigue failure. Measurements from strain gauges in the region of concrete cracks showed virtually no hysteresis or translation of the stress-strain relationship with increasing load cycles. On the other hand, strain gauges between cracks indicated a progressive increase in strain at maximum and minimum load. This translation of the stress-strain relationship indicates a loss of tension stiffening resulting from a deterioration of bond under repeated loading. A further important observation was that no fatigue fractures occurred at the location where the bar was locally milled to produce an even, flat surface for application of strain gauges. Hence, the small local reduction in cross-sectional area does not influence the fatigue strength of the bars.

Series S1		Series S2	
Test	N_u	Test	N_u
S131	98′710	S231	325′858
S132	133′960	S232	194′504
S133	105′092	S233	460′809
S134	54′088	S234	488′649
S151	138′111	S251	374′462

Tab. C.2 – Overview of reinforcing bar failures in preliminary tests.

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